

CALCULUS

Michael Spivak



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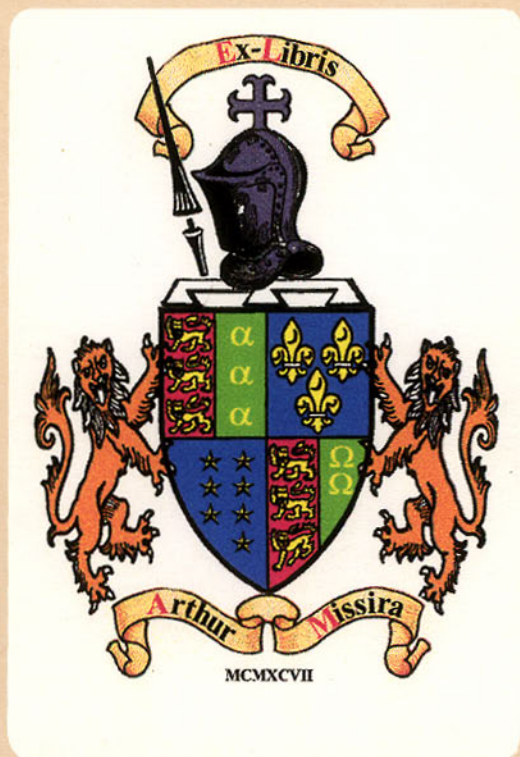
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CALCULUS

Michael Spivak

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WORLD STUDENT SERIES EDITION

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Dedicated to the Memory of Y. P.

PREFACE

*I hold every man a debtor
to his profession,
from the which as men of course
doe seeke to receive countenance and profit,
so ought they of duty to endeavour
themselves by way of amends,
to be a help and
ornament thereunto.*

FRANCIS BACON

Every aspect of this book was influenced by the desire to present calculus not merely as a prelude to but as the first real encounter with mathematics. Since the foundations of analysis provided the arena in which modern modes of mathematical thinking developed, calculus ought to be the place in which to expect, rather than avoid, the strengthening of insight with logic. In addition to developing the students' intuition about the beautiful concepts of analysis, it is surely equally important to persuade them that precision and rigor are neither deterrents to intuition, nor ends in themselves, but the natural medium in which to formulate and think about mathematical questions.

This goal implies a view of mathematics which, in a sense, the entire book attempts to defend. No matter how well particular topics may be developed, the goals of this book will be realized only if it succeeds as a whole. For this reason, it would be of little value merely to list the topics covered, or to mention pedagogical practices and other innovations. Even the cursory glance customarily bestowed on new calculus texts will probably tell more than any such extended advertisement, and the teacher with strong feelings about particular aspects of calculus will know just where to look to see if this book fulfills his requirements.

A few features do require explicit comment, however. Of the twenty-nine chapters in the book, two (starred) chapters are optional, and the three chapters comprising Part V have been included only for the benefit of those students who might want to examine on their own a construction of the real numbers. Moreover, the appendices to Chapters 3 and 11 also contain optional material.

The order of the remaining chapters is intentionally quite inflexible, since the purpose of the book is to present calculus as the evolution of one idea, not as a collection of "topics." Since the most exciting concepts of calculus do not appear until Part III, I should point out that Parts I and II will probably require less time than their length suggests—although the entire book covers a one-year course, the chapters are not meant to be covered at any uniform rate. A rather natural dividing point does occur between Parts II and III, so it is possible to reach differentiation and integration even more quickly by treating Part II very briefly, perhaps returning later for a more detailed treatment. This arrangement corresponds to the traditional organization of most calculus courses, but I feel that it will only diminish the value of the book for students who have seen a small amount of calculus previously, and for bright students with a reasonable background.

The problems have been designed with this particular audience in mind. They range from straightforward, but not overly simple, exercises which develop basic techniques and test understanding of concepts, to problems of considerable difficulty and, I hope, of comparable interest. There are about 625 problems in all. Those which emphasize manipulations usually contain many examples, numbered with small Roman numerals, while small letters

are used to label interrelated parts in other problems. Some indication of relative difficulty is provided by a system of starring and double starring, but there are so many criteria for judging difficulty, and so many hints have been provided, especially for harder problems, that this guide is not completely reliable. Many problems are so difficult, especially if the hints are not consulted, that the best of students will probably have to attempt only those which especially interest them; from the less difficult problems it should be easy to select a portion which will keep a good class busy, but not frustrated. The answer section contains solutions to about half the examples from an assortment of problems that should provide a good test of technical competence. A separate answer book contains the solutions of the other parts of these problems, and of all the other problems as well. Finally, there is a Suggested Reading list, to which the problems often refer, and an index of symbols.

I am grateful for the opportunity to mention the many people to whom I owe my thanks. Mrs. Jane Bjorkgren performed prodigious feats of typing that compensated for my fitful production of the manuscript. Mr. Richard Serkey helped collect the material which provides historical sidelights in the problems, and Mr. Richard Weiss supplied the answers appearing in the back of the book. I am especially grateful to my friends Michael Freeman, Jay Goldman, Anthony Phillips, and Robert Wells for the care with which they read, and the relentlessness with which they criticized, a preliminary version of the book. Needless to say, they are not responsible for the deficiencies which remain, especially since I sometimes rejected suggestions which would have made the book appear suitable for a larger group of students. I must express my admiration for the editors and staff of W. A. Benjamin, Inc., who were always eager to increase the appeal of the book, while recognizing the audience for which it was intended.

The inadequacies which preliminary editions always involve were gallantly endured by a rugged group of freshmen in the honors mathematics course at Brandeis University during the academic year 1965–1966. About half of this course was devoted to algebra and topology, while the other half covered calculus, with the preliminary edition as the text. It is almost obligatory in such circumstances to report that the preliminary version was a gratifying success. This is always safe—after all, the class is unlikely to rise up in a body and protest publically—but the students themselves, it seems to me, deserve the right to assign credit for the thoroughness with which they absorbed an impressive amount of mathematics. I am content to hope that some other students will be able to use the book to such good purpose, and with such enthusiasm.

Waltham, Massachusetts
February 1967

MICHAEL SPIVAK

PREFACE TO THE WORLD STUDENT SERIES EDITION

In American universities, “calculus” courses are so varied in content and viewpoint that only tradition dictates that the same title be given to all. Some of these courses cover only the most rudimentary techniques, sedulously shunning all theory. At the other extreme, so-called “honors” courses concentrate on the foundations, and emphasize the theoretical importance of the basic processes of calculus, which have often already been seen by the students in their high school classes. This book was written for just such courses. In the United States, they are so rare that I felt constrained to begin my original preface with a lengthy defense, or apology for the fact that it was really written for the very type of course which in Great Britain and Europe is quite standard, and usually called “An introduction to Real analysis”. From that long prologue I would like to salvage only one thought, and once again express my feeling that the foundations of analysis provide one of the most beautiful instances of the welding together of mathematical intuition and mathematical rigor, and that ignoring either one only weakens, rather than strengthens, the other.

A few comments about the contents and arrangement should be made, especially for the student who is learning the material on his own. Of the twenty-nine chapters in the book, two (starred) chapters are “optional”, and the three chapters comprising Part V might be thought of in the same way. Moreover, the appendices to Chapters 3 and 11 also contain “optional” material. Aside from this, the order of the remaining chapters is intentionally quite inflexible, since the purpose of the book is to present calculus as the evolution of one idea, not as a collection of “topics”. Since the most exciting concepts of calculus do not appear until Part III, I should point out that Parts I and II will probably require less time than their length suggests, and although the entire book covers a one-year course, the chapters are not meant to be covered at any uniform rate. A rather natural dividing point does occur between Parts II and III, so it is possible to reach differentiation and integration even more quickly by *reading* Part II very briefly, perhaps returning later for more detailed *study*. But I personally feel that the route set forth here, though it requires perseverance at the beginning, will be the most rewarding.

The problems, without which any mathematics text is incomplete, range from straightforward (but not overly simple) exercises which develop basic techniques and test understanding of concepts, to problems of considerable difficulty and, I hope, of comparable interest. There are about 625 problems in all. Those which *emphasize* manipulations usually contain many examples, numbered with small Roman numerals, while small letters are used to label interrelated parts in other problems. Some indication of relative difficulty is provided by a system of starring and double starring, but there are so many criteria for judging difficulty, and so many hints have been provided, especially for harder problems, that this guide is not completely reliable. Many problems are so difficult, especially if the hints are not consulted, that the best of students will probably have to attempt only those which especially interest them. From the less difficult problems it should be easy to select a portion which will be

challenging, but not frustrating. The Answer Section contains solutions to about half the examples from an assortment of problems that should provide a good test of technical competence. A separate answer book, *Supplement to Calculus*, contains the solutions of the other parts of these problems, and of all the other problems as well. Finally, there is a Suggested Reading list, to which the problems often refer, and a Glossary of Symbols.

Once again I will take the opportunity to mention the many people to whom I owe my thanks. Mrs. Jane Bjorkgren performed prodigious feats of typing that compensated for my fitful production of the manuscript. Mr. Richard Serkey helped collect the material which provided historical sidelights in the problems, and Mr. Richard Weiss supplied the answers appearing in the back of the book. I am especially grateful to my friends Michael Freeman, Jay Goldman, Anthony Phillips, and Robert Wells for the care with which they read, and the relentlessness with which they criticized, a preliminary version of the book.

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Berkeley, California

MICHAEL SPIVAK

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PART **1**
PROLOGUE

*To be conscious that
you are ignorant is a great step
to knowledge.*

BENJAMIN DISRAELI

The title of this chapter expresses in a few words the mathematical knowledge required to read this book. In fact, this short chapter is simply an explanation of what is meant by the “basic properties of numbers,” all of which—addition and multiplication, subtraction and division, solutions of equations and inequalities, factoring and other algebraic manipulations—are already familiar to us. Nevertheless, this chapter is not a review. Despite the familiarity of the subject, the survey we are about to undertake will probably seem quite novel; it does not aim to present an extended review of old material, but to condense this knowledge into a few simple and obvious properties of numbers. Some may even seem too obvious to mention, but a surprising number of diverse and important facts turn out to be consequences of the ones we shall emphasize.

Of the twelve properties which we shall study in this chapter, the first nine are concerned with the fundamental operations of addition and multiplication. For the moment we consider only addition: this operation is performed on a pair of numbers—the sum $a + b$ exists for any two given numbers a and b (which may possibly be the same number, of course). It might seem reasonable to regard addition as an operation which can be performed on several numbers at once, and consider the sum $a_1 + \cdots + a_n$ of n numbers $a_1, \dots, a_n$ as a basic concept. It is more convenient, however, to consider addition of pairs of numbers only, and to define other sums in terms of sums of this type. For the sum of three numbers a , b , and c , this may be done in two different ways. One can first add b and c , obtaining $b + c$, and then add a to this number, obtaining $a + (b + c)$; or one can first add a and b , and then add the sum $a + b$ to c , obtaining $(a + b) + c$. Of course, the two compound sums obtained are equal, and this fact is the very first property we shall list:

(P1) If a , b , and c are any numbers, then

$$a + (b + c) = (a + b) + c.$$

The statement of this property clearly renders a separate concept of the sum of three numbers superfluous; we simply agree that $a + b + c$ denotes the number $a + (b + c) = (a + b) + c$. Addition of four numbers requires similar, though slightly more involved, considerations. The symbol $a + b + c + d$ is defined to mean

- (1) $((a + b) + c) + d$,
- or (2) $(a + (b + c)) + d$,
- or (3) $a + ((b + c) + d)$,
- or (4) $a + (b + (c + d))$,
- or (5) $(a + b) + (c + d)$.

This definition is unambiguous since these numbers are all equal. Fortunately, *this* fact need not be listed separately, since it follows from the property P1 already listed. For example, we know from P1 that

$$(a + b) + c = a + (b + c),$$

and it follows immediately that (1) and (2) are equal. The equality of (2) and (3) is a direct consequence of P1, although this may not be apparent at first sight (one must let $b + c$ play the role of b in P1, and d the role of c). The equalities (3) = (4) = (5) are also simple to prove.

It is probably obvious that an appeal to P1 will also suffice to prove the equality of the 14 possible ways of summing five numbers, but it may not be so clear how we can reasonably arrange a proof that this is so without actually listing these 14 sums. Such a procedure is feasible, but would soon cease to be if we considered collections of six, seven, or more numbers; it would be totally inadequate to prove the equality of all possible sums of an arbitrary finite collection of numbers $a_1, \dots, a_n$. This fact may be taken for granted, but for those who would like to worry about the proof (and it is worth worrying about once) a reasonable approach is outlined in Problem 23. Henceforth, we shall usually make a tacit appeal to the results of this problem and write sums $a_1 + \dots + a_n$ with a blithe disregard for the arrangement of parentheses.

The number 0 has one property so important that we list it next:

(P2) If a is any number, then

$$a + 0 = 0 + a = a.$$

An important role is also played by 0 in the third property of our list:

(P3) For every number a , there is a number $-a$ such that

$$a + (-a) = (-a) + a = 0.$$

Property P2 ought to represent a distinguishing characteristic of the number 0, and it is comforting to note that we are already in a position to prove this. Indeed, if a number x satisfies

$$a + x = a$$

for any one number a , then $x = 0$ (and consequently this equation also holds for all numbers a). The proof of this assertion involves nothing more than subtracting a from both sides of the equation, in other words, adding $-a$ to both sides; as the following detailed proof shows, all three properties P1–P3 must be used to justify this operation.

$$\begin{array}{ll} \text{If} & a + x = a, \\ \text{then} & (-a) + (a + x) = (-a) + a = 0; \\ \text{hence} & ((-a) + a) + x = 0; \\ \text{hence} & 0 + x = 0; \\ \text{hence} & x = 0. \end{array}$$

As we have just hinted, it is convenient to regard subtraction as an operation derived from addition: we consider $a - b$ to be an abbreviation for $a + (-b)$. It is then possible to find the solution of certain simple equations by a series of steps (each justified by P1, P2, or P3) similar to the ones just presented for the equation $a + x = a$. For example:

$$\begin{array}{ll} \text{If} & x + 3 = 5, \\ \text{then} & (x + 3) + (-3) = 5 + (-3); \\ \text{hence} & x + (3 + (-3)) = 5 - 3 = 2; \\ \text{hence} & x + 0 = 2; \\ \text{hence} & x = 2. \end{array}$$

Naturally, such elaborate solutions are of interest only until you become convinced that they can always be supplied. In practice, it is usually just a waste of time to solve an equation by indicating so explicitly the reliance on properties P1, P2, and P3 (or any of the further properties we shall list).

Only one other property of addition remains to be listed. When considering the sums of three numbers a , b , and c , only two sums were mentioned: $(a + b) + c$ and $a + (b + c)$. Actually, several other arrangements are obtained if the order of a , b , and c is changed. That these sums are all equal depends on

(P4) If a and b are any numbers, then

$$a + b = b + a.$$

The statement of P4 is meant to emphasize that although the operation of addition of pairs of numbers might conceivably depend on the order of the two numbers, in fact it does not. It is helpful to remember that not all operations are so well behaved. For example, subtraction does not have this property: usually $a - b \neq b - a$. In passing we might ask just when $a - b$ does equal $b - a$, and it is amusing to discover how powerless we are if we rely only on properties P1–P4 to justify our manipulations. Algebra of the most elementary variety shows that $a - b = b - a$ only when $a = b$. Nevertheless, it is impossible to derive this fact from properties P1–P4; it is instructive to examine the elementary algebra carefully and determine which step(s) cannot be justified by P1–P4. We will indeed be able to justify all steps in detail when a few more properties are listed. Oddly enough, however, the crucial property involves multiplication.

The basic properties of multiplication are fortunately so similar to those for addition that little comment will be needed; both the meaning and the consequences should be clear. (As in elementary algebra, the product of a and b will be denoted by $a \cdot b$, or simply ab .)

(P5) If a , b , and c are any numbers, then

$$a \cdot (b \cdot c) = (a \cdot b) \cdot c.$$

(P6) If a is any number, then

$$a \cdot 1 = 1 \cdot a = a.$$

Moreover, $1 \neq 0$.

(The assertion that $1 \neq 0$ may seem a strange fact to list, but we have to list it, because there is no way it could possibly be proved on the basis of the other properties listed—these properties would all hold if there were only one number, namely, 0.)

(P7) For every number $a \neq 0$, there is a number a^{-1} such that

$$a \cdot a^{-1} = a^{-1} \cdot a = 1.$$

(P8) If a and b are any numbers, then

$$a \cdot b = b \cdot a.$$

One detail which deserves emphasis is the appearance of the condition $a \neq 0$ in P7. This condition is quite necessary; since $0 \cdot b = 0$ for all numbers b , there is *no* number 0^{-1} satisfying $0 \cdot 0^{-1} = 1$. This restriction has an important consequence for division. Just as subtraction was defined in terms of addition, so division is defined in terms of multiplication: the symbol a/b means $a \cdot b^{-1}$. Since 0^{-1} is meaningless, $a/0$ is also meaningless—division by 0 is *always* undefined.

Property P7 has two important consequences. If $a \cdot b = a \cdot c$, it does not necessarily follow that $b = c$; for if $a = 0$, then both $a \cdot b$ and $a \cdot c$ are 0, no matter what b and c are. However, if $a \neq 0$, then $b = c$; this can be deduced from P7 as follows:

$$\begin{array}{ll} \text{If} & a \cdot b = a \cdot c \text{ and } a \neq 0, \\ \text{then} & a^{-1} \cdot (a \cdot b) = a^{-1} \cdot (a \cdot c); \\ \text{hence} & (a^{-1} \cdot a) \cdot b = (a^{-1} \cdot a) \cdot c; \\ \text{hence} & 1 \cdot b = 1 \cdot c; \\ \text{hence} & b = c. \end{array}$$

It is also a consequence of P7 that if $a \cdot b = 0$, then either $a = 0$ or $b = 0$. In fact,

$$\begin{array}{ll} \text{if} & a \cdot b = 0 \text{ and } a \neq 0, \\ \text{then} & a^{-1} \cdot (a \cdot b) = 0; \\ \text{hence} & (a^{-1} \cdot a) \cdot b = 0; \\ \text{hence} & 1 \cdot b = 0; \\ \text{hence} & b = 0. \end{array}$$

(It may happen that $a = 0$ and $b = 0$ are both true; this possibility is not excluded when we say “either $a = 0$ or $b = 0$ ”; in mathematics “or” is always used in the sense of “one or the other, or both.”)

This latter consequence of P7 is constantly used in the solution of equations. Suppose, for example, that a number x is known to satisfy

$$(x - 1)(x - 2) = 0.$$

Then it follows that either $x - 1 = 0$ or $x - 2 = 0$; hence $x = 1$ or $x = 2$.

On the basis of the eight properties listed so far it is still possible to prove

very little. Listing the next property, which combines the operations of addition and multiplication, will alter this situation drastically.

(P9) If a , b , and c are any numbers, then

$$a \cdot (b + c) = a \cdot b + a \cdot c.$$

(Notice that the equation $(b + c) \cdot a = b \cdot a + c \cdot a$ is also true, by P8.)

As an example of the usefulness of P9 we will now determine just when $a - b = b - a$:

$$\begin{array}{ll} \text{If} & a - b = b - a, \\ \text{then} & (a - b) + b = (b - a) + b = b + (b - a); \\ \text{hence} & a = b + b - a; \\ \text{hence} & a + a = (b + b - a) + a = b + b. \\ \text{Consequently} & a \cdot (1 + 1) = b \cdot (1 + 1), \\ \text{and therefore} & a = b. \end{array}$$

A second use of P9 is the justification of the assertion $a \cdot 0 = 0$ which we have already made, and even used in a proof on page 6 (can you find where?). This fact was not listed as one of the basic properties, even though no proof was offered when it was first mentioned. With P1-P8 alone a proof was not possible, since the number 0 appears only in P2 and P3, which concern addition, while the assertion in question involves multiplication. With P9 the proof is simple, though perhaps not obvious: We have

$$\begin{aligned} a \cdot 0 + a \cdot 0 &= a \cdot (0 + 0) \\ &= a \cdot 0; \end{aligned}$$

as we have already noted, this immediately implies (by adding $(a \cdot 0)$ to both sides) that $a \cdot 0 = 0$.

A series of further consequences of P9 may help explain the somewhat mysterious rule that the product of two negative numbers is positive. To begin with, we will establish the more easily acceptable assertion that $(-a) \cdot b = -(a \cdot b)$. To prove this, note that

$$\begin{aligned} (-a) \cdot b + a \cdot b &= [(-a) + a] \cdot b \\ &= 0 \cdot b \\ &= 0. \end{aligned}$$

It follows immediately (by adding $-(a \cdot b)$ to both sides) that $(-a) \cdot b = -(a \cdot b)$. Now note that

$$\begin{aligned} (-a) \cdot (-b) + [-(a \cdot b)] &= (-a) \cdot (-b) + (-a) \cdot b \\ &= (-a) \cdot [(-b) + b] \\ &= (-a) \cdot 0 \\ &= 0. \end{aligned}$$

Consequently, adding $(a \cdot b)$ to both sides, we obtain

$$(-a) \cdot (-b) = a \cdot b.$$

The fact that the product of two negative numbers is positive is thus a consequence of P1–P9. In other words, *if we want P1 to P9 to be true, the rule for the product of two negative numbers is forced upon us.*

The various consequences of P9 examined so far, although interesting and important, do not really indicate the significance of P9; after all we could have listed each of these properties separately. Actually, P9 is the justification for almost all algebraic manipulations. For example, although we have shown how to solve the equation

$$(x - 1)(x - 2) = 0,$$

we can hardly expect to be presented with an equation in this form. We are more likely to be confronted with the equation

$$x^2 - 3x + 2 = 0.$$

The “factorization” $x^2 - 3x + 2 = (x - 1)(x - 2)$ is really a triple use of P9:

$$\begin{aligned}(x - 1) \cdot (x - 2) &= x \cdot (x - 2) + (-1) \cdot (x - 2) \\ &= x \cdot x + x \cdot (-2) + (-1) \cdot x + (-1) \cdot (-2) \\ &= x^2 + x[(-2) + (-1)] + 2 \\ &= x^2 - 3x + 2.\end{aligned}$$

A final illustration of the importance of P9 is the fact that this property is actually used every time one multiplies arabic numerals. For example, the calculation

$$\begin{array}{r} 13 \\ \times 24 \\ \hline 52 \\ 26 \\ \hline 312 \end{array}$$

is a concise arrangement for the following equations:

$$\begin{aligned}13 \cdot 24 &= 13 \cdot (2 \cdot 10 + 4) \\ &= 13 \cdot 2 \cdot 10 + 13 \cdot 4 \\ &= 26 \cdot 10 + 52.\end{aligned}$$

(Note that moving 26 to the left in the above calculation is the same as writing $26 \cdot 10$.) The multiplication $13 \cdot 4 = 52$ uses P9 also:

$$\begin{aligned}13 \cdot 4 &= (1 \cdot 10 + 3) \cdot 4 \\ &= 1 \cdot 10 \cdot 4 + 3 \cdot 4 \\ &= 4 \cdot 10 + 12 \\ &= 4 \cdot 10 + 1 \cdot 10 + 2 \\ &= (4 + 1) \cdot 10 + 2 \\ &= 5 \cdot 10 + 2 \\ &= 52.\end{aligned}$$

The properties P1–P9 have descriptive names which are not essential to remember, but which are often convenient for reference. We will take this opportunity to list properties P1–P9 together and indicate the names by which they are commonly designated.

- | | | |
|------|--|---|
| (P1) | (Associative law for addition) | $a + (b + c) = (a + b) + c.$ |
| (P2) | (Existence of an additive identity) | $a + 0 = 0 + a = a.$ |
| (P3) | (Existence of additive inverses) | $a + (-a) = (-a) + a = 0.$ |
| (P4) | (Commutative law for addition) | $a + b = b + a.$ |
| (P5) | (Associative law for multiplication) | $a \cdot (b \cdot c) = (a \cdot b) \cdot c.$ |
| (P6) | (Existence of a multiplicative identity) | $a \cdot 1 = 1 \cdot a = a; \quad 1 \neq 0.$ |
| (P7) | (Existence of multiplicative inverses) | $a \cdot a^{-1} = a^{-1} \cdot a = 1, \text{ for } a \neq 0.$ |
| (P8) | (Commutative law for multiplication) | $a \cdot b = b \cdot a.$ |
| (P9) | (Distributive law) | $a \cdot (b + c) = a \cdot b + a \cdot c.$ |

The three basic properties of numbers which remain to be listed are concerned with inequalities. Although inequalities occur rarely in elementary mathematics, they play a prominent role in calculus. The two notions of inequality, $a < b$ (a is less than b) and $a > b$ (a is greater than b), are intimately related: $a < b$ means the same as $b > a$ (thus $1 < 3$ and $3 > 1$ are merely two ways of writing the same assertion). The numbers a satisfying $a > 0$ are called **positive**, while those numbers a satisfying $a < 0$ are called **negative**. While positivity can thus be defined in terms of $<$, it is possible to reverse the procedure: $a < b$ can be defined to mean that $b - a$ is positive. In fact, it is convenient to consider the collection of all positive numbers, denoted by P , as the basic concept, and state all properties in terms of P :

- (P10) (Trichotomy law) For every number a , one and only one of the following holds:
- (i) $a = 0$,
 - (ii) a is in the collection P ,
 - (iii) $-a$ is in the collection P .
- (P11) (Closure under addition) If a and b are in P , then $a + b$ is in P .
- (P12) (Closure under multiplication) If a and b are in P , then $a \cdot b$ is in P .

These three properties should be complemented with the following definitions:

$$\begin{aligned} a > b & \text{ if } a - b \text{ is in } P; \\ a < b & \text{ if } b > a; \\ a \geq b & \text{ if } a > b \text{ or } a = b; \\ a \leq b & \text{ if } a < b \text{ or } a = b.* \end{aligned}$$

Note, in particular, that $a > 0$ if and only if a is in P .

All the familiar facts about inequalities, however elementary they may seem, are consequences of P10–P12. For example, if a and b are any two numbers, then precisely one of the following holds:

- (i) $a - b = 0$,
- (ii) $a - b$ is in the collection P ,
- (iii) $-(a - b) = b - a$ is in the collection P .

Using the definitions just made, it follows that precisely one of the following holds:

- (i) $a = b$,
- (ii) $a > b$,
- (iii) $b > a$.

A slightly more interesting fact results from the following manipulations. If $a < b$, so that $b - a$ is in P , then surely $(b + c) - (a + c)$ is in P ; thus, if $a < b$, then $a + c < b + c$. Similarly, suppose $a < b$ and $b < c$. Then

$$\begin{aligned} & b - a \text{ is in } P, \\ \text{and } & c - b \text{ is in } P, \\ \text{so } & c - a = (c - b) + (b - a) \text{ is in } P. \end{aligned}$$

This shows that if $a < b$ and $b < c$, then $a < c$. (The two inequalities $a < b$ and $b < c$ are usually written in the abbreviated form $a < b < c$, which has the third inequality $a < c$ almost built in.)

The following assertion is somewhat less obvious: If $a < 0$ and $b < 0$, then $ab > 0$. The only difficulty presented by the proof is the unraveling of definitions. The symbol $a < 0$ means, by definition, $0 > a$, which means $0 - a = -a$ is in P . Similarly $-b$ is in P , and consequently, by P12, $(-a)(-b) = ab$ is in P . Thus $ab > 0$.

The fact that $ab > 0$ if $a > 0$, $b > 0$ and also if $a < 0$, $b < 0$ has one special consequence: $a^2 > 0$ if $a \neq 0$. Thus squares of nonzero numbers are

* There is one slightly perplexing feature of the symbols $\geq$ and $\leq$. The statements

$$\begin{aligned} 1 + 1 & \leq 3 \\ 1 + 1 & \leq 2 \end{aligned}$$

are both true, even though we know that $\leq$ could be replaced by $<$ in the first, and by $=$ in the second. This sort of thing is bound to occur when $\leq$ is used with specific numbers; the usefulness of the symbol is revealed by a statement like Theorem 1—here equality holds for some values of a and b , while inequality holds for other values.

always positive, and in particular we have proved a result which might have seemed sufficiently elementary to be included in our list of properties: $1 > 0$ (since $1 = 1^2$).

The fact that $-a > 0$ if $a < 0$ is the basis of a concept which will play an extremely important role in this book. For any number a , we define the **absolute value** $|a|$ of a as follows:

$$|a| = \begin{cases} a, & a \geq 0 \\ -a, & a \leq 0. \end{cases}$$

Note that $|a|$ is always positive, except when $a = 0$. For example, we have $|-3| = 3$, $|7| = 7$, $|1 + \sqrt{2} - \sqrt{3}| = 1 + \sqrt{2} - \sqrt{3}$, and $|1 + \sqrt{2} - \sqrt{10}| = \sqrt{10} - \sqrt{2} - 1$. In general, the most straightforward approach to any problem involving absolute values requires treating several cases separately, since absolute values are defined by cases to begin with. This approach may be used to prove the following very important fact about absolute values.

THEOREM 1 For all numbers a and b , we have

$$|a + b| \leq |a| + |b|.$$

PROOF We will consider 4 cases:

- (1) $a \geq 0, b \geq 0$;
- (2) $a \geq 0, b \leq 0$;
- (3) $a \leq 0, b \geq 0$;
- (4) $a \leq 0, b \leq 0$.

In case (1) we also have $a + b \geq 0$, and the theorem is obvious; in fact,

$$|a + b| = a + b = |a| + |b|,$$

so that in this case equality holds.

In case (4) we have $a + b \leq 0$, and again equality holds:

$$|a + b| = -(a + b) = -a + (-b) = |a| + |b|.$$

In case (2), when $a \geq 0$ and $b \leq 0$, we must prove that

$$|a + b| \leq a - b.$$

This case may therefore be divided into two subcases. If $a + b \geq 0$, then we must prove that

$$\begin{aligned} a + b &\leq a - b, \\ \text{i.e.,} \quad b &\leq -b, \end{aligned}$$

which is certainly true since b is negative and $-b$ is positive. On the other hand, if $a + b \leq 0$, we must prove that

$$\begin{aligned} -a - b &\leq a - b, \\ \text{i.e.,} \quad -a &\leq a, \end{aligned}$$

which is certainly true since a is positive and $-a$ is negative.

Finally, note that case (3) may be disposed of with no additional work, by applying case (2) with a and b interchanged. ■

Although this method of treating absolute values (separate consideration of various cases) is sometimes the only approach available, there are often simpler methods which may be used. In fact, it is possible to give a much shorter proof of Theorem 1; this proof is motivated by the observation that

$$|a| = \sqrt{a^2}.$$

(Here, and throughout the book, $\sqrt{x}$ denotes the *positive* square root of x ; this symbol is defined only when $x \geq 0$.) We may now observe that

$$\begin{aligned} (|a + b|)^2 &= (a + b)^2 = a^2 + 2ab + b^2 \\ &\leq a^2 + 2|a| \cdot |b| + b^2 \\ &= |a|^2 + 2|a| \cdot |b| + |b|^2 \\ &= (|a| + |b|)^2. \end{aligned}$$

From this we can conclude that $|a + b| \leq |a| + |b|$ because $x^2 < y^2$ implies $x < y$, provided that x and y are both non-negative; a proof of *this* fact is left to the reader (Problem 5).

One final observation may be made about the theorem we have just proved: a close examination of either proof offered shows that

$$|a + b| = |a| + |b|$$

if a and b have the same sign (i.e., are both positive or both negative), or if one of the two is 0, while

$$|a + b| < |a| + |b|$$

if a and b are of opposite signs.

We will conclude this chapter with a subtle point, neglected until now, whose inclusion is required in a conscientious survey of the properties of numbers. After stating property P9, we proved that $a - b = b - a$ implies $a = b$. The proof began by establishing that

$$a \cdot (1 + 1) = b \cdot (1 + 1),$$

from which we concluded that $a = b$. This result is obtained from the equation $a \cdot (1 + 1) = b \cdot (1 + 1)$ by dividing both sides by $1 + 1$. Division by 0 should be avoided scrupulously, and it must therefore be admitted that the validity of the argument depends on knowing that $1 + 1 \neq 0$. Problem 24 is designed to convince you that this fact cannot possibly be proved from properties P1–P9 alone! Once P10, P11, and P12 are available, however, the proof is very simple: We have already seen that $1 > 0$; it follows that $1 + 1 > 0$, and in particular $1 + 1 \neq 0$.

This last demonstration has perhaps only strengthened your feeling that it is absurd to bother proving such obvious facts, but an honest assessment of our present situation will help justify serious consideration of such details. In

this chapter we have assumed that numbers are familiar objects, and that P1–P12 are merely explicit statements of obvious, well-known properties of numbers. It would be difficult, however, to justify this assumption. Although one learns how to “work with” numbers in school, just what numbers *are*, remains rather vague. A great deal of this book is devoted to elucidating the concept of numbers, and by the end of the book we will have become quite well acquainted with them. But it will be necessary to work with numbers throughout the book. It is therefore reasonable to admit frankly that we do not yet thoroughly understand numbers; we may still say that, in whatever way numbers are finally defined, they should certainly have properties P1–P12.

Most of this chapter has been an attempt to present convincing evidence that P1–P12 are indeed basic properties which we should assume in order to deduce other familiar properties of numbers. Some of the problems (which indicate the derivation of other facts about numbers from P1–P12) are offered as further evidence. It is still a crucial question whether P1–P12 actually account for *all* properties of numbers. As a matter of fact, we shall soon see that they do *not*. In the next chapter the deficiencies of properties P1–P12 will become quite clear, but the proper means for correcting these deficiencies is not so easily discovered. The crucial additional basic property of numbers which we are seeking is profound and subtle, quite unlike P1–P12. The discovery of this crucial property will require all the work of Part II of this book. In the remainder of Part I we will begin to see why some additional property is required; in order to investigate this we will have to consider a little more carefully what we mean by “numbers.”

PROBLEMS

1. Prove the following:

- (i) If $ax = a$ for some number $a \neq 0$, then $x = 1$.
- (ii) $x^2 - y^2 = (x - y)(x + y)$.
- (iii) If $x^2 = y^2$, then $x = y$ or $x = -y$.
- (iv) $x^3 - y^3 = (x - y)(x^2 + xy + y^2)$.
- (v) $x^n - y^n = (x - y)(x^{n-1} + x^{n-2}y + \cdots + xy^{n-2} + y^{n-1})$.
- (vi) $x^3 + y^3 = (x + y)(x^2 - xy + y^2)$. (There is a particularly easy way to do this, using (iv), and it will show you how to find a factorization for $x^n + y^n$ whenever n is odd.)

2. What is wrong with the following “proof”? Let $x = y$. Then

$$\begin{aligned}
 x^2 &= xy, \\
 x^2 - y^2 &= xy - y^2, \\
 (x + y)(x - y) &= y(x - y), \\
 x + y &= y, \\
 2y &= y, \\
 2 &= 1.
 \end{aligned}$$

3. Prove the following:

$$(i) \quad \frac{a}{b} = \frac{ac}{bc}, \text{ if } b, c \neq 0.$$

$$(ii) \quad \frac{a}{b} + \frac{c}{d} = \frac{ad + bc}{bd}, \text{ if } b, d \neq 0.$$

$$(iii) \quad (ab)^{-1} = a^{-1}b^{-1}, \text{ if } a, b \neq 0. \text{ (To do this you must remember the defining property of } (ab)^{-1}.)$$

$$(iv) \quad \frac{a}{b} \cdot \frac{c}{d} = \frac{ac}{db}, \text{ if } b, d \neq 0.$$

$$(v) \quad \frac{a}{b} \bigg/ \frac{c}{d} = \frac{ad}{bc}, \text{ if } b, c, d \neq 0.$$

$$(vi) \quad \text{If } b, d \neq 0, \text{ then } \frac{a}{b} = \frac{c}{d} \text{ if and only if } ad = bc. \text{ Also determine when}$$

$$\frac{a}{b} = \frac{b}{a}.$$

4. Find all numbers x for which

$$(i) \quad 4 - x < 3 - 2x.$$

$$(ii) \quad 5 - x^2 < 8.$$

$$(iii) \quad 5 - x^2 < -2.$$

$$(iv) \quad (x - 1)(x - 3) > 0. \text{ (When is a product of two numbers positive?)}$$

$$(v) \quad x^2 - 2x + 2 > 0.$$

$$(vi) \quad x^2 + x + 1 > 2.$$

$$(vii) \quad x^2 - x + 10 > 16.$$

$$(viii) \quad x^2 + x + 1 > 0.$$

$$(ix) \quad (x - \pi)(x + 5)(x - 3) > 0.$$

$$(x) \quad (x - \sqrt[3]{2})(x - \sqrt{2}) > 0.$$

$$(xi) \quad 2^x < 8.$$

$$(xii) \quad x + 3^x < 4.$$

$$(xiii) \quad \frac{1}{x} + \frac{1}{1 - x} > 0.$$

$$(xiv) \quad \frac{x - 1}{x + 1} > 0.$$

5. Prove the following:

$$(i) \quad \text{If } a < b \text{ and } c < d, \text{ then } a + c < b + d.$$

$$(ii) \quad \text{If } a < b, \text{ then } -b < -a.$$

- (iii) If $a < b$ and $c > d$, then $a - c < b - d$.
- (iv) If $a < b$ and $c > 0$, then $ac < bc$.
- (v) If $a < b$ and $c < 0$, then $ac > bc$.
- (vi) If $a > 1$, then $a^2 > a$.
- (vii) If $0 < a < 1$, then $a^2 < a$.
- (viii) If $0 \leq a < b$ and $0 \leq c < d$, then $ac < bd$.
- (ix) If $0 \leq a < b$, then $a^2 < b^2$. (Use (viii).)
- (x) If $a, b \geq 0$ and $a^2 < b^2$, then $a < b$. (Use (ix), backwards.)

6. Prove that if $0 < a < b$, then

$$a < \sqrt{ab} < \frac{a+b}{2} < b.$$

Notice that the inequality $\sqrt{ab} \leq (a+b)/2$ holds for $a, b > 0$, without the additional assumption $a < b$. A generalization of this fact occurs in Problem 2-20.

- *7. (a) Prove that if $x^n = y^n$ and n is odd, then $x = y$. Hint: First explain why it suffices to consider only the case $x, y > 0$; then show that $x < y$ and $y > x$ are both impossible.
- (b) Prove that if $x^n = y^n$ and n is even, then $x = y$ or $x = -y$.
- *8. Although the basic properties of inequalities were stated in terms of the collection P of all positive numbers, and $<$ was defined in terms of P , this procedure can be reversed. Suppose that P10–P12 are replaced by

(P'10) For any numbers a and b one, and only one, of the following holds:

- (i) $a = b$,
- (ii) $a < b$,
- (iii) $b < a$.

(P'11) For any numbers a, b , and c , if $a < b$ and $b < c$, then $a < c$.

(P'12) For any numbers a, b , and c , if $a < b$, then $a + c < b + c$.

(P'13) For any numbers a, b , and c , if $a < b$ and $0 < c$, then $ac < bc$.

Show that P10–P12 can then be deduced as theorems.

- 9. Express each of the following with at least one less pair of absolute value signs.

- (i) $|\sqrt{2} + \sqrt{3} - \sqrt{5} + \sqrt{7}|$.
- (ii) $|(|a + b| - |a| - |b|)|$.
- (iii) $|(|a + b| + |c| - |a + b + c|)|$.
- (iv) $|x^2 - 2xy + y^2|$.
- (v) $|(|\sqrt{2} + \sqrt{3}| - |\sqrt{5} - \sqrt{7}|)|$.

10. Express each of the following without absolute value signs, treating various cases separately when necessary.

- (i) $|a + b| - |b|$.
- (ii) $|(|x| - 1)|$.
- (iii) $|x| - |x^2|$.
- (iv) $a - |(a - |a|)|$.

11. Find all numbers x for which

- (i) $|x - 3| = 8$.
- (ii) $|x - 3| < 8$.
- (iii) $|x + 4| < 2$.
- (iv) $|x - 1| + |x - 2| > 1$.
- (v) $|x - 1| + |x + 1| < 2$.
- (vi) $|x - 1| + |x + 1| < 1$.
- (vii) $|x - 1| \cdot |x + 1| = 0$.
- (viii) $|x - 1| \cdot |x + 2| = 3$.

12. Prove the following:

- (i) $|xy| = |x| \cdot |y|$.
- (ii) $\left| \frac{1}{x} \right| = \frac{1}{|x|}$, if $x \neq 0$. (The best way to do this is to remember what $|x|^{-1}$ is.)
- (iii) $\frac{|x|}{|y|} = \left| \frac{x}{y} \right|$, if $y \neq 0$.
- (iv) $|x - y| \leq |x| + |y|$. (Give a very short proof.)
- (v) $||x| - |y|| \leq |x - y|$. (A very short proof is possible, if you write things in the right way.)
- (vi) $||x| - |y|| \leq |x - y|$. (Why does this follow immediately from (v)?)
- (vii) $|x + y + z| \leq |x| + |y| + |z|$. Indicate when equality holds, and prove your statement.

13. The maximum of two numbers x and y is denoted by $\max(x, y)$. Thus $\max(-1, 3) = \max(3, 3) = 3$ and $\max(-1, -4) = \max(-4, -1) = -1$. The minimum of x and y is denoted by $\min(x, y)$. Prove that

$$\max(x, y) = \frac{x + y + |y - x|}{2},$$

$$\min(x, y) = \frac{x + y - |y - x|}{2}.$$

Derive a formula for $\max(x, y, z)$ and $\min(x, y, z)$, using, for example,

$$\max(x, y, z) = \max(x, \max(y, z)).$$

14. (a) Prove that $|a| = |-a|$. (The trick is not to become confused by too many cases. First prove the statement for $a \geq 0$. Why is it then obvious for $a \leq 0$?)
- (b) Prove that $-b \leq a \leq b$ if, and only if $|a| \leq b$. In particular, it follows that $-|a| \leq a \leq |a|$.
- (c) Use this fact to give a new proof that $|a + b| \leq |a| + |b|$.
- *15. (a) Use Problem 1 and Problem 7 to prove that if x and y are not both 0, then

$$x^2 + xy + y^2 \neq 0, \\ x^4 + x^3y + x^2y^2 + xy^3 + y^4 \neq 0.$$

For every number $x \neq 0$, each of these expressions is positive for some positive number y and also for some negative y (namely, $y = \pm x$); it therefore seems reasonable that the $\neq$ signs can be replaced by $>$ signs. This maneuver is valid, but we are not yet in a position to prove this (see Problem 7-9). Parts (b) and (d) of this problem provide a direct demonstration that the $>$ signs hold.

(b) Using the fact that

$$x^2 + 2xy + y^2 = (x + y)^2 \geq 0,$$

show that the assumption $x^2 + xy + y^2 < 0$ leads to a contradiction.

(c) Show similarly that if x and y are not both 0, then

$$4x^2 + 6xy + 4y^2 > 0, \\ 3x^2 + 5xy + 3y^2 > 0.$$

** (d) Show that if x and y are not both 0, then

$$x^4 + x^3y + x^2y^2 + xy^3 + y^4 > 0.$$

*16. (a) Show that

$$(x + y)^2 = x^2 + y^2 \quad \text{only when } x = 0 \text{ or } y = 0, \\ (x + y)^3 = x^3 + y^3 \quad \text{only when } x = 0 \text{ or } y = 0 \text{ or } x = -y.$$

(b) Use Problem 15 to find out when $(x + y)^4 = x^4 + y^4$.

** (c) Find out when $(x + y)^5 = x^5 + y^5$. Hint: From the assumption $(x + y)^5 = x^5 + y^5$ you should be able to derive the equation $x^3 + 2x^2y + 2xy^2 + y^3 = 0$, if $xy \neq 0$. This implies that $(x + y)^3 = x^2y + xy^2 = xy(x + y)$.

You should now be able to make a good guess as to when $(x + y)^n = x^n + y^n$; the proof is contained in Problem 11-41.

17. (a) Suppose that $b^2 - 4c \geq 0$. Show that the numbers

$$\frac{-b + \sqrt{b^2 - 4c}}{2}, \quad \frac{-b - \sqrt{b^2 - 4c}}{2}$$

both satisfy the equation $x^2 + bx + c = 0$.

- (b) Suppose that $b^2 - 4c < 0$. Show that there are no numbers x satisfying $x^2 + bx + c = 0$; in fact, $x^2 + bx + c > 0$ for all x . Hint: "Complete the square," i.e., write $x^2 + bx + c = (x + b/2)^2 + ?$
- (c) Use this fact to give another proof that if x and y are not both 0, then $x^2 + xy + y^2 > 0$.
- (d) For which numbers α is it true that $x^2 + \alpha xy + y^2 > 0$ whenever x and y are not both 0?
- (e) Find the smallest possible value of $x^2 + bx + c$ and of $ax^2 + bx + c$, for $a \neq 0$. (Use the trick in part (b).)
18. The fact that $a^2 \geq 0$ for all numbers a , elementary as it may seem, is nevertheless the fundamental idea upon which most important inequalities are ultimately based. The great-granddaddy of all inequalities is the *Schwartz inequality*:

$$x_1 y_1 + x_2 y_2 \leq \sqrt{x_1^2 + x_2^2} \sqrt{y_1^2 + y_2^2}.$$

The three proofs of the Schwartz inequality outlined below have only one thing in common—their reliance on the fact that $a^2 \geq 0$ for all a .

- (a) Prove the Schwartz inequality by first proving that

$$(x_1^2 + x_2^2)(y_1^2 + y_2^2) = (x_1 y_1 + x_2 y_2)^2 + (x_1 y_2 - x_2 y_1)^2.$$

- (b) Prove that if $x_1 = \lambda y_1$ and $x_2 = \lambda y_2$ for some number λ , then equality holds in the Schwartz inequality. Prove the same thing if $y_1 = y_2 = 0$. Now suppose that y_1 and y_2 are not both 0, and that there is no number λ such that $x_1 = \lambda y_1$ and $x_2 = \lambda y_2$. Then

$$\begin{aligned} 0 &< (\lambda y_1 - x_1)^2 + (\lambda y_2 - x_2)^2 \\ &= \lambda^2(y_1^2 + y_2^2) - 2\lambda(x_1 y_1 + x_2 y_2) + (x_1^2 + x_2^2). \end{aligned}$$

Using Problem 17, complete the proof of the Schwartz inequality.

- (c) Prove the Schwartz inequality by using $2xy \leq x^2 + y^2$ (how is this derived?) with

$$x = \frac{x_i}{\sqrt{x_1^2 + x_2^2}}, \quad y = \frac{y_i}{\sqrt{y_1^2 + y_2^2}},$$

first for $i = 1$ and then for $i = 2$.

- (d) Deduce, from each of these three proofs, that equality holds only when $y_1 = y_2 = 0$ or when there is a number λ such that $x_1 = \lambda y_1$ and $x_2 = \lambda y_2$.

In our later work, three facts about inequalities will be crucial. Although proofs will be supplied at the appropriate point in the text, a personal assault on these problems is infinitely more enlightening than a perusal of a completely worked-out proof. The statements of these propositions involve some weird numbers, but their basic message is very simple: if x is close enough to x_0 , and y is close enough to y_0 , then $x + y$ will be close to $x_0 + y_0$, and xy will be

close to $x_0 y_0$, and $1/y$ will be close to $1/y_0$. The symbol “ ε ” which appears in these propositions is the fifth letter of the Greek alphabet (“epsilon”), and could just as well be replaced by a less intimidating Roman letter; however, tradition has made the use of ε almost sacrosanct in the contexts to which these theorems apply.

19. Prove that if

$$|x - x_0| < \frac{\varepsilon}{2} \quad \text{and} \quad |y - y_0| < \frac{\varepsilon}{2},$$

then

$$\begin{aligned} |(x + y) - (x_0 + y_0)| &< \varepsilon, \\ |(x - y) - (x_0 - y_0)| &< \varepsilon. \end{aligned}$$

*20. Prove that if

$$|x - x_0| < \min\left(\frac{\varepsilon}{2(|y_0| + 1)}, 1\right) \quad \text{and} \quad |y - y_0| < \frac{\varepsilon}{2(|x_0| + 1)},$$

then $|xy - x_0 y_0| < \varepsilon$.

(The notation “min” was defined in Problem 13, but the formula provided by that problem is irrelevant at the moment; the first inequality in the hypothesis just means that

$$|x - x_0| < \frac{\varepsilon}{2(|y_0| + 1)} \quad \text{and} \quad |x - x_0| < 1;$$

at one point in the argument you will need the first inequality, and at another point you will need the second. One more word of advice: since the hypotheses only provide information about $x - x_0$ and $y - y_0$, it is almost a foregone conclusion that the proof will depend upon writing $xy - x_0 y_0$ in a way that involves $x - x_0$ and $y - y_0$.)

*21. Prove that if $y_0 \neq 0$ and

$$|y - y_0| < \min\left(\frac{|y_0|}{2}, \frac{\varepsilon|y_0|^2}{2}\right),$$

then $y \neq 0$ and

$$\left| \frac{1}{y} - \frac{1}{y_0} \right| < \varepsilon.$$

*22. Replace the question marks in the following statement by expressions involving ε , x_0 , and y_0 so that the conclusion will be true:

If $y_0 \neq 0$ and

$$|y - y_0| < ? \quad \text{and} \quad |x - x_0| < ?$$

then $y \neq 0$ and

$$\left| \frac{x}{y} - \frac{x_0}{y_0} \right| < \varepsilon.$$

This problem is trivial in the sense that its solution follows from Problems 20 and 21 with almost no work at all (notice that $x/y = x \cdot 1/y$). The crucial point is not to become confused; decide which of the two problems should be used first, and don't panic if your answer looks unlikely.

- *23. This problem shows that the actual placement of parentheses in a sum is irrelevant. The proofs involve "mathematical induction"; if you are not familiar with such proofs, but still want to tackle this problem, it can be saved until after Chapter 2, where proofs by induction are explained.

Let us agree, for definiteness, that $a_1 + \cdots + a_n$ will denote

$$a_1 + (a_2 + (a_3 + \cdots + (a_{n-2} + (a_{n-1} + a_n))) \cdots).$$

Thus $a_1 + a_2 + a_3$ denotes $a_1 + (a_2 + a_3)$, and $a_1 + a_2 + a_3 + a_4$ denotes $a_1 + (a_2 + (a_3 + a_4))$, etc.

- (a) Prove that

$$(a_1 + \cdots + a_k) + a_{k+1} = a_1 + \cdots + a_{k+1}.$$

Hint: Use induction on k .

- (b) Prove that if $n \geq k$, then

$$(a_1 + \cdots + a_k) + (a_{k+1} + \cdots + a_n) = a_1 + \cdots + a_n.$$

Hint: Use part (a) to give a proof by induction on k .

- (c) Let $s(a_1, \dots, a_k)$ be some sum formed from $a_1, \dots, a_k$. Show that

$$s(a_1, \dots, a_k) = a_1 + \cdots + a_k.$$

Hint: There must be two sums $s'(a_1, \dots, a_l)$ and $s''(a_{l+1}, \dots, a_k)$ such that

$$s(a_1, \dots, a_k) = s'(a_1, \dots, a_l) + s''(a_{l+1}, \dots, a_k).$$

24. Suppose that we interpret "number" to mean either 0 or 1, and $+$ and $\cdot$ to be the operations defined by the following two tables.

+	0	1		·	0	1							
0	<table><tr><td>0</td><td>1</td></tr><tr><td>1</td><td>0</td></tr></table>	0	1	1	0			0	<table><tr><td>0</td><td>0</td></tr><tr><td>0</td><td>1</td></tr></table>	0	0	0	1
0	1												
1	0												
0	0												
0	1												
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Check that properties P1–P9 all hold, even though $1 + 1 = 0$.

CHAPTER 2 NUMBERS OF VARIOUS SORTS

In Chapter 1 we used the word “number” very loosely, despite our concern with the basic properties of numbers. It will now be necessary to distinguish carefully various kinds of numbers.

The simplest numbers are the “counting numbers”

1, 2, 3,

The fundamental significance of this collection of numbers is emphasized by its symbol **N** (for **natural numbers**). A brief glance at P1–P12 will show that our basic properties of “numbers” do not apply to **N**—for example, P2 and P3 do not make sense for **N**. From this point of view the system **N** has many deficiencies. Nevertheless, **N** is sufficiently important to deserve several comments before we consider larger collections of numbers.

The most basic property of **N** is the principle of “mathematical induction.” Suppose $P(x)$ means that the property P holds for the number x . Then the principle of mathematical induction states that $P(x)$ is true for all natural numbers x provided that

- (1) $P(1)$ is true.
- (2) Whenever $P(k)$ is true, $P(k + 1)$ is true.

Note that condition (2) merely asserts the truth of $P(k + 1)$ under the assumption that $P(k)$ is true; this suffices to ensure the truth of $P(x)$ for all x , if condition (1) also holds. In fact, if $P(1)$ is true, then it follows that $P(2)$ is true (by using (2) in the special case $k = 1$). Now, since $P(2)$ is true it follows that $P(3)$ is true (using (2) in the special case $k = 2$). It is clear that each number will eventually be reached by a series of steps of this sort, so that $P(k)$ is true for all numbers k .

A favorite illustration of the reasoning behind mathematical induction envisions an infinite line of people,

person number 1, person number 2, person number 3,

If each person has been instructed to tell any secret he hears to the person behind him (the one with the next largest number) and a secret is told to person number 1, then clearly every person will eventually learn the secret. If $P(x)$ is the assertion that person number x will learn the secret, then the instructions given (to tell all secrets learned to the next person) assures that condition (2) is true, and telling the secret to person number 1 makes (1) true. The following example is a less facetious use of mathematical induction. There is a useful and striking formula which expresses the sum of the first n numbers in a simple way:

$$1 + \cdots + n = \frac{n(n+1)}{2}.$$

To prove this formula, note first that it is clearly true for $n = 1$. Now *assume* that for some integer k we have

$$1 + \cdots + k = \frac{k(k+1)}{2}.$$

Then

$$\begin{aligned} 1 + \cdots + k + (k+1) &= \frac{k(k+1)}{2} + k + 1 \\ &= \frac{k(k+1) + 2k + 2}{2} \\ &= \frac{k^2 + 3k + 2}{2} \\ &= \frac{(k+1)(k+2)}{2}, \end{aligned}$$

so the formula is also true for $k+1$. By the principle of induction this proves the formula for all natural numbers n . This particular example illustrates a phenomenon that frequently occurs, especially in connection with formulas like the one just proved. Although the proof by induction is often quite straightforward, the method by which the formula was discovered remains a mystery. Problems 4 and 5 indicate how some formulas of this type may be derived.

The principle of mathematical induction may be formulated in an equivalent way without speaking of “properties” of a number, a term which is sufficiently vague to be eschewed in a mathematical discussion. A more precise formulation states that if A is any collection (or “set”—a synonymous mathematical term) of natural numbers and

- (1) 1 is in A ,
- (2) $k+1$ is in A whenever k is in A ,

then A is the set of all natural numbers. It should be clear that this formulation adequately replaces the less formal one given previously—we just consider the set A of natural numbers x which satisfy $P(x)$. For example, suppose A is the set of natural numbers n for which it is true that

$$1 + \cdots + n = \frac{n(n+1)}{2}.$$

Our previous proof of this formula showed that A contains 1, and that $k+1$ is in A , if k is. It follows that A is the set of all natural numbers, i.e., that the formula holds for all natural numbers n .

There is yet another rigorous formulation of the principle of mathematical induction, which looks quite different. If A is any collection of natural num-

bers, it is tempting to say that A must have a smallest member. Actually, this statement can fail to be true in a rather subtle way. A particularly important set of natural numbers is the collection A that contains no natural numbers at all, the “empty collection” or “null set,”* denoted by $\emptyset$. The null set $\emptyset$ is a collection of natural numbers that has no smallest member—in fact, it has no members at all. This is the only possible exception, however; if A is a nonnull set of natural numbers, then A has a least member. This “intuitively obvious” statement, known as the “well-ordering principle,” can be proved from the principle of induction as follows. Suppose that the set A has no least member. Let B be the set of natural numbers n such that $1, \dots, n$ are all *not* in A . Clearly 1 is in B (because if 1 were in A , then A would have 1 as smallest member). Moreover, if $1, \dots, k$ are not in A , surely $k + 1$ is not in A (otherwise $k + 1$ would be the smallest member of A), so $1, \dots, k + 1$ are all not in A . This shows that if k is in B , then $k + 1$ is in B . It follows that every number n is in B , i.e., the numbers $1, \dots, n$ are *not* in A for any natural number n . Thus $A = \emptyset$, which completes the proof.

It is also possible to prove the principle of induction from the well-ordering principle (Problem 9). Either principle may be considered as a basic assumption about the natural numbers.

There is still another form of induction which should be mentioned. It sometimes happens that in order to prove $P(k + 1)$ we must assume not only $P(k)$, but also $P(l)$ for all natural numbers $l \leq k$. In this case we rely on the “principle of complete induction”: If A is a set of natural numbers and

- (1) 1 is in A ,
- (2) $k + 1$ is in A if $1, \dots, k$ are in A ,

then A is the set of all natural numbers.

Although the principle of complete induction may appear much stronger than the ordinary principle of induction, it is actually a consequence of that principle. The proof of this fact is left to the reader, with a hint (Problem 10). Applications will be found in Problems 6, 16, 19, and 20.

Closely related to proofs by induction are “recursive definitions.” For example, the number $n!$ (read “ n factorial”) is defined as the product of all the natural numbers less than or equal to n :

$$n! = 1 \cdot 2 \cdot \dots \cdot (n - 1) \cdot n.$$

This can be expressed more precisely as follows:

- (1) $1! = 1$,
- (2) $n! = n \cdot (n - 1)!$.

This form of the definition exhibits the relationship between $n!$ and $(n - 1)!$

* Although it may not strike you as a collection, in the ordinary sense of the word, the null set arises quite naturally in many contexts. We frequently consider the set A , consisting of all x satisfying some property P ; often we have no guarantee that P is satisfied by *any* number, so that A might be $\emptyset$ —in fact often one proves that P is always false by showing that $A = \emptyset$.

in an explicit way that is ideally suited for proofs by induction. Problem 21 reviews a definition already familiar to you, which may be expressed more succinctly as a recursive definition; as this problem shows, the recursive definition is really necessary for a rigorous proof of some of the basic properties of the definition.

One definition which may not be familiar involves some convenient notation which we will constantly be using. Instead of writing

$$a_1 + \cdots + a_n,$$

we will usually employ the Greek letter Σ (capital sigma, for “sum”) and write

$$\sum_{i=1}^n a_i.$$

In other words, $\sum_{i=1}^n a_i$ denotes the sum of the numbers obtained by letting $i = 1, 2, \dots, n$. Thus

$$\sum_{i=1}^n i = 1 + 2 + \cdots + n = \frac{n(n+1)}{2}.$$

Notice that the letter i really has nothing to do with the number denoted by $\sum_{i=1}^n i$, and can be replaced by any convenient symbol (except n , of course!):

$$\begin{aligned} \sum_{j=1}^n j &= \frac{n(n+1)}{2}, \\ \sum_{j=1}^i j &= \frac{i(i+1)}{2}, \\ \sum_{n=1}^j n &= \frac{j(j+1)}{2}. \end{aligned}$$

To define $\sum_{i=1}^n a_i$ precisely really requires a recursive definition:

$$\begin{aligned} (1) \quad \sum_{i=1}^1 a_i &= a_1, \\ (2) \quad \sum_{i=1}^n a_i &= \sum_{i=1}^{n-1} a_i + a_n. \end{aligned}$$

But only purveyors of mathematical austerity would insist too strongly on such precision. In practice, all sorts of modifications of this symbolism are used, and no one ever considers it necessary to add any words of explanation.

The symbol

$$\sum_{\substack{i=1 \\ i \neq 4}}^n a_i,$$

for example, is an obvious way of writing

$$a_1 + a_2 + a_3 + a_5 + a_6 + \cdots + a_n,$$

or more precisely,

$$\sum_{i=1}^3 a_i + \sum_{i=5}^n a_i.$$

The deficiencies of the natural numbers which we discovered at the beginning of this chapter may be partially remedied by extending this system to the set of **integers**

$$\dots, -2, -1, 0, 1, 2, \dots$$

This set is denoted by **Z** (from German “Zahl,” number). Of properties P1–P12, only P7 fails for **Z**.

A still larger system of numbers is obtained by taking quotients m/n of integers (with $n \neq 0$). These numbers are called **rational numbers**, and the set of all rational numbers is denoted by **Q** (for “quotients”). In this system of numbers all of P1–P12 are true. It is tempting to conclude that the “properties of numbers,” which we studied in some detail in Chapter 1, refer to just one set of numbers, namely, **Q**. There is, however, a still larger collection of numbers to which properties P1–P12 apply—the set of all **real numbers**, denoted by **R**. The real numbers include not only the rational numbers, but other numbers as well (the **irrational numbers**) which can be represented by infinite decimals; π and $\sqrt{2}$ are both examples of irrational numbers. The proof that π is irrational is not easy—we shall devote all of Chapter 16 of Part III to a proof of this fact. The irrationality of $\sqrt{2}$, on the other hand, is quite simple, and was known to the Greeks. (Since the Pythagorean theorem shows that an isosceles right triangle, with sides of length 1, has a hypotenuse of length $\sqrt{2}$, it is not surprising that the Greeks should have investigated this question.) The proof depends on a few observations about the natural numbers. Every natural number n can be written either in the form $2k$ for some integer k , or else in the form $2k + 1$ for some integer k (this “obvious” fact has a simple proof by induction (Problem 7)). Those natural numbers of the form $2k$ are called **even**; those of the form $2k + 1$ are called **odd**. Note that even numbers have even squares, and odd numbers have odd squares:

$$\begin{aligned}(2k)^2 &= 4k^2 = 2 \cdot (2k^2), \\ (2k + 1)^2 &= 4k^2 + 4k + 1 = 2 \cdot (2k^2 + 2k) + 1.\end{aligned}$$

In particular it follows that the converse must also hold: if n^2 is even, then n is even; if n^2 is odd, then n is odd. The proof that $\sqrt{2}$ is irrational is now quite

simple. Suppose that $\sqrt{2}$ were rational; that is, suppose there were natural numbers p and q such that

$$\left(\frac{p}{q}\right)^2 = 2.$$

We can assume that p and q have no common divisor (since all common divisors could be divided out to begin with). Now we have

$$p^2 = 2q^2.$$

This shows that p^2 is even, and consequently p must be even; that is, $p = 2k$ for some natural number k . Then

$$p^2 = 4k^2 = 2q^2,$$

so

$$2k^2 = q^2.$$

This shows that q^2 is even, and consequently that q is even. Thus both p and q are even, contradicting the fact that p and q have no common divisor. This contradiction completes the proof.

It is important to understand precisely what this proof shows. We have demonstrated that there is no rational number x such that $x^2 = 2$. This assertion is often expressed more briefly by saying that $\sqrt{2}$ is irrational. Note, however, that the use of the symbol $\sqrt{2}$ implies the existence of *some* number (necessarily irrational) whose square is 2. We have not proved that such a number exists and we can assert confidently that, at present, a proof is *impossible* for us. Any proof at this stage would have to be based on P1–P12 (the only properties of $\mathbf{R}$ we have mentioned); since P1–P12 are also true for $\mathbf{Q}$ the exact same argument would show that there is a rational number whose square is 2, and this we know is false. (Note that the reverse argument will not work—our proof that there is no rational number whose square is 2 cannot be used to show that there is no real number whose square is 2, because our proof used not only P1–P12 but also a special property of $\mathbf{Q}$, the fact that every number in $\mathbf{Q}$ can be written p/q for integers p and q .)

This particular deficiency in our list of properties of the real numbers could, of course, be corrected by adding a new property which asserts the existence of square roots of positive numbers. Resorting to such a measure is, however, neither aesthetically pleasing nor mathematically satisfactory; we would still not know that every number has an n th root if n is odd, and that every positive number has an n th root if n is even. Even if we assumed this, we could not prove the existence of a number x satisfying $x^5 + x + 1 = 0$ (even though there does happen to be one), since we do not know how to write the solution of the equation in terms of n th roots (in fact, it is known that the solution cannot be written in this form). And, of course, we certainly do not wish to assume that all equations have solutions, since this is false (no real number x satisfies $x^2 + 1 = 0$, for example). In fact, this direction of investigation is not a fruitful one. The most useful hints about the property distinguishing $\mathbf{R}$ from $\mathbf{Q}$, the most compelling evidence for the necessity of elucidating this

property, do not come from the study of numbers alone. In order to study the properties of the real numbers in a more profound way, we must study more than the real numbers. At this point we must begin with the foundations of calculus, in particular the fundamental concept on which calculus is based—functions.

PROBLEMS

1. Prove the following formulas by induction.

$$(i) \quad 1^2 + \cdots + n^2 = \frac{n(n+1)(2n+1)}{6}.$$

$$(ii) \quad 1^3 + \cdots + n^3 = (1 + \cdots + n)^2.$$

2. Find a formula for

$$(i) \quad \sum_{i=1}^n (2i-1) = 1 + 3 + 5 + \cdots + (2n-1).$$

$$(ii) \quad \sum_{i=1}^n (2i-1)^2 = 1^2 + 3^2 + 5^2 + \cdots + (2n-1)^2.$$

Hint: What do these expressions have to do with $1 + 2 + 3 + \cdots + 2n$ and $1^2 + 2^2 + 3^2 + \cdots + (2n)^2$?

3. If $0 \leq k \leq n$, the “binomial coefficient” $\binom{n}{k}$ is defined by

$$\binom{n}{k} = \frac{n!}{k!(n-k)!} = \frac{n(n-1) \cdots (n-k+1)}{k!}, \text{ if } k \neq 0, n$$

$$\binom{n}{0} = \binom{n}{n} = 1. \text{ (This becomes a special case of the first formula if we define } 0! = 1.)$$

(a) Prove that

$$\binom{n+1}{k} = \binom{n}{k-1} + \binom{n}{k}.$$

(The proof does not require an induction argument.)

This relation gives rise to the following configuration, known as “Pascal’s triangle”—a number not on one of the sides is the sum of the two numbers above it; the binomial coefficient $\binom{n}{k}$ is the k th number in the n th row.

$$\begin{array}{cccccccc} & & & & 1 & & & \\ & & & & 1 & & 1 & \\ & & & 1 & & 2 & & 1 \\ & & 1 & & 3 & & 3 & & 1 \\ & 1 & & 4 & & 6 & & 4 & & 1 \\ 1 & & 5 & & 10 & & 10 & & 5 & & 1 \\ & & & & \dots & & & & & & \end{array}$$

- (b) Notice that all the numbers in Pascal's triangle are natural numbers. Use part (a) to prove by induction that $\binom{n}{k}$ is always a natural number. (Your entire proof by induction will, in a sense, be summed up in a glance by Pascal's triangle.)
- (c) Give another proof that $\binom{n}{k}$ is a natural number by showing that $\binom{n}{k}$ is the number of sets of exactly k integers each chosen from $1, \dots, n$.
- (d) Prove the "binomial theorem": If a and b are any numbers and n is a natural number, then

$$\begin{aligned}(a+b)^n &= a^n + \binom{n}{1} a^{n-1}b + \binom{n}{2} a^{n-2}b^2 + \cdots + \binom{n}{n-1} ab^{n-1} + b^n \\ &= \sum_{j=0}^n \binom{n}{j} a^{n-j} b^j.\end{aligned}$$

- (e) Prove that

$$\begin{aligned}\text{(i)} \quad \sum_{j=0}^n \binom{n}{j} &= \binom{n}{0} + \cdots + \binom{n}{n} = 2^n. \\ \text{(ii)} \quad \sum_{j=0}^n (-1)^j \binom{n}{j} &= \binom{n}{0} - \binom{n}{1} + \cdots \pm \binom{n}{n} = 0.\end{aligned}$$

4. (a) Prove by induction on n that

$$1 + r + r^2 + \cdots + r^n = \frac{1 - r^{n+1}}{1 - r}$$

if $r \neq 1$ (if $r = 1$, evaluating the sum certainly presents no problem).

- (b) Derive this result by setting $S = 1 + r + \cdots + r^n$, multiplying this equation by r , and solving the two equations for S .
5. The formula for $1^2 + \cdots + n^2$ may be derived as follows. We begin with the formula

$$(k+1)^3 - k^3 = 3k^2 + 3k + 1.$$

Writing this formula for $k = 1, \dots, n$ and adding, we obtain

$$\begin{array}{r} 2^3 - 1^3 = 3 \cdot 1^2 + 3 \cdot 1 + 1 \\ 3^3 - 2^3 = 3 \cdot 2^2 + 3 \cdot 2 + 1 \\ \cdot \\ \cdot \\ \cdot \\ (n+1)^3 - n^3 = 3 \cdot n^2 + 3 \cdot n + 1 \\ \hline (n+1)^3 - 1 = 3[1^2 + \cdots + n^2] + 3[1 + \cdots + n] + n. \end{array}$$

Thus we can find $\sum_{k=1}^n k^2$ if we already know $\sum_{k=1}^n k$ (which could have been found in a similar way). Use this method to find

$$(i) \quad 1^3 + \cdots + n^3.$$

$$(ii) \quad 1^4 + \cdots + n^4.$$

$$(iii) \quad \frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \cdots + \frac{1}{n(n+1)}.$$

$$(iv) \quad \frac{3}{1^2 \cdot 2^2} + \frac{5}{2^2 \cdot 3^2} + \cdots + \frac{2n+1}{n^2(n+1)^2}.$$

*6. Use the method of Problem 5 to show that $\sum_{k=1}^n k^p$ can always be written in the form

$$\frac{n^{p+1}}{p+1} + An^p + Bn^{p-1} + Cn^{p-2} + \cdots$$

(The first 10 such expressions are

$$\sum_{k=1}^n k = \frac{1}{2}n^2 + \frac{1}{2}n$$

$$\sum_{k=1}^n k^2 = \frac{1}{3}n^3 + \frac{1}{2}n^2 + \frac{1}{6}n$$

$$\sum_{k=1}^n k^3 = \frac{1}{4}n^4 + \frac{1}{2}n^3 + \frac{1}{4}n^2$$

$$\sum_{k=1}^n k^4 = \frac{1}{5}n^5 + \frac{1}{2}n^4 + \frac{1}{3}n^3 - \frac{1}{30}n$$

$$\sum_{k=1}^n k^5 = \frac{1}{6}n^6 + \frac{1}{2}n^5 + \frac{5}{12}n^4 - \frac{1}{12}n^2$$

$$\sum_{k=1}^n k^6 = \frac{1}{7}n^7 + \frac{1}{2}n^6 + \frac{1}{2}n^5 - \frac{1}{6}n^3 + \frac{1}{42}n$$

$$\sum_{k=1}^n k^7 = \frac{1}{8}n^8 + \frac{1}{2}n^7 + \frac{7}{12}n^6 - \frac{7}{24}n^4 + \frac{1}{12}n^2$$

$$\sum_{k=1}^n k^8 = \frac{1}{9}n^9 + \frac{1}{2}n^8 + \frac{2}{3}n^7 - \frac{7}{15}n^5 + \frac{2}{9}n^3 - \frac{1}{30}n$$

$$\sum_{k=1}^n k^9 = \frac{1}{10}n^{10} + \frac{1}{2}n^9 + \frac{3}{4}n^8 - \frac{7}{10}n^6 + \frac{1}{2}n^4 - \frac{1}{12}n^2$$

$$\sum_{k=1}^n k^{10} = \frac{1}{11}n^{11} + \frac{1}{2}n^{10} + \frac{5}{6}n^9 - 1n^7 + 1n^5 - \frac{1}{2}n^3 + \frac{5}{66}n.$$

Notice that the coefficients in the second column are always $\frac{1}{2}$, and that after the third column the powers of n with nonzero coefficients decrease

by 2 until n^2 or n is reached. The coefficients in all but the first two columns seem to be rather haphazard, but there actually is some sort of pattern; finding it may be regarded as a super-perspicacity test. See Problem 26-16 for the complete story.)

7. Prove that every natural number is either even or odd.
8. Prove that if a set A of natural numbers contains n_0 and contains $k + 1$ whenever it contains k , then A contains all natural numbers $\geq n_0$.
9. Prove the principle of mathematical induction from the well-ordering principle.
10. Prove the principle of complete induction from the ordinary principle of induction. Hint: If A contains 1 and A contains $n + 1$ whenever it contains $1, \dots, n$, consider the set B of all k such that $1, \dots, k$ are all in A .
11. (a) If a is rational and b is irrational, is $a + b$ necessarily irrational? What if a and b are both irrational?
 (b) If a is rational and b is irrational, is ab necessarily irrational? (Careful!)
 (c) Is there a number a such that a^2 is irrational, but a^4 is rational?
 (d) Are there two irrational numbers whose sum and product are both rational?
12. (a) Prove that $\sqrt{3}$, $\sqrt{5}$, and $\sqrt{6}$ are irrational. Hint: To treat $\sqrt{3}$, for example, use the fact that every integer is of the form $3n$ or $3n + 1$ or $3n + 2$. Why doesn't this proof work for $\sqrt{4}$?
 (b) Prove that $\sqrt[3]{2}$ and $\sqrt[3]{3}$ are irrational.
- *13. Prove that
 - (a) $\sqrt{2} + \sqrt{3}$ is irrational.
 - (b) $\sqrt{6} - \sqrt{2} - \sqrt{3}$ is irrational.
14. (a) Prove that if $x = p + \sqrt[q]{q}$ where p and q are rational, and m is a natural number, then $x^m = a + b\sqrt[q]{q}$ for some rational a and b .
 (b) Prove also that $(p - \sqrt[q]{q})^m = a - b\sqrt[q]{q}$.
15. (a) Prove that if m and n are natural numbers and $m^2/n^2 < 2$, then $(m + 2n)^2 / (m + n)^2 > 2$; show, moreover, that

$$\frac{(m + 2n)^2}{(m + n)^2} - 2 < 2 - \frac{m^2}{n^2}.$$
 - (b) Prove the same results with all inequality signs reversed.
 - (c) Prove that if $m/n < \sqrt{2}$, then there is another rational number m'/n' with $m/n < m'/n' < \sqrt{2}$.
- *16. It seems likely that $\sqrt[n]{n}$ is irrational whenever the natural number n is not the square of another natural number. Although the method of Problem 12 may actually be used to treat any particular case, it is not

clear in advance that it will always work, and a proof for the general case requires some extra information. A natural number p is called a **prime number** if it is impossible to write $p = ab$ for natural numbers a and b unless one of these is p , and the other 1; for convenience we also agree that 1 is *not* a prime number. The first few prime numbers are 2, 3, 5, 7, 11, 13, 17, 19. If $n > 1$ is not a prime, then $n = ab$, with a and b both $< n$; if either a or b is not a prime it can be factored similarly; continuing in this way proves that we can write n as a product of primes. For example, $28 = 4 \cdot 7 = 2 \cdot 2 \cdot 7$.

- (a) Turn this argument into a rigorous proof by complete induction. (To be sure, any reasonable mathematician would accept the informal argument, but this is partly because it would be obvious to him how to state it rigorously.)

A fundamental theorem about integers, which we will not prove here, states that this factorization is unique, except for the order of the factors. Thus, for example, 28 can never be written as a product of primes one of which is 3, nor can it be written in a way that involves 2 only once (now you should appreciate why 1 is not allowed as a prime).

- (b) Using this fact, prove that $\sqrt[n]{n}$ is irrational unless $n = m^2$ for some natural number m .
 (c) Prove more generally that $\sqrt[k]{n}$ is irrational unless $n = m^k$.
 (d) No discussion of prime numbers should fail to allude to Euclid's beautiful proof that there are infinitely many of them. Prove that there cannot be only finitely many prime numbers $p_1, p_2, p_3, \dots, p_n$ by considering $p_1 \cdot p_2 \cdot \dots \cdot p_n + 1$.

- *17. (a) Prove that if x satisfies

$$x^n + a_{n-1}x^{n-1} + \dots + a_0 = 0,$$

for some integers $a_{n-1}, \dots, a_0$, then x is irrational unless x is an integer. (Why is this a generalization of Problem 16?)

- (b) Prove that $\sqrt[3]{2} + \sqrt[3]{2}$ is irrational. Hint: Start by working out the first 6 powers of this number.

18. Prove Bernoulli's inequality: If $h > -1$, then

$$(1 + h)^n \geq 1 + nh.$$

Why is this trivial if $h > 0$?

19. The Fibonacci sequence $a_1, a_2, a_3, \dots$ is defined as follows:

$$\begin{aligned} a_1 &= 1, \\ a_2 &= 1, \\ a_n &= a_{n-1} + a_{n-2} \quad \text{for } n \geq 3. \end{aligned}$$

This sequence, which begins 1, 1, 2, 3, 5, 8, $\dots$, was discovered by Fibonacci (circa 1175–1250), in connection with a problem about

rabbits. Fibonacci assumed that an initial pair of rabbits gave birth to one new pair of rabbits per month, and that after two months each new pair behaved similarly. The number a_n of pairs born in the n th month is $a_{n-1} + a_{n-2}$, because a pair of rabbits is born for each pair born the previous month, and moreover each pair born two months ago now gives birth to another pair. The number of interesting results about this sequence is truly amazing—there is even a Fibonacci Association which publishes a journal, *The Fibonacci Quarterly*. Prove that

$$a_n = \frac{\left(\frac{1 + \sqrt{5}}{2}\right)^n - \left(\frac{1 - \sqrt{5}}{2}\right)^n}{\sqrt{5}}.$$

One way of deriving this astonishing formula is presented in Problem 23-8.

20. The result in Problem 1-6 has an important generalization: If $a_1, \dots, a_n \geq 0$, then

$$\sqrt[n]{a_1 \cdot \dots \cdot a_n} \leq \frac{a_1 + \dots + a_n}{n}.$$

- (a) Why is this true if $a_1 = \dots = a_n$? Suppose not all a_i are equal, say $a_i \neq a_j$. If a_i and a_j are both replaced by $(a_i + a_j)/2$ what happens to the “arithmetic mean” $A_n = (a_1 + \dots + a_n)/n$? What happens to the “geometric mean” $G_n = \sqrt[n]{a_1 \cdot \dots \cdot a_n}$? Why does repeating this process enough times eventually prove that $G_n \leq A_n$? (This is another place where it is a good exercise to provide a formal proof by induction, as well as informal reason.)

The reasoning in the previous proof is closely related to another interesting proof.

- (b) Using the fact that $G_n \leq A_n$ when $n = 2$, prove, by induction on k , that $G_n \leq A_n$ for $n = 2^k$.
 (c) For a general n , let $2^m > n$. Apply part (b) to the 2^m numbers

$$a_1, \dots, a_n, \underbrace{A_n, \dots, A_n}_{2^m - n \text{ times}}$$

to prove that $G_n \leq A_n$.

21. The following is a recursive definition of a^n :

$$\begin{aligned} a^1 &= a, \\ a^{n+1} &= a^n \cdot a. \end{aligned}$$

Prove, by induction, that

$$\begin{aligned} a^{n+m} &= a^n \cdot a^m, \\ (a^n)^m &= a^{nm}. \end{aligned}$$

(Don't try to be fancy: use either induction on n or induction on m , not both at once.)

22. Suppose we know properties P1 and P4 for the natural numbers, but that multiplication has never been mentioned. Then the following can be used as a recursive definition of multiplication:

$$\begin{aligned}1 \cdot b &= b, \\ (a + 1) \cdot b &= a \cdot b + b.\end{aligned}$$

Prove the following (in the order suggested!):

$$\begin{aligned}a \cdot (b + c) &= a \cdot b + a \cdot c \text{ (use induction on } a), \\ a \cdot 1 &= a, \\ a \cdot b &= b \cdot a \text{ (you just finished proving the case } b = 1).\end{aligned}$$

23. In this chapter we began with the natural numbers and gradually built up to the real numbers. A completely rigorous discussion of this process requires a little book in itself (see Part V). No one has ever figured out how to get to the real numbers without going through this process, but if we do accept the real numbers as given, then the natural numbers can be *defined* as the real numbers of the form 1 , $1 + 1$, $1 + 1 + 1$, etc. The whole point of this problem is to show that there is a rigorous mathematical way of saying "etc."

- (a) A set A of real numbers is called **inductive** if

- (1) 1 is in A ,
- (2) $k + 1$ is in A whenever k is in A .

Prove that

- (i) $\mathbf{R}$ is inductive.
 - (ii) The set of positive real numbers is inductive.
 - (iii) The set of positive real numbers unequal to $\frac{1}{2}$ is inductive.
 - (iv) The set of positive real numbers unequal to 5 is not inductive.
 - (v) If A and B are inductive, then the set C of real numbers which are in both A and B is also inductive.
- (b) A real number n will be called a **natural number** if n is in *every* inductive set.
- (i) Prove that 1 is a natural number.
 - (ii) Prove that $k + 1$ is a natural number if k is a natural number.

PART **2**
FOUNDATIONS

*The statement is so frequently made
that the differential calculus deals with
continuous magnitude, and yet
an explanation of this continuity is
nowhere given ;
even the most rigorous expositions
of the differential calculus do not base
their proofs upon continuity but,
with more or less consciousness of the fact,
they either appeal to geometric notions
or those suggested by geometry,
or depend upon theorems which are never
established in a purely arithmetic manner.
Among these, for example,
belongs the above-mentioned theorem,
and a more careful investigation
convinced me that this theorem, or
any one equivalent to it, can be regarded
in some way as a sufficient basis
for infinitesimal analysis.
It then only remained to discover its true
origin in the elements of arithmetic
and thus at the same time
to secure a real definition of
the essence of continuity.
I succeeded Nov. 24, 1858, and
a few days afterward I communicated
the results
of my meditations to my dear friend
Durège with whom I had a long
and lively discussion.*

RICHARD DEDEKIND

CHAPTER 3 FUNCTIONS

Undoubtedly the most important concept in all of mathematics is that of a function—in almost every branch of modern mathematics functions turn out to be the central objects of investigation. It will therefore probably not surprise you to learn that the concept of a function is one of great generality. Perhaps it will be a relief to learn that, for the present, we will be able to restrict our attention to functions of a very special kind; even this small class of functions will exhibit sufficient variety to engage our attention for quite some time. We will not even begin with a proper definition. For the moment a provisional definition will enable us to discuss functions at length, and will illustrate the intuitive notion of functions, as understood by mathematicians. Later, we will consider and discuss the advantages of the modern mathematical definition. Let us therefore begin with the following:

PROVISIONAL DEFINITION

A function is a rule which assigns, to each of certain real numbers, some other real number.

The following examples of functions are meant to illustrate and amplify this definition, which, admittedly, requires some such clarification.

Example 1 The rule which assigns to each number the square of that number.

Example 2 The rule which assigns to each number y the number

$$\frac{y^3 + 3y + 5}{y^2 + 1}.$$

Example 3 The rule which assigns to each number $c \neq 1, -1$ the number

$$\frac{c^3 + 3c + 5}{c^2 - 1}.$$

Example 4 The rule which assigns to each number x satisfying $-17 \leq x \leq \pi/3$ the number x^2 .

Example 5 The rule which assigns to each number a the number 0 if a is irrational, and the number 1 if a is rational.

Example 6 The rule which assigns

to 2 the number 5,
to 17 the number $\frac{36}{\pi}$,

to $\frac{\pi^2}{17}$ the number 28,

to $\frac{36}{\pi}$ the number 28,

and to any $y \neq 2, 17, \pi^2/17$, or $36/\pi$, the number 16 if y is of the form $a + b\sqrt{2}$ for a, b in $\mathbf{Q}$.

Example 7 The rule which assigns to each number t the number $t^3 + x$. (This rule depends, of course, on what the number x is, so we are really describing infinitely many different functions, one for each number x .)

Example 8 The rule which assigns to each number z the number of 7's in the decimal expansion of z , if this number is finite, and $-\pi$ if there are infinitely many 7's in the decimal expansion of z .

One thing should be abundantly clear from these examples—a function is *any* rule that assigns numbers to certain other numbers, not just a rule which can be expressed by an algebraic formula, or even by one uniform condition which applies to every number; nor is it necessarily a rule which you, or anybody else, can actually apply in practice (no one knows, for example, what rule 8 associates to π). Moreover, the rule may neglect some numbers and it may not even be clear to which numbers the function applies (try to determine, for example, whether the function in Example 6 applies to π). The set of numbers to which a function *does* apply is called the *domain* of the function.

Before saying anything else about functions we badly need some notation. Since throughout this book we shall frequently be talking about functions (indeed we shall hardly ever talk about anything else) we need a convenient way of naming functions, and of referring to functions in general. The standard practice is to denote a function by a letter. For obvious reasons the letter “ f ” is a favorite, thereby making “ g ” and “ h ” other obvious candidates, but any letter (or any reasonable symbol, for that matter) will do, not excluding “ x ” and “ y ,” although these letters are usually reserved for indicating numbers. If f is a function, then the number which f associates to a number x is denoted by $f(x)$ —this symbol is read “ f of x ” and is often called the **value of f at x** . Naturally, if we denote a function by x , some other letter must be chosen to denote the number (a perfectly legitimate, though perverse, choice would be “ f ,” leading to the symbol $x(f)$). Note that the symbol $f(x)$ makes sense only for x in the domain of f ; for other x the symbol $f(x)$ is not defined.

If the functions defined in Examples 1–8 are denoted by $f, g, h, r, s, \theta, \alpha_x$, and y , then we can rewrite their definitions as follows:

$$(1) \quad f(x) = x^2 \quad \text{for all } x.$$

$$(2) \quad g(y) = \frac{y^3 + 3y + 5}{y^2 + 1} \quad \text{for all } y.$$

$$(3) \quad h(c) = \frac{c^3 + 3c + 5}{c^2 - 1} \quad \text{for all } c \neq 1, -1.$$

$$(4) \quad r(x) = x^2 \quad \text{for all } x \text{ such that } -17 \leq x \leq \pi/3.$$

$$(5) \quad s(x) = \begin{cases} 0, & x \text{ irrational} \\ 1, & x \text{ rational.} \end{cases}$$

$$(6) \quad \theta(x) = \begin{cases} 5, & x = 2 \\ \frac{36}{\pi}, & x = 17 \\ 28, & x = \frac{\pi^2}{17} \\ 28, & x = \frac{36}{\pi} \\ 16, & x \neq 2, 17, \frac{\pi^2}{17}, \text{ or } \frac{36}{\pi}, \text{ and } x = a + b\sqrt{2} \text{ for } a, b \text{ in } \mathbf{Q}. \end{cases}$$

$$(7) \quad \alpha_x(t) = t^3 + x \quad \text{for all numbers } t.$$

$$(8) \quad y(x) = \begin{cases} n, & \text{exactly } n \text{ 7's appear in the decimal expansion of } x \\ -\pi, & \text{infinitely many 7's appear in the decimal expansion of } x. \end{cases}$$

These definitions illustrate the common procedure adopted for defining a function f —indicating what $f(x)$ is for every number x in the domain of f . (Notice that this is exactly the same as indicating $f(a)$ for every number a , or $f(b)$ for every number b , etc.) In practice, certain abbreviations are tolerated. Definition (1) could be written simply

$$(1) \quad f(x) = x^2,$$

the qualifying phrase “for all x ” being understood. Of course, for definition (4) the only possible abbreviation is

$$(4) \quad r(x) = x^2, \quad -17 \leq x \leq \pi/3.$$

It is usually understood that a definition such as

$$k(x) = \frac{1}{x} + \frac{1}{x-1}, \quad x \neq 0, 1$$

can be shortened to

$$k(x) = \frac{1}{x} + \frac{1}{x-1};$$

in other words, *unless the domain is explicitly restricted further, it is understood to consist of all numbers for which the definition makes any sense at all.*

You should have little difficulty checking the following assertions about the functions defined above:

$$f(x+1) = f(x) + 2x + 1;$$

$$g(x) = h(x) \text{ if } x^3 + 3x + 5 = 0;$$

$$r(x+1) = r(x) + 2x + 1 \text{ if } -17 \leq x \leq \frac{\pi}{3} - 1;$$

$$\begin{aligned}
s(x + y) &= s(x) \text{ if } y \text{ is rational;} \\
\theta\left(\frac{\pi^2}{17}\right) &= \theta\left(\frac{36}{\pi}\right); \\
\alpha_x(x) &= x \cdot [f(x) + 1]; \\
y\left(\frac{1}{3}\right) &= 0, y\left(\frac{7}{9}\right) = -\pi.
\end{aligned}$$

If the expression $f(s(a))$ looks unreasonable to you, then you are forgetting that $s(a)$ is a number like any other number, so that $f(s(a))$ makes sense. As a matter of fact, $f(s(a)) = s(a)$ for all a . Why? Even more complicated expressions than $f(s(a))$ are, after a first exposure, no more difficult to unravel. The expression

$$f(r(s(\theta(\alpha_3(y(\frac{1}{3})))))),$$

formidable as it appears, may be evaluated quite easily with a little patience:

$$\begin{aligned}
&f(r(s(\theta(\alpha_3(y(\frac{1}{3})))))) \\
&= f(r(s(\theta(\alpha_3(0))))) \\
&= f(r(s(\theta(3)))) \\
&= f(r(s(16))) \\
&= f(r(1)) \\
&= f(1) \\
&= 1.
\end{aligned}$$

The first few problems at the end of this chapter give further practice manipulating this symbolism.

The function defined in (1) is a rather special example of an extremely important class of functions, the polynomial functions. A function f is a **polynomial function** if there are real numbers $a_0, \dots, a_n$ such that

$$f(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_2 x^2 + a_1 x + a_0, \quad \text{for all } x$$

(when $f(x)$ is written in this form it is usually tacitly assumed that $a_n \neq 0$). The highest power of x with a nonzero coefficient is called the **degree** of f ; for example, the polynomial function f defined by $f(x) = 5x^6 + 137x^4 - \pi$ has degree 6.

The functions defined in (2) and (3) belong to a somewhat larger class of functions, the **rational functions**; these are the functions of the form p/q where p and q are polynomial functions (and q is not the function which is always 0). The rational functions are themselves quite special examples of an even larger class of functions, very thoroughly studied in calculus, which are simpler than many of the functions first mentioned in this chapter. The following are examples of this kind of function:

$$(9) \quad f(x) = \frac{x + x^2 + x \sin^2 x}{x \sin x + x \sin^2 x}.$$

$$(10) \quad f(x) = \sin(x^2).$$

$$(11) \quad f(x) = \sin(\sin(x^2)).$$

$$(12) \quad f(x) = \sin^2(\sin(\sin^2(x \sin^2 x^2))) \cdot \sin\left(\frac{x + \sin(x \sin x)}{x + \sin x}\right).$$

By what criterion, you may feel impelled to ask, can such functions, especially a monstrosity like (12), be considered simple? The answer is that they can be built up from a few simple functions using a few simple means of combining functions. In order to construct the functions (9)–(12) we need to start with the “identity function” I , for which $I(x) = x$, and the “sine function” $\sin$, whose value $\sin(x)$ at x is often written simply $\sin x$. The following are some of the important ways in which functions may be combined to produce new functions.

If f and g are any two functions, we can define a new function $f + g$, called the **sum** of f and g , by the equation

$$(f + g)(x) = f(x) + g(x).$$

Note that according to the conventions we have adopted, the domain of $f + g$ consists of all x for which “ $f(x) + g(x)$ ” makes sense, i.e., the set of all x in both domain f and domain g . If A and B are any two sets, then $A \cap B$ (read “ A intersect B ” or “the intersection of A and B ”) denotes the set of x in both A and B ; this notation allows us to write $\text{domain } (f + g) = \text{domain } f \cap \text{domain } g$.

In a similar vein, we define the **product** $f \cdot g$ and the **quotient** $\frac{f}{g}$ (or f/g) of f and g by

$$(f \cdot g)(x) = f(x) \cdot g(x)$$

and

$$\left(\frac{f}{g}\right)(x) = \frac{f(x)}{g(x)}.$$

Moreover, if g is a function and c is a number, we define a new function $c \cdot g$ by

$$(c \cdot g)(x) = c \cdot g(x).$$

This becomes a special case of the notation $f \cdot g$ if we agree that the symbol c should also represent the function f defined by $f(x) = c$; such a function, which has the same value for all numbers x , is called a **constant function**.

The domain of $f \cdot g$ is $\text{domain } f \cap \text{domain } g$, and the domain of $c \cdot g$ is simply the domain of g . On the other hand, the domain of f/g is rather complicated—it may be written $\text{domain } f \cap \text{domain } g \cap \{x: g(x) \neq 0\}$, the symbol $\{x: g(x) \neq 0\}$ denoting the set of numbers x such that $g(x) \neq 0$. In general, $\{x: \dots\}$ denotes the set of all x such that “ $\dots$ ” is true. Thus $\{x: x^3 + 3 < 11\}$ denotes the set of all numbers x such that $x^3 < 8$, and consequently $\{x: x^3 + 3 < 11\} = \{x: x < 2\}$. Either of these symbols could just as well have been written using y everywhere instead of x . Variations of this notation are common, but hardly require any discussion. Any one can guess that $\{x > 0: x^3 < 8\}$ denotes the set of positive numbers whose cube is less than 8; it could be expressed more formally as $\{x: x > 0 \text{ and } x^3 < 8\}$.

Incidentally, this set is equal to the set $\{x: 0 < x < 2\}$. One variation is slightly less transparent, but very standard. The set $\{1, 3, 2, 4\}$, for example, contains just the four numbers 1, 2, 3, and 4; it can also be denoted by $\{x: x = 1 \text{ or } x = 3 \text{ or } x = 2 \text{ or } x = 4\}$.

Certain facts about the sum, product, and quotient of functions are obvious consequences of facts about sums, products, and quotients of numbers. For example, it is very easy to prove that

$$(f + g) + h = f + (g + h).$$

The proof is characteristic of almost every proof which demonstrates that two functions are equal—the two functions must be shown to have the same domain, and the same value at any number in the domain. For example, to prove that $(f + g) + h = f + (g + h)$, note that unraveling the definition of the two sides gives

$$\begin{aligned} [(f + g) + h](x) &= (f + g)(x) + h(x) \\ &= [f(x) + g(x)] + h(x) \end{aligned}$$

and

$$\begin{aligned} [f + (g + h)](x) &= f(x) + (g + h)(x) \\ &= f(x) + [g(x) + h(x)], \end{aligned}$$

and the equality of $[f(x) + g(x)] + h(x)$ and $f(x) + [g(x) + h(x)]$ is a fact about numbers. In this proof the equality of the two domains was not explicitly mentioned because this is obvious, as soon as we begin to write down these equations; the domain of $(f + g) + h$ and of $f + (g + h)$ is clearly domain $f \cap \text{domain } g \cap \text{domain } h$. We naturally write $f + g + h$ for $(f + g) + h = f + (g + h)$, precisely as we did for numbers.

It is just as easy to prove that $(f \cdot g) \cdot h = f \cdot (g \cdot h)$, and this function is denoted by $f \cdot g \cdot h$. The equations $f + g = g + f$ and $f \cdot g = g \cdot f$ should also present no difficulty.

Using the operations $+$, $\cdot$, $/$ we can now express the function f defined in (9) by

$$f = \frac{I + I \cdot I + I \cdot \sin \cdot \sin}{I \cdot \sin + I \cdot \sin \cdot \sin}.$$

It should be clear, however, that we cannot express function (10) this way. We require yet another way of combining functions. This combination, the composition of two functions, is by far the most important.

If f and g are any two functions, we define a new function $f \circ g$, the **composition** of f and g , by

$$(f \circ g)(x) = f(g(x));$$

the domain of $f \circ g$ is $\{x: x \text{ is in domain } g \text{ and } g(x) \text{ is in domain } f\}$. The symbol " $f \circ g$ " is often read " f circle g ." Compared to the phrase "the composition of f and g " this has the advantage of brevity, of course, but there is another advantage of far greater import: there is much less chance of confusing

$f \circ g$ with $g \circ f$, and these must *not* be confused, since they are not usually equal; in fact, almost any f and g chosen at random will illustrate this point (try $f = I \cdot I$ and $g = \sin$, for example). Lest you become too apprehensive about the operation of composition, let us hasten to point out that composition *is* associative:

$$(f \circ g) \circ h = f \circ (g \circ h)$$

(and the proof is a triviality); this function is denoted by $f \circ g \circ h$. We can now write the functions (10), (11), (12) as

$$(10) \quad f = \sin \circ (I \cdot I),$$

$$(11) \quad f = \sin \circ \sin \circ (I \cdot I),$$

$$(12) \quad f = (\sin \cdot \sin) \circ \sin \circ (\sin \cdot \sin) \circ (I \cdot [(\sin \cdot \sin) \circ (I \cdot I)]) \cdot \sin \circ \left(\frac{I + \sin \circ (I \cdot \sin)}{I + \sin} \right).$$

One fact has probably already become clear. Although this method of writing functions reveals their “structure” very clearly, it is hardly short or convenient. The shortest name for the function f such that $f(x) = \sin(x^2)$ for all x unfortunately seems to be “the function f such that $f(x) = \sin(x^2)$ for all x .” The need for abbreviating this clumsy description has been clear for two hundred years, but no reasonable abbreviation has received universal acclaim. At present the strongest contender for this honor is something like

$$x \rightarrow \sin(x^2)$$

(read “ x goes to $\sin(x^2)$ ” or just “ x arrow $\sin(x^2)$ ”), but it is hardly popular among writers of calculus textbooks. In this book we will tolerate a certain amount of ellipsis, and speak of “the function $f(x) = \sin(x^2)$.” Even more popular is the quite drastic abbreviation: “the function $\sin(x^2)$.” For the sake of precision we will never use this description, which, strictly speaking, confuses a number and a function, but it is so convenient that you will probably end up adopting it for personal use. As with any convention, utility is the motivating factor, and this criterion is reasonable so long as the slight logical deficiencies cause no confusion. On occasion, confusion *will* arise unless a more precise description is used. For example, “the function $x + t^3$ ” is an ambiguous phrase; it could mean either

$$x \rightarrow x + t^3, \text{ i.e., the function } f \text{ such that } f(x) = x + t^3 \text{ for all } x$$

or

$$t \rightarrow x + t^3, \text{ i.e., the function } f \text{ such that } f(t) = x + t^3 \text{ for all } t.$$

As we shall see, however, for many important concepts associated with functions, calculus has a notation which contains the “ $x \rightarrow$ ” built in.

By now we have made a sufficiently extensive investigation of functions to warrant reconsidering our definition. We have defined a function as a “rule,” but it is hardly clear what this means. If we ask “What happens if you break this rule?” it is not easy to say whether this question is merely facetious or

actually profound. A more substantial objection to the use of the word “rule” is that

$$f(x) = x^2$$

and

$$f(x) = x^2 + 3x + 3 - 3(x + 1)$$

are certainly *different* rules, if by a rule we mean the actual instructions given for determining $f(x)$; nevertheless, we want

$$f(x) = x^2$$

and

$$f(x) = x^2 + 3x + 3 - 3(x + 1)$$

to define the same function. For this reason, a function is sometimes defined as an “association” between numbers; unfortunately the word “association” escapes the objections raised against “rule” only because it is even more vague.

There is, of course, a satisfactory way of defining functions, or we should never have gone to the trouble of criticizing our original definition. But a satisfactory definition can never be constructed by finding synonyms for English words which are troublesome. The definition which mathematicians have finally accepted for “function” is a beautiful example of the means by which intuitive ideas have been incorporated into rigorous mathematics. The correct question to ask about a function is not “What is a rule?” or “What is an association?” but “What does one have to know about a function in order to know all about it?” The answer to the last question is easy—for each number x one needs to know the number $f(x)$; we can imagine a table which would display all the information one could desire about the function $f(x) = x^2$:

x	$f(x)$
1	1
−1	1
2	4
−2	4
$\sqrt{2}$	2
$-\sqrt{2}$	2
π	π^2
$-\pi$	π^2

It is not even necessary to arrange the numbers in a table (which would actually be impossible if we wanted to list all of them). Instead of a two column array we can consider various pairs of numbers

$$(1, 1), (-1, 1), (2, 4), (-2, 4), (\pi, \pi^2), (\sqrt{2}, 2), \dots$$

simply collected together into a set.* To find $f(1)$ we simply take the second number of the pair whose first member is 1; to find $f(\pi)$ we take the second number of the pair whose first member is π . We seem to be saying that a function might as well be defined as a collection of pairs of numbers. For example, if we were given the following collection (which contains just 5 pairs):

$$f = \{(1, 7), (3, 7), (5, 3), (4, 8), (8, 4)\},$$

then $f(1) = 7, f(3) = 7, f(5) = 3, f(4) = 8, f(8) = 4$ and 1, 3, 4, 5, 8 are the only numbers in the domain of f . If we consider the collection

$$f = \{(1, 7), (3, 7), (2, 5), (1, 8), (8, 4)\},$$

then $f(3) = 7, f(2) = 5, f(8) = 4$; but it is impossible to decide whether $f(1) = 7$ or $f(1) = 8$. In other words, a function cannot be defined to be any old collection of pairs of numbers; we must rule out the possibility which arose in this case. We are therefore led to the following definition.

DEFINITION

A **function** is a collection of pairs of numbers with the following property: if (a, b) and (a, c) are both in the collection, then $b = c$; in other words, the collection must not contain two different pairs with the same first element.

This is our first full-fledged definition, and illustrates the format we shall always use to define significant new concepts. These definitions are so important (at least as important as theorems) that it is essential to know when one is actually at hand, and to distinguish them from comments, motivating remarks, and casual explanations. They will be preceded by the word **DEFINITION**, contain the term being defined in boldface letters, and constitute a paragraph unto themselves.

There is one more definition (actually defining two things at once) which can now be made rigorously:

DEFINITION

If f is a function, the **domain** of f is the set of all a for which there is some b such that (a, b) is in f . If a is in the domain of f , it follows from the definition of a function that there is, in fact, a *unique* number b such that (a, b) is in f . This unique b is denoted by $f(a)$.

With this definition we have reached our goal: the important thing about a function f is that a number $f(x)$ is determined for each number x in its domain. You may feel that we have also reached the point where an intuitive definition has been replaced by an abstraction with which the mind can hardly grapple. Two consolations may be offered. First, although a function

* The pairs occurring here are often called "ordered pairs," to emphasize that, for example, $(2, 4)$ is not the same pair as $(4, 2)$. It is only fair to warn that we are going to define functions in terms of ordered pairs, another undefined term. Ordered pairs *can* be defined, however, and an appendix to this chapter has been provided for skeptics.

has been defined as a collection of pairs, there is nothing to stop you from *thinking* of a function as a rule. Second, neither the intuitive nor the formal definition indicates the best way of thinking about functions. The best way is to draw pictures; but this requires a chapter all by itself.

PROBLEMS

1. Let $f(x) = 1/(1+x)$. What is
 - (i) $f(f(x))$ (for which x does this make sense?).
 - (ii) $f\left(\frac{1}{x}\right)$.
 - (iii) $f(cx)$.
 - (iv) $f(x+y)$.
 - (v) $f(x) + f(y)$.
 - (vi) For which numbers c is there a number x such that $f(cx) = f(x)$.
Hint: There are a lot more than you might think at first glance.
 - (vii) For which numbers c is it true that $f(cx) = f(x)$ for two different numbers x ?

2. Let $g(x) = x^2$, and let

$$h(x) = \begin{cases} 0, & x \text{ rational} \\ 1, & x \text{ irrational.} \end{cases}$$

- (i) For which y is $h(y) \leq y$?
 - (ii) For which y is $h(y) \leq g(y)$?
 - (iii) What is $g(h(z)) - h(z)$?
 - (iv) For which w is $g(w) \leq w$?
 - (v) For which ε is $g(g(\varepsilon)) = g(\varepsilon)$?
3. Find the domain of the functions defined by the following formulas.
 - (i) $f(x) = \sqrt{1-x^2}$.
 - (ii) $f(x) = \sqrt{1-\sqrt{1-x^2}}$.
 - (iii) $f(x) = \frac{1}{x-1} + \frac{1}{x-2}$.
 - (iv) $f(x) = \sqrt{1-x^2} + \sqrt{x^2-1}$.
 - (v) $f(x) = \sqrt{1-x} + \sqrt{x-2}$.
4. Let $S(x) = x^2$, let $P(x) = 2^x$, and let $s(x) = \sin x$. Find each of the following. In each case your answer should be a *number*.
 - (i) $(S \circ P)(y)$.
 - (ii) $(S \circ s)(y)$.
 - (iii) $(S \circ P \circ s)(t) + (s \circ P)(t)$.
 - (iv) $s(t^3)$.
5. Express each of the following functions in terms of S , P , s , using only $+$, $\cdot$, and $\circ$ (for example, the answer to (i) is $P \circ s$). In each case your

answer should be a *function*.

- (i) $f(x) = 2^{\sin x}$.
- (ii) $f(x) = \sin 2^x$.
- (iii) $f(x) = \sin x^2$.
- (iv) $f(x) = \sin^2 x$ (remember that $\sin^2 x$ is an abbreviation for $(\sin x)^2$).
- (v) $f(t) = 2^t$. (Note: a^{b^c} always means $a^{(b^c)}$; this convention is adopted because $(a^b)^c$ can be written more simply as a^{bc} .)
- (vi) $f(u) = \sin(2^u + 2^{u^2})$.
- (vii) $f(y) = \sin(\sin(\sin(2^{2^{\sin y}})))$.
- (viii) $f(a) = 2^{\sin^2 a} + \sin(a^2) + 2^{\sin(a^2 + \sin a)}$.

Polynomial functions, because they are simple, yet flexible, occupy a favored role in most investigations of functions. The following two problems illustrate their flexibility, and guide you through a derivation of their most important elementary properties.

6. (a) If $x_1, \dots, x_n$ are distinct numbers, find a polynomial function f_i of degree $n - 1$ which is 1 at x_i and 0 at x_j for $j \neq i$. Hint: The product of all $(x - x_i)$ for $j \neq i$, is 0 at x_j if $j \neq i$. (This product is usually denoted by

$$\prod_{\substack{j=1 \\ j \neq i}}^n (x - x_j),$$

the symbol Π (capital pi) playing the same role for products that Σ plays for sums.)

- (b) Now find a polynomial function f of degree $n - 1$ such that $f(x_i) = a_i$, where $a_1, \dots, a_n$ are given numbers. (You should use the functions f_i from part (a). The formula you will obtain is called the "Lagrange interpolation formula.")
7. (a) Prove that for any polynomial function f , and any number a , there is a polynomial function g , and a number b , such that $f(x) = (x - a)g(x) + b$ for all x . (The idea is simply to divide $(x - a)$ into $f(x)$ by long division, until a constant remainder is left. For example, the calculation

$$\begin{array}{r}
 \begin{array}{r} x^2 \quad +x \quad -2 \\ x-1 \overline{) x^3 } \end{array} \\
 \underline{-x^2 } \\
 \begin{array}{r} x^2 \quad -3x \\ x-1 \overline{) x^2 } \end{array} \\
 \underline{-x^2 } \\
 \begin{array}{r} -2x + 1 \\ x-1 \overline{) -2x + 1} \end{array} \\
 \underline{-2x + 2} \\
 -1
 \end{array}$$

shows that $x^3 - 3x + 1 = (x - 1)(x^2 + x - 2) - 1$. A formal proof is possible by induction on the degree of f .)

- (b) Prove that if $f(a) = 0$, then $f(x) = (x - a)g(x)$ for some polynomial function g . (The converse is obvious.)
- (c) Prove that if f is a polynomial function of degree n , then f has at most n roots, i.e., there are at most n numbers a with $f(a) = 0$.
- (d) Show that for each n there is a polynomial function of degree n with n roots. If n is even find a polynomial function of degree n with no roots, and if n is odd find one with only one root.
8. For which numbers a , b , c , and d will the function

$$f(x) = \frac{ax + d}{cx + b}$$

satisfy $f(f(x)) = x$ for all x ?

9. (a) If A is any set of real numbers, define a function C_A as follows:

$$C_A(x) = \begin{cases} 1, & x \text{ in } A \\ 0, & x \text{ not in } A. \end{cases}$$

Find expressions for $C_{A \cap B}$ and $C_{A \cup B}$ and $C_{\mathbf{R} - A}$, in terms of C_A and C_B . (The symbol $A \cap B$ was defined in this chapter, but the other two may be new to you. They can be defined as follows:

$$\begin{aligned} A \cup B &= \{x: x \text{ is in } A \text{ or } x \text{ is in } B\}, \\ \mathbf{R} - A &= \{x: x \text{ is in } \mathbf{R} \text{ but } x \text{ is not in } A\}. \end{aligned}$$

- (b) Suppose f is a function such that $f(x) = 0$ or 1 for each x . Prove that there is a set A such that $f = C_A$.
- (c) Show that $f = f^2$ if and only if $f = C_A$ for some set A .
10. (a) For which functions f is there a function g such that $f = g^2$? Hint: You can certainly answer this question if "function" is replaced by "number."
- (b) For which functions f is there a function g such that $f = 1/g$?
- *(c) For which functions b and c can we find a function x such that

$$(x(t))^2 + b(t)x(t) + c(t) = 0$$

for all numbers t ?

- *(d) What conditions must the functions a and b satisfy if there is to be a function x such that

$$a(t)x(t) + b(t) = 0$$

for all numbers t ? How many such functions x will there be?

11. (a) Suppose that H is a function and y is a number such that $H(H(y)) = y$. What is

$$\underbrace{H(H(H(\cdots (H(y) \cdots)))}_{80 \text{ times}}?$$

- (b) Same question if 80 is replaced by 81.
- (c) Same question if $H(H(y)) = H(y)$.
- * (d) Find a function H such that $H(H(x)) = H(x)$ for all numbers x , and such that $H(1) = 36$, $H(2) = \pi/3$, $H(13) = 47$, $H(36) = 36$, $H(\pi/3) = \pi/3$, $H(47) = 47$. (Don't try to "solve" for $H(x)$; there are many functions H with $H(H(x)) = H(x)$. The extra conditions on H are supposed to suggest a way of finding a suitable H .)
- * (e) Find a function H such that $H(H(x)) = H(x)$ for all x , and such that $H(1) = 7$, $H(17) = 18$.
12. A function f is **even** if $f(x) = f(-x)$ and **odd** if $f(x) = -f(-x)$. For example, f is even if $f(x) = x^2$ or $f(x) = |x|$ or $f(x) = \cos x$, while f is odd if $f(x) = x$ or $f(x) = \sin x$.
- (a) Determine whether $f + g$ is even, odd, or not necessarily either, in the four cases obtained by choosing f even or odd, and g even or odd. (Your answers can most conveniently be displayed in a 2×2 table.)
- (b) Do the same for $f \cdot g$.
- (c) Do the same for $f \circ g$.
- (d) Prove that every even function f can be written $f(x) = g(|x|)$, for infinitely many functions g .
- *13. (a) Prove that any function f with domain $\mathbf{R}$ can be written $f = E + O$, where E is even and O is odd.
- (b) Prove that this way of writing f is unique. (If you try to do part (b) first, by "solving" for E and O you will probably find the solution to part (a).)
14. If f is any function, define a new function $|f|$ by $|f|(x) = |f(x)|$. If f and g are functions, define two new functions, $\max(f, g)$ and $\min(f, g)$, by

$$\begin{aligned}\max(f, g)(x) &= \max(f(x), g(x)), \\ \min(f, g)(x) &= \min(f(x), g(x)).\end{aligned}$$

Find an expression for $\max(f, g)$ and $\min(f, g)$ in terms of $|f|$.

15. (a) Show that $f = \max(f, 0) + \min(f, 0)$. This particular way of writing f is fairly useful; the functions $\max(f, 0)$ and $\min(f, 0)$ are called the **positive** and **negative parts** of f .
- (b) A function f is called **nonnegative** if $f(x) \geq 0$ for all x . Prove that any function f can be written $f = g - h$, where g and h are nonnegative, in infinitely many ways. (The "standard way" is $g = \max(f, 0)$ and $h = -\min(f, 0)$.) Hint: Any number can certainly be written as the difference of two nonnegative numbers in infinitely many ways.
- *16. Suppose f satisfies $f(x + y) = f(x) + f(y)$ for all x and y .
- (a) Prove that $f(x_1 + \cdots + x_n) = f(x_1) + \cdots + f(x_n)$.

- (b) Prove that there is some number c such that $f(x) = cx$ for all *rational* numbers x (at this point we're not trying to say anything about $f(x)$ for irrational x). Hint: First figure out what c must be. Now prove that $f(x) = cx$, first when x is a natural number, then when x is an integer, then when x is the reciprocal of an integer and, finally, for all rational x .
- *17.** If $f(x) = 0$ for all x , then f satisfies $f(x + y) = f(x) + f(y)$ for all x and y , and also $f(x \cdot y) = f(x) \cdot f(y)$ for all x and y . Now suppose that f satisfies these two properties, but that $f(x)$ is not always 0. Prove that $f(x) = x$ for all x , as follows:
- Prove that $f(1) = 1$.
 - Prove that $f(x) = x$ if x is rational.
 - Prove that $f(x) > 0$ if $x > 0$. (This part is tricky, but if you have been paying attention to the philosophical remarks accompanying the problems in the last two chapters, you will know what to do.)
 - Prove that $f(x) > f(y)$ if $x > y$.
 - Prove that $f(x) = x$ for all x . Hint: Use the fact that between any two numbers there is a rational number.
- *18.** Precisely what conditions must f , g , h , and k satisfy in order that $f(x)g(y) = h(x)k(y)$ for all x and y ?
- *19.** (a) Prove that there do *not* exist functions f and g with either of the following properties:
- $f(x) + g(y) = xy$ for all x and y .
 - $f(x + y) = g(x) - y$ for all x and y .
- Hint: Try to get some information about f or g by choosing particular values of x and y .
- (b) Find functions f and g such that $f(x + y) = g(xy)$ for all x and y .
- *20.** (a) Find a function f , other than a constant function, such that $|f(y) - f(x)| \leq |y - x|$.
- (b) Suppose that $f(y) - f(x) \leq (y - x)^2$ for all x and y . (Why does this imply that $|f(y) - f(x)| \leq (y - x)^2$?) Prove that f is a constant function. Hint: Divide the interval $[x, y]$ into n equal pieces.
- 21.** Prove or give a counterexample for each of the following assertions:
- $f \circ (g + h) = f \circ g + f \circ h$.
 - $(g + h) \circ f = g \circ f + h \circ f$.
 - $\frac{1}{f \circ g} = \frac{1}{f} \circ g$.
 - $\frac{1}{f \circ g} = f \circ \left(\frac{1}{g}\right)$.

22. (a) Suppose $g = h \circ f$. Prove that if $f(x) = f(y)$, then $g(x) = g(y)$.
 (b) Conversely, suppose that f and g are two functions such that $g(x) = g(y)$ whenever $f(x) = f(y)$. Prove that $g = h \circ f$ for some function h . Hint: Just try to define $h(z)$ when z is of the form $z = f(x)$ (these are the only z that matter) and use the hypotheses to show that your definition will not run into trouble.
23. Suppose that $f \circ g = I$, where $I(x) = x$. Prove that
 (a) if $x \neq y$, then $g(x) \neq g(y)$;
 (b) every number b can be written $b = f(a)$ for some number a .
- *24. (a) Suppose g is a function with the property that $g(x) \neq g(y)$ if $x \neq y$. Prove that there is a function f such that $f \circ g = I$.
 (b) Suppose that f is a function such that every number b can be written $b = f(a)$ for some number a . Prove that there is a function g such that $f \circ g = I$.
- *25. Find a function f such that $g \circ f = I$ for some g , but such that there is no function h with $f \circ h = I$.
- *26. Suppose $f \circ g = I$ and $h \circ f = I$. Prove that $g = h$. Hint: Use the fact that composition is associative.
27. (a) Suppose $f(x) = x + 1$. Are there any functions g such that $f \circ g = g \circ f$?
 (b) Suppose f is a constant function. For which functions g does $f \circ g = g \circ f$?
 (c) Suppose that $f \circ g = g \circ f$ for all functions g . Show that f is the identity function, $f(x) = x$.
28. (a) Let F be the set of all functions whose domain is $\mathbf{R}$. Prove that, using $+$ and $\cdot$ as defined in this chapter, all of properties P1–P9 except P7 hold for F , provided 0 and 1 are interpreted as constant functions.
 (b) Show that P7 does not hold.
 *(c) Show that P10–P12 cannot hold. In other words, show that there is no collection P of functions in F , such that P10–P12 hold for P . (It is sufficient, and will simplify things, to consider only functions which are 0 except at two points x_0 and x_1 .)
 (d) Suppose we define $f < g$ to mean that $f(x) < g(x)$ for all x . Which of P'10–P'13 (in Problem 1-8) now hold?
 (e) If $f < g$, is $h \circ f < h \circ g$? Is $f \circ h < g \circ h$?

APPENDIX. ORDERED PAIRS

Not only in the definition of a function, but in other parts of the book as well, it is necessary to use the notion of an ordered pair of objects. A definition has not yet been given, and we have never even stated explicitly what properties an ordered pair is supposed to have. The one property which we will require states formally that the ordered pair (a, b) should be determined by a and b , and the order in which they are given:

$$\text{if } (a, b) = (c, d), \text{ then } a = c \text{ and } b = d.$$

Ordered pairs may be treated most conveniently by simply introducing (a, b) as an undefined term and adopting the basic property as an axiom—since this property is the only significant fact about ordered pairs, there is not much point worrying about what an ordered pair “really” is. Those who find this treatment satisfactory need read no further.

The rest of this short appendix is for the benefit of those readers who will feel uncomfortable unless ordered pairs are somehow defined so that this basic property becomes a theorem. There is no point in restricting our attention to ordered pairs of numbers; it is just as reasonable, and just as important, to have available the notion of an ordered pair of any two mathematical objects. This means that our definition ought to involve only concepts common to all branches of mathematics. The one common concept which pervades all areas of mathematics is that of a set, and ordered pairs (like everything else in mathematics) can be defined in this context; an ordered pair will turn out to be a set of a rather special sort.

The set $\{a, b\}$, containing the two elements a and b , is an obvious first choice, but will not do as a definition for (a, b) , because there is no way of determining from $\{a, b\}$ which of a or b is meant to be the first element. A more promising candidate is the rather startling set:

$$\{\{a\}, \{a, b\}\}.$$

This set has two members, both of which are *themselves* sets; one member is the set $\{a\}$, containing the single member a , the other is the set $\{a, b\}$. Shocking as it may seem, we are going to define (a, b) to be this set. The justification for this choice is given by the theorem immediately following the definition—the definition works, and there really isn’t anything else worth saying.

DEFINITION

$$(a, b) = \{\{a\}, \{a, b\}\}.$$

THEOREM 1

If $(a, b) = (c, d)$, then $a = c$ and $b = d$.

PROOF

The hypothesis means that

$$\{\{a\}, \{a, b\}\} = \{\{c\}, \{c, d\}\}.$$

Now $\{\{a\}, \{a, b\}\}$ contains just two members, $\{a\}$ and $\{a, b\}$; and a is the only common element of these two members of $\{\{a\}, \{a, b\}\}$. Similarly, c is the unique common member of both members of $\{\{c\}, \{c, d\}\}$. Therefore $a = c$. We therefore have

$$\{\{a\}, \{a, b\}\} = \{\{a\}, \{a, d\}\},$$

and only the proof that $b = d$ remains. It is convenient to distinguish 2 cases.

Case 1. $b = a$. In this case, $\{a, b\} = \{a\}$, so the set $\{\{a\}, \{a, b\}\}$ really has only one member, namely, $\{a\}$. The same must be true of $\{\{a\}, \{a, d\}\}$, so $\{a, d\} = \{a\}$, which implies that $d = a = b$.

Case 2. $b \neq a$. In this case, b is in one member of $\{\{a\}, \{a, b\}\}$ but not in the other. It must therefore be true that b is in one member of $\{\{a\}, \{a, d\}\}$ but not in the other. This can happen only if b is in $\{a, d\}$, but b is not in $\{a\}$; thus $b = a$ or $b = d$, but $b \neq a$; so $b = d$. ■

4

CHAPTER GRAPHS

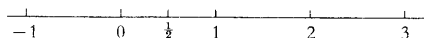


FIGURE 1

Mention the real numbers to a mathematician and the image of a straight line will probably form in his mind, quite involuntarily. And most likely he will neither banish nor too eagerly embrace this mental picture of the real numbers. “Geometric intuition” will allow him to interpret statements about numbers in terms of this picture, and may even suggest methods of proving them. Although the properties of the real numbers which were studied in Part I are not greatly illuminated by a geometric picture, such an interpretation will be a great aid in Part II.

You are probably already familiar with the conventional method of considering the straight line as a picture of the real numbers, i.e., of associating to each real number a point on a line. To do this (Figure 1) we pick, arbitrarily, a point which we label 0, and a point to the right, which we label 1. The point twice as far to the right is labeled 2, the point the same distance from 0 to 1, but to the left of 0, is labeled -1 , etc. With this arrangement, if $a < b$, then the point corresponding to a lies to the left of the point corresponding to b . We can also draw rational numbers, such as $\frac{1}{2}$, in the obvious way. It is usually taken for granted that the irrational numbers also somehow fit into this scheme, so that every real number can be drawn as a point on the line. We will not make too much fuss about justifying this assumption, since this method of “drawing” numbers is intended solely as a method of picturing certain abstract ideas, and our proofs will never rely on these pictures (although we will frequently use a picture to suggest or help explain a proof). Because this geometric picture plays such a prominent, albeit inessential role, geometric terminology is frequently employed when speaking of numbers—thus a number is sometimes called a *point*, and $\mathbf{R}$ is often called the *real line*.

The number $|a - b|$ has a simple interpretation in terms of this geometric picture: it is the distance between a and b , the length of the line segment which has a as one end point and b as the other. This means, to choose an example whose frequent occurrence justifies special consideration, that the set of numbers x which satisfy $|x - a| < \epsilon$ may be pictured as the collection of points whose distance from a is less than ϵ . This set of points is the “interval” from $a - \epsilon$ to $a + \epsilon$, which may also be described as the points corresponding to numbers x with $a - \epsilon < x < a + \epsilon$ (Figure 2).

Sets of numbers which correspond to intervals arise so frequently that it is desirable to have special names for them. The set $\{x: a < x < b\}$ is denoted by (a, b) and called the **open interval** from a to b . This notation naturally creates some ambiguity, since (a, b) is also used to denote a pair of numbers, but in context it is always clear (or can easily be made clear) whether one is talking about a pair or an interval. Note that if $a \geq b$, then $(a, b) = \emptyset$, the set



FIGURE 2

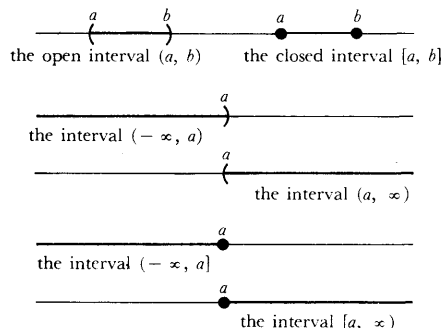


FIGURE 3

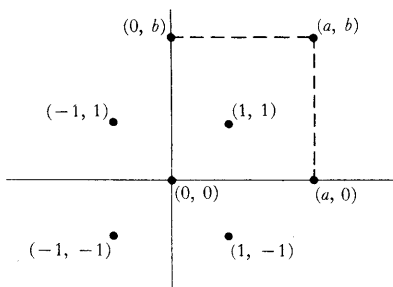


FIGURE 4

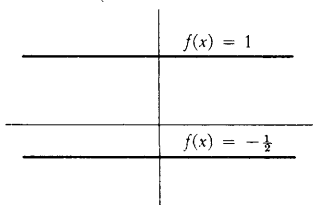


FIGURE 5

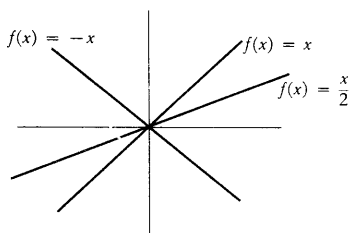


FIGURE 6

with no elements; in practice, however, it is almost always assumed (explicitly if one has been careful, and implicitly otherwise), that whenever an interval (a, b) is mentioned, the number a is less than b .

The set $\{x: a \leq x \leq b\}$ is denoted by $[a, b]$ and is called the **closed interval** from a to b . This symbol is usually reserved for the case $a < b$, but it is sometimes used for $a = b$, also. The usual pictures for the intervals (a, b) and $[a, b]$ are shown in Figure 3; since no reasonably accurate picture could ever indicate the difference between the two intervals, various conventions have been adopted. Figure 3 also shows certain "infinite" intervals. The set $\{x: x > a\}$ is denoted by (a, ∞) , while the set $\{x: x \geq a\}$ is denoted by $[a, \infty)$; the sets $(-\infty, a)$ and $(-\infty, a]$ are defined similarly. At this point a standard warning must be issued: the symbols ∞ and $-\infty$, though usually read "infinity" and "minus infinity," are *purely* suggestive; there is no number " ∞ " which satisfies $\infty \geq a$ for all numbers a . While the symbols ∞ and $-\infty$ will appear in many contexts, it is always necessary to define these uses in ways that refer only to numbers. The set $\mathbf{R}$ of all real numbers is also considered to be an "interval," and is sometimes denoted by $(-\infty, \infty)$.

Of even greater interest to us than the method of drawing numbers is a method of drawing pairs of numbers. This procedure, probably also familiar to you, requires a "coordinate system," two straight lines intersecting at right angles. To distinguish these straight lines, we call one the *horizontal axis*, and one the *vertical axis*. (More prosaic terminology, such as the "first" and "second" axes, is probably preferable from a logical point of view, but most people hold their books, or at least their blackboards, in the same way, so that "horizontal" and "vertical" are more descriptive.) Each of the two axes could be labeled with real numbers, but we can also label points on the horizontal axis with pairs $(a, 0)$ and points on the vertical axis with pairs $(0, b)$, so that the intersection of the two axes, the "origin" of the coordinate system, is labeled $(0, 0)$. Any point (a, b) can now be drawn as in Figure 4, lying at the vertex of the rectangle whose other three vertices are labeled $(0, 0)$, $(a, 0)$, and $(0, b)$. The numbers a and b are called the *first* and *second coordinates*, respectively, of the point determined in this way.

Our real concern, let us recall, is a method of drawing functions. Since a function is just a collection of pairs of numbers, we can draw a function by drawing each of the pairs in the function. The drawing obtained in this way is called the **graph** of the function. In other words, the graph of f contains all the points corresponding to pairs $(x, f(x))$. Since most functions contain infinitely many pairs, drawing the graph promises to be a laborious undertaking, but, in fact, many functions have graphs which are quite easy to draw.

Not surprisingly, the simplest functions of all, the constant functions $f(x) = c$, have the simplest graphs. It is easy to see that the graph of the function $f(x) = c$ is a straight line parallel to the horizontal axis, at distance c from it (Figure 5).

The functions $f(x) = cx$ also have particularly simple graphs—straight lines through $(0, 0)$, as in Figure 6. A proof of this fact is indicated in Figure 7:

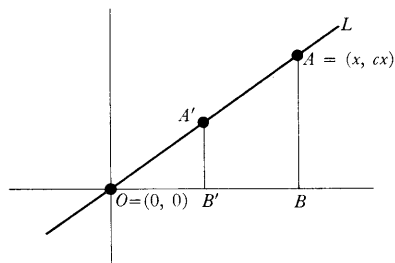


FIGURE 7

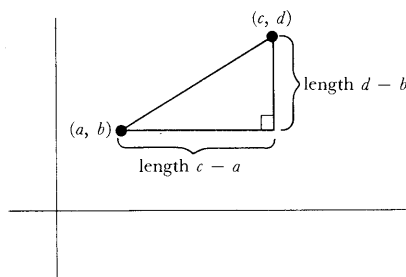


FIGURE 8

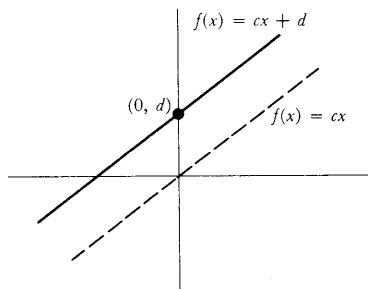


FIGURE 9

Let x be some number not equal to 0, and let L be the straight line which passes through the origin O , corresponding to $(0, 0)$, and through the point A , corresponding to (x, cx) . A point A' , with first coordinate y , will lie on L when the triangle $A'B'O$ is similar to the triangle ABO , thus when

$$\frac{A'B'}{OB'} = \frac{AB}{OB} = c;$$

this is precisely the condition that A' correspond to the pair (y, cy) , i.e., that A' lies on the graph of f . The argument has implicitly assumed that $c > 0$, but the other cases are treated easily enough. The number c , which measures the ratio of the sides of the triangles appearing in the proof, is called the *slope* of the straight line, and a line parallel to this line is also said to have slope c .

This demonstration has neither been labeled nor treated as a formal proof. Indeed, a rigorous demonstration would necessitate a digression which we are not at all prepared to follow. The rigorous proof of *any* statement connecting geometric and algebraic concepts would first require a real proof (or a precisely stated assumption) that the points on a straight line correspond in an exact way to the real numbers. Aside from this, it would be necessary to develop plane geometry as precisely as we intend to develop the properties of real numbers. Now the detailed development of plane geometry is a beautiful subject, but it is by no means a prerequisite for the study of calculus. We shall use geometric pictures only as an aid to intuition; for our purposes (and for most of mathematics) it is perfectly satisfactory to *define* the plane to be the set of all pairs of real numbers, and to *define* straight lines as certain collections of pairs, including, among others, the collections $\{(x, cx): x \text{ a real number}\}$. To provide this artificially constructed geometry with all the structure of geometry studied in high school, one more definition is required. If (a, b) and (c, d) are two points in the plane, i.e., pairs of real numbers, we *define* the **distance** between (a, b) and (c, d) to be

$$\sqrt{(a - c)^2 + (b - d)^2}.$$

If the motivation for this definition is not clear, Figure 8 should serve as adequate explanation—with this definition the Pythagorean theorem has been built into our geometry.*

Reverting once more to our informal geometric picture, it is not hard to see (Figure 9) that the graph of the function $f(x) = cx + d$ is a straight line with slope c , passing through the point $(0, d)$. For this reason, the functions $f(x) = cx + d$ are called **linear functions**. Simple as they are, linear functions occur frequently, and you should feel comfortable working with them. The following is a typical problem whose solution should not cause any trouble. Given two distinct points (a, b) and (c, d) , find the linear function f whose

* The fastidious reader might object to this definition on the grounds that nonnegative numbers are not yet known to have square roots. This objection is really unanswerable at the moment—the definition will just have to be accepted with reservations, until this little point is settled.

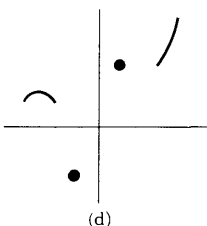
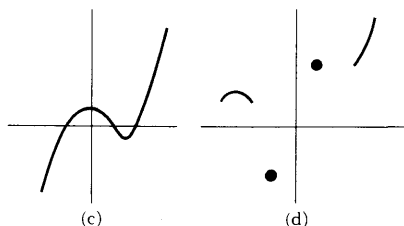
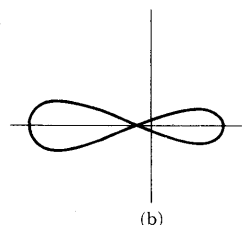
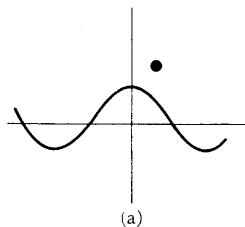


FIGURE 10

graph goes through (a, b) and (c, d) . This amounts to saying that $f(a) = b$ and $f(c) = d$. If f is to be of the form $f(x) = \alpha x + \beta$, then we must have

$$\begin{aligned}\alpha a + \beta &= b, \\ \alpha c + \beta &= d;\end{aligned}$$

therefore $\alpha = (d - b)/(c - a)$ and $\beta = b - [(d - b)/(c - a)]a$, so

$$f(x) = \frac{d - b}{c - a}x + b - \frac{d - b}{c - a}a = \frac{d - b}{c - a}(x - a) + b,$$

a formula most easily remembered by using the “point-slope form” (see Problem 5).

Of course, this solution is possible only if $a \neq c$; the graphs of linear functions account only for the straight lines which are not parallel to the vertical axis. The vertical straight lines are not the graph of *any* function at all; in fact, the graph of a function can never contain even two distinct points on the same vertical line. This conclusion is immediate from the definition of a function—two points on the same vertical line correspond to pairs of the form (a, b) and (a, c) and, by definition, a function cannot contain (a, b) and (a, c) if $b \neq c$. Conversely, if a set of points in the plane has the property that no two points lie on the same vertical line, then it is surely the graph of a function. Thus the first 2 sets in Figure 10 are not graphs of functions and the last 2 are; notice that the fourth is the graph of a function whose domain is not all of $\mathbf{R}$, since some vertical lines have no points on them at all.

After the linear functions the simplest is perhaps the function $f(x) = x^2$. If we draw some of the pairs in f , i.e., some of the pairs of the form (x, x^2) , we obtain a picture like Figure 11.

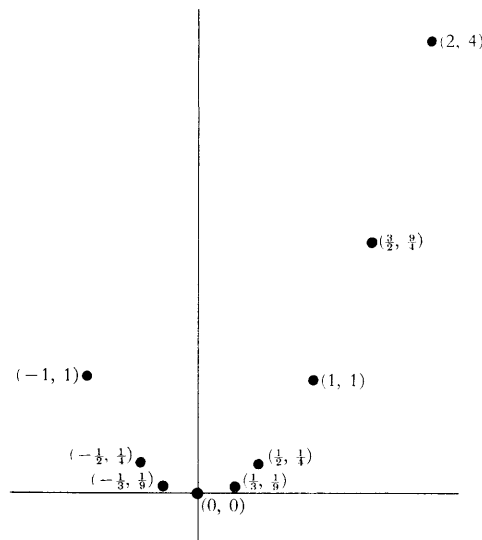


FIGURE 11

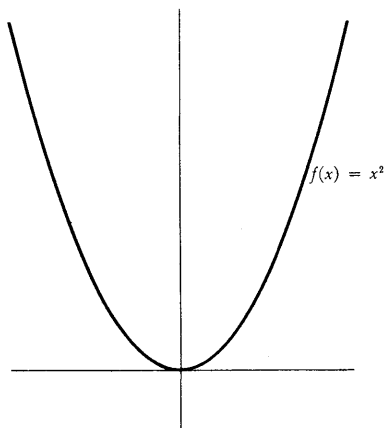
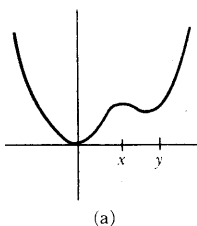
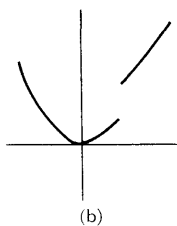


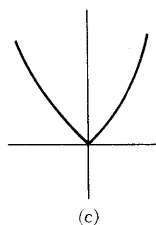
FIGURE 12



(a)



(b)



(c)

FIGURE 13

It is not hard to convince yourself that all the pairs (x, x^2) lie along a curve like the one shown in Figure 12; this curve is known as a **parabola**.

Since a graph is just a drawing on paper, made (in this case) with printer's ink, the question "Is this what the graph really looks like?" is hard to phrase in any sensible manner. No drawing is ever *really* correct since the line has thickness. Nevertheless, there are some questions which one *can* ask: for example, how can you be sure that the graph does not look like one of the drawings in Figure 13? It is easy to see, and even to prove, that the graph cannot look like (a); for if $0 < x < y$, then $x^2 < y^2$, so the graph should be higher at y than at x , which is not the case in (a). It is also easy to see, simply by drawing a very accurate graph, first plotting many pairs (x, x^2) , that the graph cannot have a large "jump" as in (b) or a "corner" as in (c). In order to prove these assertions, however, we first need to say, in a mathematical way, what it means for a function not to have a "jump" or "corner"; these ideas already involve some of the fundamental concepts of calculus. Eventually we will be able to define them rigorously, but meanwhile you may amuse yourself by attempting to define these concepts, and then examining your definitions critically. Later these definitions may be compared with the ones mathematicians have agreed upon. If they compare favorably, you are certainly to be congratulated!

The functions $f(x) = x^n$, for various natural numbers n , are sometimes called **power functions**. Their graphs are most easily compared as in Figure 14, by drawing several at once.

The power functions are only special cases of polynomial functions, introduced in the previous chapter. Two particular polynomial functions are

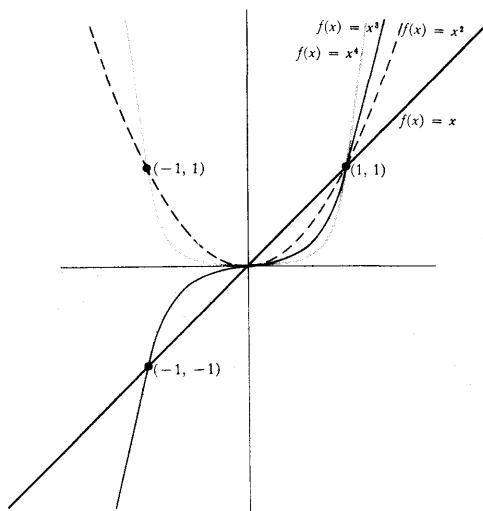
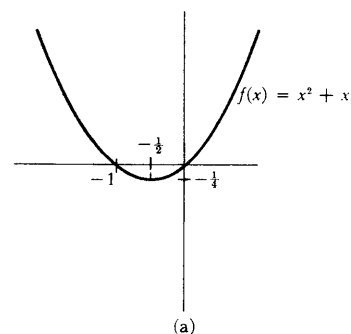
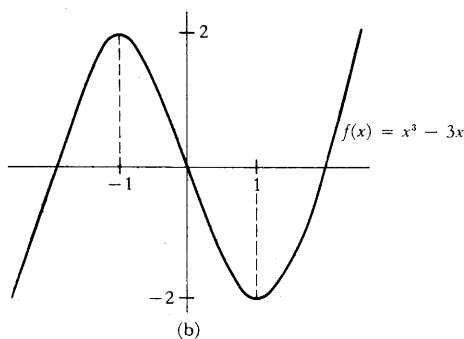


FIGURE 14



(a)



(b)

FIGURE 15

graphed in Figure 15, while Figure 16 is meant to give a general idea of the graph of the polynomial function

$$f(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_0,$$

in the case $a_n > 0$.

In general, the graph of f will have at most $n - 1$ “peaks” or “valleys” (a “peak” is a point like $(x, f(x))$ in Figure 16, while a “valley” is a point like $(y, f(y))$). The number of peaks and valleys may actually be much smaller (the power functions, for example, have at most one valley). Although these assertions are easy to make, we will not even contemplate giving proofs until Part III (once the powerful methods of Part III are available, the proofs will be very easy).

Figure 17 illustrates the graphs of several rational functions. The rational functions exhibit even greater variety than the polynomial functions, but their behavior will also be easy to analyze once we can use the derivative, the basic tool of Part III.

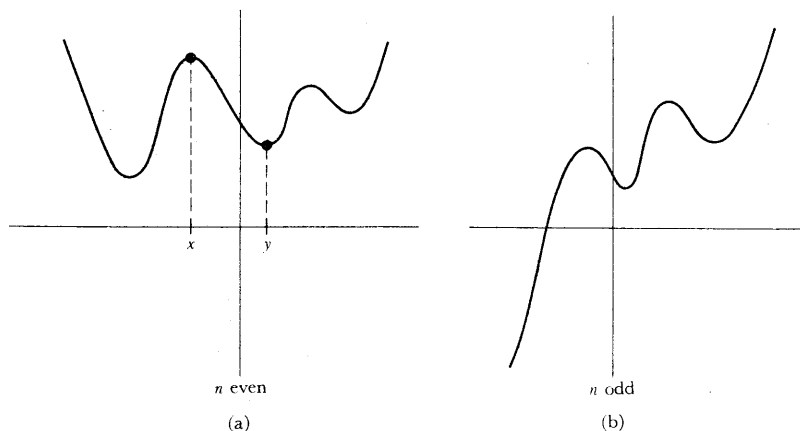
Many interesting graphs can be constructed by “piecing together” the graphs of functions already studied. The graph in Figure 18 is made up entirely of straight lines. The function f with this graph satisfies

$$f\left(\frac{1}{n}\right) = (-1)^{n+1},$$

$$f\left(\frac{-1}{n}\right) = (-1)^{n+1},$$

$$f(x) = 1, \quad |x| \geq 1,$$

and is a linear function on each interval $[1/(n+1), 1/n]$ and $[-1/n, -1/(n+1)]$. (The number 0 is not in the domain of f .) Of course, one can write out an explicit formula for $f(x)$, when x is in $[1/(n+1), 1/n]$; this is a good exercise in the use of linear functions, and will also convince you that a picture is worth a thousand words.



(a)

(b)

FIGURE 16

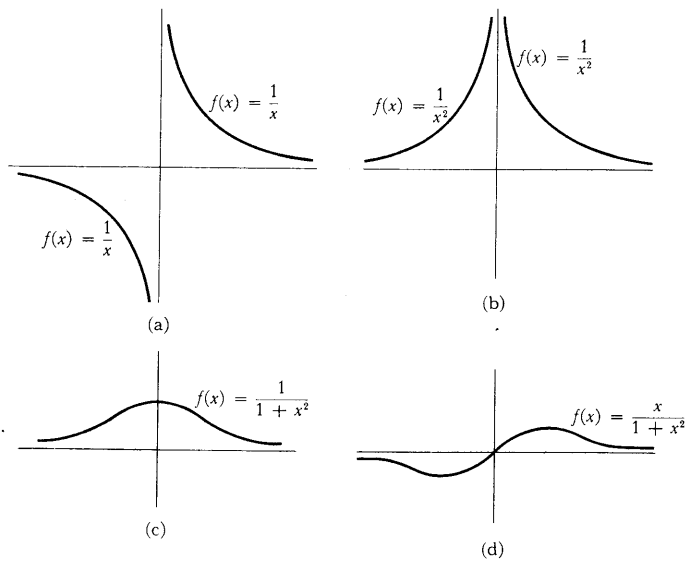


FIGURE 17

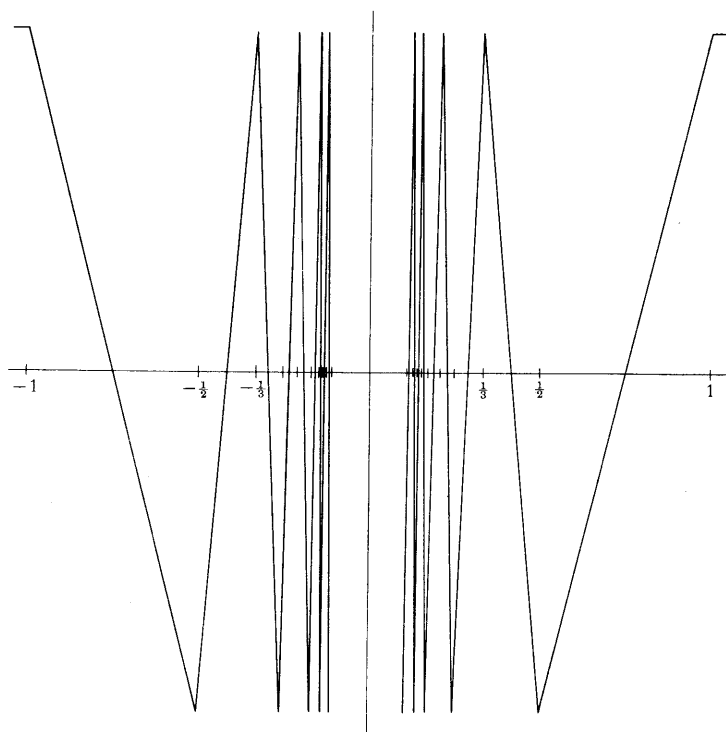


FIGURE 18

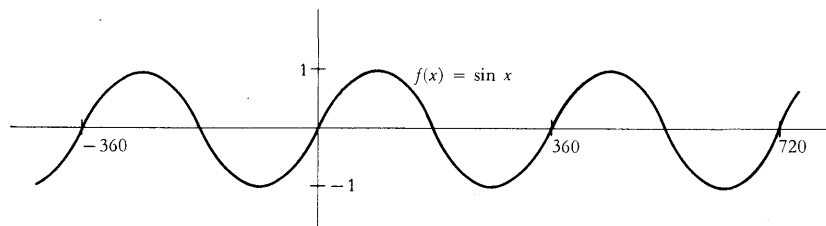


FIGURE 19

It is actually possible to define, in a much simpler way, a function which exhibits this same property of oscillating infinitely often near 0, by using the sine function. In Chapter 15 we will discuss this function in detail, and radian measure in particular; for the time being it will be easiest to use degree measurements for angles. The graph of the sine function is shown in Figure 19 (the scale on the horizontal axis has been altered so that the graph will be clearer; radian measure has, besides important mathematical properties, the additional advantage that such changes are unnecessary).

Now consider the function $f(x) = \sin 1/x$. The graph of f is shown in Figure 20. (To draw this graph it helps to first observe that

$$\begin{aligned} f(x) &= 0 & \text{for } x &= \frac{1}{180}, \frac{1}{360}, \frac{1}{540}, \dots, \\ f(x) &= 1 & \text{for } x &= \frac{1}{90}, \frac{1}{90 + 360}, \frac{1}{90 + 720}, \dots, \\ f(x) &= -1 & \text{for } x &= \frac{1}{270}, \frac{1}{270 + 360}, \frac{1}{270 + 720}, \dots \end{aligned}$$

Notice that when x is large, so that $1/x$ is small, $f(x)$ is also small; when x

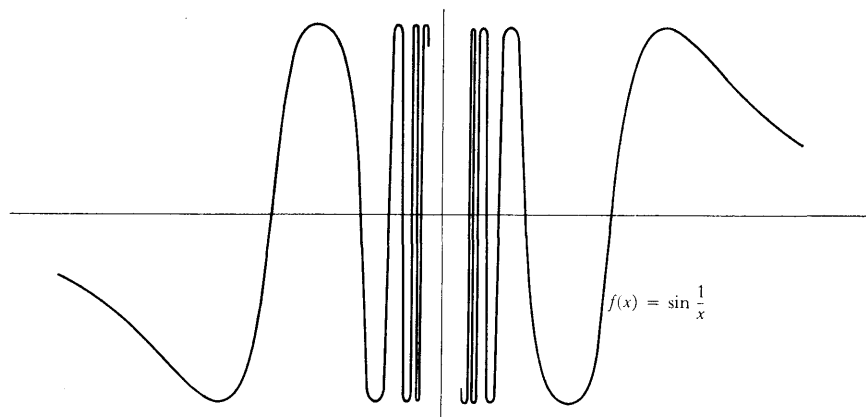


FIGURE 20

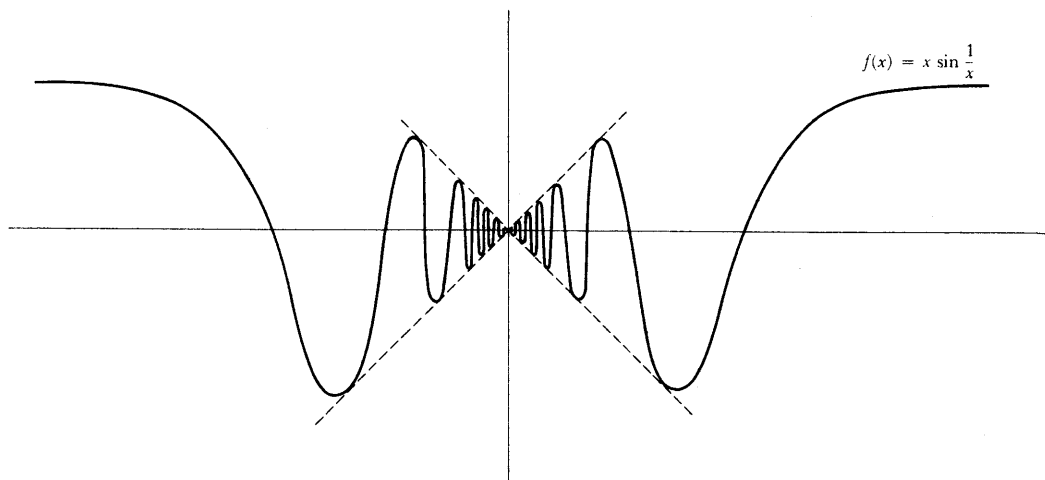


FIGURE 21

is “large negative,” that is, when $|x|$ is large for negative x , again $f(x)$ is close to 0, although $f(x) < 0$.

An interesting modification of this function is $f(x) = x^2 \sin 1/x$. The graph of this function is sketched in Figure 22. Since $\sin 1/x$ oscillates infinitely often near 0 between 1 and -1 , the function $f(x) = x^2 \sin 1/x$ oscillates infinitely often between x and $-x$. The behavior of the graph for x large or large nega-

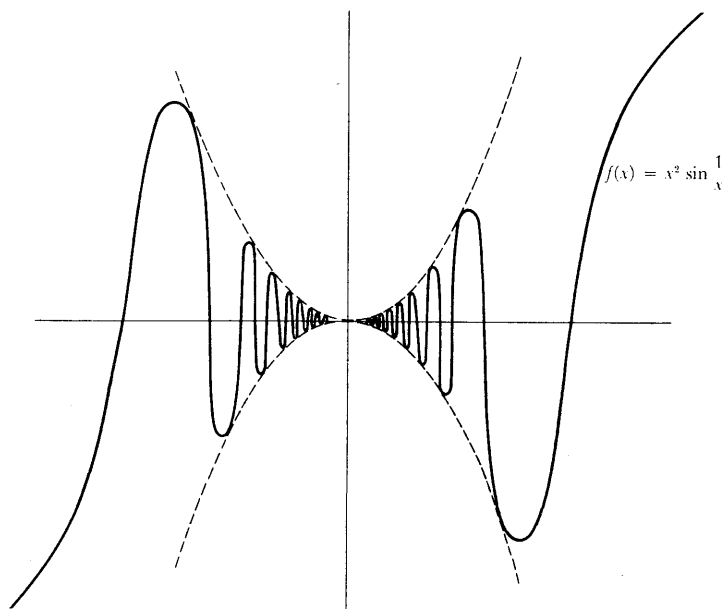


FIGURE 22

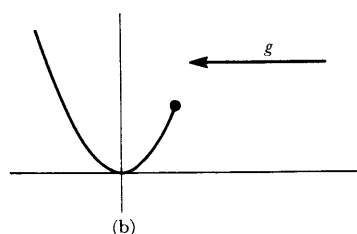
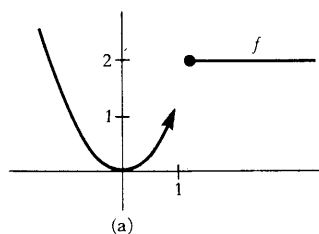


FIGURE 23

tive is harder to analyze. Since $\sin 1/x$ is getting close to 0, while x is getting larger and larger, there seems to be no telling what the product will do. It is possible to decide, but this is another question that is best deferred to Part III. The graph of $f(x) = x^2 \sin 1/x$ has also been illustrated (Figure 22).

For these infinitely oscillating functions, it is clear that the graph cannot hope to be really “accurate.” The best we can do is to show part of it, and leave out the part near 0 (which is the interesting part). Actually, it is easy to find much simpler functions whose graph cannot be “accurately” drawn. The graphs of

$$f(x) = \begin{cases} x^2, & x < 1 \\ 2, & x \geq 1 \end{cases} \quad \text{and} \quad g(x) = \begin{cases} x^2, & x \leq 1 \\ 2, & x > 1 \end{cases}$$

can only be distinguished by some convention similar to that used for open and closed intervals (Figure 23).

Our last example is a function whose graph is spectacularly nondrawable:

$$f(x) = \begin{cases} 0, & x \text{ irrational} \\ 1, & x \text{ rational} \end{cases}$$

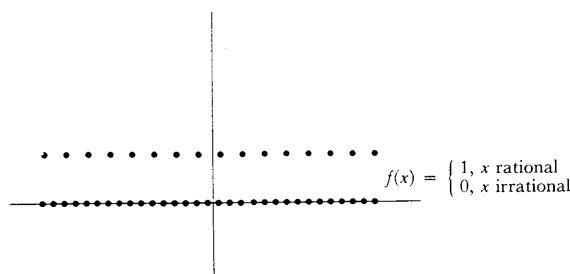


FIGURE 24

The graph of f must contain infinitely many points on the horizontal axis and also infinitely many points on a line parallel to the horizontal axis, but it must not contain either of these lines entirely. Figure 24 shows the usual textbook picture of the graph. To distinguish the two parts of the graph, the dots are placed closer together on the line corresponding to irrational x . (There is actually a mathematical reason behind this convention, but it depends on some sophisticated ideas, introduced in Problems 20-6' and 20-7.)

The peculiarities exhibited by some functions are so engrossing that it is easy to forget some of the simplest, and most important, subsets of the plane, which are not the graphs of functions. The most important example of all is the **circle**. A circle with center (a, b) and radius $r > 0$ contains, by definition, all the points (x, y) whose distance from (a, b) is equal to r . The circle thus consists (Figure 25) of all points (x, y) with

$$\sqrt{(x-a)^2 + (y-b)^2} = r$$

or

$$(x-a)^2 + (y-b)^2 = r^2.$$

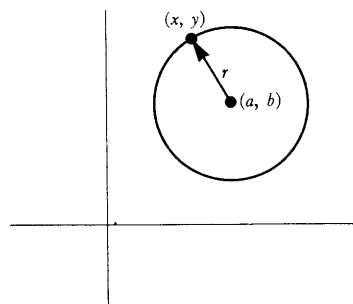


FIGURE 25

The circle with center $(0, 0)$ and radius 1, often regarded as a sort of standard copy, is called the *unit circle*.

A close relative of the circle is the **ellipse**. This is defined as the set of points, the *sum* of whose distances from two fixed points is a constant. (When the two fixed points are the same, we obtain a circle.) If, for convenience, the two points are taken to be $(-c, 0)$ and $(c, 0)$, and the sum of the distances is taken to be $2a$ (the factor 2 simplifies some algebra), then (x, y) is on the ellipse if and only if

$$\sqrt{(x - (-c))^2 + y^2} + \sqrt{(x - c)^2 + y^2} = 2a$$

or

$$\sqrt{(x + c)^2 + y^2} = 2a - \sqrt{(x - c)^2 + y^2}$$

or

$$x^2 + 2cx + c^2 + y^2 = 4a^2 - 4a\sqrt{(x - c)^2 + y^2} + x^2 - 2cx + c^2 + y^2$$

or

$$4(cx - a^2) = -4a\sqrt{(x - c)^2 + y^2}$$

or

$$c^2x^2 - 2cxa^2 + a^4 = a^2(x^2 - 2cx + c^2 + y^2)$$

or

$$(c^2 - a^2)x^2 - a^2y^2 = a^2(c^2 - a^2)$$

or

$$\frac{x^2}{a^2} + \frac{y^2}{a^2 - c^2} = 1.$$

This is usually written simply

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1,$$

where $b = \sqrt{a^2 - c^2}$ (since we must clearly choose $a > c$, it follows that $a^2 - c^2 > 0$). A picture of an ellipse is shown in Figure 26. The ellipse intersects the horizontal axis when $y = 0$, so that

$$\frac{x^2}{a^2} = 1, \quad x = \pm a,$$

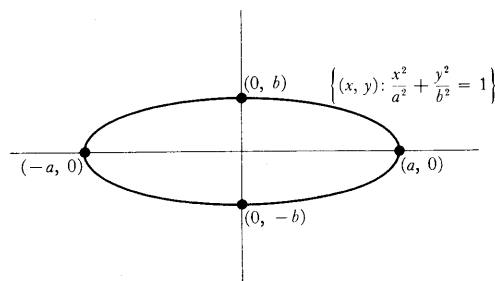


FIGURE 26

and it intersects the vertical axis when $x = 0$, so that

$$\frac{y^2}{b^2} = 1, \quad y = \pm b.$$

The **hyperbola** is defined analogously, except that we require the *difference* of the two distances to be constant. Choosing the points $(-c, 0)$ and $(c, 0)$ once again, and the constant difference as $2a$, we obtain, as the condition that (x, y) be on the hyperbola,

$$\sqrt{(x+c)^2 + y^2} - \sqrt{(x-c)^2 + y^2} = \pm 2a,$$

which may be simplified to

$$\frac{x^2}{a^2} + \frac{y^2}{a^2 - c^2} = 1.$$

In this case, however, we must clearly choose $c > a$, so that $a^2 - c^2 < 0$. If $b = \sqrt{c^2 - a^2}$, then (x, y) is on the hyperbola if and only if

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1.$$

The picture is shown in Figure 27. It contains two pieces, because the difference between the distances of (x, y) from $(-c, 0)$ and $(c, 0)$ may be taken in two different orders. The hyperbola intersects the horizontal axis when $y = 0$, so that $x = \pm a$, but it never intersects the vertical axis.

It is interesting to compare (Figure 28) the hyperbola with $a = b = \sqrt{2}$ and the graph of the function $f(x) = 1/x$. The drawings look quite similar, and the two sets are actually identical, except for a rotation through an angle of 45° (Problem 21).

Clearly no rotation of the plane will change circles or ellipses into the graphs of functions. Nevertheless, the study of these important geometric figures can often be reduced to the study of functions. Ellipses, for example, are

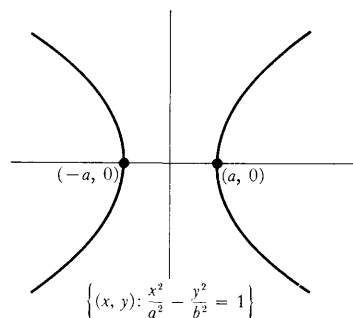


FIGURE 27

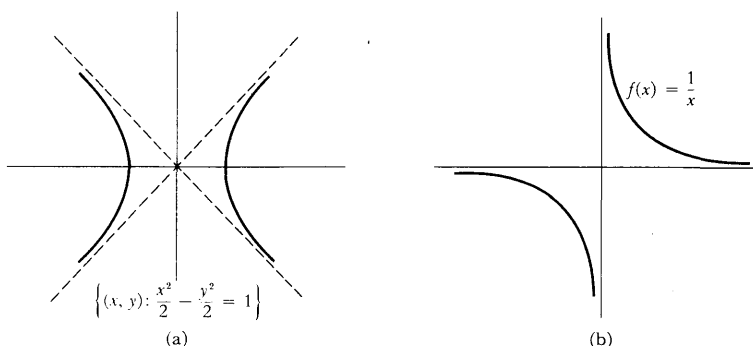


FIGURE 28

made up of the graphs of two functions,

$$f(x) = b \sqrt{1 - (x^2/a^2)}, \quad -a \leq x \leq a$$

and

$$g(x) = -b \sqrt{1 - (x^2/a^2)}, \quad -a \leq x \leq a.$$

Of course, there are many other pairs of functions with this same property. For example, we can take

$$f(x) = \begin{cases} b \sqrt{1 - (x^2/a^2)}, & 0 < x \leq a \\ -b \sqrt{1 - (x^2/a^2)}, & -a \leq x \leq 0 \end{cases}$$

and

$$g(x) = \begin{cases} -b \sqrt{1 - (x^2/a^2)}, & 0 < x \leq a \\ b \sqrt{1 - (x^2/a^2)}, & -a \leq x \leq 0. \end{cases}$$

We could also choose

$$f(x) = \begin{cases} b \sqrt{1 - (x^2/a^2)}, & x \text{ rational}, \quad -a \leq x \leq a \\ -b \sqrt{1 - (x^2/a^2)}, & x \text{ irrational}, \quad -a \leq x \leq a \end{cases}$$

and

$$g(x) = \begin{cases} -b \sqrt{1 - (x^2/a^2)}, & x \text{ rational}, \quad -a \leq x \leq a \\ b \sqrt{1 - (x^2/a^2)}, & x \text{ irrational}, \quad -a \leq x \leq a. \end{cases}$$

But all these other pairs necessarily involve unreasonable functions which jump around. A proof, or even a precise statement of this fact, is too difficult at present. Although you have probably already begun to make a distinction between those functions with reasonable graphs, and those with unreasonable graphs, you may find it very difficult to state a reasonable definition of reasonable functions. A mathematical definition of this concept is by no means easy, and a great deal of this book may be viewed as successive attempts to impose more and more conditions that a "reasonable" function must satisfy. As we define some of these conditions, we will take time out to ask if we have really succeeded in isolating the functions which deserve to be called reasonable. The answer, unfortunately, will always be "no," or at best, a qualified "yes."

PROBLEMS

1. Indicate on a straight line the set of all x satisfying the following conditions. Also name each set, using the notation for intervals (in some cases you will also need the $\cup$ sign).

- (i) $|x - 3| < 1.$
- (ii) $|x - 3| \leq 1.$
- (iii) $|x - a| < \epsilon.$
- (iv) $|x^2 - 1| < \frac{1}{2}.$

- (v) $\frac{1}{1+x^2} \geq \frac{1}{5}$.
- (vi) $\frac{1}{1+x^2} \leq a$ (give an answer in terms of a , distinguishing various cases).
- (vii) $x^2 + 1 \geq 2$.
- (viii) $(x+1)(x-1)(x-2) > 0$.
2. Draw the set of all points (x, y) satisfying the following conditions. (In most cases your picture will be a sizable portion of a plane, not just a line or curve.)
- (i) $x > y$.
- (ii) $x + a > y + b$.
- (iii) $y < x^2$.
- (iv) $y \leq x^2$.
- (v) $|x - y| < 1$.
- (vi) $|x + y| < 1$.
- (vii) $x + y$ is an integer.
- (viii) $\frac{1}{x + y}$ is an integer.
- (ix) $(x - 1)^2 + (y - 2)^2 < 1$.
- (x) $x^2 < y < x^4$.
3. Draw the set of all points (x, y) satisfying the following conditions.
- (i) $|x| + |y| = 1$.
- (ii) $|x| - |y| = 1$.
- (iii) $|x - 1| = |y - 1|$.
- (iv) $|1 - x| = |y - 1|$.
- (v) $x^2 + y^2 = 0$.
- (vi) $xy = 0$.
- (vii) $x^2 - 2x + y^2 = 4$.
- (viii) $x^2 = y^2$.
4. Draw the set of all points (x, y) satisfying the following conditions:
- (i) $x = y^2$.
- (ii) $\frac{y^2}{a^2} - \frac{x^2}{b^2} = 1$.
- (iii) $x = |y|$.
- (iv) $x = \sin y$.

Hint: You already know the answers when x and y are interchanged.

5. (a) Show that the straight line through (a, b) with slope m is the graph of the function $f(x) = m(x - a) + b$. This formula, known as the

“point-slope form” is far more convenient than the equivalent expression $f(x) = mx + (b - ma)$; it is immediately clear from the point-slope form that the slope is m , and that the value of f at a is b .

- (b) For $a \neq c$, show that the straight line through (a, b) and (c, d) is the graph of the function

$$f(x) = \frac{d - b}{c - a}(x - a) + b.$$

- (c) When are the graphs of $f(x) = mx + b$ and $g(x) = m'x + b'$ parallel straight lines?
6. (a) For any numbers A, B , and C , with A and B not both 0, show that the set of all (x, y) satisfying $Ax + By + C = 0$ is a straight line (possibly a vertical one). Hint: First decide when a vertical straight line is described.
- (b) Show conversely that every straight line, including vertical ones, can be described as the set of all (x, y) satisfying $Ax + By + C = 0$.
7. (a) Prove that the graphs of the functions

$$\begin{aligned} f(x) &= mx + b, \\ g(x) &= nx + c, \end{aligned}$$

are perpendicular if $mn = -1$, by computing the squares of the lengths of the sides of the triangle in Figure 29. (Why is this special case, where the lines intersect at the origin, as good as the general case?)

- (b) Prove that the two straight lines consisting of all (x, y) satisfying the conditions

$$\begin{aligned} Ax + By + C &= 0, \\ A'x + B'y + C' &= 0, \end{aligned}$$

are perpendicular if and only if $AA' + BB' = 0$.

8. (a) Prove, using Problem 1-18, that

$$\sqrt{(x_1 + y_1)^2 + (x_2 + y_2)^2} \leq \sqrt{x_1^2 + x_2^2} + \sqrt{y_1^2 + y_2^2}.$$

- (b) Prove that

$$\begin{aligned} \sqrt{(x_3 - x_1)^2 + (y_3 - y_1)^2} &\leq \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} \\ &\quad + \sqrt{(x_3 - x_2)^2 + (y_3 - y_2)^2}. \end{aligned}$$

Interpret this inequality geometrically (it is called the “triangle inequality”). When does strict inequality hold?

9. Sketch the graphs of the following functions, plotting enough points to get a good idea of the general appearance. (Part of the problem is to make a reasonable decision how many is “enough”; the queries posed below are meant to show that a little thought will often be more valuable than hundreds of individual points.)

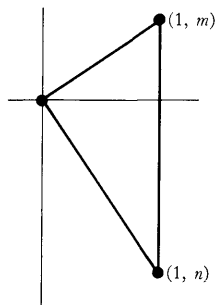


FIGURE 29

- (i) $f(x) = x + \frac{1}{x}$. (What happens for x near 0, and for large x ? Where does the graph lie in relation to the graph of the identity function? Why does it suffice to consider only positive x at first?)

(ii) $f(x) = x - \frac{1}{x}$.

(iii) $f(x) = x^2 + \frac{1}{x^2}$.

(iv) $f(x) = x^2 - \frac{1}{x^2}$.

10. Describe the general features of the graph of f if

- (i) f is even.
 (ii) f is odd.
 (iii) f is nonnegative.
 (iv) $f(x) = f(x + a)$ for all x (a function with this property is called **periodic**, with **period** a).

11. Graph the functions $f(x) = \sqrt[m]{x}$ for $m = 1, 2, 3, 4$. (There is an easy way to do this, using Figure 14. Be sure to remember, however, that $\sqrt[m]{x}$ means the *positive* m th root of x when m is even; you should also note that there will be an important difference between the graph when m is even and when m is odd.)

12. (a) Graph $f(x) = |x|$ and $f(x) = x^2$.
 (b) Graph $f(x) = |\sin x|$ and $f(x) = \sin^2 x$. (There is an important difference between the graphs, which we cannot yet even describe rigorously. See if you can discover what it is; part (a) is meant to be a clue.)

13. Describe the graph of g in terms of the graph of f if

- (i) $g(x) = f(x) + c$.
 (ii) $g(x) = f(x + c)$. (It is easy to make a mistake here.)
 (iii) $g(x) = cf(x)$. (Distinguish the cases $c = 0$, $c > 0$, $c < 0$.)
 (iv) $g(x) = f(cx)$.
 (v) $g(x) = f(1/x)$.
 (vi) $g(x) = f(|x|)$.
 (vii) $g(x) = |f(x)|$.
 (viii) $g(x) = \max(f, 0)$.
 (ix) $g(x) = \min(f, 0)$.
 (x) $g(x) = \max(f, 1)$.

14. Draw the graph of $f(x) = ax^2 + bx + c$. Hint: Use the methods of Problem 1-17.

15. The symbol $[x]$ denotes the largest integer which is $\leq x$. Thus $[2.1] = [2] = 2$ and $[-0.9] = [-1.2] = -1$. Draw the graph of the following functions (they are all quite interesting, and several will reappear frequently in other problems).

(i) $f(x) = [x]$.

(ii) $f(x) = x - [x]$.

(iii) $f(x) = \sqrt{x - [x]}$.

(iv) $f(x) = [x] + \sqrt{x - [x]}$.

(v) $f(x) = \left[\frac{1}{x} \right]$.

(vi) $f(x) = \frac{1}{\left[\frac{1}{x} \right]}$.

16. Graph the following functions.

(i) $f(x) = \{x\}$, where $\{x\}$ is defined to be the distance from x to the nearest integer.

(ii) $f(x) = \{2x\}$.

(iii) $f(x) = \{x\} + \frac{1}{2}\{2x\}$.

(iv) $f(x) = \{4x\}$.

(v) $f(x) = \{x\} + \frac{1}{2}\{2x\} + \frac{1}{4}\{4x\}$.

Many functions may be described in terms of the decimal expansion of a number. Although we will not be in a position to describe infinite decimals rigorously until Chapter 22, your intuitive notion of infinite decimals should suffice to carry you through the following problem, and others which occur before Chapter 22. There is one ambiguity about infinite decimals which must be eliminated: Every decimal ending in a string of 9's is equal to another ending in a string of 0's (e.g., $1.23999 \dots = 1.24000 \dots$). We will always use the one ending in 9's

- *17. Describe as best you can the graphs of the following functions (a complete picture is usually out of the question).

(i) $f(x)$ = the 1st number in the decimal expansion of x .

(ii) $f(x)$ = the 2nd number in the decimal expansion of x .

(iii) $f(x)$ = the number of 7's in the decimal expansion of x if this number is finite, and 0 otherwise.

(iv) $f(x) = 0$ if the number of 7's in the decimal expansion of x is finite, and 1 otherwise.

(v) $f(x)$ = the number obtained by replacing all digits in the decimal expansion of x which come after the first 7 (if any) by 0.

(vi) $f(x) = 0$ if 1 never appears in the decimal expansion of x , and n if 1 first appears in the n th place.

*18. Let

$$f(x) = \begin{cases} 0, & x \text{ irrational} \\ \frac{1}{q}, & x = \frac{p}{q} \text{ rational in lowest terms.} \end{cases}$$

(A number p/q is in **lowest terms** if p and q are integers with no common factor, and $q > 0$). Draw the graph of f as well as you can (don't sprinkle points randomly on the paper; consider first the rational numbers with $q = 2$, then those with $q = 3$, etc.).

19. (a) The points on the graph of $f(x) = x^2$ are the ones of the form (x, x^2) . Prove that each such point is equidistant from the point $(0, \frac{1}{4})$ and the graph of $g(x) = -\frac{1}{4}$. (See Figure 30.)

- (b) Given a point $P = (\alpha, \beta)$ and a horizontal line L , the graph of $g(x) = \gamma$, show that the set of all points (x, y) equidistant from P and L is the graph of a function of the form $f(x) = ax^2 + bx + c$.

- *20. (a) Show that the square of the distance from (c, d) to (x, mx) is

$$x^2(m^2 + 1) + x(-2md - 2c) + d^2 + c^2.$$

Using Problem 1-17 to find the minimum of these numbers, show that the distance from (c, d) to the graph of $f(x) = mx$ is

$$|cm - d|/\sqrt{m^2 + 1}.$$

- (b) Find the distance from (c, d) to the graph of $f(x) = mx + b$. (Reduce this case to part (a).)

- *21. (a) Using Problem 20, show that the numbers x' and y' indicated in Figure 31 are given by

$$x' = \frac{1}{\sqrt{2}}x + \frac{1}{\sqrt{2}}y,$$

$$y' = -\frac{1}{\sqrt{2}}x + \frac{1}{\sqrt{2}}y.$$

- (b) Show that the set of all (x, y) with $(x'/\sqrt{2})^2 - (y'/\sqrt{2})^2 = 1$ is the same as the set of all (x, y) with $xy = 1$.

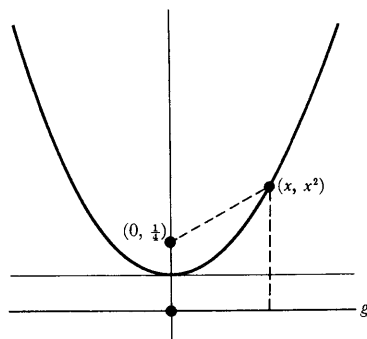


FIGURE 30

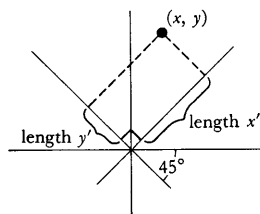


FIGURE 31

CHAPTER 5 LIMITS

The concept of a limit is surely the most important, and probably the most difficult one in all of calculus. The goal of this chapter is the definition of limits, but we are, once more, going to begin with a provisional definition; what we shall define is not the word “limit” but the notion of a function approaching a limit.

PROVISIONAL DEFINITION

The function f approaches the limit l near a , if we can make $f(x)$ as close as we like to l by requiring that x be sufficiently close to, but unequal to, a .

Of the six functions graphed in Figure 1, only the first three approach l at a . Notice that although $g(a)$ is not defined, and $h(a)$ is defined “the wrong way,” it is still true that g and h approach l near a . This is because we explicitly ruled out, in our definition, the necessity of ever considering the value of the function at a —it is only necessary that $f(x)$ should be close to l for x close to a , but *unequal to* a . We are simply not interested in the value of $f(a)$, or even in the question of whether $f(a)$ is defined.

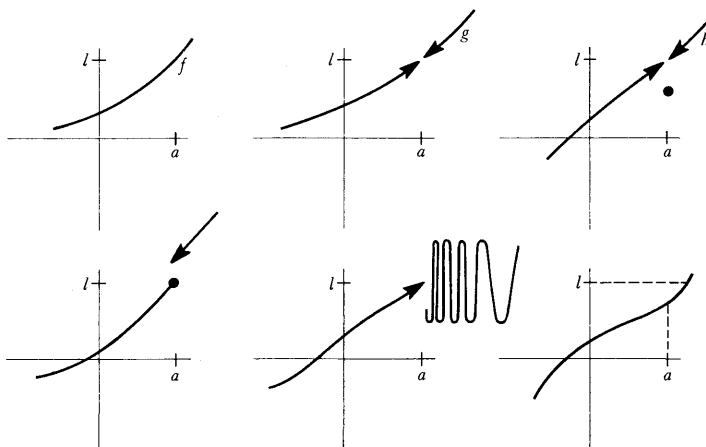


FIGURE 1

One convenient way of picturing the assertion that f approaches l near a is provided by a method of drawing functions that was not mentioned in Chapter 4. In this method, we draw two straight lines, each representing $\mathbf{R}$, and arrows from a point x in one, to $f(x)$ in the other. Figure 2 illustrates such a picture for two different functions.

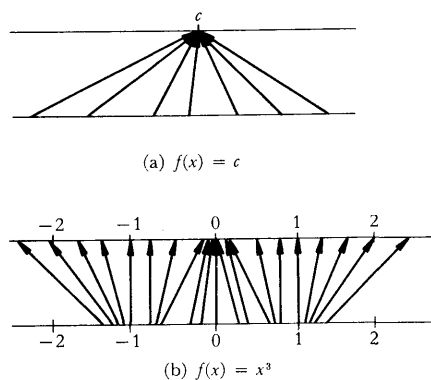


FIGURE 2

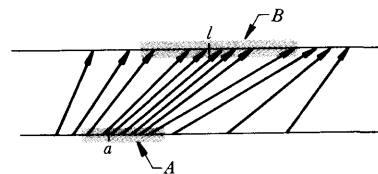


FIGURE 3

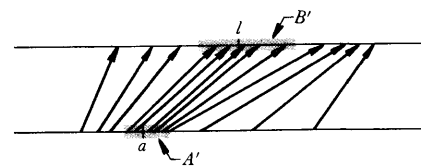


FIGURE 4

Now consider a function f whose drawing looks like Figure 3. Suppose we ask that $f(x)$ be close to l , say within the open interval B which has been drawn in Figure 3. This can be guaranteed if we consider only the numbers x in the interval A of Figure 3. (In this diagram we have chosen the largest interval which will work; any smaller interval containing a could have been chosen instead.) If we choose a smaller interval B' (Figure 4) we will, usually, have to choose a smaller A' , but no matter how small we choose the open interval B , there is always supposed to be some open interval A which works.

A similar pictorial interpretation is possible in terms of the graph of f , but in this case the interval B must be drawn on the vertical axis, and the set A on the horizontal axis. The fact that $f(x)$ is in B when x is in A means that the part of the graph lying over A is contained in the region which is bounded by the horizontal lines through the end points of B ; compare Figure 5(a), where a valid interval A has been chosen, with Figure 5(b), where A is too large.

In order to apply our definition to a particular function, let us consider $f(x) = x \sin 1/x$ (Figure 6). Despite the erratic behavior of this function near 0 it is clear, at least intuitively, that f approaches 0 near 0, and it is certainly to be hoped that our definition will allow us to reach the same conclusion. In the case we are considering, both a and l of the definition are 0, so we must ask if we can get $f(x) = x \sin 1/x$ as close to 0 as desired if we require that x be sufficiently close to 0, but $\neq 0$. To be specific, suppose we wish to get $x \sin 1/x$ within $\frac{1}{10}$ of 0. This means we want

$$-\frac{1}{10} < x \sin \frac{1}{x} < \frac{1}{10},$$

or, more succinctly, $|x \sin 1/x| < \frac{1}{10}$. Now this is easy. Since

$$\left| \sin \frac{1}{x} \right| \leq 1, \quad \text{for all } x \neq 0,$$

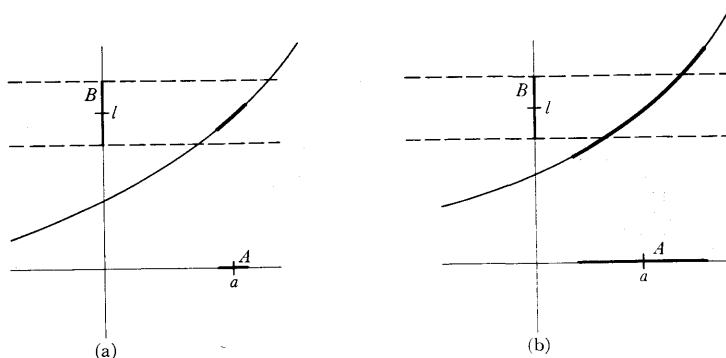


FIGURE 5

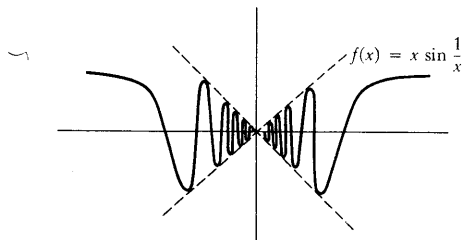


FIGURE 6

we have

$$\left| x \sin \frac{1}{x} \right| \leq |x|, \quad \text{for all } x \neq 0.$$

This means that if $|x| < \frac{1}{10}$ and $x \neq 0$, then $|x \sin 1/x| < \frac{1}{10}$; in other words, $x \sin 1/x$ is within $\frac{1}{10}$ of 0 provided that x is within $\frac{1}{10}$ of 0, but $\neq 0$. There is nothing special about the number $\frac{1}{10}$; it is just as easy to guarantee that $|f(x) - 0| < \frac{1}{100}$ —simply require that $|x| < \frac{1}{100}$, but $x \neq 0$. In fact, if we take any positive number ε we can make $|f(x) - 0| < \varepsilon$ simply by requiring that $|x| < \varepsilon$, and $x \neq 0$.

For the function $f(x) = x^2 \sin 1/x$ (Figure 7) it seems even clearer that f approaches 0 near 0. If, for example, we want

$$\left| x^2 \sin \frac{1}{x} \right| < \frac{1}{10},$$

then we certainly need only require that $|x| < \frac{1}{10}$ and $x \neq 0$, since this implies that $|x^2| < \frac{1}{100}$ and consequently

$$\left| x^2 \sin \frac{1}{x} \right| \leq |x^2| < \frac{1}{100} < \frac{1}{10}.$$

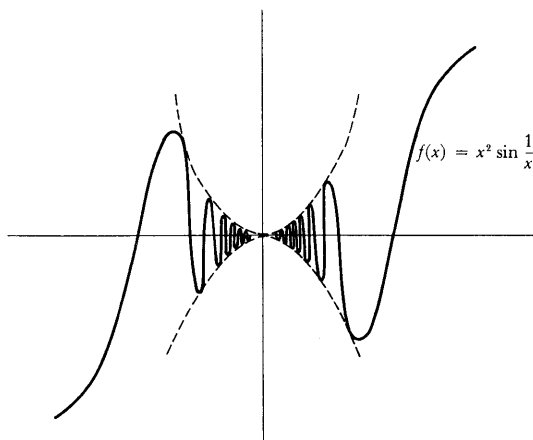


FIGURE 7

(We could do even better, and allow $|x| < 1/\sqrt{10}$ and $x \neq 0$, but there is no particular virtue in being as economical as possible.) In general, if $\varepsilon > 0$, to ensure that

$$\left| x^2 \sin \frac{1}{x} \right| < \varepsilon,$$

we need only require that

$$|x| < \varepsilon \quad \text{and} \quad x \neq 0,$$

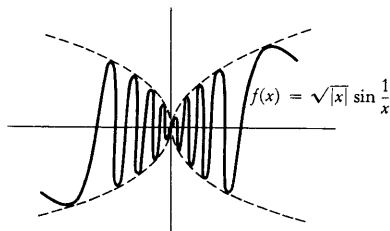


FIGURE 8

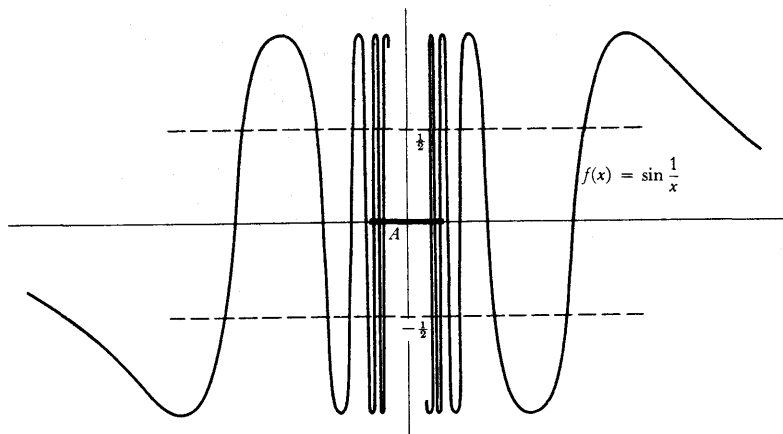


FIGURE 9

provided that $\varepsilon \leq 1$. If we are given an ε which is greater than 1 (it might be, even though it is “small” ε 's which are of interest), then it does not suffice to require that $|x| < \varepsilon$, but it certainly suffices to require that $|x| < 1$ and $x \neq 0$.

As a third example, consider the function $f(x) = \sqrt{|x|} \sin 1/x$ (Figure 8). In order to make $|\sqrt{|x|} \sin 1/x| < \varepsilon$ we can require that

$$|x| < \varepsilon^2 \quad \text{and} \quad x \neq 0,$$

if $\varepsilon \leq 1$, or that $|x| < 1$ and $x \neq 0$ if $\varepsilon > 1$ (the algebra is left to you).

Finally, let us consider the function $f(x) = \sin 1/x$ (Figure 9). For this function it is *false* that f approaches 0 near 0. This amounts to saying that it is not true for every number $\varepsilon > 0$ that we can get $|f(x) - 0| < \varepsilon$ by choosing x sufficiently small, and $\neq 0$. To show this we simply have to find *one* $\varepsilon > 0$ for which the condition $|f(x) - 0| < \varepsilon$ cannot be guaranteed, no matter how small we require $|x|$ to be. In fact, $\varepsilon = \frac{1}{2}$ will do: it is impossible to ensure that $|f(x)| < \frac{1}{2}$ no matter how small we require $|x|$ to be; for if A is any interval containing 0, there is some number $x = 1/(90 + 360n)$ which is in this interval, and for this x we have $f(x) = 1$.

This same argument can be used (Figure 10) to show that f does not approach *any* number near 0. To show this we must again find, for any particular number l , some number $\varepsilon > 0$ so that $|f(x) - l| < \varepsilon$ is *not* true, no matter how small x is required to be. The choice $\varepsilon = \frac{1}{2}$ works for any number l ; that is, no matter how small we require $|x|$ to be, we cannot ensure that $|f(x) - l| < \frac{1}{2}$. The reason is, that for any interval A containing 0 there is some $x_1 = 1/(90 + 360n)$ in this interval, so that

$$f(x_1) = 1,$$

and also some $x_2 = 1/(270 + 360m)$ in this interval, so that

$$f(x_2) = -1.$$

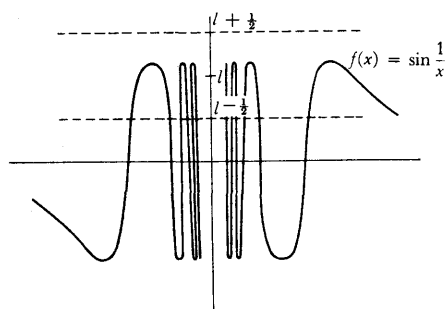


FIGURE 10

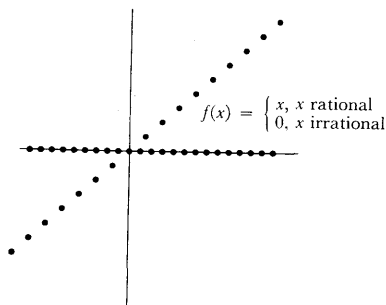


FIGURE 11

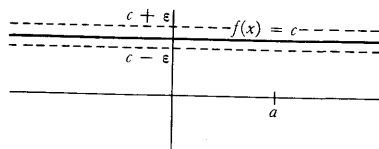


FIGURE 12

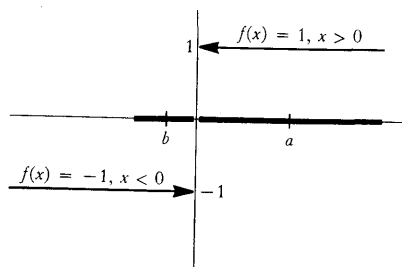


FIGURE 13

But the interval from $l - \frac{1}{2}$ to $l + \frac{1}{2}$ cannot contain both -1 and 1 , since its total length is only 1 ; so we cannot have

$$|1 - l| < \frac{1}{2} \text{ and also } |-1 - l| < \frac{1}{2},$$

no matter what l is.

The phenomenon exhibited by $f(x) = \sin 1/x$ near 0 can occur in many ways. If we consider the function

$$f(x) = \begin{cases} 0, & x \text{ irrational} \\ 1, & x \text{ rational} \end{cases}$$

then, no matter what a is, f does not approach any number l near a . In fact, we cannot make $|f(x) - l| < \frac{1}{4}$ no matter how close we bring x to a , because in any interval around a there are numbers x with $f(x) = 0$, and also numbers x with $f(x) = 1$, so that we would need $|0 - l| < \frac{1}{4}$ and also $|1 - l| < \frac{1}{4}$.

An amusing variation on this behavior is presented by the function shown in Figure 11:

$$f(x) = \begin{cases} x, & x \text{ rational} \\ 0, & x \text{ irrational} \end{cases}$$

The behavior of this function is "opposite" to that of $g(x) = \sin 1/x$; it approaches 0 at 0 , but does not approach any number at a , if $a \neq 0$. By now you should have no difficulty convincing yourself that this is true.

As a contrast to the functions considered so far, which have been quite pathological, we will now examine some of the simplest functions.

If $f(x) = c$, then f approaches c near a , for every number a . In fact, to ensure that $|f(x) - c| < \epsilon$ one does not need to restrict x to be near a at all; the condition is automatically satisfied (Figure 12).

As a slight variation, let f be the function shown in Figure 13:

$$f(x) = \begin{cases} -1, & x < 0 \\ 1, & x > 0 \end{cases}$$

If $a > 0$, then f approaches 1 near a : indeed, to ensure that $|f(x) - 1| < \epsilon$ it certainly suffices to require that $|x - a| < a$, since this implies

$$\begin{aligned} -a &< x - a \\ \text{or} \quad 0 &< x, \end{aligned}$$

so that $f(x) = 1$. Similarly, if $b < 0$, then f approaches -1 near b : to ensure that $|f(x) - (-1)| < \epsilon$ it suffices to require that $|x - b| < -b$. Finally, as you may easily check, f does not approach any number near 0 .

The function $f(x) = x$ is easily dealt with. Clearly f approaches a near a : to ensure that $|f(x) - a| < \epsilon$ we just have to require that $|x - a| < \epsilon$.

The function $f(x) = x^2$ requires a little more work. To show that f approaches a^2 near a , we must decide how to ensure that

$$|x^2 - a^2| < \epsilon.$$

Factoring looks like the most promising procedure: we want

$$|x - a| \cdot |x + a| < \epsilon.$$

Obviously the factor $|x + a|$ is the one that will cause trouble. On the other hand, there is no need to make $|x + a|$ particularly small; as long as we know *some* bound on the values of $|x + a|$ we will be in good shape. For example, if $|x + a| < 1,000,000$, then we will just need to require that $|x - a| < \epsilon/1,000,000$. Therefore, to begin with, let us require that $|x - a| < 1$ (any positive number other than 1 would do just as well); presumably this will ensure that x is not too large, and consequently that $|x + a|$ is not too large. As a matter of fact, Problem 1-12 shows that

$$|x| - |a| \leq |x - a| < 1,$$

so

$$|x| < 1 + |a|,$$

and consequently

$$|x + a| \leq |x| + |a| < 2|a| + 1.$$

Now we need only the additional requirement that $|x - a| < \epsilon/(2|a| + 1)$. In other words,

$$\text{if } |x - a| < \min\left(1, \frac{\epsilon}{2|a| + 1}\right), \text{ then } |x^2 - a^2| < \epsilon.$$

Naturally, $\min(1, \epsilon/(2|a| + 1))$ will just be $\epsilon/(2|a| + 1)$ for small ϵ .

Precisely the same sort of trick will show that if $f(x) = x^3$, then f approaches a^3 near a . In fact,

$$\text{if } |x - a| < \min\left(1, \frac{\epsilon}{(1 + |a|)^2 + |a|(1 + |a|) + |a|^2}\right), \text{ then } |x^3 - a^3| < \epsilon.$$

The proof of this assertion will show where the weird denominator comes from: If $|x - a| < 1$, then $|x| < |a| + 1$, and consequently

$$\begin{aligned} |x^2 + ax + a^2| &\leq |x|^2 + |a| \cdot |x| + |a|^2 \\ &< (1 + |a|)^2 + |a|(1 + |a|) + |a|^2. \end{aligned}$$

Therefore

$$\begin{aligned} |x^3 - a^3| &= |x - a| \cdot |x^2 + ax + a^2| \\ &< \frac{\epsilon}{(1 + |a|)^2 + |a|(1 + |a|) + |a|^2} \cdot [(1 + |a|)^2 + |a|(1 + |a|) + |a|^2] \\ &= \epsilon. \end{aligned}$$

The time has now come to point out that of the many demonstrations about limits which we have given, not one has been a real proof. The fault lies not with our reasoning, but with our definition. If our provisional definition of a function was open to criticism, our provisional definition of approaching a limit is even more vulnerable. This definition is simply not sufficiently precise to be used in proofs. It is hardly clear how one "makes" $f(x)$ close to l (whatever "close" means) by "requiring" x to be sufficiently close to a (however

close “sufficiently” close is supposed to be). Despite the criticisms of our definition you may feel (I certainly hope you do) that our arguments were nevertheless quite convincing. In order to present any sort of argument at all, we have been practically forced to invent the real definition. It is possible to arrive at this definition in several steps, each one clarifying some obscure phrase which still remains. Let us begin, once again, with the provisional definition:

The function f approaches the limit l near a , if we can make $f(x)$ as close as we like to l by requiring that x be sufficiently close to, but unequal to, a .

The very first change which we made in this definition was to note that making $f(x)$ close to l meant making $|f(x) - l|$ small, and similarly for x and a :

The function f approaches the limit l near a , if we can make $|f(x) - l|$ as small as we like by requiring that $|x - a|$ be sufficiently small, and $x \neq a$.

The second, more crucial, change was to note that making $|f(x) - l|$ “as small as we like” means making $|f(x) - l| < \epsilon$ for any $\epsilon > 0$ that happens to be given us:

The function f approaches the limit l near a , if for every number $\epsilon > 0$ we can make $|f(x) - l| < \epsilon$ by requiring that $|x - a|$ be sufficiently small, and $x \neq a$.

There is a common pattern to all the demonstrations about limits which we have given. For each number $\epsilon > 0$ we found some other positive number, δ say, with the property that if $x \neq a$ and $|x - a| < \delta$, then $|f(x) - l| < \epsilon$. For the function $f(x) = x \sin 1/x$ (with $a = 0$, $l = 0$), the number δ was just the number ϵ ; for $f(x) = \sqrt{|x|} \sin 1/x$, it was ϵ^2 if $\epsilon \leq 1$, and it was 1 if $\epsilon > 1$; for $f(x) = x^2$ it was the minimum of 1 and $\epsilon/(2|a| + 1)$. In general, it may not be at all clear how to find the number δ , given ϵ , but it is the condition $|x - a| < \delta$ which expresses how small “sufficiently” small must be:

The function f approaches the limit l near a , if for every $\epsilon > 0$ there is some $\delta > 0$ such that, for all x , if $|x - a| < \delta$ and $x \neq a$, then $|f(x) - l| < \epsilon$.

This is practically the definition we will adopt. We will make only one trivial change, noting that “ $|x - a| < \delta$ and $x \neq a$ ” can just as well be expressed “ $0 < |x - a| < \delta$.”

DEFINITION

The function f approaches the limit l near a means: for every $\epsilon > 0$ there is some $\delta > 0$ such that, for all x , if $0 < |x - a| < \delta$, then $|f(x) - l| < \epsilon$.

This definition is so important (*everything* we do from now on depends on it) that proceeding any further without knowing it is hopeless. If necessary memorize it, like a poem! That, at least, is better than stating it incorrectly; if you do this you are doomed to give incorrect proofs. A good exercise in giving correct proofs is to review every fact already demonstrated about functions approaching limits, giving real proofs of each. This requires writing down the correct definition of what you are proving, but not much more—all the algebraic work has been done already. When proving that f does *not* approach l at a , be sure to negate the definition correctly:

If it is *not* true that

for every $\epsilon > 0$ there is some $\delta > 0$ such that, for all x , if $0 < |x - a| < \delta$, then $|f(x) - l| < \epsilon$,

then

there is *some* $\epsilon > 0$ such that for *every* $\delta > 0$ there is *some* x which satisfies $0 < |x - a| < \delta$ but not $|f(x) - l| < \epsilon$.

Thus, to show that the function $f(x) = \sin 1/x$ does not approach 0 near 0, we consider $\epsilon = \frac{1}{2}$ and note that for every $\delta > 0$ there is some x with $0 < |x - 0| < \delta$ but not $|\sin 1/x - 0| < \frac{1}{2}$ —namely, an x of the form $1/(90 + 360n)$, where n is so large that $1/(90 + 360n) < \delta$.

As an illustration of the use of the definition of a function approaching a limit, we have reserved the function shown in Figure 14, a standard example, but one of the most complicated:

$$f(x) = \begin{cases} 0, & x \text{ irrational}, 0 < x < 1 \\ 1/q, & x = p/q \text{ in lowest terms}, 0 < x < 1. \end{cases}$$

(Recall that p/q is in lowest terms if p and q are integers with no common factor and $q > 0$.)

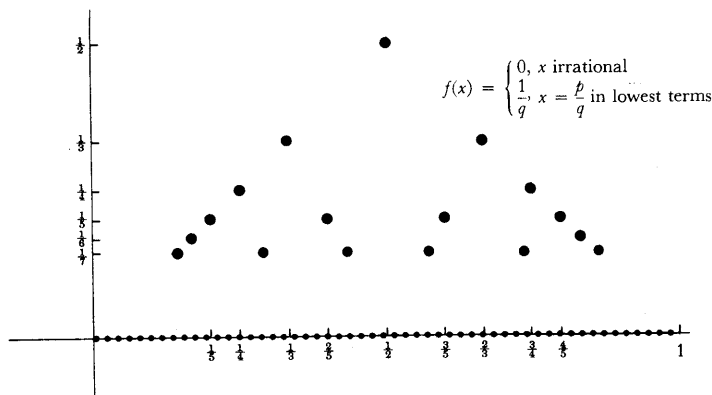


FIGURE 14

For any number a , with $0 < a < 1$, the function f approaches 0 at a . To prove this, consider any number $\epsilon > 0$. Let n be a natural number so large that $1/n \leq \epsilon$. Notice that the only numbers x for which $|f(x) - 0| < \epsilon$ could be false are:

$$\frac{1}{2}, \frac{1}{3}, \frac{2}{3}, \frac{1}{4}, \frac{3}{4}, \frac{1}{5}, \frac{2}{5}, \frac{3}{5}, \frac{4}{5}, \dots; \frac{1}{n}, \dots, \frac{n-1}{n}.$$

(If a is rational, then a might be one of these numbers.) However many of these numbers there may be, there are, at any rate, only finitely many. Therefore, of all these numbers, one is closest to a ; that is, $|p/q - a|$ is smallest for one p/q among these numbers. (If a happens to be one of these numbers, then consider only the values $|p/q - a|$ for $p/q \neq a$.) This closest distance may be chosen as the δ . For if $0 < |x - a| < \delta$, then x is *not* one of

$$\frac{1}{2}, \dots, \frac{n-1}{n}$$

and therefore $|f(x) - 0| < \epsilon$ is true. This completes the proof. Note that our description of the δ which works for a given ϵ is completely adequate—there is no reason why we must give a formula for δ in terms of ϵ .

Armed with our definition, we are now prepared to prove our first theorem; you have probably assumed the result all along, which is a very reasonable thing to do. This theorem is really a test case for our definition: if the theorem could not be proved, our definition would be useless.

THEOREM 1 A function cannot approach two different limits near a . In other words, if f approaches l near a , and f approaches m near a , then $l = m$.

PROOF Since this is our first theorem about limits it will certainly be necessary to translate the hypotheses according to the definition.

Since f approaches l near a , we know that for any $\epsilon > 0$ there is some number $\delta_1 > 0$ such that, for all x ,

$$\text{if } 0 < |x - a| < \delta_1, \text{ then } |f(x) - l| < \epsilon.$$

We also know, since f approaches m near a , that there is some $\delta_2 > 0$ such that, for all x ,

$$\text{if } 0 < |x - a| < \delta_2, \text{ then } |f(x) - m| < \epsilon.$$

We have had to use two numbers, δ_1 and δ_2 , since there is no guarantee that the δ which works in one definition will work in the other. But, in fact, it is now easy to conclude that for any $\epsilon > 0$ there is some $\delta > 0$ such that, for all x ,

$$\text{if } 0 < |x - a| < \delta, \text{ then } |f(x) - l| < \epsilon \text{ and } |f(x) - m| < \epsilon;$$

we simply choose $\delta = \min(\delta_1, \delta_2)$.

To complete the proof we just have to pick a particular $\varepsilon > 0$ for which the two conditions

$$|f(x) - l| < \varepsilon \quad \text{and} \quad |f(x) - m| < \varepsilon$$

cannot both hold, if $l \neq m$. The proper choice is suggested by Figure 15. If $l \neq m$, so that $|l - m| > 0$, we can choose $|l - m|/2$ as our ε . It follows that there is a $\delta > 0$ such that, for all x ,

$$\text{if } 0 < |x - a| < \delta, \text{ then } |f(x) - l| < \frac{|l - m|}{2}$$

$$\text{and } |f(x) - m| < \frac{|l - m|}{2}.$$

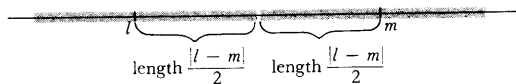


FIGURE 15

This implies that for $0 < |x - a| < \delta$ we have

$$\begin{aligned} |l - m| &= |l - f(x) + f(x) - m| \leq |l - f(x)| + |f(x) - m| \\ &< \frac{|l - m|}{2} + \frac{|l - m|}{2} \\ &= |l - m|, \end{aligned}$$

a contradiction. ■

The number l which f approaches near a is denoted by $\lim_{x \rightarrow a} f(x)$ (read: the limit of $f(x)$ as x approaches a). This definition is possible only because of Theorem 1, which ensures that $\lim_{x \rightarrow a} f(x)$ never has to stand for two different numbers. The equation

$$\lim_{x \rightarrow a} f(x) = l$$

has exactly the same meaning as the phrase

f approaches l near a .

The possibility still remains that f does not approach l near a , for any l , so that $\lim_{x \rightarrow a} f(x) = l$ is false for every number l . This is usually expressed by saying that “ $\lim_{x \rightarrow a} f(x)$ does not exist.”

Notice that our new notation introduces an extra, utterly irrelevant letter x , which could be replaced by t , y , or any other letter which does not already appear—the symbols

$$\lim_{x \rightarrow a} f(x), \quad \lim_{t \rightarrow a} f(t), \quad \lim_{y \rightarrow a} f(y),$$

all denote precisely the same number, which depends on f and a , and has nothing to do with x , t , or y (these letters, in fact, do not denote anything at all). A more logical symbol would be something like $\lim_a f$, but this notation, despite its brevity, is so infuriatingly rigid that almost no one has seriously tried to use it. The notation $\lim_{x \rightarrow a} f(x)$ is much more useful because a function

f often has no simple name, even though it might be possible to express $f(x)$ by a simple formula involving x . Thus, the short symbol

$$\lim_{x \rightarrow a} (x^2 + \sin x)$$

could be paraphrased only by the awkward expression

$$\lim_a f, \text{ where } f(x) = x^2 + \sin x.$$

Another advantage of the standard symbolism is illustrated by the expressions

$$\lim_{x \rightarrow a} x + t^3,$$

$$\lim_{t \rightarrow a} x + t^3.$$

The first means the number which f approaches near a when

$$f(x) = x + t^3, \text{ for all } x;$$

the second means the number which f approaches near a when

$$f(t) = x + t^3, \text{ for all } t.$$

You should have little difficulty (especially if you consult Theorem 2) proving that

$$\lim_{x \rightarrow a} x + t^3 = a + t^3,$$

$$\lim_{t \rightarrow a} x + t^3 = x + a^3.$$

These examples illustrate the main advantage of our notation, which is its flexibility. In fact, the notation $\lim_{x \rightarrow a} f(x)$ is so flexible that there is some danger of forgetting what it really means. Here is a simple exercise in the use of this notation, which will be important later: first interpret precisely, and then prove the equality of the expressions

$$\lim_{x \rightarrow a} f(x) \quad \text{and} \quad \lim_{h \rightarrow 0} f(a + h).$$

An important part of this chapter is the proof of a theorem which will make it easy to find many limits. The proof depends upon certain properties of inequalities and absolute values, hardly surprising when one considers the definition of limit. Although these facts have already been stated in Problems 1-19, 1-20, and 1-21, because of their importance they will be presented once again, in the form of a lemma (a lemma is an auxiliary theorem, a result that justifies its existence only by virtue of its prominent role in the proof of another theorem). The lemma says, roughly, that if x is close to x_0 , and y is close to y_0 , then $x + y$ will be close to $x_0 + y_0$, and xy will be close to $x_0 y_0$, and $1/y$ will be close to $1/y_0$. This intuitive statement is much easier to remember than the precise estimates of the lemma, and it is not unreasonable to read the proof of Theorem 2 first, in order to see just how these estimates are used.

LEMMA (1) If

$$|x - x_0| < \frac{\varepsilon}{2} \text{ and } |y - y_0| < \frac{\varepsilon}{2},$$

then

$$|(x + y) - (x_0 + y_0)| < \varepsilon.$$

(2) If

$$|x - x_0| < \min\left(1, \frac{\varepsilon}{2(|y_0| + 1)}\right) \text{ and } |y - y_0| < \frac{\varepsilon}{2(|x_0| + 1)},$$

then

$$|xy - x_0y_0| < \varepsilon.$$

(3) If $y_0 \neq 0$ and

$$|y - y_0| < \min\left(\frac{|y_0|}{2}, \frac{\varepsilon|y_0|^2}{2}\right),$$

then $y \neq 0$ and

$$\left|\frac{1}{y} - \frac{1}{y_0}\right| < \varepsilon.$$

PROOF

$$(1) |(x + y) - (x_0 + y_0)| = |(x - x_0) + (y - y_0)|$$

$$\leq |x - x_0| + |y - y_0| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

(2) Since $|x - x_0| < 1$ we have

$$|x| - |x_0| \leq |x - x_0| < 1,$$

so that

$$|x| < 1 + |x_0|.$$

Thus

$$\begin{aligned} |xy - x_0y_0| &= |x(y - y_0) + y_0(x - x_0)| \\ &\leq |x| \cdot |y - y_0| + |y_0| \cdot |x - x_0| \\ &< (1 + |x_0|) \cdot \frac{\varepsilon}{2(|x_0| + 1)} + |y_0| \cdot \frac{\varepsilon}{2(|y_0| + 1)} \\ &< \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon. \end{aligned}$$

(3) We have

$$|y_0| - |y| < |y - y_0| < \frac{|y_0|}{2},$$

so $|y| > |y_0|/2$. In particular, $y \neq 0$, and

$$\frac{1}{|y|} < \frac{2}{|y_0|}.$$

Thus

$$\left|\frac{1}{y} - \frac{1}{y_0}\right| = \frac{|y_0 - y|}{|y| \cdot |y_0|} < \frac{2}{|y_0|} \cdot \frac{1}{|y_0|} \cdot \frac{\varepsilon|y_0|^2}{2} = \varepsilon. \blacksquare$$

THEOREM 2 If $\lim_{x \rightarrow a} f(x) = l$ and $\lim_{x \rightarrow a} g(x) = m$, then

$$(1) \lim_{x \rightarrow a} (f + g)(x) = l + m;$$

$$(2) \lim_{x \rightarrow a} (f \cdot g)(x) = l \cdot m.$$

Moreover, if $m \neq 0$, then

$$(3) \lim_{x \rightarrow a} \left(\frac{1}{g} \right)(x) = \frac{1}{m}.$$

PROOF The hypothesis means that for every $\varepsilon > 0$ there are $\delta_1, \delta_2 > 0$ such that, for all x ,

$$\begin{aligned} &\text{if } 0 < |x - a| < \delta_1, \text{ then } |f(x) - l| < \varepsilon, \\ &\text{and if } 0 < |x - a| < \delta_2, \text{ then } |g(x) - m| < \varepsilon. \end{aligned}$$

This means (since, after all, $\varepsilon/2$ is also a positive number) that there are $\delta_1, \delta_2 > 0$ such that, for all x ,

$$\text{if } 0 < |x - a| < \delta_1, \text{ then } |f(x) - l| < \frac{\varepsilon}{2},$$

$$\text{and if } 0 < |x - a| < \delta_2, \text{ then } |g(x) - m| < \frac{\varepsilon}{2}.$$

Now let $\delta = \min(\delta_1, \delta_2)$. If $0 < |x - a| < \delta$, then $0 < |x - a| < \delta_1$ and $0 < |x - a| < \delta_2$ are both true, so both

$$|f(x) - l| < \frac{\varepsilon}{2} \quad \text{and} \quad |g(x) - m| < \frac{\varepsilon}{2}$$

are true. But by part (1) of the lemma this implies that $|(f + g)(x) - (l + m)| < \varepsilon$. This proves (1).

To prove (2) we proceed similarly, after consulting part (2) of the lemma. If $\varepsilon > 0$ there are $\delta_1, \delta_2 > 0$ such that, for all x ,

$$\text{if } 0 < |x - a| < \delta_1, \text{ then } |f(x) - l| < \min \left(1, \frac{\varepsilon}{2(|m| + 1)} \right),$$

$$\text{and if } 0 < |x - a| < \delta_2, \text{ then } |g(x) - m| < \frac{\varepsilon}{2(|l| + 1)}.$$

Again let $\delta = \min(\delta_1, \delta_2)$. If $0 < |x - a| < \delta$, then

$$|f(x) - l| < \min \left(1, \frac{\varepsilon}{2(|m| + 1)} \right) \quad \text{and} \quad |g(x) - m| < \frac{\varepsilon}{2(|l| + 1)}.$$

So, by the lemma, $|(f \cdot g)(x) - l \cdot m| < \varepsilon$, and this proves (2).

Finally, if $\varepsilon > 0$ there is a $\delta > 0$ such that, for all x ,

$$\text{if } 0 < |x - a| < \delta, \text{ then } |g(x) - m| < \min \left(\frac{|m|}{2}, \frac{\varepsilon|m|^2}{2} \right).$$

But according to part (3) of the lemma this means, first, that $g(x) \neq 0$, so $(1/g)(x)$ makes sense, and second that

$$\left| \left(\frac{1}{g} \right)(x) - \frac{1}{m} \right| < \varepsilon.$$

This proves (3). ■

Using Theorem 2 we can prove, trivially, such facts as

$$\lim_{x \rightarrow a} \frac{x^3 + 7x^5}{x^2 + 1} = \frac{a^3 + 7a^5}{a^2 + 1},$$

without going through the laborious process of finding a δ , given an ε . We must begin with

$$\lim_{x \rightarrow a} 7 = 7,$$

$$\lim_{x \rightarrow a} 1 = 1,$$

$$\lim_{x \rightarrow a} x = a,$$

but these are easy to prove directly. If we *want* to find the δ , however, the proof of Theorem 2 amounts to a prescription for doing this. Suppose, to take a simpler example, that we want to find a δ such that, for all x ,

$$\text{if } 0 < |x - a| < \delta, \text{ then } |x^2 + x - (a^2 + a)| < \varepsilon.$$

Consulting the proof of Theorem 2(1), we see that we must first find δ_1 and $\delta_2 > 0$ such that, for all x ,

$$\text{if } 0 < |x - a| < \delta_1, \text{ then } |x^2 - a^2| < \frac{\varepsilon}{2}$$

$$\text{and if } 0 < |x - a| < \delta_2, \text{ then } |x - a| < \frac{\varepsilon}{2}.$$

Since we have already given proofs that $\lim_{x \rightarrow a} x^2 = a^2$ and $\lim_{x \rightarrow a} x = a$, we know how to do this:

$$\delta_1 = \min \left(1, \frac{\frac{\varepsilon}{2}}{2(|a| + 1)} \right),$$

$$\delta_2 = \frac{\varepsilon}{2}.$$

Thus we can take

$$\delta = \min(\delta_1, \delta_2) = \min \left(\min \left(1, \frac{\frac{\varepsilon}{2}}{2(|a| + 1)} \right), \frac{\varepsilon}{2} \right).$$

If $a \neq 0$, the same method can be used to find a $\delta > 0$ such that, for all x ,

$$\text{if } 0 < |x - a| < \delta, \text{ then } \left| \frac{1}{x^2} - \frac{1}{a^2} \right| < \epsilon.$$

The proof of Theorem 2(3) shows that the second condition will follow if we find a $\delta > 0$ such that, for all x ,

$$\text{if } 0 < |x - a| < \delta, \text{ then } |x^2 - a^2| < \min\left(\frac{|a|^2}{2}, \frac{\epsilon|a|^4}{2}\right).$$

Thus we can take

$$\delta = \min\left(1, \frac{\min\left(\frac{|a|^2}{2}, \frac{\epsilon|a|^4}{2}\right)}{2(|a| + 1)}\right).$$

Naturally, these complicated expressions for δ can be simplified considerably, after they have been derived.

One technical detail in the proof of Theorem 2 deserves some discussion. In order for $\lim_{x \rightarrow a} f(x)$ to be defined it is, as we know, not necessary for f to be defined at a , nor is it necessary for f to be defined at all points $x \neq a$. However, there must be some $\delta > 0$ such that $f(x)$ is defined for x satisfying $0 < |x - a| < \delta$; otherwise the clause

$$\text{"if } 0 < |x - a| < \delta, \text{ then } |f(x) - l| < \epsilon"$$

would make no sense at all, since the symbol $f(x)$ would make no sense for some x 's. If f and g are two functions for which the definition makes sense, it is easy to see that the same is true for $f + g$ and $f \cdot g$. But this is not so clear for $1/g$, since $1/g$ is undefined for x with $g(x) = 0$. However, this fact gets established in the proof of Theorem 2(3).

There are times when we would like to speak of the limit which f approaches at a , even though there is no $\delta > 0$ such that $f(x)$ is defined for x satisfying $0 < |x - a| < \delta$. For example, we want to distinguish the behavior of the two functions shown in Figure 16, even though they are not defined for numbers less than a . For the function of Figure 16(a) we write

$$\lim_{x \rightarrow a^+} f(x) = l \quad \text{or} \quad \lim_{x \downarrow a} f(x) = l.$$

(The symbols on the left are read: the limit of $f(x)$ as x approaches a from above). These "limits from above" are obviously closely related to ordinary limits, and the definition is very similar: $\lim_{x \rightarrow a^+} f(x) = l$ means that for every $\epsilon > 0$ there is a $\delta > 0$ such that, for all x ,

$$\text{if } 0 < x - a < \delta, \text{ then } |f(x) - l| < \epsilon.$$

(The condition " $0 < x - a < \delta$ " is equivalent to " $0 < |x - a| < \delta$ and $x > a$.")

"Limits from below" (Figure 17) are defined similarly: $\lim_{x \rightarrow a^-} f(x) = l$

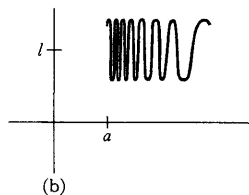
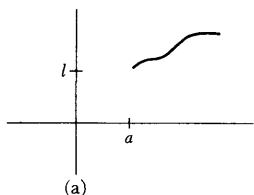


FIGURE 16

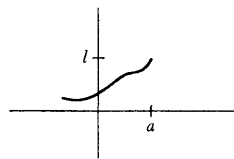


FIGURE 17

(or $\lim_{x \uparrow a} f(x) = l$) means that for every $\varepsilon > 0$ there is a $\delta > 0$ such that, for all x ,

$$\text{if } 0 < a - x < \delta, \text{ then } |f(x) - l| < \varepsilon.$$

It is quite possible to consider limits from above and below even if f is defined for numbers both greater and less than a . Thus, for the function f of Figure 13, we have

$$\lim_{x \rightarrow 0^+} f(x) = 1 \quad \text{and} \quad \lim_{x \rightarrow 0^-} f(x) = -1.$$

It is an easy exercise (Problem 27) to show that $\lim_{x \rightarrow a} f(x)$ exists if and only if $\lim_{x \rightarrow a^+} f(x)$ and $\lim_{x \rightarrow a^-} f(x)$ both exist and are equal.

Like the definitions of limits from above and below, which have been smuggled into the text informally, there are other modifications of the limit concept which will be found useful. In Chapter 4 it was claimed that if x is large, then $\sin 1/x$ is close to 0. This assertion is usually written

$$\lim_{x \rightarrow \infty} \sin 1/x = 0.$$

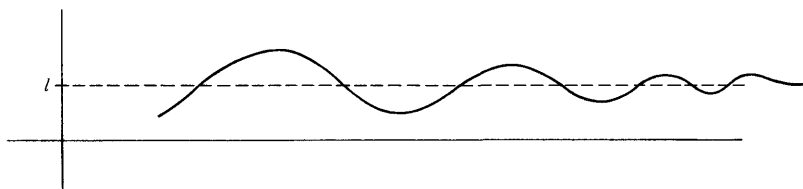


FIGURE 18

The symbol $\lim_{x \rightarrow \infty} f(x)$ is read “the limit of $f(x)$ as x approaches ∞ ,” or “as x becomes infinite,” and a limit of the form $\lim_{x \rightarrow \infty} f(x)$ is often called a limit at infinity. Figure 18 illustrates a general situation where $\lim_{x \rightarrow \infty} f(x) = l$. Formally, $\lim_{x \rightarrow \infty} f(x) = l$ means that for every $\varepsilon > 0$ there is a number N such that, for all x ,

$$\text{if } x > N, \text{ then } |f(x) - l| < \varepsilon.$$

The analogy with the definition of ordinary limits should be clear: whereas the condition “ $0 < |x - a| < \delta$ ” expresses the fact that x is close to a , the condition “ $x > N$ ” expresses the fact that x is large.

We have spent so little time on limits from above and below, and at infinity, because the general philosophy behind the definitions should be clear if you understand the definition of ordinary limits (which are by far the most important). Many exercises on these definitions are provided in the Problems, which also contain several other types of limits which are occasionally useful.

PROBLEMS

1. Find the following limits. (These limits all follow, after some algebraic manipulations, from the various parts of Theorem 2; be sure you know which ones are used in each case, but don't bother listing them.)

$$(i) \lim_{x \rightarrow 1} \frac{x^2 - 1}{x + 1}.$$

$$(ii) \lim_{x \rightarrow 2} \frac{x^3 - 8}{x - 2}.$$

$$(iii) \lim_{x \rightarrow 3} \frac{x^3 - 8}{x - 2}.$$

$$(iv) \lim_{x \rightarrow y} \frac{x^n - y^n}{x - y}.$$

$$(v) \lim_{y \rightarrow x} \frac{x^n - y^n}{x - y}.$$

$$(vi) \lim_{h \rightarrow 0} \frac{\sqrt{a+h} - \sqrt{a}}{h}.$$

2. In each of the following cases, find a δ such that $|f(x) - l| < \epsilon$ for all x satisfying $0 < |x - a| < \delta$.

$$(i) f(x) = x^4; l = a^4.$$

$$(ii) f(x) = \frac{1}{x}; a = 1, l = 1.$$

$$(iii) f(x) = x^4 + \frac{1}{x}; a = 1, l = 2.$$

$$(iv) f(x) = \frac{x}{1 + \sin^2 x}; a = 0, l = 0.$$

$$(v) f(x) = \sqrt{|x|}; a = 0, l = 0.$$

$$(vi) f(x) = \sqrt{x}; a = 1, l = 1.$$

3. For each of the functions in Problem 4-15, decide for which numbers a the limit $\lim_{x \rightarrow a} f(x)$ exists.

- *4. (a) Do the same for each of the functions in Problem 4-17.

- (b) Same problem, if we use infinite decimals ending in a string of 0's instead of those ending in a string of 9's.

5. Suppose the functions f and g have the following property: for all $\epsilon > 0$ and all x ,

$$\text{if } 0 < |x - 2| < \sin^2\left(\frac{\epsilon^2}{9}\right) + \epsilon, \text{ then } |f(x) - 2| < \epsilon,$$

$$\text{if } 0 < |x - 2| < \epsilon^2, \text{ then } |g(x) - 4| < \epsilon.$$

For each $\epsilon > 0$ find a $\delta > 0$ such that, for all x ,

- (i) if $0 < |x - 2| < \delta$, then $|f(x) + g(x) - 6| < \epsilon$.

- (ii) if $0 < |x - 2| < \delta$, then $|f(x)g(x) - 8| < \varepsilon$.
- (iii) if $0 < |x - 2| < \delta$, then $\left| \frac{1}{g(x)} - \frac{1}{4} \right| < \varepsilon$.
- (iv) if $0 < |x - 2| < \delta$, then $\left| \frac{f(x)}{g(x)} - \frac{1}{2} \right| < \varepsilon$.
6. Give an example of a function f for which the following assertion is false: If $|f(x) - l| < \varepsilon$ when $0 < |x - a| < \delta$, then $|f(x) - l| < \varepsilon/2$ when $0 < |x - a| < \delta/2$.
7. (a) If $\lim_{x \rightarrow a} f(x)$ and $\lim_{x \rightarrow a} g(x)$ do not exist, can $\lim_{x \rightarrow a} [f(x) + g(x)]$ or $\lim_{x \rightarrow a} f(x)g(x)$ exist?
- (b) If $\lim_{x \rightarrow a} f(x)$ exists and $\lim_{x \rightarrow a} [f(x) + g(x)]$ exists, must $\lim_{x \rightarrow a} g(x)$ exist?
- (c) If $\lim_{x \rightarrow a} f(x)$ exists and $\lim_{x \rightarrow a} g(x)$ does not exist, can $\lim_{x \rightarrow a} [f(x) + g(x)]$ exist?
- (d) If $\lim_{x \rightarrow a} f(x)$ exists and $\lim_{x \rightarrow a} f(x)g(x)$ exists, does it follow that $\lim_{x \rightarrow a} g(x)$ exists?
8. Prove that $\lim_{x \rightarrow a} f(x) = \lim_{h \rightarrow 0} f(a + h)$. (This is mainly an exercise in understanding what the terms mean.)
9. (a) Prove that $\lim_{x \rightarrow a} f(x) = l$ if and only if $\lim_{x \rightarrow a} [f(x) - l] = 0$. (First see why the assertion is obvious; then provide a rigorous proof. In this chapter most problems which ask for proofs should be treated in the same way.)
- (b) Prove that $\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a+0} f(x)$.
- (c) Prove that $\lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow 0} f(x^3)$.
- (d) Give an example where $\lim_{x \rightarrow 0} f(x^2)$ exists, but $\lim_{x \rightarrow 0} f(x)$ does not.
10. Suppose there is a $\delta > 0$ such that $f(x) = g(x)$ when $0 < |x - a| < \delta$. Prove that $\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} g(x)$. In other words, $\lim_{x \rightarrow a} f(x)$ depends only on the values of $f(x)$ for x near a —this fact is often expressed by saying that limits are a “local property.” (It will clearly help to use δ' , or some other letter, instead of δ , in the definition of limits.)
11. (a) Suppose that $f(x) \leq g(x)$ for all x . Prove that $\lim_{x \rightarrow a} f(x) \leq \lim_{x \rightarrow a} g(x)$, provided that these limits exist.
- (b) How can the hypotheses be weakened?
- (c) If $f(x) < g(x)$ for all x , does it necessarily follow that $\lim_{x \rightarrow a} f(x) < \lim_{x \rightarrow a} g(x)$?
12. Suppose that $f(x) \leq g(x) \leq h(x)$ and that $\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} h(x)$. Prove that $\lim_{x \rightarrow a} g(x)$ exists, and that $\lim_{x \rightarrow a} g(x) = \lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} h(x)$. (Draw a picture!)

- *13.** (a) Prove that if $\lim_{x \rightarrow 0} f(x)/x = l$ and $b \neq 0$, then $\lim_{x \rightarrow 0} f(bx)/x = bl$.
 Hint: Write $f(bx)/x = b[f(bx)/bx]$.
 (b) What happens if $b = 0$?
 (c) Part (a) enables us to find $\lim_{x \rightarrow 0} (\sin 2x)/x$ in terms of $\lim_{x \rightarrow 0} (\sin x)/x$.
 Find this limit in another way.
- 14.** (a) Prove that if $\lim_{x \rightarrow a} f(x) = l$, then $\lim_{x \rightarrow a} |f(x)| = |l|$.
 (b) Prove that if $\lim_{x \rightarrow a} f(x) = l$ and $\lim_{x \rightarrow a} g(x) = m$, then $\lim_{x \rightarrow a} \max(f, g)(x) = \max(l, m)$ and similarly for $\min$.
- 15.** (a) Prove that $\lim_{x \rightarrow 0} 1/x$ does not exist, i.e., show that $\lim_{x \rightarrow 0} 1/x = l$ is false for every number l .
 (b) Prove that $\lim_{x \rightarrow 1} 1/(x - 1)$ does not exist.
- 16.** Prove that if $\lim_{x \rightarrow a} f(x) = l$, then there is a number $\delta > 0$ and a number M such that $|f(x)| < M$ if $0 < |x - a| < \delta$. (What does this mean pictorially?) Hint: Why does it suffice to prove that $l - 1 < f(x) < l + 1$ for $0 < |x - a| < \delta$?
- 17.** Prove that if $f(x) = 0$ for irrational x and $f(x) = 1$ for rational x , then $\lim_{x \rightarrow a} f(x)$ does not exist for any a .
- *18.** Prove that if $f(x) = x$ for rational x , and $f(x) = -x$ for irrational x , then $\lim_{x \rightarrow a} f(x)$ does not exist if $a \neq 0$.
- 19.** (a) Prove that if $\lim_{x \rightarrow 0} g(x) = 0$, then $\lim_{x \rightarrow 0} g(x) \sin 1/x = 0$.
 (b) Generalize this fact as follows: If $\lim_{x \rightarrow 0} g(x) = 0$ and $|h(x)| \leq M$ for all x , then $\lim_{x \rightarrow 0} g(x)h(x) = 0$. (Naturally it is unnecessary to do part (a) if you succeed in doing part (b); actually the statement of part (b) may make it easier than (a)—that's one of the values of generalization.)
- 20.** Consider a function f with the following property: if g is any function for which $\lim_{x \rightarrow 0} g(x)$ does not exist, then $\lim_{x \rightarrow 0} [f(x) + g(x)]$ also does not exist. Prove that this happens if and only if $\lim_{x \rightarrow 0} f(x)$ does exist. Hint: This is actually very easy: the assumption that $\lim_{x \rightarrow 0} f(x)$ does not exist leads to an immediate contradiction if you consider the right g .
- **21.** This problem is the analogue of Problem 20 when $f + g$ is replaced by $f \cdot g$. In this case the situation is considerably more complex, and the analysis requires several steps (those in search of an especially challenging problem can attempt an independent solution).
- (a) Suppose that $\lim_{x \rightarrow 0} f(x)$ exists and is $\neq 0$. Prove that if $\lim_{x \rightarrow 0} g(x)$ does not exist, then $\lim_{x \rightarrow 0} f(x)g(x)$ also does not exist.

- (b) Prove the same result if $\lim_{x \rightarrow 0} |f(x)| = \infty$. (The precise definition of this sort of limit is given in Problem 33.)
- (c) Prove that if neither of these two conditions holds, then there is a function g such that $\lim_{x \rightarrow 0} g(x)$ does not exist, but $\lim_{x \rightarrow 0} f(x)g(x)$ does exist. Hint: Consider separately the following two cases: (1) For some $\varepsilon > 0$ we have $|f(x)| > \varepsilon$ for all sufficiently small x . (2) For every $\varepsilon > 0$, there are arbitrarily small x with $|f(x)| < \varepsilon$. In the second case, begin by choosing points x_n with $|x_n| < 1/n$ and $|f(x_n)| < 1/n$.
- *22.** Suppose that A_n is, for each natural number n , some *finite* set of numbers in $[0, 1]$, and that A_n and A_m have no members in common if $m \neq n$. Define f as follows:

$$f(x) = \begin{cases} 1/n, & x \text{ in } A_n \\ 0, & x \text{ not in } A_n \text{ for any } n. \end{cases}$$

Prove that $\lim_{x \rightarrow a} f(x) = 0$ for all a in $[0, 1]$.

- 23.** Explain why the following definitions of $\lim_{x \rightarrow a} f(x) = l$ are all correct: For every $\delta > 0$ there is an $\varepsilon > 0$ such that, for all x ,

- (i) if $0 < |x - a| < \varepsilon$, then $|f(x) - l| < \delta$.
- (ii) if $0 < |x - a| < \varepsilon$, then $|f(x) - l| \leq \delta$.
- (iii) if $0 < |x - a| < \varepsilon$, then $|f(x) - l| < 5\delta$.
- (iv) if $0 < |x - a| < \varepsilon/10$, then $|f(x) - l| < \delta$.

- *24.** Give examples to show that the following definitions of $\lim_{x \rightarrow a} f(x) = l$ are *not* correct.

- (a) For all $\delta > 0$ there is an $\varepsilon > 0$ such that if $0 < |x - a| < \delta$, then $|f(x) - l| < \varepsilon$.
- (b) For all $\varepsilon > 0$ there is a $\delta > 0$ such that if $|f(x) - l| < \varepsilon$, then $0 < |x - a| < \delta$.

- 25.** For each of the functions in Problem 4-15 indicate for which numbers a the one-sided limits $\lim_{x \rightarrow a^+} f(x)$ and $\lim_{x \rightarrow a^-} f(x)$ exist.

- *26.** (a) Do the same for each of the functions in Problem 4-17.
 (b) Also consider what happens if decimals ending in 0's are used instead of decimals ending in 9's.
- 27.** Prove that $\lim_{x \rightarrow a} f(x)$ exists if $\lim_{x \rightarrow a^+} f(x) = \lim_{x \rightarrow a^-} f(x)$.
- 28.** Prove that

- (i) $\lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^-} f(-x)$.
- (ii) $\lim_{x \rightarrow 0} f(|x|) = \lim_{x \rightarrow 0^+} f(x)$.

$$(iii) \lim_{x \rightarrow 0} f(x^2) = \lim_{x \rightarrow 0^+} f(x).$$

(These equations, and others like them, are open to several interpretations. They might mean only that the two limits are equal if they both exist; or that if a certain one of the limits exists, the other also exists and is equal to it; or that if either limit exists, then the other exists and is equal to it. Decide for yourself which interpretations are suitable.)

- *29. Suppose that $\lim_{x \rightarrow a^-} f(x) < \lim_{x \rightarrow a^+} f(x)$. (Draw a picture to illustrate this assertion.) Prove that there is some $\delta > 0$ such that $f(x) < f(y)$ whenever $x < a < y$ and $|x - a| < \delta$ and $|y - a| < \delta$. Is the converse true?
- *30. Prove that $\lim_{x \rightarrow \infty} (a_n x^n + \cdots + a_0) / (b_m x^m + \cdots + b_0)$ exists if and only if $m \geq n$. What is the limit when $m = n$? When $m > n$? Hint: The one easy limit is $\lim_{x \rightarrow \infty} 1/x^k = 0$; do some algebra so that this is the only information you need.

$$31. \text{ Prove that } \lim_{x \rightarrow 0^+} f(1/x) = \lim_{x \rightarrow \infty} f(x).$$

$$32. \text{ Define " } \lim_{x \rightarrow -\infty} f(x) = l \text{."}$$

$$(a) \text{ Find } \lim_{x \rightarrow -\infty} (a_n x^n + \cdots + a_0) / (b_m x^m + \cdots + b_0).$$

$$(b) \text{ Prove that } \lim_{x \rightarrow \infty} f(x) = \lim_{x \rightarrow -\infty} f(-x).$$

$$(c) \text{ Prove that } \lim_{x \rightarrow 0^-} f(1/x) = \lim_{x \rightarrow -\infty} f(x).$$

33. We define $\lim_{x \rightarrow a} f(x) = \infty$ to mean that for all N there is a $\delta > 0$ such that, for all x , if $0 < |x - a| < \delta$, then $f(x) > N$. (Draw an appropriate picture!)

$$(a) \text{ Show that } \lim_{x \rightarrow 3} 1/(x - 3)^2 = \infty.$$

$$(b) \text{ Prove that if } f(x) > \varepsilon > 0 \text{ for all } x, \text{ and } \lim_{x \rightarrow a} g(x) = 0, \text{ then}$$

$$\lim_{x \rightarrow a} f(x)/g(x) = \infty.$$

34. (a) Define $\lim_{x \rightarrow a^+} f(x) = \infty$, $\lim_{x \rightarrow a^-} f(x) = \infty$, and $\lim_{x \rightarrow \infty} f(x) = \infty$. (Or at least convince yourself that you could write down the definitions if you had the energy. How many other such symbols can you define?)

$$(b) \text{ Prove that } \lim_{x \rightarrow 0^+} 1/x = \infty.$$

$$(c) \text{ Prove that } \lim_{x \rightarrow 0^+} f(x) = \infty \text{ if and only if } \lim_{x \rightarrow \infty} f(1/x) = \infty.$$

6

CHAPTER 6 CONTINUOUS FUNCTIONS

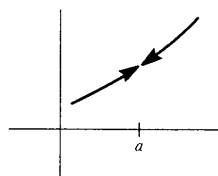


FIGURE 1

If f is an arbitrary function, it is not necessarily true that

$$\lim_{x \rightarrow a} f(x) = f(a).$$

In fact, there are many ways this can fail to be true. For example, f might not even be defined at a , in which case the equation makes no sense (Figure 1).

Again, $\lim_{x \rightarrow a} f(x)$ might not exist (Figure 2). Finally, as illustrated in Figure 3, even if f is defined at a and $\lim_{x \rightarrow a} f(x)$ exists, the limit might not equal $f(a)$.

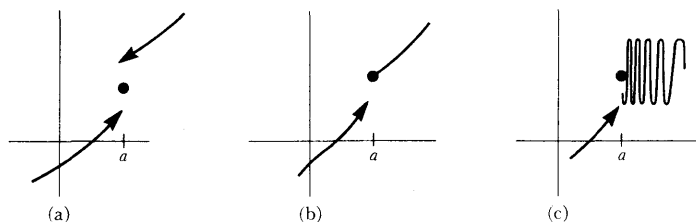


FIGURE 2

We would like to regard all behavior of this type as abnormal and honor, with some complimentary designation, functions which do not exhibit such peculiarities. The term which has been adopted is “continuous.” Intuitively, a function f is continuous if the graph contains no breaks, jumps, or wild oscillations. Although this description will usually enable you to decide whether a function is continuous simply by looking at its graph (a skill well worth cultivating) it is easy to be fooled, and the precise definition is *very* important.

DEFINITION

The function f is **continuous at a** if

$$\lim_{x \rightarrow a} f(x) = f(a).$$

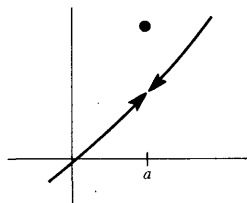


FIGURE 3

We will have no difficulty finding many examples of functions which are, or are not, continuous at some number a —every example involving limits provides an example about continuity, and Chapter 5 certainly provides enough of these.

The function $f(x) = \sin 1/x$ is not continuous at 0, because it is not even defined at 0, and the same is true of the function $g(x) = x \sin 1/x$. On the other hand, if we are willing to extend the second of these functions, that is, if

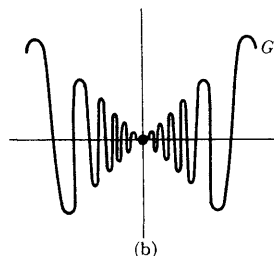
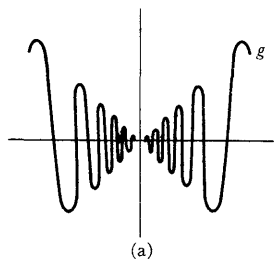


FIGURE 4

we wish to define a new function G by

$$G(x) = \begin{cases} x \sin 1/x, & x \neq 0 \\ a, & x = 0, \end{cases}$$

then the choice of $a = G(0)$ can be made in such a way that G will be continuous at 0—to do this we can (in fact, we must) define $G(0) = 0$ (Figure 4). This sort of extension is not possible for f ; if we define

$$F(x) = \begin{cases} \sin 1/x, & x \neq 0 \\ a, & x = 0, \end{cases}$$

then F will not be continuous at 0, no matter what a is, because $\lim_{x \rightarrow 0} f(x)$ does not exist.

The function

$$f(x) = \begin{cases} x, & x \text{ rational} \\ 0, & x \text{ irrational} \end{cases}$$

is not continuous at a , if $a \neq 0$, since $\lim_{x \rightarrow a} f(x)$ does not exist. However, $\lim_{x \rightarrow 0} f(x) = 0 = f(0)$, so f is continuous at precisely one point, 0.

The functions $f(x) = c$, $g(x) = x$, and $h(x) = x^2$ are continuous at all numbers a , since

$$\begin{aligned} \lim_{x \rightarrow a} f(x) &= \lim_{x \rightarrow a} c = c = f(a), \\ \lim_{x \rightarrow a} g(x) &= \lim_{x \rightarrow a} x = a = g(a), \\ \lim_{x \rightarrow a} h(x) &= \lim_{x \rightarrow a} x^2 = a^2 = h(a). \end{aligned}$$

Finally, consider the function

$$f(x) = \begin{cases} 0, & x \text{ irrational} \\ 1/q, & x = p/q \text{ in lowest terms.} \end{cases}$$

In Chapter 5 we showed that $\lim_{x \rightarrow a} f(x) = 0$ for all a . Since $0 = f(a)$ only when a is irrational, this function is continuous at a if a is irrational, but not if a is rational.

It is even easier to give examples of continuity if we prove two simple theorems.

THEOREM 1 If f and g are continuous at a , then

- (1) $f + g$ is continuous at a ,
- (2) $f \cdot g$ is continuous at a .

Moreover, if $g(a) \neq 0$, then

- (3) $1/g$ is continuous at a .

PROOF Since f and g are continuous at a ,

$$\lim_{x \rightarrow a} f(x) = f(a) \quad \text{and} \quad \lim_{x \rightarrow a} g(x) = g(a).$$

By Theorem 2(1) of Chapter 5 this implies that

$$\lim_{x \rightarrow a} (f + g)(x) = f(a) + g(a) = (f + g)(a),$$

which is just the assertion that $f + g$ is continuous at a . The proofs of parts (2) and (3) are left to you. ■

Starting with the functions $f(x) = c$ and $f(x) = x$, which are continuous at a , for every a , we can use Theorem 1 to conclude that a function

$$f(x) = \frac{b_n x^n + b_{n-1} x^{n-1} + \cdots + b_0}{c_m x^m + c_{m-1} x^{m-1} + \cdots + c_0}$$

is continuous at every point in its domain. But it is harder to get much further than that. When we discuss the sine function in detail it will be easy to prove that $\sin$ is continuous at a for all a ; let us assume this fact meanwhile. A function like

$$f(x) = \frac{\sin^2 x + x^2 + x^4 \sin x}{\sin^{27} x + 4x^2 \sin^2 x}$$

can now be proved continuous at every point in its domain. But we are still unable to prove the continuity of a function like $f(x) = \sin(x^2)$; we obviously need a theorem about the composition of continuous functions. Before stating this theorem, the following point about the definition of continuity is worth noting. If we translate the equation $\lim_{x \rightarrow a} f(x) = f(a)$ according to the definition of limits, we obtain

$$\begin{aligned} &\text{for every } \varepsilon > 0 \text{ there is } \delta > 0 \text{ such that, for all } x, \\ &\text{if } 0 < |x - a| < \delta, \text{ then } |f(x) - f(a)| < \varepsilon. \end{aligned}$$

But in this case, where the limit is $f(a)$, the phrase

$$0 < |x - a| < \delta$$

may be changed to the simpler condition

$$|x - a| < \delta,$$

since if $x = a$ it is certainly true that $|f(x) - f(a)| < \varepsilon$.

THEOREM 2 If g is continuous at a , and f is continuous at $g(a)$, then $f \circ g$ is continuous at a . (Notice that f is required to be continuous at $g(a)$, not at a .)

PROOF Let $\varepsilon > 0$. We wish to find a $\delta > 0$ such that for all x ,

$$\begin{aligned} &\text{if } |x - a| < \delta, \text{ then } |(f \circ g)(x) - (f \circ g)(a)| < \varepsilon, \\ &\text{i.e., } |f(g(x)) - f(g(a))| < \varepsilon. \end{aligned}$$

We first use continuity of f to estimate how close $g(x)$ must be to $g(a)$ in order for this inequality to hold. Since f is continuous at $g(a)$, there is a $\delta' > 0$ such that for all y ,

$$(1) \quad \text{if } |y - g(a)| < \delta', \text{ then } |f(y) - f(g(a))| < \varepsilon.$$

In particular, this means that

$$(2) \quad \text{if } |g(x) - g(a)| < \delta', \text{ then } |f(g(x)) - f(g(a))| < \varepsilon.$$

We now use continuity of g to estimate how close x must be to a in order for the inequality $|g(x) - g(a)| < \delta'$ to hold. The number δ' is a positive number just like any other positive number; we can therefore take δ' as the ε (!) in the definition of continuity of g at a . We conclude that there is a $\delta > 0$ such that, for all x ,

$$(3) \quad \text{if } |x - a| < \delta, \text{ then } |g(x) - g(a)| < \delta'.$$

Combining (2) and (3) we see that for all x ,

$$\text{if } |x - a| < \delta, \text{ then } |f(g(x)) - f(g(a))| < \varepsilon. \blacksquare$$

We can now reconsider the function

$$f(x) = \begin{cases} x \sin 1/x, & x \neq 0 \\ 0, & x = 0. \end{cases}$$

We have already noted that f is continuous at 0. A few applications of Theorems 1 and 2, together with the continuity of $\sin$, show that f is also continuous at a , for $a \neq 0$. Functions like $f(x) = \sin(x^2 + \sin(x + \sin^2(x^3)))$ should be equally easy for you to analyze.

The few theorems of this chapter have all been related to continuity of functions at a single point, but the concept of continuity doesn't begin to be really interesting until we focus our attention on functions which are continuous at all points of some interval. If f is continuous at x for all x in (a, b) , then f is called **continuous on** (a, b) . Continuity on a closed interval must be defined a little differently; a function f is called **continuous on** $[a, b]$ if

- (1) f is continuous at x for all x in (a, b) ,
- (2) $\lim_{x \rightarrow a^+} f(x) = f(a)$ and $\lim_{x \rightarrow b^-} f(x) = f(b)$.

Functions which are continuous on an interval are usually regarded as especially well behaved; indeed continuity might be specified as the first condition which a "reasonable" function ought to satisfy. A continuous function is sometimes described, intuitively, as one whose graph can be drawn without lifting your pencil from the paper. Consideration of the function

$$f(x) = \begin{cases} x \sin 1/x, & x \neq 0 \\ 0, & x = 0 \end{cases}$$

shows that this description is a little too optimistic, but it is nevertheless true that there are many important results involving functions which are continuous on an interval. These theorems are generally much harder than the ones in this chapter, but there is a simple theorem which forms a bridge between the two kinds of results. The hypothesis of this theorem requires continuity at only a single point, but the conclusion describes the behavior of the function on some interval containing the point. Although this theorem is really a lemma for later arguments, it is included here as a preview of things to come.

THEOREM 3 Suppose f is continuous at a , and $f(a) > 0$. Then there is a number $\delta > 0$ such that $f(x) > 0$ for all x satisfying $|x - a| < \delta$. Similarly, if $f(a) < 0$, then there is a number $\delta > 0$ such that $f(x) < 0$ for all x satisfying $|x - a| < \delta$.

PROOF Consider the case $f(a) > 0$. Since f is continuous at a , if $\varepsilon > 0$ there is a $\delta > 0$ such that, for all x ,

$$\text{if } |x - a| < \delta, \text{ then } |f(x) - f(a)| < \varepsilon.$$

Since $f(a) > 0$ we can take $f(a)$ as the ε . Thus there is $\delta > 0$ so that for all x ,

$$\text{if } |x - a| < \delta, \text{ then } |f(x) - f(a)| < f(a),$$

and this last inequality implies $f(x) > 0$.

A similar proof can be given in the case $f(a) < 0$; take $\varepsilon = -f(a)$. Or one can apply the first case to the function $-f$. ■

PROBLEMS

- For which of the following functions f is there a continuous function F with domain $\mathbf{R}$ such that $F(x) = f(x)$ for all x in the domain of f ?
 - $f(x) = \frac{x^2 - 4}{x - 2}$.
 - $f(x) = \frac{|x|}{x}$.
 - $f(x) = 0$, x irrational.
 - $f(x) = 1/q$, $x = p/q$ rational in lowest terms.
- At which points are the functions of Problems 4-15 and 4-17 continuous?
- Suppose that f is a function satisfying $|f(x)| \leq |x|$ for all x . Show that f is continuous at 0. (Notice that $f(0)$ must equal 0.)
 - Give an example of such a function f which is not continuous at any $a \neq 0$.
 - Suppose that g is continuous at 0 and $g(0) = 0$, and $|f(x)| \leq |g(x)|$. Prove that f is continuous at 0.
- Give an example of a function f such that f is continuous nowhere, but $|f|$ is continuous everywhere.

5. For each number a , find a function which is continuous at a , but not at any other points.
6. (a) Find a function f which is discontinuous at $1, \frac{1}{2}, \frac{1}{3}, \dots$ but continuous at all other points.
 (b) Find a function f which is discontinuous at $1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots$, and at 0 , but continuous at all other points.
7. Suppose that f satisfies $f(x + y) = f(x) + f(y)$, and that f is continuous at 0 . Prove that f is continuous at a for all a .
8. Suppose that f is continuous at a and $f(a) = 0$. Prove that if $\alpha \neq 0$, then $f + \alpha$ is nonzero in some open interval containing a .
9. (a) Suppose f is *not* continuous at a . Prove that for some number $\varepsilon > 0$ there are numbers x arbitrarily close to a with $|f(x) - f(a)| > \varepsilon$. Illustrate graphically.
 (b) Conclude that for some number $\varepsilon > 0$ *either* there are numbers x arbitrarily close to a with $f(x) < f(a) - \varepsilon$ *or* there are numbers x arbitrarily close to a with $f(x) > f(a) + \varepsilon$.
10. (a) Prove that if f is continuous at a , then so is $|f|$.
 (b) Prove that every continuous f can be written $f = E + O$, where E is even and continuous and O is odd and continuous.
 (c) Prove that if f and g are continuous, then so are $\max(f, g)$ and $\min(f, g)$.
 (d) Prove that every continuous f can be written $f = g - h$, where g and h are nonnegative and continuous.
11. Prove Theorem 1(3) by using Theorem 2 and continuity of the function $f(x) = 1/x$.
- *12. (a) Prove that if f is continuous at l and $\lim_{x \rightarrow a} g(x) = l$, then $\lim_{x \rightarrow a} f(g(x)) = f(l)$. (You can go right back to the definitions, but it is easier to consider the function G with $G(x) = g(x)$ for $x \neq a$, and $G(a) = l$.)
 (b) Show that if continuity of f at l is not assumed, then it is not generally true that $\lim_{x \rightarrow a} f(g(x)) = f(\lim_{x \rightarrow a} g(x))$. Hint: Try $f(x) = 0$ for $x \neq l$, and $f(l) = 1$.
13. (a) Prove that if f is continuous on $[a, b]$, then there is a function g which is continuous on $\mathbf{R}$, and which satisfies $g(x) = f(x)$ for all x in $[a, b]$. Hint: Since you obviously have a great deal of choice, try making g constant on $(-\infty, a]$ and $[b, \infty)$.
 (b) Give an example to show that this assertion is false if $[a, b]$ is replaced by (a, b) .
14. (a) Suppose that g and h are continuous at a , and that $g(a) = h(a)$. Define $f(x)$ to be $g(x)$ if $x \geq a$ and $h(x)$ if $x \leq a$. Prove that f is continuous at a .
 (b) Suppose g is continuous on $[a, b]$ and h is continuous on $[b, c]$ and $g(b) = h(b)$. Let $f(x)$ be $g(x)$ for x in $[a, b]$ and $h(x)$ for x in $[b, c]$. Show that f is continuous on $[a, c]$. (Thus, continuous functions can be "pasted together.")

15. (a) Prove the following version of Theorem 3 for “right-hand continuity”: Suppose that $\lim_{x \rightarrow a^+} f(x) = f(a)$, and $f(a) > 0$. Then there is a number $\delta > 0$ such that $f(x) > 0$ for all x satisfying $0 \leq x - a < \delta$. Similarly, if $f(a) < 0$, then there is a number $\delta > 0$ such that $f(x) < 0$ for all x satisfying $0 \leq x - a < \delta$.
- (b) Prove a version of Theorem 3 when $\lim_{x \rightarrow b^-} f(x) = f(b)$.
16. If $\lim_{x \rightarrow a} f(x)$ exists, but is $\neq f(a)$, then f is said to have a **removable discontinuity** at a .
- (a) If $f(x) = \sin 1/x$ for $x \neq 0$ and $f(0) = 1$, does f have a removable discontinuity at 0? What if $f(x) = x \sin 1/x$ for $x \neq 0$, and $f(0) = 1$?
- (b) Suppose f has a removable discontinuity at a . Let $g(x) = f(x)$ for $x \neq a$, and let $g(a) = \lim_{x \rightarrow a} f(x)$. Prove that g is continuous at a . (Don’t work very hard; this is quite easy.)
- (c) Let $f(x) = 0$ if x is irrational, and let $f(p/q) = 1/q$ if p/q is in lowest terms. What is the function g defined by $g(x) = \lim_{y \rightarrow x} f(y)$?
- *(d) Let f be a function with the property that every point of discontinuity is a removable discontinuity. This means that $\lim_{y \rightarrow x} f(y)$ exists for all x , but f may be discontinuous at some (even infinitely many) numbers x . Define $g(x) = \lim_{y \rightarrow x} f(y)$. Prove that g is continuous. (This is not quite so easy as part (b).)
- ** (e) Is there a function f which is discontinuous at every point, and which has only removable discontinuities? (It is worth thinking about this problem now, but mainly as a test of intuition; even if you suspect the correct answer, you will almost certainly be unable to prove it at the present time. See Problem 21-24.)

CHAPTER 7 THREE HARD THEOREMS

This chapter is devoted to three theorems about continuous functions, and some of their consequences. The proofs of the three theorems themselves will not be given until the next chapter, for reasons which are explained at the end of this chapter.

THEOREM 1 If f is continuous on $[a, b]$ and $f(a) < 0 < f(b)$, then there is some x in $[a, b]$ such that $f(x) = 0$.

(Geometrically, this means that the graph of a continuous function which starts below the horizontal axis and ends above it must cross this axis at some point, as in Figure 1.)

THEOREM 2 If f is continuous on $[a, b]$, then f is bounded above on $[a, b]$, that is, there is some number N such that $f(x) \leq N$ for all x in $[a, b]$.

(Geometrically, this theorem means that the graph of f lies below some line parallel to the horizontal axis, as in Figure 2).

THEOREM 3 If f is continuous on $[a, b]$, then there is some number y in $[a, b]$ such that $f(y) \geq f(x)$ for all x in $[a, b]$ (Figure 3).

These three theorems differ markedly from the theorems of Chapter 6. The hypotheses of those theorems always involved continuity at a single point,

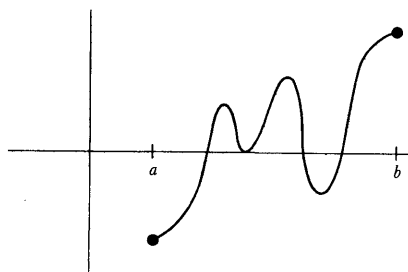


FIGURE 1

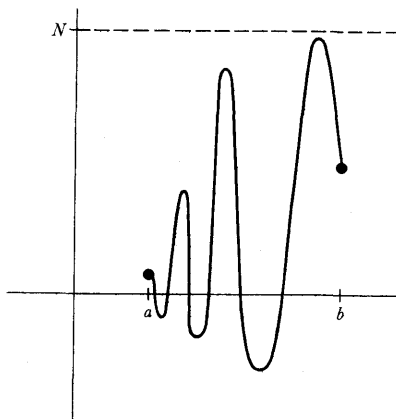


FIGURE 2

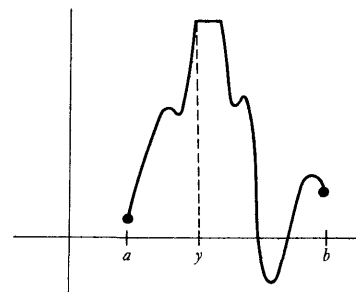


FIGURE 3

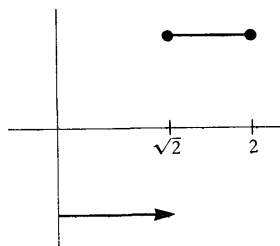


FIGURE 4

while the hypotheses of the present theorems require continuity on a whole interval $[a, b]$ —if continuity fails to hold at a single point, the conclusions may fail. For example, let f be the function shown in Figure 4,

$$f(x) = \begin{cases} -1, & 0 \leq x < \sqrt{2} \\ 1, & \sqrt{2} \leq x \leq 2. \end{cases}$$

Then f is continuous at every point of $[0, 2]$ except $\sqrt{2}$, and $f(0) < 0 < f(2)$, but there is no point x in $[0, 2]$ such that $f(x) = 0$; the discontinuity at the single point $\sqrt{2}$ is sufficient to destroy the conclusion of Theorem 1.

Similarly, suppose that f is the function shown in Figure 5,

$$f(x) = \begin{cases} 1/x, & x \neq 0 \\ 0, & x = 0. \end{cases}$$

Then f is continuous at every point of $[0, 1]$ except 0, but f is not bounded above on $[0, 1]$. In fact for any number $N > 0$, we have $f(\frac{1}{2N}) = 2N > N$.

This example also shows that the closed interval $[a, b]$ in Theorem 2 cannot be replaced by the open interval (a, b) , for the function f is continuous on $(0, 1)$, but is not bounded there.

Finally, consider the function shown in Figure 6,

$$f(x) = \begin{cases} x^2, & x < 1 \\ 0, & x \geq 1. \end{cases}$$

On the interval $[0, 1]$ the function f is bounded above, so f does satisfy the conclusion of Theorem 2, even though f is not continuous on $[0, 1]$. But f does not satisfy the conclusion of Theorem 3—there is no y in $[0, 1]$ such that $f(y) \geq f(x)$ for all x in $[0, 1]$; in fact, it is certainly not true that $f(1) \geq f(x)$ for all x in $[0, 1]$ so we cannot choose $y = 1$, nor can we choose $0 \leq y < 1$ because $f(y) < f(x)$ if x is any number with $y < x < 1$.

This example shows that Theorem 3 is considerably stronger than Theorem 2. Theorem 3 is often paraphrased by saying that a continuous function on a closed interval “takes on its maximum value” on that interval.

As a compensation for the stringency of the hypotheses of our three theorems, the conclusions are of a totally different order than those of previous theorems. They describe the behavior of a function, not just near a point, but on a whole interval; such “global” properties of a function are always significantly more difficult to prove than “local” properties, and are correspondingly of much greater power. To illustrate the usefulness of Theorems 1, 2, and 3, we will soon deduce some important consequences, but it will help to first mention some simple generalizations of these theorems.

THEOREM 4

If f is continuous on $[a, b]$ and $f(a) < c < f(b)$, then there is some x in $[a, b]$ such that $f(x) = c$.

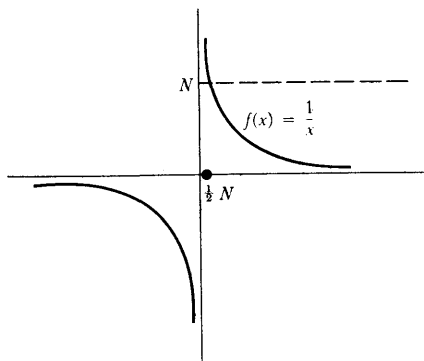


FIGURE 5

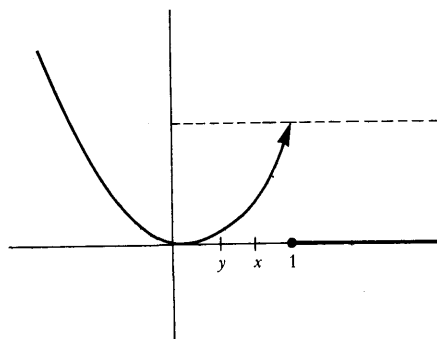


FIGURE 6

PROOF Let $g = f - c$. Then g is continuous, and $g(a) < 0 < g(b)$. By Theorem 1, there is some x in $[a, b]$ such that $g(x) = 0$. But this means that $f(x) = c$. ■

THEOREM 5 If f is continuous on $[a, b]$ and $f(a) > c > f(b)$, then there is some x in $[a, b]$ such that $f(x) = c$.

PROOF The function $-f$ is continuous on $[a, b]$ and $-f(a) < -c < -f(b)$. By Theorem 4 there is some x in $[a, b]$ such that $-f(x) = -c$, which means that $f(x) = c$. ■

Theorems 4 and 5 together show that f takes on any value between $f(a)$ and $f(b)$. We can do even better than this: if c and d are in $[a, b]$, then f takes on any value between $f(c)$ and $f(d)$. The proof is simple; if, for example, $c < d$, then just apply Theorems 4 and 5 to the interval $[c, d]$. Summarizing, if a continuous function on an interval takes on two values, it takes on every value in between; this slight generalization of Theorem 1 is often called the Intermediate Value Theorem.

THEOREM 6 If f is continuous on $[a, b]$, then f is bounded below on $[a, b]$, that is, there is some number N such that $f(x) \geq N$ for all x in $[a, b]$.

PROOF The function $-f$ is continuous on $[a, b]$, so by Theorem 2 there is a number M such that $-f(x) \leq M$ for all x in $[a, b]$. But this means that $f(x) \geq -M$ for all x in $[a, b]$, so we can let $N = -M$. ■

Theorems 2 and 6 together show that a continuous function f on $[a, b]$ is bounded on $[a, b]$, that is, there is a number N such that $|f(x)| \leq N$ for all x in $[a, b]$. In fact, since Theorem 2 ensures the existence of a number N_1 such that $f(x) \leq N_1$ for all x in $[a, b]$, and Theorem 6 ensures the existence of a number N_2 such that $f(x) \geq N_2$ for all x in $[a, b]$, we can take $N = \max(|N_1|, |N_2|)$.

THEOREM 7 If f is continuous on $[a, b]$, then there is some y in $[a, b]$ such that $f(y) \leq f(x)$ for all x in $[a, b]$.
(A continuous function on a closed interval takes on its minimum value on that interval.)

PROOF The function $-f$ is continuous on $[a, b]$; by Theorem 3 there is some y in $[a, b]$ such that $-f(y) \geq -f(x)$ for all x in $[a, b]$, which means that $f(y) \leq f(x)$ for all x in $[a, b]$. ■

Now that we have derived the trivial consequences of Theorems 1, 2, and 3, we can begin proving a few interesting things.

THEOREM 8 Every positive number has a square root. In other words, if $\alpha > 0$, then there is some number x such that $x^2 = \alpha$.

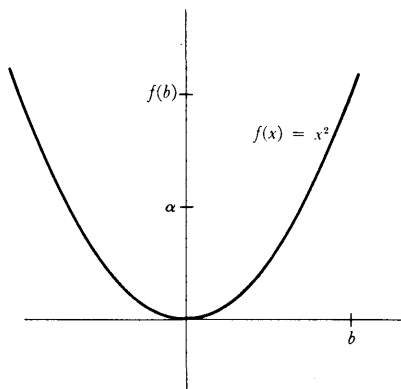


FIGURE 7

PROOF

Consider the function $f(x) = x^2$, which is certainly continuous. Notice that the statement of the theorem can be expressed in terms of f : “the number α has a square root” means that f takes on the value α . The proof of this fact about f will be an easy consequence of Theorem 4.

There is obviously a number $b > 0$ such that $f(b) > \alpha$ (as illustrated in Figure 7); in fact, if $\alpha > 1$ we can take $b = \alpha$, while if $\alpha < 1$ we can take $b = 1$. Since $f(0) < \alpha < f(b)$, Theorem 4 applied to $[0, b]$ implies that for some x (in $[0, b]$), we have $f(x) = \alpha$, i.e., $x^2 = \alpha$. ■

Precisely the same argument can be used to prove that a positive number has an n th root, for any natural number n . If n happens to be odd, one can do better: *every* number has an n th root. To prove this we just note that if the positive number α has the n th root x , i.e., if $x^n = \alpha$, then $(-x)^n = -\alpha$ (since n is odd), so $-\alpha$ has the n th root $-x$. The assertion, that for odd n any number α has an n th root, is equivalent to the statement that the equation

$$x^n - \alpha = 0$$

has a root if n is odd. Expressed in this way the result is susceptible of great generalization.

THEOREM 9

If n is odd, then any equation

$$x^n + a_{n-1}x^{n-1} + \cdots + a_0 = 0$$

has a root.

PROOF

We obviously want to consider the function

$$f(x) = x^n + a_{n-1}x^{n-1} + \cdots + a_0;$$

we would like to prove that f is sometimes positive and sometimes negative. The intuitive idea is that for large $|x|$, the function is very much like $g(x) = x^n$ and, since n is odd, this function is positive for large positive x and negative for large negative x . A little algebra is all we need to make this intuitive idea work.

The proper analysis of the function f depends on writing

$$f(x) = x^n + a_{n-1}x^{n-1} + \cdots + a_0 = x^n \left(1 + \frac{a_{n-1}}{x} + \cdots + \frac{a_0}{x^n} \right).$$

Note that

$$\left| \frac{a_{n-1}}{x} + \frac{a_{n-2}}{x^2} + \cdots + \frac{a_0}{x^n} \right| \leq \frac{|a_{n-1}|}{|x|} + \cdots + \frac{|a_0|}{|x^n|}.$$

Consequently, if we choose x satisfying

$$(*) \quad |x| > 1, 2n|a_{n-1}|, \dots, 2n|a_0|,$$

then $|x^k| \geq |x|$ and

$$\frac{|a_{n-k}|}{|x^k|} < \frac{|a_{n-k}|}{|x|} < \frac{|a_{n-k}|}{2n|a_{n-k}|} = \frac{1}{2n},$$

so

$$\left| \frac{a_{n-1}}{x} + \frac{a_{n-2}}{x^2} + \cdots + \frac{a_0}{x^n} \right| \leq \underbrace{\frac{1}{2n} + \cdots + \frac{1}{2n}}_{n \text{ terms}} = \frac{1}{2}.$$

In other words,

$$-\frac{1}{2} \leq \frac{a_{n-1}}{x} + \cdots + \frac{a_0}{x^n} \leq \frac{1}{2},$$

which implies that

$$\frac{1}{2} \leq 1 + \frac{a_{n-1}}{x} + \cdots + \frac{a_0}{x^n}.$$

Therefore, if we choose an $x_1 > 0$ which satisfies (*), then

$$\frac{x_1^n}{2} \leq x_1^n \left(1 + \frac{a_{n-1}}{x_1} + \cdots + \frac{a_0}{x_1^n} \right) = f(x_1),$$

so that $f(x_1) > 0$. On the other hand, if $x_2 < 0$ satisfies (*), then $x_2^n < 0$ and

$$\frac{x_2^n}{2} \geq x_2^n \left(1 + \frac{a_{n-1}}{x_2} + \cdots + \frac{a_0}{x_2^n} \right) = f(x_2),$$

so that $f(x_2) < 0$.

Now applying Theorem 1 to the interval $[x_2, x_1]$ we conclude that there is an x in $[x_2, x_1]$ such that $f(x) = 0$. ■

Theorem 9 disposes of the problem of odd degree equations so happily that it would be frustrating to leave the problem of even degree equations completely undiscussed. At first sight, however, the problem seems insuperable. Some equations, like $x^2 - 1 = 0$, have a solution, and some, like $x^2 + 1 = 0$, do not—what more is there to say? If we are willing to consider a more general question, however, something interesting *can* be said. Instead of trying to solve the equation

$$x^n + a_{n-1}x^{n-1} + \cdots + a_0 = 0,$$

let us ask about the possibility of solving the equations

$$x^n + a_{n-1}x^{n-1} + \cdots + a_0 = c$$

for all possible numbers c . This amounts to allowing the constant term a_0 to vary. The information which can be given concerning the solution of these equations depends on a fact which is illustrated in Figure 8.

The graph of the function $f(x) = x^n + a_{n-1}x^{n-1} + \cdots + a_0$, with n even, contains, at least the way we have drawn it, a lowest point. In other words, there is a number y such that $f(y) \leq f(x)$ for all numbers x —the function f takes on a minimum value, not just on each closed interval, but on the whole line. (Notice that this is false if n is odd.) The proof depends on Theorem 7, but a tricky application will be required. We can apply Theorem 7 to any interval

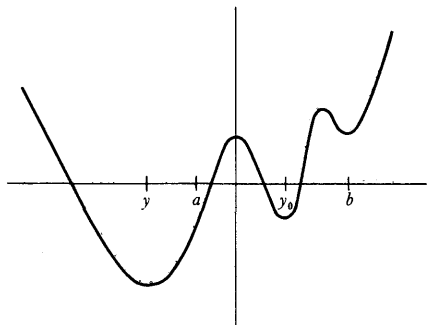


FIGURE 8

$[a, b]$, and obtain a point y_0 such that $f(y_0)$ is the minimum value of f on $[a, b]$; but if $[a, b]$ happens to be the interval shown in Figure 8, for example, then the point y_0 will not be the place where f has its minimum value for the whole line. In the next theorem the entire point of the proof is to choose an interval $[a, b]$ in such a way that this cannot happen.

THEOREM 10 If n is even and $f(x) = x^n + a_{n-1}x^{n-1} + \cdots + a_0$, then there is a number y such that $f(y) \leq f(x)$ for all x .

PROOF As in the proof of Theorem 9, if

$$M = \max(1, 2n|a_{n-1}|, \dots, 2n|a_0|),$$

then for all x with $|x| \geq M$, we have

$$\frac{1}{2} \leq 1 + \frac{a_{n-1}}{x} + \cdots + \frac{a_0}{x^n}.$$

Since n is even, $x^n \geq 0$ for all x , so

$$\frac{x^n}{2} \leq x^n \left(1 + \frac{a_{n-1}}{x} + \cdots + \frac{a_0}{x^n} \right) = f(x),$$

provided that $|x| \geq M$. Now consider the number $f(0)$. Let $b > 0$ be a number such that $b^n \geq 2f(0)$ and also $b > M$. Then, if $x \geq b$, we have (Figure 9)

$$f(x) \geq \frac{x^n}{2} \geq \frac{b^n}{2} \geq f(0).$$

Similarly, if $x \leq -b$, then

$$f(x) \geq \frac{x^n}{2} \geq \frac{(-b)^n}{2} = \frac{b^n}{2} \geq f(0).$$

Summarizing:

$$\text{if } x \geq b \text{ or } x \leq -b, \text{ then } f(x) \geq f(0).$$

Now apply Theorem 7 to the function f on the interval $[-b, b]$. We conclude that there is a number y such that

$$(1) \quad \text{if } -b \leq x \leq b, \text{ then } f(y) \leq f(x).$$

In particular, $f(y) \leq f(0)$. Thus

$$(2) \quad \text{if } x \leq -b \text{ or } x \geq b, \text{ then } f(x) \geq f(0) \geq f(y).$$

Combining (1) and (2) we see that $f(y) \leq f(x)$ for all x . ■

Theorem 10 now allows us to prove the following result.

THEOREM 11 Consider the equation

$$(*) \quad x^n + a_{n-1}x^{n-1} + \cdots + a_0 = c,$$

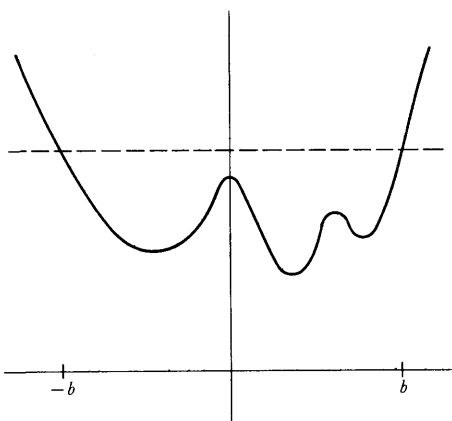


FIGURE 9

and suppose n is even. Then there is a number m such that $(*)$ has a solution for $c \geq m$ and has no solution for $c < m$.

PROOF Let $f(x) = x^n + a_{n-1}x^{n-1} + \cdots + a_0$ (Figure 10).

According to Theorem 10 there is a number y such that $f(y) \leq f(x)$ for all x . Let $m = f(y)$. If $c < m$, then the equation $(*)$ obviously has no solution, since the left side always has a value $\geq m$. If $c = m$, then $(*)$ has y as a solution. Finally, suppose $c > m$. Let b be a number such that $b > y$ and $f(b) > c$. Then $f(y) = m < c < f(b)$. Consequently, by Theorem 4, there is some number x in $[y, b]$ such that $f(x) = c$, so x is a solution of $(*)$. ■

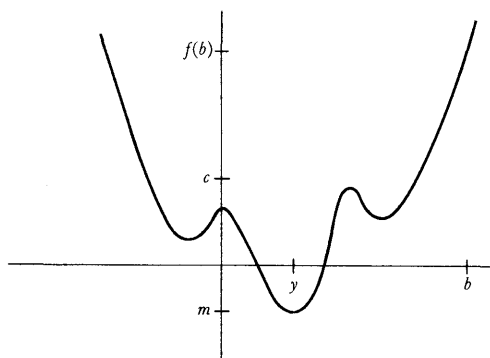


FIGURE 10

These consequences of Theorems 1, 2, and 3 are the only ones we will derive now (these theorems will play a fundamental role in everything we do later, however). Only one task remains—to prove Theorems 1, 2, and 3. Unfortunately, we cannot hope to do this—on the basis of our present knowledge about the real numbers (namely, P1–P12) a proof is *impossible*. There are several ways of convincing ourselves that this gloomy conclusion is actually the case. For example, the proof of Theorem 8 relies only on the proof of Theorem 1; if we could prove Theorem 1, then the proof of Theorem 8 would be complete, and we would have a proof that every positive number has a square root. As pointed out in Part I, it is impossible to prove this on the basis of P1–P12. Again, suppose we consider the function

$$f(x) = \frac{1}{x^2 - 2}.$$

If there were no number x with $x^2 = 2$, then f would be continuous, since the denominator would never $= 0$. But f is not bounded on $[0, 2]$. So Theorem 2 depends essentially on the existence of numbers other than rational numbers, and therefore on some property of the reals other than P1–P12.

Despite our inability to prove Theorems 1, 2, and 3, they are certainly results which we want to be true. If the pictures we have been drawing have any connection with the mathematics we are doing, if our notion of continuous function corresponds to any degree with our intuitive notion, Theorems 1, 2, and 3 have got to be true. Since a proof of any of these theorems must require some new property of $\mathbf{R}$ which has so far been overlooked, our present difficulties suggest a way to discover that property: let us try to construct a proof of Theorem 1, for example, and see what goes wrong.

One idea which seems promising is to locate the first point where $f(x) = 0$, that is, the smallest x in $[a, b]$ such that $f(x) = 0$. To find this point, first consider the set A which contains all numbers x in $[a, b]$ such that f is negative on $[a, x]$. In Figure 11, x is such a point, while x' is not. The set A itself is indicated by a heavy line. Since f is negative at a , and positive at b , the set A contains some points greater than a , while all points sufficiently close to b are not in A . (We are here using the continuity of f on $[a, b]$, as well as Problem 6-15.)

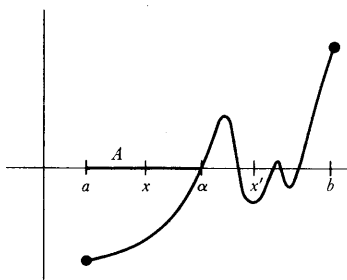


FIGURE 11

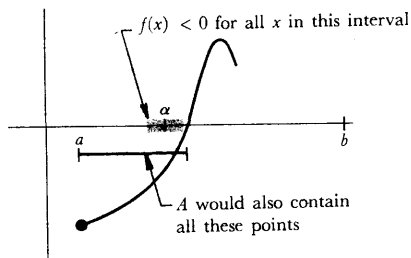


FIGURE 12

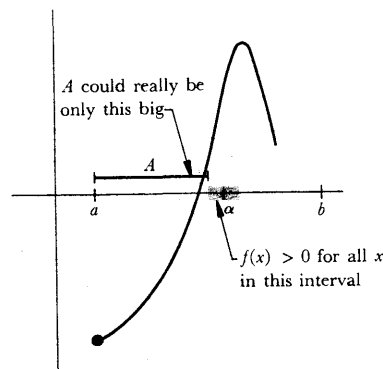


FIGURE 13

Now suppose α is the smallest number which is greater than all members of A ; clearly $a < \alpha < b$. We claim that $f(\alpha) = 0$, and to prove this we only have to eliminate the possibilities $f(\alpha) < 0$ and $f(\alpha) > 0$.

Suppose first that $f(\alpha) < 0$. Then, by Theorem 6-3, $f(x)$ would be less than 0 for all x in a small interval containing α , in particular for some numbers bigger than α (Figure 12); but this contradicts the fact that α is bigger than every member of A , since the larger numbers would also be in A . Consequently, $f(\alpha) < 0$ is false.

On the other hand, suppose $f(\alpha) > 0$. Again applying Theorem 6-3, we see that $f(x)$ would be positive for all x in a small interval containing α , in particular for some numbers smaller than α (Figure 13). This means that these smaller numbers are all *not* in A . Consequently, one could have chosen an even smaller α which would be greater than all members of A . Once again we have a contradiction; $f(\alpha) > 0$ is also false. Hence $f(\alpha) = 0$ and, we are tempted to say, Q.E.D.

We know, however, that something must be wrong, since no new properties of $\mathbf{R}$ were ever used, and it does not require much scrutiny to find the dubious point. It is clear that we can choose a number α which is greater than all members of A (for example, we can choose $\alpha = b$), but it is not so clear that we can choose a *smallest* one. In fact, suppose A consists of all numbers $x \geq 0$ such that $x^2 < 2$. If the number $\sqrt{2}$ did not exist, there would not be a least number greater than all the members of A ; for any $y > \sqrt{2}$ we chose, we could always choose a still smaller one.

Now that we have discovered the fallacy, it is almost obvious what additional property of the real numbers we need. All we must do is say it properly and use it. That is the business of the next chapter.

PROBLEMS

1. For each of the following functions, decide which are bounded above or below on the indicated interval, and which take on their maximum or minimum value. (Notice that f *might* have these properties even if f is not continuous, and even if the interval is not a closed interval.)
 - (i) $f(x) = x^2$ on $(-1, 1)$.
 - (ii) $f(x) = x^3$ on $(-1, 1)$.
 - (iii) $f(x) = x^2$ on $\mathbf{R}$.
 - (iv) $f(x) = x^2$ on $[0, \infty)$.
 - (v) $f(x) = \begin{cases} x^2, & x \leq a \\ a + 2, & x > a \end{cases}$ on $(-a - 1, a + 1)$. (It will be necessary to consider several possibilities for a .)
 - (vi) $f(x) = \begin{cases} x^2, & x < a \\ a + 2, & x \geq a \end{cases}$ on $[-a - 1, a + 1]$.
 - (vii) $f(x) = \begin{cases} 0, & x \text{ irrational} \\ 1/q, & x = p/q \text{ in lowest terms} \end{cases}$ on $[0, 1]$.

- (viii) $f(x) = \begin{cases} 1, & x \text{ irrational} \\ 1/q, & x = p/q \text{ in lowest terms} \end{cases} \quad \text{on } [0, 1].$
- (ix) $f(x) = \begin{cases} 1, & x \text{ irrational} \\ -1/q, & x = p/q \text{ in lowest terms} \end{cases} \quad \text{on } [0, 1].$
- (x) $f(x) = \begin{cases} x, & x \text{ rational} \\ 0, & x \text{ irrational} \end{cases} \quad \text{on } [0, a].$
- (xi) $f(x) = \sin^2(\cos x + \sqrt{1 + a^2}) \quad \text{on } [0, a^3].$
- (xii) $f(x) = [x] \quad \text{on } [0, a].$
2. For each of the following polynomial functions f , find an integer n such that $f(x) = 0$ for some x between n and $n + 1$.
- $f(x) = x^3 - x + 3.$
 - $f(x) = x^5 + 5x^4 + 2x + 1.$
 - $f(x) = x^5 + x + 1.$
 - $f(x) = 4x^2 - 4x + 1.$
3. Prove that there is some number x such that
- $x^{179} + \frac{163}{1 + x^2 + \sin^2 x} = 119.$
 - $\sin x = x - 1.$
4. This problem is a continuation of Problem 3-7.
- If $n - k$ is even, and ≥ 0 , find a polynomial function of degree n with exactly k roots.
 - A root a of the polynomial function f is said to have **multiplicity** m if $f(x) = (x - a)^m g(x)$, where g is a polynomial function that does *not* have a as a root. Let f be a polynomial function of degree n . Suppose that f has k roots, counting multiplicities, i.e., suppose that k is the sum of the multiplicities of all the roots. Show that $n - k$ is even.
5. Suppose that f is continuous on $[a, b]$ and that $f(x)$ is always rational. What can be said about f ?
6. Suppose that f is a continuous function on $[-1, 1]$ such that $x^2 + (f(x))^2 = 1$ for all x . (This means that $(x, f(x))$ always lies on the unit circle.) Show that either $f(x) = \sqrt{1 - x^2}$ for all x , or else $f(x) = -\sqrt{1 - x^2}$ for all x .
7. How many continuous functions f are there which satisfy $(f(x))^2 = x^2$ for all x ?
8. Suppose that f and g are continuous, that $f^2 = g^2$, and that $f(x) \neq 0$ for all x . Prove that either $f(x) = g(x)$ for all x , or else $f(x) = -g(x)$ for all x .
9. (a) Suppose that f is continuous, that $f(x) = 0$ only for $x = a$, and that $f(x) > 0$ for some $x > a$ as well as for some $x < a$. What can be said about $f(x)$ for all $x \neq a$?
- *(b) Using only the fact that $x^2 + xy + y^2 \neq 0$ if x and y are not both 0, show that $x^2 + xy + y^2 > 0$ if x and y are not both 0. (The trick is to consider various *fixed* $y \neq 0$, so that you can define a function.)

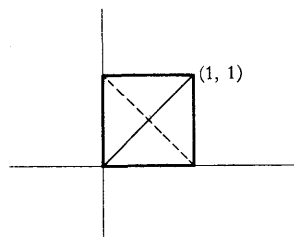


FIGURE 14

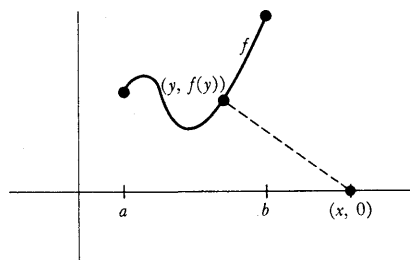


FIGURE 15

- *(c) Discuss similarly the sign of $x^3 + x^2y + xy^2 + y^3$ when x and y are not both 0.
10. Suppose f and g are continuous on $[a, b]$ and that $f(a) < g(a)$, but $f(b) > g(b)$. Prove that $f(x) = g(x)$ for some x in $[a, b]$. (If your proof isn't very short, it's not the right one.)
11. Suppose that f is a continuous function on $[0, 1]$ and that $f(x)$ is in $[0, 1]$ for each x (draw a picture). Prove that $f(x) = x$ for some number x .
12. (a) Problem 11 shows that f intersects the diagonal of the square in Figure 14 (solid line). Show that f must also intersect the other (dashed) diagonal.
- (b) Prove the following more general fact: If g is continuous on $[0, 1]$ and $g(0) = 0, g(1) = 1$ or $g(0) = 1, g(1) = 0$, then $f(x) = g(x)$ for some x .
13. (a) Let $f(x) = \sin 1/x$ for $x \neq 0$ and let $f(0) = 0$. Is f continuous on $[-1, 1]$? Show that f satisfies the conclusion of the Intermediate Value Theorem on $[-1, 1]$; in other words, if f takes on two values somewhere on $[-1, 1]$, it also takes on every value in between.
- *(b) Suppose that f satisfies the conclusion of the Intermediate Value Theorem, and that f takes on each value *only once*. Prove that f is continuous.
- *(c) Generalize to the case where f takes on each value only finitely many times.
14. If f is a continuous function on $[0, 1]$, let $\|f\|$ be the maximum value of $|f|$ on $[0, 1]$.
- (a) Prove that for any number c we have $\|cf\| = |c| \cdot \|f\|$.
- *(b) Prove that $\|f + g\| \leq \|f\| + \|g\|$. Give an example where $\|f + g\| \neq \|f\| + \|g\|$.
- (c) Prove that $\|h - f\| \leq \|h - g\| + \|g - f\|$.
- *15. Suppose that ϕ is continuous and $\lim_{x \rightarrow \infty} \phi(x)/x^n = 0 = \lim_{x \rightarrow -\infty} \phi(x)/x^n$.
- (a) Prove that if n is odd, then there is a number x such that $x^n + \phi(x) = 0$.
- (b) Prove that if n is even, then there is a number y such that $y^n + \phi(y) \leq x^n + \phi(x)$ for all x .
- Hint: Of which proofs does this problem test your understanding?
- *16. Let f be any polynomial function. Prove that there is some number y such that $|f(y)| \leq |f(x)|$ for all x .
- *17. Suppose that f is a continuous function with $f(x) > 0$ for all x , and $\lim_{x \rightarrow \infty} f(x) = 0 = \lim_{x \rightarrow -\infty} f(x)$. (Draw a picture.) Prove that there is some number y such that $f(y) \geq f(x)$ for all x .
- *18. (a) Suppose that f is continuous on $[a, b]$, and let x be any number. Prove that there is a point on the graph of f which is closest to $(x, 0)$; in other words there is some y in $[a, b]$ such that the distance from $(x, 0)$ to $(y, f(y))$ is $\leq$ distance from $(x, 0)$ to $(z, f(z))$ for all z in $[a, b]$. (See Figure 15.)

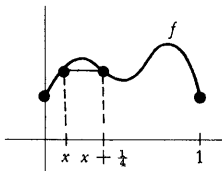


FIGURE 16

- (b) Show that this same assertion is not necessarily true if $[a, b]$ is replaced by (a, b) throughout.
- (c) Show that the assertion *is* true if $[a, b]$ is replaced by $\mathbf{R}$ throughout.
- (d) In cases (a) and (c), let $g(x)$ be the minimum distance from $(x, 0)$ to a point on the graph of f . Prove that $g(y) \leq g(x) + |x - y|$, and conclude that g is continuous.
- (e) Prove that there are numbers x_0 and x_1 in $[a, b]$ such that the distance from $(x_0, 0)$ to $(x_1, f(x_1))$ is $\leq$ the distance from $(x'_0, 0)$ to $(x'_1, f(x'_1))$ for any x'_0, x'_1 in $[a, b]$.
- **19.** (a) Suppose that f is continuous on $[0, 1]$ and $f(0) = f(1)$. Let n be any natural number. Prove that there is some number x such that $f(x) = f(x + 1/n)$, as shown in Figure 16 for $n = 4$. Hint: Consider the function $g(x) = f(x) - f(x + 1/n)$; what would be true if $g(x) \neq 0$ for all x ?
- (b) Suppose $0 < a < 1$, but that a is not equal to $1/n$ for any natural number n . Find a function f which is continuous on $[0, 1]$ and which satisfies $f(0) = f(1)$, but which does not satisfy $f(x) = f(x + a)$ for any x .
- **20.** (a) Prove that there does not exist a continuous function f defined on $\mathbf{R}$ which takes on every value exactly twice. Hint: If $f(a) = f(b)$ for $a < b$, then either $f(x) > f(a)$ for all x in (a, b) or $f(x) < f(a)$ for all x in (a, b) . Why? In the first case all values close to $f(a)$, but slightly larger than $f(a)$, are taken on somewhere in (a, b) ; this implies that $f(x) < f(a)$ for $x < a$ and $x > b$.
- (b) Refine part (a) by proving that there is no continuous function f which takes on each value either 0 times or 2 times, i.e., which takes on exactly twice each value that it does take on. Hint: The previous hint implies that f has either a maximum or a minimum value (which must be taken on twice). What can be said about values close to the maximum value?
- (c) Find a continuous function f which takes on every value exactly 3 times. More generally, find one which takes on every value exactly n times, if n is odd.
- (d) Prove that if n is even, then there is no continuous f which takes on every value exactly n times. Hint: To treat the case $n = 4$, for example, let $f(x_1) = f(x_2) = f(x_3) = f(x_4)$. Then either $f(x) > 0$ for all x in two of the three intervals (x_1, x_2) , (x_2, x_3) , (x_3, x_4) , or else $f(x) < 0$ for all x in two of these three intervals.

This chapter reveals the most important property of the real numbers. Nevertheless, it is merely a sequel to Chapter 7; the path which must be followed has already been indicated, and further discussion would be useless delay.

DEFINITION

A set A of real numbers is **bounded above** if there is a number x such that

$$x \geq a \quad \text{for every } a \text{ in } A.$$

Such a number x is called an **upper bound** for A .

Obviously A is bounded above if and only if there is a number x which is an upper bound for A (and in this case there will be lots of upper bounds for A); we often say, as a concession to idiomatic English, that “ A has an upper bound” when we mean that there is a number which is an upper bound for A .

Notice that the term “bounded above” has now been used in two ways—first, in Chapter 7, in reference to functions, and now in reference to sets. This dual usage should cause no confusion, since it will always be clear whether we are talking about a set of numbers or a function. Moreover, the two definitions are closely connected: if A is the set $\{f(x): a \leq x \leq b\}$, then the function f is bounded above on $[a, b]$ if and only if the set A is bounded above.

The entire collection $\mathbf{R}$ of real numbers, and the natural numbers $\mathbf{N}$, are both examples of sets which are *not* bounded above. An example of a set which *is* bounded above is

$$A = \{x: 0 \leq x < 1\}.$$

To show that A is bounded above we need only name some upper bound for A , which is easy enough; for example, 138 is an upper bound for A , and so are 2, $1\frac{1}{2}$, $1\frac{1}{4}$, and 1. Clearly, 1 is the least upper bound of A ; although the phrase just introduced is self-explanatory, in order to avoid any possible confusion (in particular, to ensure that we all know what the superlative of “less” means), we define this explicitly.

DEFINITION

A number x is a **least upper bound** of A if

- (1) x is an upper bound of A ,
and (2) if y is an upper bound of A , then $x \leq y$.

The use of the indefinite article “a” in this definition was merely a concession to temporary ignorance. Now that we have made a precise definition, it is easily seen that if x and y are both least upper bounds of A , then $x = y$. Indeed, in this case

$$\begin{aligned} x &\leq y, && \text{since } y \text{ is an upper bound, and } x \text{ is a least upper bound,} \\ \text{and } y &\leq x, && \text{since } x \text{ is an upper bound, and } y \text{ is a least upper bound;} \end{aligned}$$

it follows that $x = y$. For this reason we speak of *the* least upper bound of A . The term **supremum** of A is synonymous and has one advantage. It abbreviates quite nicely to

$$\sup A \quad (\text{pronounced “soup } A\text{”})$$

and saves us from the abbreviation

$$\text{lub } A$$

(which is nevertheless used by some authors).

There is a series of important definitions, analogous to those just given, which can now be treated more briefly. A set A of real numbers is **bounded below** if there is a number x such that

$$x \leq a \quad \text{for every } a \text{ in } A.$$

Such a number x is called a **lower bound** for A . A number x is the **greatest lower bound** of A if

- (1) x is a lower bound of A ,
- and (2) if y is a lower bound of A , then $x \geq y$.

The greatest lower bound of A is also called the **infimum** of A , abbreviated

$$\inf A;$$

some authors use the abbreviation

$$\text{glb } A.$$

One detail has been omitted from our discussion so far—the question of which sets have at least one, and hence exactly one, least upper bound or greatest lower bound. We will consider only least upper bounds, since the question for greatest lower bounds can then be answered easily (Problem 2).

If A is not bounded above, then A has no upper bound at all, so A certainly cannot be expected to have a least upper bound. It is tempting to say that A does have a least upper bound if it has *some* upper bound, but, like the principle of mathematical induction, this assertion can fail to be true in a rather special way. If $A = \emptyset$, then A is bounded above. Indeed, any number x is an upper bound for $\emptyset$:

$$x \geq y \quad \text{for every } y \text{ in } \emptyset,$$

simply because there is no y in $\emptyset$. Since *every* number is an upper bound for $\emptyset$, there is surely no least upper bound for $\emptyset$. With this trivial exception however,

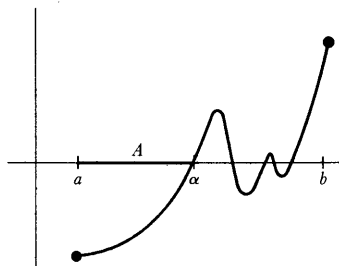


FIGURE 1

THEOREM 7-1

If f is continuous on $[a, b]$ and $f(a) < 0 < f(b)$, then there is some number x in $[a, b]$ such that $f(x) = 0$.

PROOF

Our proof is merely a rigorous version of the outline developed at the end of Chapter 7—we will locate the smallest number x in $[a, b]$ with $f(x) = 0$.

Define the set A , shown in Figure 1, as follows:

$$A = \{x: a \leq x \leq b, \text{ and } f \text{ is negative on the interval } [a, x]\}.$$

Clearly $A \neq \emptyset$, since a is in A ; in fact, there is some $\delta > 0$ such that A contains all points x satisfying $a \leq x < a + \delta$; this follows from Problem 6-15, since f is continuous on $[a, b]$ and $f(a) < 0$. Similarly, b is an upper bound for A and, in fact, there is a $\delta > 0$ such that all points x satisfying $b - \delta < x \leq b$ are upper bounds for A ; this also follows from Problem 6-15, since $f(b) > 0$.

From these remarks it follows that A has a least upper bound α and that $a < \alpha < b$. We now wish to show that $f(\alpha) = 0$, by eliminating the possibilities $f(\alpha) < 0$ and $f(\alpha) > 0$.

Suppose first that $f(\alpha) < 0$. By Theorem 6-3, there is a $\delta > 0$ such that $f(x) < 0$ for $\alpha - \delta < x < \alpha + \delta$ (Figure 2). Now there is some number x_0 in A which satisfies $\alpha - \delta < x_0 < \alpha$ (because otherwise α would not be the *least* upper bound of A). This means that f is negative on the whole interval $[a, x_0]$. But if x_1 is a number between α and $\alpha + \delta$, then f is also negative on the whole interval $[x_0, x_1]$. Therefore f is negative on the interval $[a, x_1]$, so x_1 is in A . But this contradicts the fact that α is an upper bound for A ; our original assumption that $f(\alpha) < 0$ must be false.

Suppose, on the other hand, that $f(\alpha) > 0$. Then there is a number $\delta > 0$ such that $f(x) > 0$ for $\alpha - \delta < x < \alpha + \delta$ (Figure 3). Once again we know that there is an x_0 in A satisfying $\alpha - \delta < x_0 < \alpha$; but this means that f is negative on $[a, x_0]$, which is impossible, since $f(x_0) > 0$. Thus the assumption

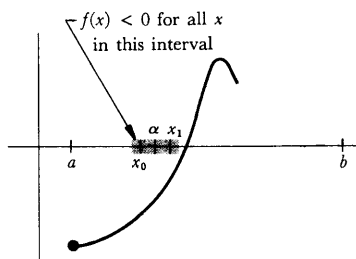


FIGURE 2

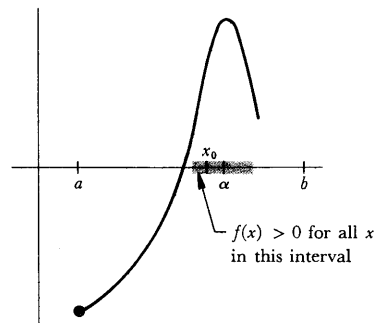


FIGURE 3

$f(\alpha) > 0$ also leads to a contradiction, leaving $f(\alpha) = 0$ as the only possible alternative.

The proofs of Theorems 2 and 3 of Chapter 7 require a simple preliminary result, which will play much the same role as Theorem 6-3 played in the previous proof.

THEOREM 1 If f is continuous at a , then there is a number $\delta > 0$ such that f is bounded above on the interval $(a - \delta, a + \delta)$ (see Figure 4).

PROOF Since $\lim_{x \rightarrow a} f(x) = f(a)$, there is, for every $\epsilon > 0$, a $\delta > 0$ such that, for all x ,

$$\text{if } |x - a| < \delta, \text{ then } |f(x) - f(a)| < \epsilon.$$

It is only necessary to apply this statement to some particular ϵ (any one will do), for example, $\epsilon = 1$. We conclude that there is a $\delta > 0$ such that, for all x ,

$$\text{if } |x - a| < \delta, \text{ then } |f(x) - f(a)| < 1.$$

It follows, in particular, that if $|x - a| < \delta$, then $f(x) - f(a) < 1$. This completes the proof: on the interval $(a - \delta, a + \delta)$ the function f is bounded above by $f(a) + 1$. ■

It should hardly be necessary to add that we can now also prove that f is bounded below on some interval $(a - \delta, a + \delta)$, and, finally, that f is bounded on some open interval containing a .

A more significant point is the observation that if $\lim_{x \rightarrow a^+} f(x) = f(a)$, then

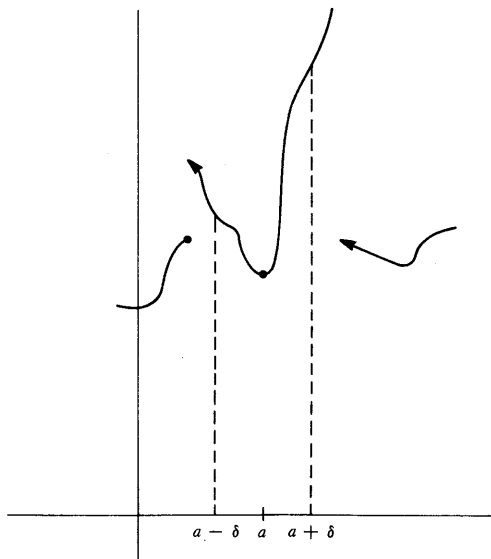


FIGURE 4

there is a $\delta > 0$ such that f is bounded on the set $\{x: a \leq x < a + \delta\}$, and a similar observation holds if $\lim_{x \rightarrow b^-} f(x) = f(b)$. Having made these observations (and assuming that you will supply the proofs), we tackle our second major theorem.

THEOREM 7-2 If f is continuous on $[a, b]$, then f is bounded above on $[a, b]$.

PROOF Let

$$A = \{x: a \leq x \leq b \text{ and } f \text{ is bounded above on } [a, x]\}.$$

Clearly $A \neq \emptyset$ (since a is in A), and A is bounded above (by b), so A has a least upper bound α . Notice that we are here applying the term “bounded above” both to the set A , which can be visualized as lying on the horizontal axis, and to f , i.e., to the sets $\{f(y): a \leq y \leq x\}$, which can be visualized as lying on the vertical axis (Figure 5).

Our first step is to prove that we actually have $\alpha = b$. Suppose, instead, that $\alpha < b$. By Theorem 1 there is $\delta > 0$ such that f is bounded on $(\alpha - \delta, \alpha + \delta)$. Since α is the least upper bound of A there is some x_0 in A satisfying $\alpha - \delta < x_0 < \alpha$. This means that f is bounded on $[a, x_0]$. But if x_1 is any number with $\alpha < x_1 < \alpha + \delta$, then f is also bounded on $[x_0, x_1]$. Therefore f is bounded on $[a, x_1]$, so x_1 is in A , contradicting the fact that α is an upper bound for A . This contradiction shows that $\alpha = b$. One detail should be mentioned: this demonstration implicitly assumed that $a < \alpha$ (so that f would be defined on some interval $(\alpha - \delta, \alpha + \delta)$); the possibility $a = \alpha$ can be ruled out similarly, using the existence of a $\delta > 0$ such that f is bounded on $\{x: a \leq x < a + \delta\}$.

The proof is not quite complete—we only know that f is bounded on $[a, x]$ for every $x < b$, not necessarily that f is bounded on $[a, b]$. However, only one small argument needs to be added.

There is a $\delta > 0$ such that f is bounded on $\{x: b - \delta < x \leq b\}$. There is x_0 in A such that $b - \delta < x_0 < b$. Thus f is bounded on $[a, x_0]$ and also on $[x_0, b]$, so f is bounded on $[a, b]$. ■

To prove the third important theorem we resort to a trick.

THEOREM 7-3 If f is continuous on $[a, b]$, then there is a number y in $[a, b]$ such that $f(y) \geq f(x)$ for all x in $[a, b]$.

PROOF We already know that f is bounded on $[a, b]$, which means that the set

$$\{f(x): x \text{ in } [a, b]\}$$

is bounded. This set is obviously not $\emptyset$, so it has a least upper bound α . Since $\alpha \geq f(x)$ for x in $[a, b]$ it suffices to show that $\alpha = f(y)$ for some y in $[a, b]$.

Suppose instead that $\alpha \neq f(y)$ for all y in $[a, b]$. Then the function g defined by

$$g(x) = \frac{1}{\alpha - f(x)}, \quad x \text{ in } [a, b]$$

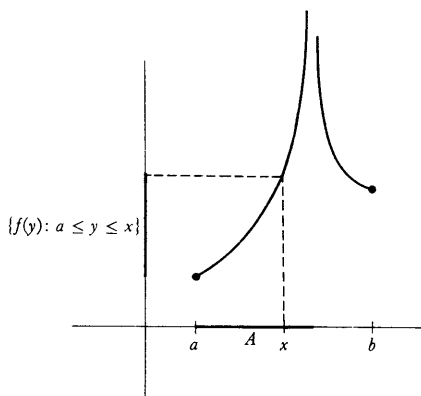


FIGURE 5

is continuous on $[a, b]$, since the denominator of the right side is never 0. On the other hand, α is the least upper bound of $\{f(x): x \text{ in } [a, b]\}$; this means that

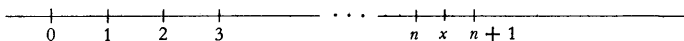
for every $\varepsilon > 0$ there is x in $[a, b]$ with $\alpha - f(x) < \varepsilon$.

This, in turn, means that

for every $\varepsilon > 0$ there is x in $[a, b]$ with $g(x) > 1/\varepsilon$.

But *this* means that g is not bounded on $[a, b]$, contradicting the previous theorem. ■

At the beginning of this chapter the set of natural numbers $\mathbf{N}$ was given as an example of an unbounded set. We are now going to *prove* that $\mathbf{N}$ is unbounded. After the difficult theorems proved in this chapter you may be startled to find such an “obvious” theorem winding up our proceedings. If so, you are, perhaps, allowing the geometrical picture of $\mathbf{R}$ to influence you too strongly. “Look,” you may say, “the real numbers look like



so every number x is between two integers $n, n + 1$ (unless x is itself an integer).” Basing the argument on a geometric picture is not a proof, however, and even the geometric picture contains an assumption: that if you place unit segments end-to-end you will eventually get a segment larger than any given segment. This axiom, often omitted from a first introduction to geometry, is usually attributed (not quite justly) to Archimedes, and the corresponding property for numbers, that $\mathbf{N}$ is not bounded, is called the *Archimedian property* of the reals. This property is *not* a consequence of P1–P12 (see reference [17] of the Suggested Reading), although it does hold for $\mathbf{Q}$, of course. Once we have P13 however, there are no longer any problems.

THEOREM 2 $\mathbf{N}$ is not bounded above.

PROOF Suppose $\mathbf{N}$ were bounded above. Since $\mathbf{N} \neq \emptyset$, there would be a least upper bound α for $\mathbf{N}$. Then

$$\alpha \geq n \quad \text{for all } n \text{ in } \mathbf{N}.$$

Consequently,

$$\alpha \geq n + 1 \quad \text{for all } n \text{ in } \mathbf{N},$$

since $n + 1$ is in $\mathbf{N}$ if n is in $\mathbf{N}$. But this means that

$$\alpha - 1 \geq n \quad \text{for all } n \text{ in } \mathbf{N},$$

and *this* means that $\alpha - 1$ is also an upper bound for $\mathbf{N}$, contradicting the fact that α is the least upper bound. ■

There is a consequence of Theorem 2 (actually an equivalent formulation) which we have very often assumed implicitly.

THEOREM 3 For any $\varepsilon > 0$ there is a natural number n with $1/n < \varepsilon$.

PROOF Suppose not; then $1/n \geq \varepsilon$ for all n in $\mathbf{N}$. Thus $n \leq 1/\varepsilon$ for all n in $\mathbf{N}$. But this means that $1/\varepsilon$ is an upper bound for $\mathbf{N}$, contradicting Theorem 2. ■

A brief glance through Chapter 6 will show you that the result of Theorem 3 was used in the discussion of many examples. Of course, Theorem 3 was not available at the time, but the examples were so important that in order to give them some cheating was tolerated. As partial justification for this dishonesty we can claim that this result was never used in the proof of a *theorem*, but if your faith has been shaken, a review of all the proofs given so far is in order. Fortunately, such deception will not be necessary again. We have now stated every property of the real numbers that we will ever need. Henceforth, no more lies.

PROBLEMS

1. Find the least upper bound and the greatest lower bound (if they exist) of the following sets. Also decide which sets have greatest and least elements (i.e., decide when the least upper bound and greatest lower bound happens to belong to the set).

(i) $\left\{\frac{1}{n} : n \text{ in } \mathbf{N}\right\}.$

(ii) $\left\{\frac{1}{n} : n \text{ in } \mathbf{Z} \text{ and } n \neq 0\right\}.$

(iii) $\{x : x = 0 \text{ or } x = 1/n \text{ for some } n \text{ in } \mathbf{N}\}.$

(iv) $\{x : 0 \leq x \leq \sqrt{2} \text{ and } x \text{ is rational}\}.$

(v) $\{x : x^2 + x + 1 \geq 0\}.$

(vi) $\{x : x^2 + x - 1 < 0\}.$

(vii) $\{x : x < 0 \text{ and } x^2 + x - 1 < 0\}.$

(viii) $\left\{\frac{1}{n} + (-1)^n : n \text{ in } \mathbf{N}\right\}.$

2. (a) Suppose $A \neq \emptyset$ is bounded below. Let $-A$ denote the set of all $-x$ for x in A . Prove that $-A \neq \emptyset$, that $-A$ is bounded above, and that $-\sup(-A)$ is the greatest lower bound of A .
- (b) If $A \neq \emptyset$ is bounded below, let B be the set of all lower bounds of A . Show that $B \neq \emptyset$, that B is bounded above, and that $\sup B$ is the greatest lower bound of A .

3. Let f be a continuous function on $[a, b]$ with $f(a) < 0 < f(b)$.
 - (a) The proof of Theorem 1 showed that there is a smallest x in $[a, b]$ with $f(x) = 0$. Is there necessarily a second smallest x in $[a, b]$ with $f(x) = 0$? Show that there is a largest x in $[a, b]$ with $f(x) = 0$. (Try to give an easy proof by considering a new function closely related to f .)
 - (b) The proof of Theorem 1 depended upon consideration of $A = \{x: a \leq x \leq b \text{ and } f \text{ is negative on } [a, x]\}$. Give another proof of Theorem 1, which depends upon consideration of $B = \{x: a \leq x \leq b \text{ and } f(x) < 0\}$. Which point x in $[a, b]$ with $f(x) = 0$ will this proof locate? Give an example where the sets A and B are not the same.
4. Suppose that f is continuous on $[a, b]$ and that $f(a) = f(b) = 0$. Suppose also that $f(x_0) > 0$ for some x_0 in $[a, b]$. Prove that there are numbers c and d with $a \leq c < x_0 < d \leq b$ such that $f(c) = f(d) = 0$, but $f(x) > 0$ for all x in (c, d) . Hint: The previous problem can be used to good advantage.
5.
 - (a) Suppose that $y - x > 1$. Prove that there is an integer k such that $x < k < y$. Hint: Let l be the largest integer satisfying $l \leq x$, and consider $l + 1$.
 - (b) Suppose $x < y$. Prove that there is a rational number r such that $x < r < y$. Hint: If $1/n < y - x$, then $ny - nx > 1$. (Query: Why have parts (a) and (b) been postponed until this problem set?)
 - (c) Suppose that $r < s$ are rational numbers. Prove that there is an irrational number between r and s . Hint: As a start, you know that there is an irrational number between 0 and 1.
 - (d) Suppose that $x < y$. Prove that there is an irrational number between x and y . Hint: It is unnecessary to do any more work; this follows from (b) and (c).
- *6. A set A of real numbers is said to be **dense** if every open interval contains a point of A . For example, Problem 5 shows that the set of rational numbers and the set of irrational numbers are each dense.
 - (a) Prove that if f is continuous and $f(x) = 0$ for all numbers x in a dense set A , then $f(x) = 0$ for all x .
 - (b) Prove that if f and g are continuous and $f(x) = g(x)$ for all x in a dense set A , then $f(x) = g(x)$ for all x .
 - (c) If we assume instead that $f(x) \geq g(x)$ for all x in A , show that $f(x) \geq g(x)$ for all x . Can $\geq$ be replaced by $>$ throughout?
7. Prove that if f is continuous and $f(x + y) = f(x) + f(y)$ for all x and y , then there is a number c such that $f(x) = cx$ for all x . (This conclusion can be demonstrated simply by combining the results of two previous problems.) Point of information: There *do* exist *noncontinuous* functions f satisfying $f(x + y) = f(x) + f(y)$ for all x and y , but we cannot prove this now; in fact, this simple question involves ideas that are usually never

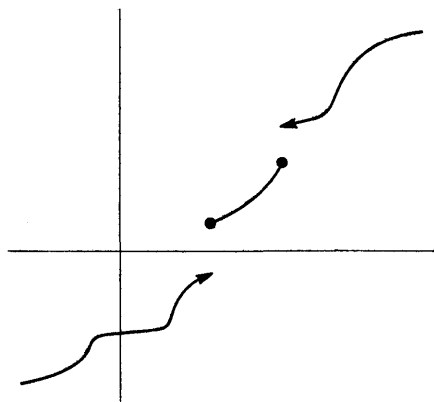


FIGURE 6

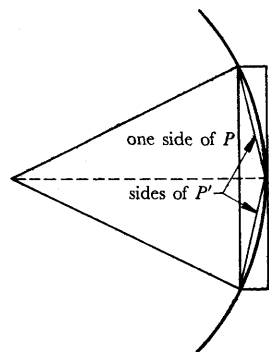


FIGURE 7

mentioned in any undergraduate course. The Suggested Reading contains references.

- *8. Suppose that f is a function such that $f(a) \leq f(b)$ whenever $a < b$ (Figure 6).

- Prove that $\lim_{x \rightarrow a^-} f(x)$ and $\lim_{x \rightarrow a^+} f(x)$ both exist. Hint: Why is this problem in this chapter?
- Prove that f never has a removable discontinuity (this terminology comes from Problem 6-16).
- Prove that if f satisfies the conclusions of the Intermediate Value Theorem, then f is continuous.

- *9. If f is a bounded function on $[0, 1]$, let $\|f\| = \sup \{|f(x)| : x \text{ in } [0, 1]\}$. Prove analogues of the properties of $\| \cdot \|$ in Problem 7-14.

10. Suppose $\alpha > 0$. Prove that every number x can be written uniquely in the form $x = k\alpha + x'$, where k is an integer, and $0 \leq x' < \alpha$.

11. (a) Suppose that $a_1, a_2, a_3, \dots$ is a sequence of positive numbers with $a_{n+1} \leq a_n/2$. Prove that for any $\epsilon > 0$ there is some n with $a_n < \epsilon$.

- (b) Suppose P is a regular polygon inscribed inside a circle. If P' is the inscribed regular polygon with twice as many sides, show that the difference between the area of the circle and the area of P' is less than half the difference between the area of the circle and the area of P (use Figure 7).

- (c) Prove that there is a regular polygon P inscribed in a circle with area as close as desired to the area of the circle. In order to do part (c) you will need part (a). This was clear to the Greeks, who used part (a) as the basis for their entire treatment of proportion and area. By calculating the areas of polygons, this method ("the method of exhaustion") allows computations of π to any desired accuracy; Archimedes used it to show that $\frac{22}{7} < \pi < \frac{22}{7}$. But it has far greater theoretical importance:

- (d) Using the fact that the areas of two regular polygons with the same number of sides have the same ratio as the square of their sides, prove that the areas of two circles have the same ratios as the square of their radii. Hint: Deduce a contradiction from the assumption that the ratio of the areas is greater, or less, than the ratio of the square of the radii by inscribing appropriate polygons.

12. Suppose that A and B are two nonempty sets of numbers such that $x \leq y$ for all x in A and all y in B .

- Prove that $\sup A \leq y$ for all y in B .
- Prove that $\sup A \leq \inf B$.

13. Let A and B be two nonempty sets of numbers which are bounded above, and let $A + B$ denote the set of all numbers $x + y$ with x in A and y in B . Prove that $\sup(A + B) = \sup A + \sup B$. Hint: The inequality

$\sup(A + B) \leq \sup A + \sup B$ is easy. Why? To prove that $\sup A + \sup B \leq \sup(A + B)$ it suffices to prove that $\sup A + \sup B \leq \sup(A + B) + \varepsilon$ for all $\varepsilon > 0$; begin by choosing x in A and y in B with $\sup A - x < \varepsilon/2$ and $\sup B - y < \varepsilon/2$.



14. (a) Consider a sequence of closed intervals $I_1 = [a_1, b_1]$, $I_2 = [a_2, b_2]$, $\dots$. Suppose that $a_n \leq a_{n+1}$ and $b_{n+1} \leq b_n$ for all n (Figure 8). Prove that there is a point x which is in every I_n .
- (b) Show that this conclusion is false if we consider open intervals instead of closed intervals.

The simple result of Problem 14(a) is called the “Nested Interval Theorem.” It may be used to give alternative proofs of Theorems 1 and 2. The appropriate reasoning, outlined in the next two problems, illustrates a general method, called a “bisection argument.”

- *15. Suppose f is continuous on $[a, b]$ and $f(a) < 0 < f(b)$. Then either $f((a+b)/2) = 0$, or f has different signs at the end points of $[a, (a+b)/2]$, or f has different signs at the end points of $[(a+b)/2, b]$. Why? If $f((a+b)/2) \neq 0$, let I_1 be one of the two intervals on which f changes sign. Now bisect I_1 . Either f is 0 at the midpoint, or f changes sign on one of the two intervals. Let I_2 be such an interval. Continue in this way, to define I_n for each n (unless f is 0 at some midpoint). Use the Nested Interval Theorem to find a point x where $f(x) = 0$.
- *16. Suppose f were continuous on $[a, b]$, but not bounded on $[a, b]$. Then f would be unbounded on either $[a, (a+b)/2]$ or $[(a+b)/2, b]$. Why? Let I_1 be one of these intervals on which f is unbounded. Proceed as in Problem 15 to obtain a contradiction.
17. (a) Let $A = \{x: x < \alpha\}$. Prove the following (they are all easy):
- (i) If x is in A and $y < x$, then y is in A .
 - (ii) $A \neq \emptyset$.
 - (iii) $A \neq \mathbf{R}$.
 - (iv) If x is in A , then there is some number x' in A such that $x < x'$.
- (b) Suppose, conversely, that A satisfies (i)–(iv). Prove that $A = \{x: x < \sup A\}$.
- *18. A number x is called an **almost upper bound** for A if there are only finitely many numbers y in A with $y \geq x$. An **almost lower bound** is defined similarly.
- (a) Find all almost upper bounds and almost lower bounds of the sets in Problem 1.
 - (b) Suppose that A is a bounded infinite set. Prove that the set B of all almost upper bounds of A is nonempty, and bounded below.

- (c) It follows from part (b) that $\inf B$ exists; this number is called the **limit superior** of A , and denoted by $\overline{\lim} A$ or $\limsup A$. Find $\overline{\lim} A$ for each set A in Problem 1.
- (d) Define $\underline{\lim} A$, and find it for all A in Problem 1.
- *19. If A is a bounded infinite set prove
- $\underline{\lim} A \leq \overline{\lim} A$.
 - $\overline{\lim} A \leq \sup A$.
 - If $\overline{\lim} A < \sup A$, then A contains a largest element.
 - The analogues of parts (b) and (c) for $\underline{\lim}$.

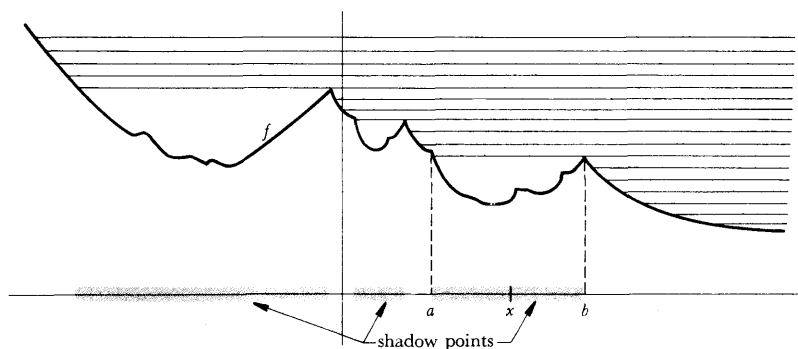


FIGURE 9

- *20. Let f be a continuous function on $\mathbf{R}$. A point x is called a **shadow point** of f if there is a number $y > x$ with $f(y) > f(x)$. The rationale for this terminology is indicated in Figure 9; the parallel lines are the rays of the sun rising in the east (you are facing north). Suppose that all points of (a, b) are shadow points, but that a and b are not shadow points.
- For x in (a, b) , prove that $f(x) \leq f(b)$. Hint: Let $A = \{y: x \leq y \leq b \text{ and } f(x) \leq f(y)\}$. If $\sup A$ were less than b , then $\sup A$ would be a shadow point. Use this fact to obtain a contradiction to the fact that b is not a shadow point.
 - Now prove that $f(a) \leq f(b)$. (This is a simple consequence of continuity.)
 - Finally, using the fact that a is not a shadow point, prove that $f(a) = f(b)$.

This result is known as the Rising Sun Lemma. Aside from serving as a good illustration of the use of least upper bounds, it is instrumental in proving several beautiful theorems that do not appear in this book; see page 371.

PART **3**
DERIVATIVES
AND
INTEGRALS

*In 1604, at the height of
his scientific career, Galileo argued
that for a rectilinear motion
in which speed increases proportionally
to distance covered,
the law of motion should be
just that ($x = ct^2$)
which he had discovered
in the investigation of falling bodies.
Between 1695 and 1700
not a single one of the monthly issues
of Leipzig's *Acta Eruditorum* was published
without articles of Leibniz,
the Bernoulli brothers
or the Marquis de l'Hôpital treating,
with notation only slightly different from
that which we use today,
the most varied problems of
differential calculus, integral calculus
and the calculus of variations.
Thus in the space of almost precisely
one century
infinitesimal calculus or,
as we now call it in English,
The Calculus,
the calculating tool par excellence,
had been forged;
and nearly three centuries of
constant use have not completely dulled
this incomparable instrument.*

NICHOLAS BOURBAKI

The derivative of a function is the first of the two major concepts of this section. Together with the integral, it constitutes the source from which calculus derives its particular flavor. While it is true that the concept of a function is fundamental, that you cannot do anything without limits or continuity, and that least upper bounds are essential, everything we have done until now has been preparation—if adequate, this section will be easier than the preceding ones—for the really exciting ideas to come, the powerful concepts that are truly characteristic of calculus.

Perhaps (some would say “certainly”) the interest of the ideas to be introduced in this section stems from the intimate connection between the mathematical concepts and certain physical ideas. Many definitions, and even some theorems, may be described in terms of physical problems, often in a revealing way. In fact, the demands of physics were the original inspiration for these fundamental ideas of calculus, and we shall frequently mention the physical interpretations. But we shall always first define the ideas in precise mathematical form, and discuss their significance in terms of mathematical problems.

The collection of all functions exhibits such diversity that there is almost no hope of discovering any interesting general properties pertaining to all. Because continuous functions form such a restricted class, we might expect to find some nontrivial theorems pertaining to them, and the sudden abundance of theorems after Chapter 6 shows that this expectation is justified. But the most interesting and most powerful results about functions will be obtained only when we restrict our attention even further, to functions which have even greater claim to be called “reasonable,” which are even better behaved than most continuous functions.

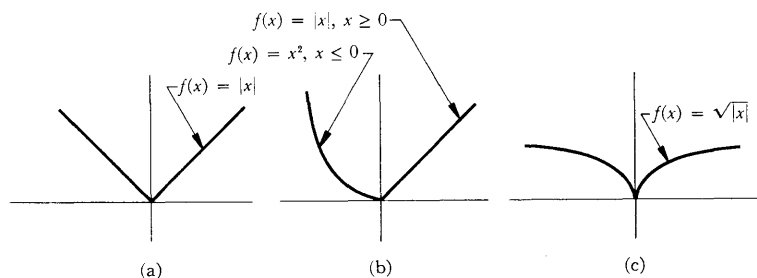


FIGURE 1

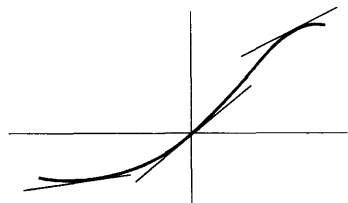


FIGURE 2

Figure 1 illustrates certain types of misbehavior which continuous functions can display. The graphs of these functions are “bent” at $(0, 0)$, unlike the graph of Figure 2, where it is possible to draw a “tangent line” at each point. The quotation marks have been used to avoid the suggestion that we have

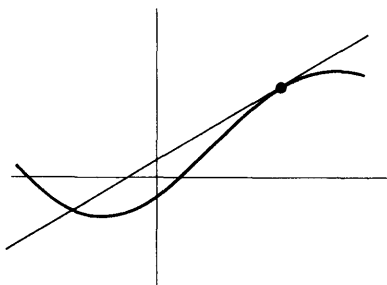


FIGURE 3

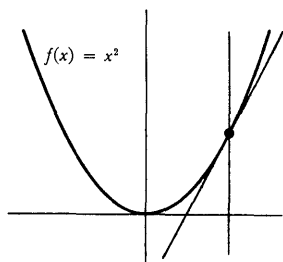


FIGURE 4

defined “bent” or “tangent line,” although we are suggesting that the graph might be “bent” at a point where a “tangent line” cannot be drawn. You have probably already noticed that a tangent line cannot be defined as a line which intersects the graph only once—such a definition would be both too restrictive and too permissive. With such a definition, the straight line shown in Figure 3 would not be a tangent line to the graph in that picture, while the parabola would have two tangent lines at each point (Figure 4), and the three functions in Figure 5 would have more than one tangent line at the points where they are “bent.”

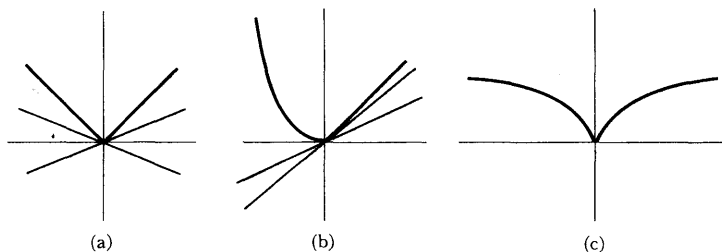


FIGURE 5

A more promising approach to the definition of a tangent line might start with “secant lines,” and use the notion of limits. If $h \neq 0$, then the two distinct points $(a, f(a))$ and $(a + h, f(a + h))$ determine, as in Figure 6, a straight line whose slope is

$$\frac{f(a + h) - f(a)}{h}.$$

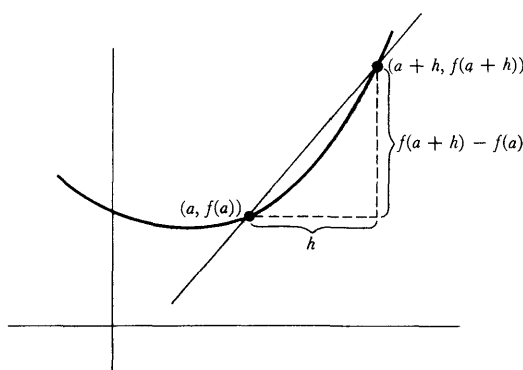


FIGURE 6

As Figure 7 illustrates, the “tangent line” at $(a, f(a))$ seems to be the limit, in some sense, of these “secant lines,” as h approaches 0. We have never before talked about a “limit” of lines, but we *can* talk about the limit of their slopes:

the slope of the tangent line through $(a, f(a))$ should be

$$\lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}.$$

We are ready for a definition, and some comments.

DEFINITION

The function f is **differentiable at a** if

$$\lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h} \text{ exists.}$$

In this case the limit is denoted by $f'(a)$ and is called the **derivative of f at a** . (We also say that f is **differentiable** if f is differentiable at a for every a in the domain of f .)

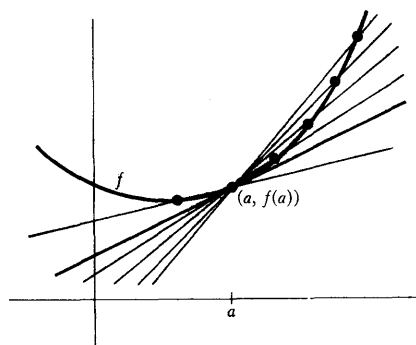


FIGURE 7

The first comment on our definition is really an addendum; we define the **tangent line** to the graph of f at $(a, f(a))$ to be the line through $(a, f(a))$ with slope $f'(a)$. This means that the tangent line at $(a, f(a))$ is defined only if f is differentiable at a .

The second comment refers to notation. The symbol $f'(a)$ is certainly reminiscent of functional notation. In fact, for any function f , we denote by f' the function whose domain is the set of all numbers a such that f is differentiable at a , and whose value at such a number a is

$$\lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}.$$

(To be very precise: f' is the collections of all pairs

$$\left(a, \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h} \right)$$

for which $\lim_{h \rightarrow 0} [f(a+h) - f(a)]/h$ exists.) The function f' is called the **derivative of f** .

Our third comment, somewhat longer than the previous two, refers to the physical interpretation of the derivative. Consider a particle which is moving along a straight line (Figure 8(a)) on which we have chosen an “origin” point O , and a direction in which distances from O shall be written as positive numbers, the distance from O of points in the other direction being written as negative numbers. Let $s(t)$ denote the distance of the particle from O , at time t . The suggestive notation $s(t)$ has been chosen purposely; since a distance $s(t)$ is

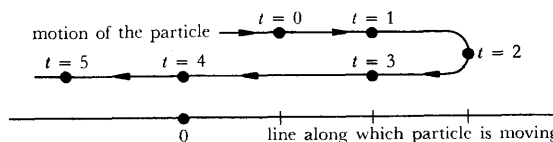


FIGURE 8(a)

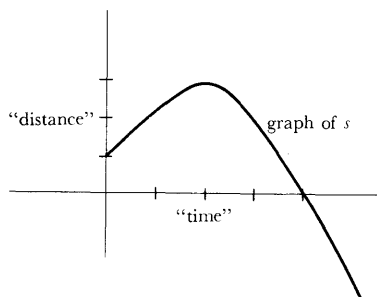


FIGURE 8(b)

determined for each number t , the physical situation automatically supplies us with a certain function s . The graph of s indicates the distance of the particle from O , on the vertical axis, in terms of the time, indicated on the horizontal axis (Figure 8(b)).

The quotient

$$\frac{s(a+h) - s(a)}{h}$$

has a natural physical interpretation. It is the “average velocity” of the particle during the time interval from a to $a+h$. For any particular a , this average speed depends on h , of course. On the other hand, the limit

$$\lim_{h \rightarrow 0} \frac{s(a+h) - s(a)}{h}$$

depends only on a (as well as the particular function s) and there are important physical reasons for considering this limit. We would like to speak of the “velocity of the particle at time a ,” but the usual definition of velocity is really a definition of average velocity; the only reasonable definition of “velocity at time a ” (so-called “instantaneous velocity”) is the limit

$$\lim_{h \rightarrow 0} \frac{s(a+h) - s(a)}{h}.$$

Thus we *define* the **(instantaneous) velocity** of the particle at a to be $s'(a)$. Notice that $s'(a)$ could easily be negative; the absolute value $|s'(a)|$ is sometimes called the **(instantaneous) speed**.

It is important to realize that instantaneous velocity is a theoretical concept, an abstraction which does not correspond precisely to any observable quantity. While it would not be fair to say that instantaneous velocity has nothing to do with average velocity, remember that $s'(t)$ is not

$$\frac{s(t+h) - s(t)}{h}$$

for any particular h , but merely the limit of these average velocities as h approaches 0. Thus, when velocities are measured in physics, what a physicist really measures is an average velocity over some (very small) time interval; such a procedure cannot be expected to give an exact answer, but this is really no defect, because physical measurements can never be exact anyway.

The velocity of a particle is often called the “rate of change of its position.” This notion of the derivative, as a rate of change, applies to any other physical situation in which some quantity varies with time. For example, the “rate of change of mass” of a growing object means the derivative of the function m , where $m(t)$ is the mass at time t .

In order to become familiar with the basic definitions of this chapter, we will spend quite some time examining the derivatives of particular functions. Before proving the important theoretical results of Chapter 11, we want to

have a good idea of what the derivative of a function looks like. The next chapter is devoted exclusively to one aspect of this problem—calculating the derivative of complicated functions. In this chapter we will emphasize the concepts, rather than the calculations, by considering a few simple examples. Simplest of all is a constant function, $f(x) = c$. In this case

$$\lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h} = \lim_{h \rightarrow 0} \frac{c - c}{h} = 0.$$

Thus f is differentiable at a for every number a , and $f'(a) = 0$. This means that the tangent line to the graph of f always has slope 0, so the tangent line always coincides with the graph.

Constant functions are not the only ones whose graphs coincide with their tangent lines—this happens for any linear function $f(x) = cx + d$. Indeed

$$\begin{aligned} f'(a) &= \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h} \\ &= \lim_{h \rightarrow 0} \frac{c(a+h) + d - [ca + d]}{h} \\ &= \lim_{h \rightarrow 0} \frac{ch}{h} = c; \end{aligned}$$

the slope of the tangent line is c , the same as the slope of the graph of f .

A refreshing difference occurs for $f(x) = x^2$. Here

$$\begin{aligned} f'(a) &= \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h} \\ &= \lim_{h \rightarrow 0} \frac{(a+h)^2 - a^2}{h} \\ &= \lim_{h \rightarrow 0} \frac{a^2 + 2ah + h^2 - a^2}{h} \\ &= \lim_{h \rightarrow 0} (2a + h) \\ &= 2a. \end{aligned}$$

Some of the tangent lines to the graph of f are shown in Figure 9. In this picture each tangent line appears to intersect the graph only once, and this fact can be checked fairly easily: Since the tangent line through (a, a^2) has slope $2a$, it is the graph of the function

$$\begin{aligned} g(x) &= 2a(x - a) + a^2 \\ &= 2ax - a^2. \end{aligned}$$

Now, if the graph of f and g intersect at a point $(x, f(x)) = (x, g(x))$, then

$$\begin{aligned} x^2 &= 2ax - a^2 \\ \text{or } x^2 - 2ax + a^2 &= 0; \\ \text{so } (x - a)^2 &= 0 \\ \text{or } x &= a. \end{aligned}$$

In other words, (a, a^2) is the only point of intersection.

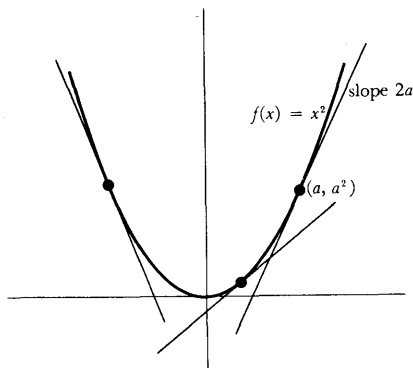


FIGURE 9

The function $f(x) = x^2$ happens to be quite special in this regard; usually a tangent line will intersect the graph more than once. Consider, for example, the function $f(x) = x^3$. In this case

$$\begin{aligned} f'(a) &= \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h} \\ &= \lim_{h \rightarrow 0} \frac{(a+h)^3 - a^3}{h} \\ &= \lim_{h \rightarrow 0} \frac{a^3 + 3a^2h + 3ah^2 + h^3 - a^3}{h} \\ &= \lim_{h \rightarrow 0} \frac{3a^2h + 3ah^2 + h^3}{h} \\ &= \lim_{h \rightarrow 0} 3a^2 + 3ah + h^2 \\ &= 3a^2. \end{aligned}$$

Thus the tangent line to the graph of f at (a, a^3) has slope $3a^2$. This means that the tangent line is the graph of

$$\begin{aligned} g(x) &= 3a^2(x - a) + a^3 \\ &= 3a^2x - 2a^3. \end{aligned}$$

The graphs of f and g intersect at the point $(x, f(x)) = (x, g(x))$ when

$$\begin{aligned} x^3 &= 3a^2x - 2a^3 \\ \text{or } x^3 - 3a^2x + 2a^3 &= 0. \end{aligned}$$

This equation is easily solved if we remember that one solution of the equation has got to be $x = a$, so that $(x - a)$ is a factor of the left side; the other factor can then be found by dividing. We obtain

$$(x - a)(x^2 + ax - 2a^2) = 0.$$

It so happens that $x^2 + ax - 2a^2$ also has $x - a$ as a factor; we obtain finally

$$(x - a)(x - a)(x + 2a) = 0.$$

Thus, as illustrated in Figure 10, the tangent line through (a, a^3) also intersects the graph at the point $(-2a, -8a^3)$. These two points are always distinct, except when $a = 0$.

We have already found the derivative of sufficiently many functions to illustrate the classical, and still very popular, notation for derivatives. For a given function f , the derivative f' is often denoted by

$$\frac{df(x)}{dx}.$$

For example, the symbol

$$\frac{dx^2}{dx}$$

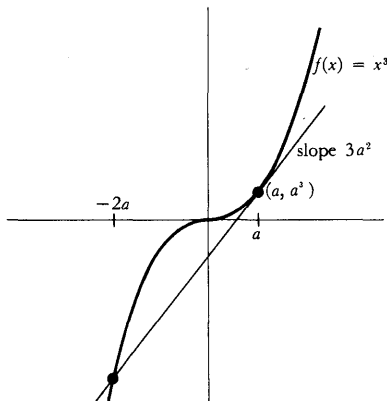


FIGURE 10

denotes the derivative of the function $f(x) = x^2$. Needless to say, the separate parts of the expression

$$\frac{df(x)}{dx}$$

are not supposed to have any sort of independent existence—the d 's are *not* numbers, they *cannot* be canceled, and the entire expression is *not* the quotient of two other numbers “ $df(x)$ ” and “ dx .” This notation is due to Leibniz (generally considered an independent co-discoverer of calculus, along with Newton), and is affectionately referred to as Leibnizian notation.* Although the notation $df(x)/dx$ seems very complicated, in concrete cases it may be shorter; after all, the symbol dx^2/dx is actually more concise than the phrase “the derivative of the function $f(x) = x^2$.”

The following formulas state in standard Leibnizian notation all the information that we have found so far:

$$\begin{aligned}\frac{dc}{dx} &= 0, \\ \frac{d(ax + b)}{dx} &= a, \\ \frac{dx^2}{dx} &= 2x, \\ \frac{dx^3}{dx} &= 3x^2.\end{aligned}$$

Although the meaning of these formulas is clear enough, attempts at literal interpretation are hindered by the reasonable stricture that an equation should not contain a function on one side and a number on the other. For example, if the third equation is to be true, then either $df(x)/dx$ must denote $f'(x)$, rather than f' , or else $2x$ must denote, not a number, but the function whose value at x is $2x$. It is really impossible to assert that one or the other of these alternatives is intended; in practice $df(x)/dx$ sometimes means f' and sometimes means $f'(x)$, while $2x$ may denote either a number or a function. Because of this ambiguity, most authors are reluctant to denote $f'(a)$ by

$$\frac{df(x)}{dx}(a);$$

instead $f'(a)$ is usually denoted by the barbaric, but unambiguous, symbol

$$\left. \frac{df(x)}{dx} \right|_{x=a}.$$

* Leibniz was led to this symbol by his intuitive notion of the derivative, which he considered to be, not the limit of quotients $[f(x+h) - f(x)]/h$, but the “value” of this quotient when h is an “infinitely small” number. This “infinitely small” quantity was denoted by dx and the corresponding “infinitely small” difference $f(x+dx) - f(x)$ by $df(x)$. Although this point of view is impossible to reconcile with properties (P1)–(P13) of the real numbers, some people find this notion of the derivative congenial.

In addition to these difficulties, Leibnizian notation is associated with one more ambiguity. Although the notation dx^2/dx is absolutely standard, the notation $df(x)/dx$ is often replaced by df/dx . This, of course, is in conformity with the practice of confusing a function with its value at x . So strong is this tendency that functions are often indicated by a phrase like the following: "consider the function $y = x^2$." We will sometimes follow classical practice to the extent of using y as the name of a function, but we will nevertheless carefully distinguish between the function and its values—thus we will always say something like "consider the function (defined by) $y(x) = x^2$."

Despite the many ambiguities of Leibnizian notation, it is used almost exclusively in older mathematical writing, and is still used very frequently today. The staunchest opponents of Leibnizian notation admit that it will be around for quite some time, while its most ardent admirers would say that it will be around forever, and a good thing too! In any case, Leibnizian notation cannot be ignored completely.

The policy adopted in this book is to disallow Leibnizian notation within the text, but to include it in the Problems; several chapters contain a few (immediately recognizable) problems which are expressly designed to illustrate the vagaries of Leibnizian notation. Trusting that these problems will provide ample practice in this notation, we return to our basic task of examining some simple examples of derivatives.

The few functions examined so far have all been differentiable. To fully appreciate the significance of the derivative it is equally important to know some examples of functions which are *not* differentiable. The obvious candidates are the three functions first discussed in this chapter, and illustrated in Figure 1; if they turn out to be differentiable at 0 something has clearly gone wrong.

Consider first $f(x) = |x|$. In this case

$$\frac{f(0+h) - f(0)}{h} = \frac{|h|}{h}.$$

Now $|h|/h = 1$ for $h > 0$, and $|h|/h = -1$ for $h < 0$. This shows that

$$\lim_{h \rightarrow 0} \frac{f(h) - f(0)}{h} \text{ does not exist.}$$

In fact,

$$\lim_{h \rightarrow 0^+} \frac{f(h) - f(0)}{h} = 1$$

$$\text{and } \lim_{h \rightarrow 0^-} \frac{f(h) - f(0)}{h} = -1.$$

(These two limits are sometimes called the **right-hand derivative** and the **left-hand derivative**, respectively, of f at 0.)

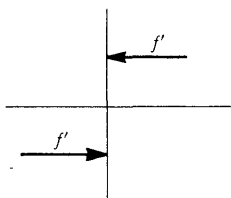
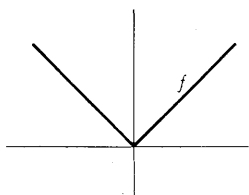


FIGURE 11

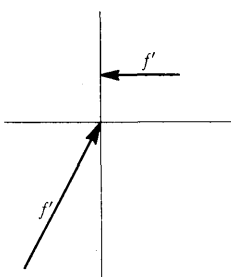
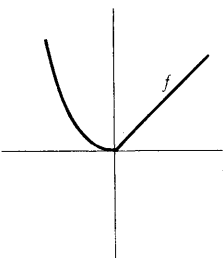


FIGURE 12

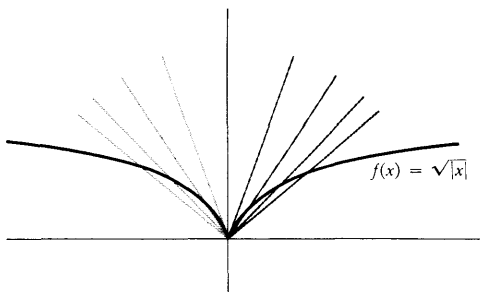


FIGURE 13

If $a \neq 0$, then $f'(a)$ does exist. In fact,

$$\begin{aligned} f'(x) &= 1 & \text{if } x > 0, \\ f'(x) &= -1 & \text{if } x < 0. \end{aligned}$$

The proof of this fact is left to you (it is easy if you remember the derivative of a linear function). The graphs of f and of f' are shown in Figure 11.

For the function

$$f(x) = \begin{cases} x^2, & x \leq 0 \\ x, & x \geq 0, \end{cases}$$

a similar difficulty arises in connection with $f'(0)$. We have

$$\frac{f(h) - f(0)}{h} = \begin{cases} \frac{h^2}{h} = h, & h < 0 \\ \frac{h}{h} = 1, & h > 0. \end{cases}$$

Therefore,

$$\begin{aligned} \lim_{h \rightarrow 0^-} \frac{f(h) - f(0)}{h} &= 0, \\ \text{but } \lim_{h \rightarrow 0^+} \frac{f(h) - f(0)}{h} &= 1. \end{aligned}$$

Thus $f'(0)$ does not exist; f is not differentiable at 0. Once again, however, $f'(x)$ exists for $x \neq 0$ —it is easy to see that

$$f'(x) = \begin{cases} 2x, & x < 0 \\ 1, & x > 0. \end{cases}$$

The graphs of f and f' are shown in Figure 12.

Even worse things happen for $f(x) = \sqrt{|x|}$. For this function

$$\frac{f(h) - f(0)}{h} = \begin{cases} \frac{\sqrt{h}}{h} = \frac{1}{\sqrt{h}}, & h > 0 \\ \frac{\sqrt{-h}}{h} = -\frac{1}{\sqrt{-h}}, & h < 0. \end{cases}$$

In this case the right-hand limit

$$\lim_{h \rightarrow 0^+} \frac{f(h) - f(0)}{h} = \lim_{h \rightarrow 0^+} \frac{1}{\sqrt{h}}$$

does not exist; instead $1/\sqrt{h}$ becomes arbitrarily large as h approaches 0. And, what's more, $-1/\sqrt{-h}$ becomes arbitrarily large in absolute value, but negative (Figure 13).

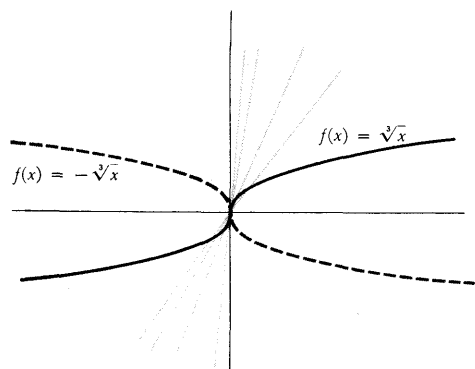


FIGURE 14

The function $f(x) = \sqrt[3]{x}$, although not differentiable at 0, is at least a little better behaved than this. The quotient

$$\frac{f(h) - f(0)}{h} = \frac{\sqrt[3]{h}}{h} = \frac{h^{1/3}}{h} = \frac{1}{h^{2/3}} = \frac{1}{(\sqrt[3]{h})^2}$$

simply becomes arbitrarily large as h goes to 0. Sometimes one says that f has an “infinite” derivative at 0. Geometrically this means that the graph of f has a “tangent line” which is parallel to the vertical axis (Figure 14). Of course, $f(x) = -\sqrt[3]{x}$ has the same geometric property, but one would say that f has a derivative of “negative infinity” at 0.

Remember that differentiability is supposed to be an improvement over mere continuity. This idea is supported by the many examples of functions which are continuous, but not differentiable; however, one important point remains to be noted:

THEOREM 1 If f is differentiable at a , then f is continuous at a .

PROOF

$$\begin{aligned} \lim_{h \rightarrow 0} f(a+h) - f(a) &= \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h} \cdot h \\ &= \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h} \cdot \lim_{h \rightarrow 0} h \\ &= f'(a) \cdot 0 \\ &= 0. \end{aligned}$$

As we pointed out in Chapter 5, the equation $\lim_{h \rightarrow 0} f(a+h) - f(a) = 0$ is equivalent to $\lim_{x \rightarrow a} f(x) = f(a)$; thus f is continuous at a . ■

It is very important to remember Theorem 1, and just as important to remember that the converse is not true. A differentiable function is continuous, but a continuous function need not be differentiable (keep in mind the function $f(x) = |x|$, and you will never forget which statement is true and which false).

The continuous functions examined so far have been differentiable at all points with at most one exception, but it is easy to give examples of continuous functions which are not differentiable at several points, even an infinite number (Figure 15). Actually, one can do much worse than this. There is a function

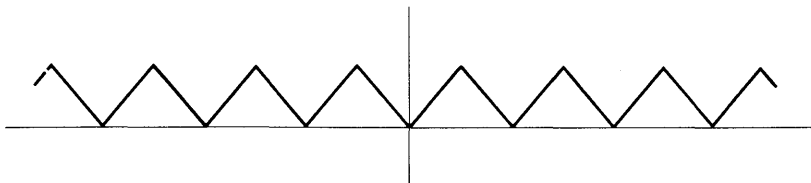


FIGURE 15

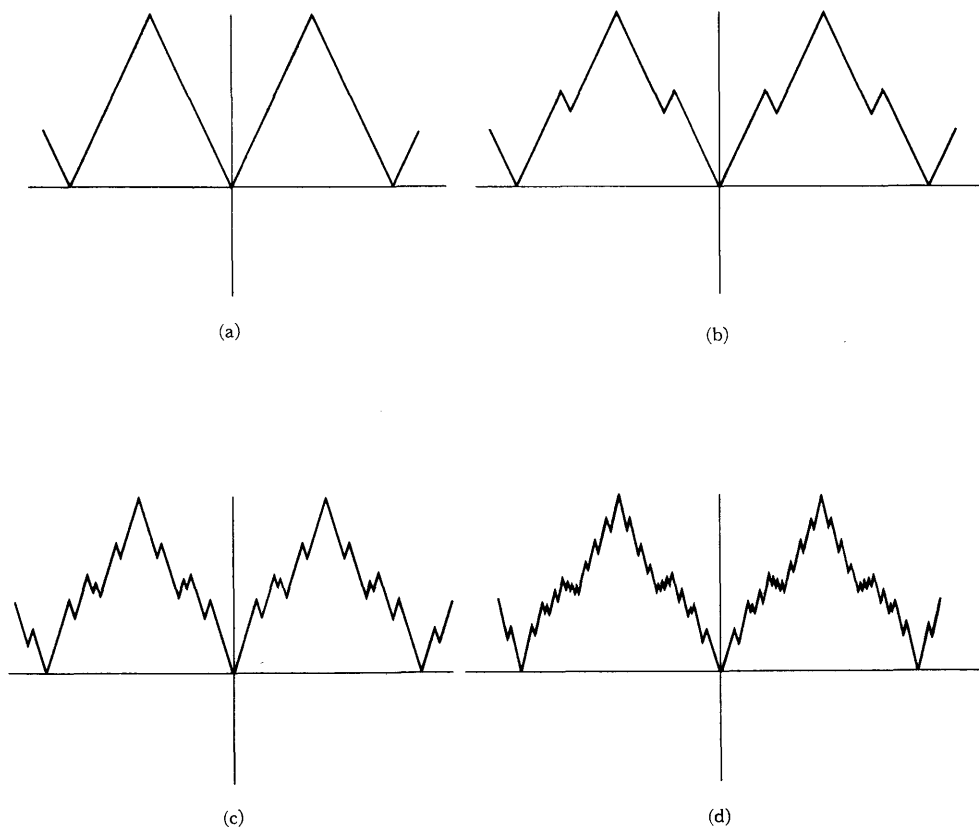


FIGURE 16

which is *continuous everywhere* and *differentiable nowhere*! Unfortunately, the definition of this function will be inaccessible to us until Chapter 23, and I have been unable to persuade the artist to draw it (consider carefully what the graph should look like and you will sympathize with his point of view). It is possible to draw some rough approximations to the graph, however; several successively better approximations are shown in Figure 16.

Although such spectacular examples of nondifferentiability must be postponed, we can, with a little ingenuity, find a continuous function which is not differentiable at infinitely many points, *all of which are in* $[0, 1]$. One such function is illustrated in Figure 17. The reader is given the problem of defining it precisely; it is a straight line version of the function

$$f(x) = \begin{cases} x \sin \frac{1}{x}, & x \neq 0 \\ 0, & x = 0. \end{cases}$$

This particular function f is itself quite sensitive to the question of differentiability. Indeed, for $h \neq 0$ we have

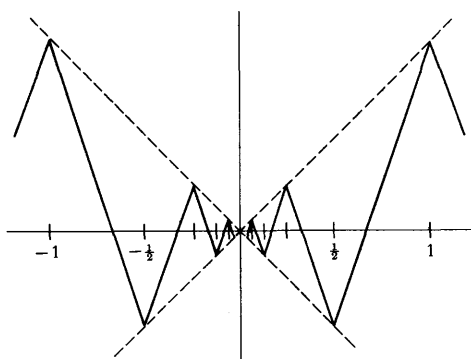


FIGURE 17

$$\frac{f(h) - f(0)}{h} = \frac{h \sin \frac{1}{h} - 0}{h} = \sin \frac{1}{h}.$$

Long ago we proved that $\lim_{h \rightarrow 0} \sin 1/h$ does not exist, so f is not differentiable at 0. Geometrically, one can see that a tangent line cannot exist, by noting that the secant line through $(0, 0)$ and $(h, f(h))$ in Figure 18 can have any slope between -1 and 1 , no matter how small we require h to be.

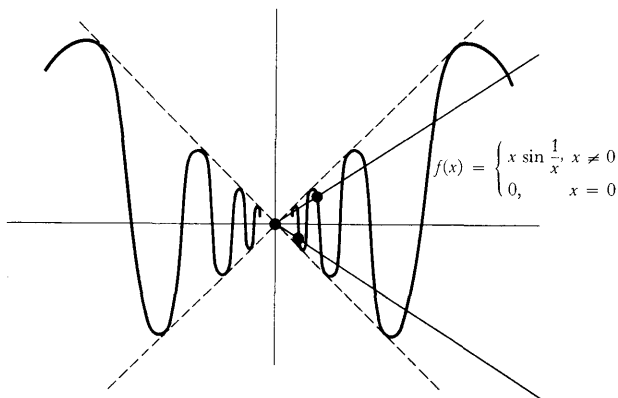


FIGURE 18

This finding represents something of a triumph; although continuous, the function f seems somehow quite unreasonable, and we can now enunciate one mathematically undesirable feature of this function—it is not differentiable at 0. Nevertheless, one should not become too enthusiastic about the criterion of differentiability. For example, the function

$$g(x) = \begin{cases} x^2 \sin \frac{1}{x}, & x \neq 0 \\ 0, & x = 0 \end{cases}$$

is differentiable at 0; in fact $g'(0) = 0$:

$$\begin{aligned} \lim_{h \rightarrow 0} \frac{g(h) - g(0)}{h} &= \lim_{h \rightarrow 0} \frac{h^2 \sin \frac{1}{h}}{h} \\ &= \lim_{h \rightarrow 0} h \sin \frac{1}{h} \\ &= 0. \end{aligned}$$

The tangent line to the graph of g at $(0, 0)$ is therefore the horizontal axis (Figure 19).

This example suggests that we should seek even more restrictive conditions on a function than mere differentiability. We can actually use the derivative to

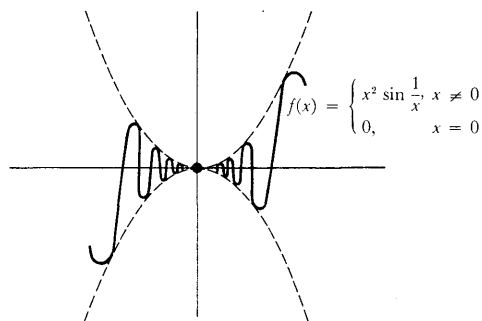


FIGURE 19

formulate such conditions if we introduce another set of definitions, the last of this chapter.

For any function f , we obtain, by taking the derivative, a new function f' (whose domain may be considerably smaller than that of f). The notion of differentiability can be applied to the function f' , or course, yielding another function $(f')'$, whose domain consists of all points a such that f' is differentiable at a . The function $(f')'$ is usually written simply f'' and is called the **second derivative** of f . If $f''(a)$ exists, then f is said to be 2-times differentiable at a , and the number $f''(a)$ is called the **second derivative of f at a** .

In physics the second derivative is particularly important. If $s(t)$ is the position at time t of a particle moving along a straight line, then $s''(t)$ is called the **acceleration** at time t . Acceleration plays a special role in physics, because, as stated in Newton's laws of motion, the force on a particle is the product of its mass and its acceleration. Consequently you can feel the second derivative when you sit in an accelerating car.

There is no reason to stop at the second derivative—we can define $f''' = (f'')'$, $f'''' = (f''')'$, etc. This notation rapidly becomes unwieldy, so the following abbreviation is usually adopted (it is really a recursive definition):

$$\begin{aligned} f^{(1)} &= f', \\ f^{(k+1)} &= (f^{(k)})'. \end{aligned}$$

Thus

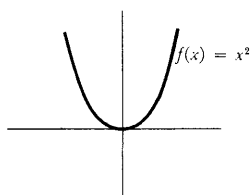
$$\begin{aligned} f^{(1)} &= f', \\ f^{(2)} &= f'' = (f')', \\ f^{(3)} &= f''' = (f'')', \\ f^{(4)} &= f'''' = (f''')', \\ &\text{etc.} \end{aligned}$$

The various functions $f^{(k)}$, for $k \geq 2$, are sometimes called **higher-order derivatives** of f .

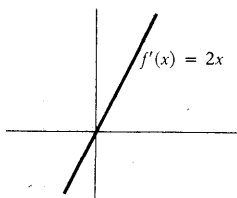
Usually, we resort to the notation $f^{(k)}$ only for $k \geq 4$, but it is convenient to have $f^{(k)}$ defined for smaller k also. In fact, a reasonable definition can be made for $f^{(0)}$, namely,

$$f^{(0)} = f.$$

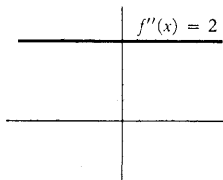
Leibnizian notation for higher-order derivatives should also be mentioned.



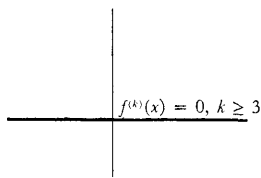
(a)



(b)



(c)



(d)

FIGURE 20

The natural Leibnizian symbol for $f''(x)$, namely,

$$\frac{d\left(\frac{df(x)}{dx}\right)}{dx},$$

is abbreviated to

$$\frac{d^2f(x)}{(dx)^2}, \quad \text{or more frequently to} \quad \frac{d^2f(x)}{dx^2}.$$

Similar notation is used for $f^{(n)}(x)$.

The following example illustrates the notation $f^{(k)}$, and also shows, in one very simple case, how various higher-order derivatives are related to the original function. Let $f(x) = x^2$. Then, as we have already checked,

$$\begin{aligned} f'(x) &= 2x, \\ f''(x) &= 2, \\ f'''(x) &= 0, \\ f^{(k)}(x) &= 0, \quad \text{if } k \geq 3. \end{aligned}$$

Figure 20 shows the function f , together with its various derivatives.

A rather more illuminating example is presented by the following function, whose graph is shown in Figure 21(a):

$$f(x) = \begin{cases} x^2, & x \geq 0 \\ -x^2, & x \leq 0. \end{cases}$$

It is easy to see that

$$\begin{aligned} f'(a) &= 2a & \text{if } a > 0, \\ f'(a) &= -2a & \text{if } a < 0. \end{aligned}$$

Moreover,

$$\begin{aligned} f'(0) &= \lim_{h \rightarrow 0} \frac{f(h) - f(0)}{h} \\ &= \lim_{h \rightarrow 0} \frac{f(h)}{h}. \end{aligned}$$

Now

$$\lim_{h \rightarrow 0^+} \frac{f(h)}{h} = \lim_{h \rightarrow 0^+} \frac{h^2}{h} = 0$$

$$\text{and } \lim_{h \rightarrow 0^-} \frac{f(h)}{h} = \lim_{h \rightarrow 0^-} \frac{-h^2}{h} = 0,$$

so

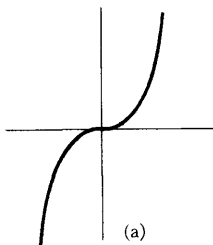
$$f'(0) = \lim_{h \rightarrow 0} \frac{f(h)}{h} = 0.$$

This information can all be summarized as follows:

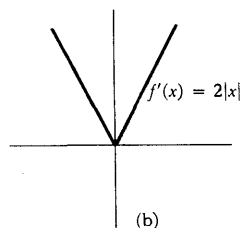
$$f'(x) = 2|x|.$$

It follows that $f''(0)$ does not exist! Existence of the second derivative is thus a

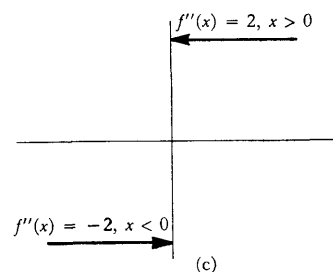
$$f(x) = \begin{cases} x^2, & x \geq 0 \\ -x^2, & x \leq 0 \end{cases}$$



(a)



(b)



(c)

FIGURE 21

rather strong criterion for a function to satisfy. Even a “smooth looking” function like f reveals some irregularity when examined with the second derivative. This suggests that the irregular behavior of the function

$$g(x) = \begin{cases} x^2 \sin \frac{1}{x}, & x \neq 0 \\ 0, & x = 0 \end{cases}$$

might also be revealed by the second derivative. At the moment we know that $g'(0) = 0$, but we do not know $g'(a)$ for any $a \neq 0$, so it is hopeless to begin computing $g''(0)$. We will return to this question at the end of the next chapter, after we have perfected the technique of finding derivatives.

PROBLEMS

- Prove, working directly from the definition, that if $f(x) = 1/x$, then $f'(a) = -1/a^2$, for $a \neq 0$.
 - Prove that the tangent line to the graph of f at $(a, 1/a)$ does not intersect the graph of f , except at $(a, 1/a)$.
- Prove that if $f(x) = 1/x^2$, then $f'(a) = -2/a^3$ for $a \neq 0$.
 - Prove that the tangent line to f at $(a, 1/a^2)$ intersects f at one other point, which lies on the opposite side of the vertical axis.
- Prove that if $f(x) = \sqrt{x}$, then $f'(a) = 1/2\sqrt{a}$, for $a > 0$. (The expression you obtain for $[f(a+h) - f(a)]/h$ will require some algebraic face lifting, but the answer should suggest the right trick.)
- For each natural number n , let $S_n(x) = x^n$. Remembering that $S_1'(x) = 1$, $S_2'(x) = 2x$, and $S_3'(x) = 3x^2$, conjecture a formula for $S_n'(x)$. Prove your conjecture. (The expression $(x+h)^n$ may be expanded by the binomial theorem.)
- Find f' if $f(x) = [x]$.
- Prove, starting from the definition (and drawing a picture to illustrate):
 - if $g(x) = f(x) + c$, then $g'(x) = f'(x)$;
 - if $g(x) = cf(x)$, then $g'(x) = cf'(x)$.
- Suppose that $f(x) = x^3$.
 - What is $f'(9)$, $f'(25)$, $f'(36)$?
 - What is $f'(3^2)$, $f'(5^2)$, $f'(6^2)$?
 - What is $f'(a^2)$, $f'(x^2)$?

If you do not find this problem silly, you are missing a very important point: $f'(x^2)$ means the derivative of f at the number which we happen to be calling x^2 ; it is *not* the derivative at x of the function $g(x) = f(x^2)$. Just to drive the point home:

- For $f(x) = x^3$, compare $f'(x^2)$ and $g'(x)$ where $g(x) = f(x^2)$.

8. (a) Suppose $g(x) = f(x + c)$. Prove (starting from the definition) that $g'(x) = f'(x + c)$. Draw a picture to illustrate this. To do this problem you must write out the definitions of $g'(x)$ and $f'(x + c)$ correctly. The purpose of Problem 7 was to convince you that although this problem is easy, it is not an utter triviality, and there is something to prove: you cannot simply put prime marks into the equation $g(x) = f(x + a)$. To emphasize this point:
- (b) Prove that if $g(x) = f(cx)$, then $g'(x) = c \cdot f'(cx)$. Try to see pictorially why this should be true, also.
- (c) Suppose that f is differentiable and periodic, with period a (i.e., $f(x + a) = f(x)$ for all x). Prove that f' is also periodic.
9. Find $f'(x)$ and also $f'(x + 3)$ in the following cases. Be very methodical, or you will surely slip up somewhere. Consult the answers (after you do the problem, naturally).
- (i) $f(x) = (x + 3)^5$.
- (ii) $f(x + 3) = x^5$.
- (iii) $f(x + 3) = (x + 5)^7$.
10. Find $f'(x)$ if $f(x) = g(t + x)$, and if $f(t) = g(t + x)$. The answers will *not* be the same.
11. (a) Prove that Galileo was wrong: if a body falls a distance $s(t)$ in t seconds, and s' is proportional to s , then s cannot be a function of the form $s(t) = ct^2$.
- (b) Prove that the following facts are true about s if $s(t) = (a/2)t^2$ (the first fact will show why we switched from c to $a/2$):
- (i) $s''(t) = a$ (the acceleration is constant).
- (ii) $[s'(t)]^2 = 2as(t)$.
- (c) If s is measured in feet, the value of a is 32. How many seconds do you have to get out of the way of a chandelier which falls from a 400-foot ceiling? If you don't make it, how fast will the chandelier be going when it hits you? Where was the chandelier when it was moving with half that speed?
12. Imagine a road on which the speed limit is specified at every single point. In other words, there is a certain function L such that the speed limit x miles from the beginning of the road is $L(x)$. Two cars, A and B , are driving along this road; car A 's position at time t is $a(t)$, and car B 's is $b(t)$.
- (a) What equation expresses the fact that car A always travels at the speed limit? (The answer is *not* $a'(t) = L(t)$.)
- (b) Suppose that A always goes at the speed limit, and that B 's position at time t is A 's position at time $t - 1$. Show that B is also going at the speed limit at all times.
- (c) Suppose B always stays a constant distance behind A . Under what conditions will B still always travel at the speed limit?

13. Suppose that $f(a) = g(a)$ and that the left-hand derivative of f at a equals the right-hand derivative of g at a . Define $h(x) = f(x)$ for $x \leq a$, and $h(x) = g(x)$ for $x \geq a$. Prove that h is differentiable at a .
14. Let $f(x) = x^2$ if x is rational, and $f(x) = 0$ if x is irrational. Prove that f is differentiable at 0. (Don't be scared by this function. Just write out the definition of $f'(0)$.)
- *15. (a) Let f be a function such that $|f(x)| \leq x^2$ for all x . Prove that f is differentiable at 0. (If you have done Problem 14 you should be able to do this.)
 (b) This result can be generalized if x^2 is replaced by $|g(x)|$, where g has what property?
16. Let $\alpha > 1$. If f satisfies $|f(x)| \leq |x|^\alpha$, prove that f is differentiable at 0.
17. Let $0 < \beta < 1$. Prove that if f satisfies $|f(x)| \geq |x|^\beta$ and $f(0) = 0$, then f is not differentiable at 0.
- *18. Let $f(x) = 0$ for irrational x , and $1/q$ for $x = p/q$ in lowest terms. Prove that f is not differentiable at a for any a . Hint: It obviously suffices to prove this for irrational a . Why? If $a = n.a_1a_2a_3 \dots$ is the decimal expansion of a , consider $[f(a+h) - f(a)]/h$ for h rational, and also for

$$h = -0.00 \dots 0a_{n+1}a_{n+2} \dots$$

19. (a) Suppose that $f(a) = g(a) = h(a)$, that $f(x) \leq g(x) \leq h(x)$ for all x , and that $f'(a) = h'(a)$. Prove that g is differentiable at a , and that $f'(a) = g'(a) = h'(a)$. (Begin with the definition of $g'(a)$.)
 (b) Show that the conclusion does not follow if we omit the hypothesis $f(a) = g(a) = h(a)$.
20. Let f be any polynomial function; we will see in the next chapter that f is differentiable. The tangent line to f at $(a, f(a))$ is the graph of $g(x) = f'(a)(x - a) + f(a)$. Thus $f(x) - g(x)$ is the polynomial function $d(x) = f(x) - f'(a)(x - a) - f(a)$. We have already seen that if $f(x) = x^2$, then $d(x) = (x - a)^2$, and if $f(x) = x^3$, then $d(x) = (x - a)^2(x + 2a)$.

- (a) Find $d(x)$ when $f(x) = x^4$, and show that it is divisible by $(x - a)^2$.
 (b) There certainly seems to be some evidence that $d(x)$ is always divisible by $(x - a)^2$. Figure 22 provides an intuitive argument: usually, lines parallel to the tangent line will intersect the graph at two points; the tangent line intersects the graph only once near the point, so the intersection should be a "double intersection." To give a rigorous proof, first note that

$$\frac{d(x)}{x - a} = \frac{f(x) - f(a)}{x - a} - f'(a).$$

Now answer the following questions. Why is $f(x) - f(a)$ divisible by $(x - a)$? Why is there a polynomial function h such that $h(x) = d(x)/(x - a)$ for $x \neq a$? Why is $\lim_{x \rightarrow a} h(x) = 0$? Why is $h(a) = 0$? Why does this solve the problem?

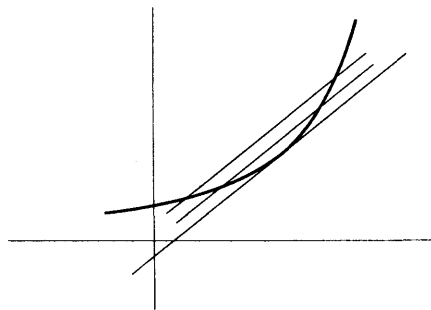


FIGURE 22

21. (a) Show that $f'(a) = \lim_{x \rightarrow a} [f(x) - f(a)]/(x - a)$. (Nothing deep here.)
 (b) Show that derivatives are a "local property": if $f(x) = g(x)$ for all x in some open interval containing a , then $f'(a) = g'(a)$. (This means that in computing $f'(a)$, you can ignore $f(x)$ for any particular $x \neq a$. Of course you can't ignore $f(x)$ for all such x at once!)

- *22. (a) Suppose that f is differentiable at x . Prove that

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x-h)}{2h}.$$

Hint: Remember an old algebraic trick—a number is not changed if the same quantity is added to and then subtracted from it.

- ** (b) Prove, more generally, that

$$f'(x) = \lim_{h, k \rightarrow 0^+} \frac{f(x+h) - f(x-k)}{h+k}.$$

- *23. Prove that if f is even, then $f'(x) = -f'(-x)$. (In order to minimize confusion, let $g(x) = f(-x)$; find $g'(x)$ and *then* remember what other thing g is.) Draw a picture!
- *24. Prove that if f is odd, then $f'(x) = f'(-x)$. Once again, draw a picture.
25. Problems 23 and 24 say that f' is even if f is odd, and odd if f is even. What can therefore be said about $f^{(k)}$?
26. Find $f''(x)$ if
- (i) $f(x) = x^3$.
 - (ii) $f(x) = x^5$.
 - (iii) $f'(x) = x^4$.
 - (iv) $f(x+3) = x^5$.
27. If $S_n(x) = x^n$, and $0 \leq k \leq n$, prove that

$$\begin{aligned} S_n^{(k)}(x) &= \frac{n!}{(n-k)!} x^{n-k} \\ &= k! \binom{n}{k} x^{n-k}. \end{aligned}$$

- *28. (a) Find $f'(x)$ if $f(x) = |x|^3$. Find $f''(x)$. Does $f'''(x)$ exist for all x ?
 (b) Analyze f similarly if $f(x) = x^4$ for $x \geq 0$ and $f(x) = -x^4$ for $x \leq 0$.
- *29. Let $f(x) = x^n$ for $x \geq 0$ and let $f(x) = 0$ for $x \leq 0$. Prove that $f^{(n-1)}$ exists (and find a formula for it), but that $f^{(n)}(0)$ does not exist.
30. Interpret the following specimens of Leibnizian notation; each is a restatement of some fact occurring in a previous problem.
- (i) $\frac{dx^n}{dx} = nx^{n-1}$.

$$(ii) \quad \frac{dz}{dy} = -\frac{1}{y^2} \text{ if } z = \frac{1}{y}.$$

$$(iii) \quad \frac{d[f(x) + c]}{dx} = \frac{df(x)}{dx}.$$

$$(iv) \quad \frac{d[cf(x)]}{dx} = c \frac{df(x)}{dx}.$$

$$(v) \quad \frac{dz}{dx} = \frac{dy}{dx} \text{ if } z = y + c.$$

$$(vi) \quad \left. \frac{dx^3}{dx} \right|_{x=a^3} = 3a^4.$$

$$(vii) \quad \left. \frac{df(x+a)}{dx} \right|_{x=b} = \left. \frac{df(x)}{dx} \right|_{x=b+a}.$$

$$(viii) \quad \left. \frac{df(cx)}{dx} \right|_{x=b} = c \cdot \left. \frac{df(x)}{dx} \right|_{x=cb}.$$

$$(ix) \quad \frac{df(cx)}{dx} = c \cdot \left. \frac{df(y)}{dy} \right|_{y=cx}.$$

$$(x) \quad \frac{d^k x^n}{dx^k} = k! \binom{n}{k} x^{n-k}.$$

The process of finding the derivative of a function is called *differentiation*. From the previous chapter you may have the impression that this process is usually laborious, requires recourse to the definition of the derivative, and depends upon successfully recognizing some limit. It is true that such a procedure is often the only possible approach—if you forget the definition of the derivative you are likely to be lost. Nevertheless, in this chapter we will learn to differentiate a large number of functions, without the necessity of even recalling the definition. A few theorems will provide a mechanical process for differentiating a large class of functions, which are formed from a few simple functions by the process of addition, multiplication, division, and composition. This description should suggest what theorems will be proved. We will first find the derivative of a few simple functions, and then prove theorems about the sum, products, quotients, and compositions of differentiable functions. The first theorem is merely a formal recognition of a computation carried out in the previous chapter.

THEOREM 1 If f is a constant function, $f(x) = c$, then

$$f'(a) = 0 \quad \text{for all numbers } a.$$

PROOF

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h} = \lim_{h \rightarrow 0} \frac{c - c}{h} = 0. \blacksquare$$

The second theorem is also a special case of a computation in the last chapter.

THEOREM 2 If f is the identity function, $f(x) = x$, then

PROOF

$$f'(a) = 1 \quad \text{for all numbers } a.$$

$$\begin{aligned} f'(a) &= \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h} \\ &= \lim_{h \rightarrow 0} \frac{a+h-a}{h} \\ &= \lim_{h \rightarrow 0} \frac{h}{h} = 1. \blacksquare \end{aligned}$$

The derivative of the sum of two functions is just what one would hope—the sum of the derivatives.

THEOREM 3 If f and g are differentiable at a , then $f + g$ is also differentiable at a , and

$$(f + g)'(a) = f'(a) + g'(a).$$

PROOF

$$\begin{aligned}(f + g)'(a) &= \lim_{h \rightarrow 0} \frac{(f + g)(a + h) - (f + g)(a)}{h} \\&= \lim_{h \rightarrow 0} \frac{f(a + h) + g(a + h) - [f(a) + g(a)]}{h} \\&= \lim_{h \rightarrow 0} \left[\frac{f(a + h) - f(a)}{h} + \frac{g(a + h) - g(a)}{h} \right] \\&= \lim_{h \rightarrow 0} \frac{f(a + h) - f(a)}{h} + \lim_{h \rightarrow 0} \frac{g(a + h) - g(a)}{h} \\&= f'(a) + g'(a). \blacksquare\end{aligned}$$

The formula for the derivative of a product is not as simple as one might wish, but it is nevertheless pleasantly symmetric, and the proof requires only a simple algebraic trick, which we have found useful before—a number is not changed if the same quantity is added to and subtracted from it.

THEOREM 4 If f and g are differentiable at a , then $f \cdot g$ is also differentiable at a , and

$$(f \cdot g)'(a) = f'(a) \cdot g(a) + f(a) \cdot g'(a).$$

PROOF

$$\begin{aligned}(f \cdot g)'(a) &= \lim_{h \rightarrow 0} \frac{(f \cdot g)(a + h) - (f \cdot g)(a)}{h} \\&= \lim_{h \rightarrow 0} \frac{f(a + h)g(a + h) - f(a)g(a)}{h} \\&= \lim_{h \rightarrow 0} \left[\frac{f(a + h)[g(a + h) - g(a)]}{h} + \frac{[f(a + h) - f(a)]g(a)}{h} \right] \\&= \lim_{h \rightarrow 0} f(a + h) \cdot \lim_{h \rightarrow 0} \frac{g(a + h) - g(a)}{h} + \lim_{h \rightarrow 0} \frac{f(a + h) - f(a)}{h} \cdot \lim_{h \rightarrow 0} g(a) \\&= f(a) \cdot g'(a) + f'(a) \cdot g(a).\end{aligned}$$

(Notice that we have used Theorem 9-1 to conclude that $\lim_{h \rightarrow 0} f(a + h) = f(a)$.) $\blacksquare$

In one special case Theorem 4 simplifies considerably:

THEOREM 5 If $g(x) = cf(x)$ and f is differentiable at a , then g is differentiable at a , and

$$g'(a) = c \cdot f'(a).$$

PROOF

If $h(x) = c$, so that $g = h \cdot f$, then by Theorem 4,

$$\begin{aligned}g'(a) &= (h \cdot f)'(a) \\&= h(a) \cdot f'(a) + h'(a) \cdot f(a) \\&= c \cdot f'(a) + 0 \cdot f(a) \\&= c \cdot f'(a). \blacksquare\end{aligned}$$

Notice, in particular, that $(-f)'(a) = -f'(a)$, and consequently $(f - g)'(a) = (f + [-g])'(a) = f'(a) - g'(a)$.

To demonstrate what we have already achieved, we will compute the derivative of some more special functions.

THEOREM 6 If $f(x) = x^n$ for some natural number n , then

$$f'(a) = na^{n-1} \quad \text{for all } a.$$

PROOF The proof will be by induction on n . For $n = 1$ this is simply Theorem 2. Now assume that the theorem is true for n , so that if $f(x) = x^n$, then

$$f'(a) = na^{n-1} \quad \text{for all } a.$$

Let $g(x) = x^{n+1}$. If $I(x) = x$, the equation $x^{n+1} = x^n \cdot x$ can be written

$$g(x) = f(x) \cdot I(x) \quad \text{for all } x;$$

thus $g = f \cdot I$. It follows from Theorem 4 that

$$\begin{aligned} g'(a) &= (f \cdot I)'(a) = f'(a) \cdot I(a) + f(a) \cdot I'(a) \\ &= na^{n-1} \cdot a + a^n \cdot 1 \\ &= na^n + a^n \\ &= (n+1)a^n, \quad \text{for all } a. \end{aligned}$$

This is precisely the case $n+1$ which we wished to prove. ■

Putting together the theorems proved so far we can now find f' for f of the form

$$f(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_2 x^2 + a_1 x + a_0.$$

We obtain

$$f'(x) = na_n x^{n-1} + (n-1)a_{n-1} x^{n-2} + \cdots + 2a_2 x + a_1.$$

We can also find f'' :

$$f''(x) = n(n-1)a_n x^{n-2} + (n-1)(n-2)a_{n-1} x^{n-3} + \cdots + 2a_2.$$

This process can be continued easily. Each differentiation reduces the highest power of x by 1, and eliminates one more a_i . It is a good idea to work out the derivatives f''' , $f^{(4)}$, and perhaps $f^{(5)}$, until the pattern becomes quite clear. The last interesting derivative is

$$f^{(n)}(x) = n!a_n;$$

for $k > n$ we have

$$f^{(k)}(x) = 0.$$

Clearly, the next step in our program is to find the derivative of a quotient f/g . It is quite a bit simpler, and, because of Theorem 4, obviously sufficient to find the derivative of $1/g$.

THEOREM 7 If g is differentiable at a , and $g(a) \neq 0$, then $1/g$ is differentiable at a , and

$$\left(\frac{1}{g}\right)'(a) = \frac{-g'(a)}{[g(a)]^2}.$$

PROOF Before we even write

$$\frac{\left(\frac{1}{g}\right)(a+h) - \left(\frac{1}{g}\right)(a)}{h}$$

we must be sure that this expression makes sense—it is necessary to check that $(1/g)(a+h)$ is defined for sufficiently small h . This requires only two observations. Since g is, by hypothesis, differentiable at a , it follows from Theorem 9-1 that g is continuous at a . Since $g(a) \neq 0$, it follows from Theorem 6-3 that there is some $\delta > 0$ such that $g(a+h) \neq 0$ for $|h| < \delta$. Therefore $(1/g)(a+h)$ *does* make sense for small enough h , and we can write

$$\begin{aligned} \lim_{h \rightarrow 0} \frac{\left(\frac{1}{g}\right)(a+h) - \left(\frac{1}{g}\right)(a)}{h} &= \lim_{h \rightarrow 0} \frac{\frac{1}{g(a+h)} - \frac{1}{g(a)}}{h} \\ &= \lim_{h \rightarrow 0} \frac{g(a) - g(a+h)}{h[g(a) \cdot g(a+h)]} \\ &= \lim_{h \rightarrow 0} \frac{-[g(a+h) - g(a)]}{h} \cdot \frac{1}{g(a)g(a+h)} \\ &= \lim_{h \rightarrow 0} \frac{-[g(a+h) - g(a)]}{h} \cdot \lim_{h \rightarrow 0} \frac{1}{g(a) \cdot g(a+h)} \\ &= -g'(a) \cdot \frac{1}{[g(a)]^2}. \end{aligned}$$

(Notice that we have used continuity of g at a once again.) ■

The general formula for the derivative of a quotient is now easy to derive. Though not particularly appealing, it is important, and must simply be memorized (I always use the incantation: “bottom times derivative of top, minus top times derivative of bottom, over bottom squared.”)

THEOREM 8 If f and g are differentiable at a and $g(a) \neq 0$, then f/g is differentiable at a , and

$$\left(\frac{f}{g}\right)'(a) = \frac{g(a) \cdot f'(a) - f(a) \cdot g'(a)}{[g(a)]^2}.$$

PROOF Since $f/g = f \cdot (1/g)$ we have

$$\begin{aligned}\left(\frac{f}{g}\right)'(a) &= \left(f \cdot \frac{1}{g}\right)'(a) \\ &= f'(a) \cdot \left(\frac{1}{g}\right)'(a) + f(a) \cdot \left(\frac{1}{g}\right)'(a) \\ &= \frac{f'(a)}{g(a)} + \frac{f(a)(-g'(a))}{[g(a)]^2} \\ &= \frac{f'(a) \cdot g(a) - f(a) \cdot g'(a)}{[g(a)]^2}. \blacksquare\end{aligned}$$

We can now differentiate a few more functions. For example,

$$\text{if } f(x) = \frac{x^2 - 1}{x^2 + 1}, \text{ then } f'(x) = \frac{(x^2 + 1)(2x) - (x^2 - 1)(2x)}{(x^2 + 1)^2} = \frac{4x}{(x^2 + 1)^2};$$

$$\text{if } f(x) = \frac{x}{x^2 + 1}, \text{ then } f'(x) = \frac{(x^2 + 1) - x(2x)}{(x^2 + 1)^2} = \frac{1 - x^2}{(x^2 + 1)^2};$$

$$\text{if } f(x) = \frac{1}{x}, \quad \text{then } f'(x) = -\frac{1}{x^2} = (-1)x^{-2}.$$

Notice that the last example can be generalized: if

$$f(x) = x^{-n} = \frac{1}{x^n}, \quad \text{for some natural number } n,$$

then

$$f'(x) = \frac{-nx^{n-1}}{x^{2n}} = (-n)x^{-n-1};$$

thus Theorem 6 actually holds both for positive and negative integers. If we interpret $f(x) = x^0$ to mean $f(x) = 1$, and $f'(x) = 0 \cdot x^{-1}$ to mean $f'(x) = 0$, then Theorem 6 is true for $n = 0$ also. (The word “interpret” is necessary because it is not clear how 0^0 should be defined and, in any case, $0 \cdot 0^{-1}$ is meaningless.)

Further progress in differentiation requires the knowledge of the derivatives of certain special functions to be studied later. One of these is the sine function. For the moment we shall divulge, and use, the following information, without proof:

$$\begin{aligned}\sin'(a) &= \cos a & \text{for all } a, \\ \cos'(a) &= -\sin a & \text{for all } a.\end{aligned}$$

This information allows us to differentiate many other functions. For example, if

$$f(x) = x \sin x,$$

then

$$\begin{aligned}f'(x) &= x \cos x + \sin x, \\ f''(x) &= -x \sin x + \cos x + \cos x \\ &= -x \sin x + 2 \cos x;\end{aligned}$$

if

$$g(x) = \sin^2 x = \sin x \cdot \sin x,$$

then

$$\begin{aligned} g'(x) &= \sin x \cos x + \cos x \sin x \\ &= 2 \sin x \cos x, \\ g''(x) &= 2[(\sin x)(-\sin x) + \cos x \cos x] \\ &= 2[\cos^2 x - \sin^2 x]; \end{aligned}$$

if

$$h(x) = \cos^2 x = \cos x \cdot \cos x,$$

then

$$\begin{aligned} h'(x) &= (\cos x)(-\sin x) + (-\sin x) \cos x \\ &= -2 \sin x \cos x, \\ h''(x) &= -2[\cos^2 x - \sin^2 x]. \end{aligned}$$

Notice that

$$g'(x) + h'(x) = 0,$$

hardly surprising, since $(g + h)(x) = \sin^2 x + \cos^2 x = 1$. As we would expect, we also have $g''(x) + h''(x) = 0$.

The examples above involved only products of two functions. A function involving triple products can be handled by Theorem 4 also; in fact it can be handled in two ways. Remember that $f \cdot g \cdot h$ is an abbreviation for

$$(f \cdot g) \cdot h \quad \text{or} \quad f \cdot (g \cdot h).$$

Choosing the first of these, for example, we have

$$\begin{aligned} (f \cdot g \cdot h)'(x) &= (f \cdot g)'(x) \cdot h(x) + (f \cdot g)(x)h'(x) \\ &= [f'(x)g(x) + f(x)g'(x)]h(x) + f(x)g(x)h'(x) \\ &= f'(x)g(x)h(x) + f(x)g'(x)h(x) + f(x)g(x)h'(x). \end{aligned}$$

The choice of $f \cdot (g \cdot h)$ would, of course, have given the same result, with a different intermediate step. The final answer is completely symmetric and easily remembered:

$(f \cdot g \cdot h)'$ is the sum of the three terms obtained by differentiating each of f , g , and h and multiplying by the other two.

For example, if

$$f(x) = x^3 \sin x \cos x,$$

then

$$f'(x) = 3x^2 \sin x \cos x + x^3 \cos x \cos x + x^3 (\sin x)(-\sin x).$$

Products of more than 3 functions can be handled similarly. For example, you should have little difficulty deriving the formula

$$\begin{aligned} (f \cdot g \cdot h \cdot k)'(x) &= f'(x)g(x)h(x)k(x) + f(x)g'(x)h(x)k(x) \\ &\quad + f(x)g(x)h'(x)k(x) + f(x)g(x)h(x)k'(x). \end{aligned}$$

You might even try to prove (by induction) the general formula:

$$(f_1 \cdot \dots \cdot f_n)'(x) = \sum_{i=1}^n f_1(x) \cdot \dots \cdot f_{i-1}(x) f_i'(x) f_{i+1}(x) \cdot \dots \cdot f_n(x).$$

Differentiating the most interesting functions obviously requires a formula for $(f \circ g)'(x)$ in terms of f' and g' . To ensure that $f \circ g$ be differentiable at a , one reasonable hypothesis would seem to be that g be differentiable at a . Since the behavior of $f \circ g$ near a depends on the behavior of f near $g(a)$ (not near a), it also seems reasonable to assume that f is differentiable at $g(a)$. Indeed we shall prove that if g is differentiable at a and f is differentiable at $g(a)$, then $f \circ g$ is differentiable at a , and

$$(f \circ g)'(a) = f'(g(a)) \cdot g'(a).$$

This extremely important formula is called the *Chain Rule*, presumably because a composition of functions might be called a “chain” of functions. Notice that $(f \circ g)'$ is practically the product of f' and g' , but not quite: f' must be evaluated at $g(a)$ and g' at a . Before attempting to prove this theorem we will try a few applications. Suppose

$$f(x) = \sin x^2.$$

Let us, temporarily, use S to denote the (“squaring”) function $S(x) = x^2$. Then

$$f = \sin \circ S.$$

Therefore we have

$$\begin{aligned} f'(x) &= \sin'(S(x)) \cdot S'(x) \\ &= \cos x^2 \cdot 2x. \end{aligned}$$

Quite a different result is obtained if

$$f(x) = \sin^2 x.$$

In this case

$$f = S \circ \sin,$$

so

$$\begin{aligned} f'(x) &= S'(\sin x) \cdot \sin'(x) \\ &= 2 \sin x \cdot \cos x. \end{aligned}$$

Notice that this agrees (as it should) with the result obtained by writing $f = \sin \cdot \sin$ and using the product formula.

Although we have invented a special symbol, S , to name the “squaring” function, it does not take much practice to do problems like this without bothering to write down special symbols for functions, and without even bothering to write down the particular composition which f is—one soon becomes accustomed to taking f apart in one’s head. The following differentiations may be used as practice for such mental gymnastics—if you find it necessary to work a few out on paper, by all means do so, but try to develop the knack of writing f' immediately after seeing the definition of f ; problems of this sort are so simple that, if you just remember the Chain Rule, there is no thought necessary.

$$\begin{array}{ll} \text{if } f(x) = \sin x^3 & \text{then } f'(x) = \cos x^3 \cdot 3x^2 \\ f(x) = \sin^3 x & f'(x) = 3 \sin^2 x \cdot \cos x \end{array}$$

$$\begin{array}{ll}
 f(x) = \sin \frac{1}{x} & f'(x) = \cos \frac{1}{x} \cdot \left(\frac{-1}{x^2} \right) \\
 f(x) = \sin(\sin x) & f'(x) = \cos(\sin x) \cdot \cos x \\
 f(x) = \sin(x^3 + 3x^2) & f'(x) = \cos(x^3 + 3x^2) \cdot (3x^2 + 6x) \\
 f(x) = (x^3 + 3x^2)^{53} & f'(x) = 53(x^3 + 3x^2)^{52} \cdot (3x^2 + 6x).
 \end{array}$$

A function like

$$f(x) = \sin^2 x^2 = [\sin x^2]^2,$$

which is the composition of three functions,

$$f = S \circ \sin \circ S,$$

can also be differentiated by the Chain Rule. It is only necessary to remember that a triple composition $f \circ g \circ h$ means $(f \circ g) \circ h$ or $f \circ (g \circ h)$. Thus if

$$f(x) = \sin^2 x^2$$

we can write

$$\begin{aligned}
 f &= (S \circ \sin) \circ S, \\
 f &= S \circ (\sin \circ S).
 \end{aligned}$$

The derivative of either expression can be found by applying the Chain Rule twice; the only doubtful point is whether the two expressions lead to equally simple calculations. As a matter of fact, as any experienced differentiator knows, it is much better to use the second:

$$f = S \circ (\sin \circ S).$$

We can now write down $f'(x)$ in one fell swoop. To begin with, note that the first function to be differentiated is S , so the formula for $f'(x)$ begins

$$f'(x) = 2(\quad) \cdot \quad .$$

Inside the parentheses we must put $\sin x^2$, the value at x of the second function, $\sin \circ S$. Thus we begin by writing

$$f'(x) = 2 \sin x^2 \cdot \quad$$

(the parentheses weren't really necessary, after all). We must now multiply this much of the answer by the derivative of $\sin \circ S$ at x ; this part is easy—it involves a composition of two functions, which we already know how to handle. We obtain, for the final answer,

$$f'(x) = 2 \sin x^2 \cdot \cos x^2 \cdot 2x.$$

The following example is handled similarly. Suppose

$$f(x) = \sin(\sin x^2).$$

Without even bothering to write down f as a composition $g \circ h \circ k$ of three functions, we can see that the left-most one will be $\sin$, so our expression for $f'(x)$ begins

$$f'(x) = \cos(\quad) \cdot \quad .$$

Inside the parenthesis we must put the value of $h \circ k(x)$; this is simply $\sin x^2$ (what you get from $\sin(\sin x^2)$ by deleting the first $\sin$). So our expression for $f'(x)$ begins

$$f'(x) = \cos(\sin x^2) \cdot \text{[redacted]}.$$

We can now forget about the first $\sin$ in $\sin(\sin x^2)$; we have to multiply what we have so far by the derivative of the function whose value at x is $\sin x^2$ —which is again a problem we already know how to solve:

$$f'(x) = \cos(\sin x^2) \cdot \cos x^2 \cdot 2x.$$

Finally, here are the derivatives of some other functions which are the composition of $\sin$ and S , as well as some other triple compositions. You can probably just “see” that the answers are correct—if not, try writing out f as a composition:

$f(x) = \sin((\sin x)^2)$ $f(x) = [\sin(\sin x)]^2$ $f(x) = \sin(\sin(\sin x))$ $f(x) = \sin^2(x \sin x)$	$\text{then } f'(x) = \cos((\sin x)^2) \cdot 2 \sin x \cdot \cos x$ $f'(x) = 2 \sin(\sin x) \cdot \cos(\sin x) \cdot \cos x$ $f'(x) = \cos(\sin(\sin x)) \cdot \cos(\sin x) \cdot \cos x$ $f'(x) = 2 \sin(x \sin x) \cdot \cos(x \sin x)$ $\cdot [\sin x + x \cos x]$ $f(x) = \sin(\sin(x^2 \sin x))$ $f'(x) = \cos(\sin(x^2 \sin x))$ $\cdot \cos(x^2 \sin x) \cdot [2x \sin x + x^2 \cos x].$
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The rule for treating compositions of four (or even more) functions is easy—always (mentally) put in parentheses starting from the right,

$$f \circ (g \circ (h \circ k)),$$

and start reducing the calculation to the derivative of a composition of a smaller number of functions:

$$f'(g(h(k(x)))) \cdot \text{[redacted]}.$$

For example, if

$$f(x) = \sin^2(\sin^2(x)) \quad [f = S \circ \sin \circ S \circ \sin \\ = S \circ (\sin \circ (S \circ \sin))]$$

then

$$f'(x) = 2 \sin(\sin^2 x) \cdot \cos(\sin^2 x) \cdot 2 \sin x \cdot \cos x;$$

if

$$f(x) = \sin((\sin x^2)^2) \quad [f = \sin \circ S \circ \sin \circ S \\ = \sin \circ (S \circ (\sin \circ S))]$$

then

$$f'(x) = \cos((\sin x^2)^2) \cdot 2 \sin x^2 \cdot \cos x^2 \cdot 2x;$$

if

$$f(x) = \sin^2(\sin(\sin x)) \quad [\text{fill in yourself, if necessary}]$$

then

$$f'(x) = 2 \sin(\sin(\sin x)) \cdot \cos(\sin(\sin x)) \cdot \cos(\sin x) \cdot \cos x.$$

With these examples as reference, you require only one thing to become a master differentiator—practice. You can be safely turned loose on the exercises at the end of the chapter, and it is now high time that we proved the Chain Rule.

The following argument, while not a proof, indicates some of the tricks one might try, as well as some of the difficulties encountered. We begin, of course, with the definition—

$$\begin{aligned}(f \circ g)'(a) &= \lim_{h \rightarrow 0} \frac{(f \circ g)(a+h) - (f \circ g)(a)}{h} \\ &= \lim_{h \rightarrow 0} \frac{f(g(a+h)) - f(g(a))}{h}.\end{aligned}$$

Somewhere in here we would like the expression for $g'(a)$. One approach is to put it in by fiat:

$$\lim_{h \rightarrow 0} \frac{f(g(a+h)) - f(g(a))}{h} = \lim_{h \rightarrow 0} \frac{f(g(a+h)) - f(g(a))}{g(a+h) - g(a)} \cdot \frac{g(a+h) - g(a)}{h}.$$

This does not look bad, and it looks even better if we write

$$\begin{aligned}\lim_{h \rightarrow 0} \frac{(f \circ g)(a+h) - (f \circ g)(a)}{h} \\ = \lim_{h \rightarrow 0} \frac{f(g(a) + [g(a+h) - g(a)]) - f(g(a))}{g(a+h) - g(a)} \cdot \lim_{h \rightarrow 0} \frac{g(a+h) - g(a)}{h}.\end{aligned}$$

The second limit is the factor $g'(a)$ which we want. If we let $g(a+h) - g(a) = k$ (to be precise we should write $k(h)$), then the first limit is

$$\lim_{h \rightarrow 0} \frac{f(g(a) + k) - f(g(a))}{k}.$$

It looks as if this limit should be $f'(g(a))$, since continuity of g at a implies that k goes to 0 as h does. In fact, one can, and we soon will, make this sort of reasoning precise. There is already a problem, however, which you will have noticed if you are the kind of person who does not divide blindly. Even for $h \neq 0$ we might have $g(a+h) - g(a) = 0$, making the division and multiplication by $g(a+h) - g(a)$ meaningless. True, we only care about small h , but $g(a+h) - g(a)$ could be 0 for arbitrarily small h . The easiest way this can happen is for g to be a constant function, $g(x) = c$. Then $g(a+h) - g(a) = 0$ for all h . In this case, $f \circ g$ is also a constant function, $(f \circ g)(x) = f(c)$, so the Chain Rule does indeed hold:

$$(f \circ g)'(a) = 0 = f'(g(a)) \cdot g'(a).$$

However, there are also nonconstant functions g for which $g(a+h) - g(a) = 0$ for arbitrarily small h . For example, if $a = 0$, the function g might be

$$g(x) = \begin{cases} x^2 \sin \frac{1}{x}, & x \neq 0 \\ 0, & x = 0. \end{cases}$$

In this case, $g'(0) = 0$, as we showed in Chapter 9. If the Chain Rule is correct, we must have $(f \circ g)'(0) = 0$ for any differentiable f , and this is not exactly obvious. A proof of the Chain Rule can be found by considering such recalcitrant functions separately, but it is easier simply to abandon this approach, and use a trick.

THEOREM 9 (THE CHAIN RULE)

If g is differentiable at a , and f is differentiable at $g(a)$, then $f \circ g$ is differentiable at a , and

$$(f \circ g)'(a) = f'(g(a)) \cdot g'(a).$$

PROOF Define a function ϕ as follows.

$$\phi(h) = \begin{cases} \frac{f(g(a+h)) - f(g(a))}{g(a+h) - g(a)}, & \text{if } g(a+h) - g(a) \neq 0 \\ f'(g(a)), & \text{if } g(a+h) - g(a) = 0. \end{cases}$$

It should be intuitively clear that ϕ is continuous at 0: When h is small, $g(a+h) - g(a)$ is also small, so if $g(a+h) - g(a)$ is not zero, then $\phi(h)$ will be close to $f'(g(a))$; and if it is zero, then $\phi(h)$ actually equals $f'(g(a))$, which is even better. Since the continuity of ϕ is the crux of the whole proof we will provide a careful translation of this intuitive argument.

We know that f is differentiable at $g(a)$. This means that

$$\lim_{k \rightarrow 0} \frac{f(g(a) + k) - f(g(a))}{k} = f'(g(a)).$$

Thus, if $\varepsilon > 0$ there is some number $\delta' > 0$ such that, for all k ,

$$(1) \quad \text{if } 0 < |k| < \delta', \text{ then } \left| \frac{f(g(a) + k) - f(g(a))}{k} - f'(g(a)) \right| < \varepsilon.$$

Now g is differentiable at a , hence continuous at a , so there is a $\delta > 0$ such that, for all h ,

$$(2) \quad \text{if } |h| < \delta, \text{ then } |g(a+h) - g(a)| < \delta'.$$

Consider now any h with $|h| < \delta$. If $k = g(a+h) - g(a) \neq 0$, then

$$\phi(h) = \frac{f(g(a+h)) - f(g(a))}{g(a+h) - g(a)} = \frac{f(g(a) + k) - f(g(a))}{k};$$

it follows from (2) that $|k| < \delta'$, and hence from (1) that

$$|\phi(h) - f'(g(a))| < \varepsilon.$$

On the other hand, if $g(a+h) - g(a) = 0$, then $\phi(h) = f'(g(a))$, so it is surely true that

$$|\phi(h) - f'(g(a))| < \varepsilon.$$

We have therefore proved that

$$\lim_{h \rightarrow 0} \phi(h) = f'(g(a)),$$

so ϕ is continuous at 0. The rest of the proof is easy. If $h \neq 0$, then we have

$$\frac{f(g(a+h)) - f(g(a))}{h} = \phi(h) \cdot \frac{g(a+h) - g(a)}{h}$$

even if $g(a+h) - g(a) = 0$ (because in that case both sides are 0). Therefore

$$\begin{aligned}(f \circ g)'(a) &= \lim_{h \rightarrow 0} \frac{f(g(a+h)) - f(g(a))}{h} = \lim_{h \rightarrow 0} \phi(h) \cdot \lim_{h \rightarrow 0} \frac{g(a+h) - g(a)}{h} \\ &= f'(g(a)) \cdot g'(a) \cdot \blacksquare\end{aligned}$$

Now that we can differentiate so many functions so easily we can take another look at the function

$$f(x) = \begin{cases} x^2 \sin \frac{1}{x}, & x \neq 0 \\ 0, & x = 0. \end{cases}$$

In Chapter 9 we showed that $f'(0) = 0$, working straight from the definition (the only possible way). For $x \neq 0$ we can use the methods of this chapter. We have

$$f'(x) = 2x \sin \frac{1}{x} + x^2 \cos \frac{1}{x} \cdot \left(-\frac{1}{x^2}\right);$$

Thus

$$f'(x) = \begin{cases} 2x \sin \frac{1}{x} - \cos \frac{1}{x}, & x \neq 0 \\ 0, & x = 0. \end{cases}$$

As this formula reveals, the first derivative f' is indeed badly behaved at 0—it is not even continuous there. If we consider instead

$$f(x) = \begin{cases} x^3 \sin \frac{1}{x}, & x \neq 0 \\ 0, & x = 0, \end{cases}$$

then

$$f'(x) = \begin{cases} 3x^2 \sin \frac{1}{x} - x \cos \frac{1}{x}, & x \neq 0 \\ 0, & x = 0. \end{cases}$$

In this case f' is continuous at 0, but $f''(0)$ does not exist (because the expression $3x^2 \sin 1/x$ defines a function which is differentiable at 0 but the expression $-x \cos 1/x$ does not).

As you may suspect, increasing the power of x yet again produces another improvement. If

$$f(x) = \begin{cases} x^4 \sin \frac{1}{x}, & x \neq 0 \\ 0, & x = 0, \end{cases}$$

then

$$f'(x) = \begin{cases} 4x^3 \sin \frac{1}{x} - x^2 \cos \frac{1}{x}, & x \neq 0 \\ 0, & x = 0. \end{cases}$$

It is easy to compute, right from the definition, that $(f')'(0) = 0$, and $f''(x)$ is easy to find for $x \neq 0$:

$$f''(x) = \begin{cases} 12x^2 \sin \frac{1}{x} - 4x \cos \frac{1}{x} - 2x \cos \frac{1}{x} - \sin \frac{1}{x}, & x \neq 0 \\ 0, & x = 0. \end{cases}$$

In this case, the *second* derivative f'' is not continuous at 0. By now you may have guessed the pattern, which two of the problems asks you to establish: if

$$f(x) = \begin{cases} x^{2n} \sin \frac{1}{x}, & x \neq 0 \\ 0, & x = 0, \end{cases}$$

then $f'(0), \dots, f^{(n)}(0)$ exist, but $f^{(n)}$ is not continuous at 0; if

$$f(x) = \begin{cases} x^{2n+1} \sin \frac{1}{x}, & x \neq 0 \\ 0, & x = 0, \end{cases}$$

then $f'(0), \dots, f^{(n)}(0)$ exist, and $f^{(n)}$ is continuous at 0, but $f^{(n)}$ is not differentiable at 0. These examples may suggest that “reasonable” functions can be characterized by the possession of higher-order derivatives—no matter how hard we try to mask the infinite oscillation of $f(x) = \sin 1/x$, a derivative of sufficiently high order seems able to reveal the underlying irregularity. Unfortunately, we will see later that much worse things can happen.

After all these involved calculations, we will bring this chapter to a close with a minor remark. It is often tempting, and seems more elegant, to write some of the theorems in this chapter as equations about functions, rather than about their values. Thus Theorem 3 might be written

$$(f + g)' = f' + g',$$

Theorem 4 might be written as

$$(f \cdot g)' = f \cdot g' + f' \cdot g,$$

and Theorem 9 often appears in the form

$$(f \circ g)' = (f' \circ g) \cdot g'.$$

Strictly speaking, these equations may be false, because the functions on the left-hand side might have a larger domain than those on the right. Nevertheless, this is hardly worth worrying about. If f and g are differentiable everywhere in their domains, then these equations, and others like them, *are* true, and this is the only case any one cares about.

PROBLEMS

1. As a warm up exercise, find $f'(x)$ for each of the following f . (Don't worry about the domain of f or f' ; just get a formula for $f'(x)$ that gives the right answer when it makes sense.)

(i) $f(x) = \sin(x + x^2)$.

(ii) $f(x) = \sin x + \sin x^2$.

(iii) $f(x) = \sin(\cos x)$.

(iv) $f(x) = \sin(\sin x)$.

(v) $f(x) = \sin\left(\frac{\cos x}{x}\right)$.

(vi) $f(x) = \frac{\sin(\cos x)}{x}$.

(vii) $f(x) = \sin(x + \sin x)$.

(viii) $f(x) = \sin(\cos(\sin x))$.

2. Find $f'(x)$ for each of the following functions f . (It took the author 20 minutes to compute the derivatives for the answer section, and it should not take you much longer. Although rapid calculation is not the goal of mathematics, if you hope to treat theoretical applications of the Chain Rule with aplomb, these concrete applications should be child's play—mathematicians like to pretend that they can't even add, but most of them can when they have to.)

(i) $f(x) = \sin((x + 1)^2(x + 2))$.

(ii) $f(x) = \sin^3(x^2 + \sin x)$.

(iii) $f(x) = \sin^2((x + \sin x)^2)$.

(iv) $f(x) = \sin\left(\frac{x^3}{\cos x^3}\right)$.

(v) $f(x) = \sin(x \sin x) + \sin(\sin x^2)$.

(vi) $f(x) = (\cos x)^{31}$.

(vii) $f(x) = \sin^2 x \sin x^2 \sin^2 x^2$.

(viii) $f(x) = \sin^3(\sin^2(\sin x))$.

(ix) $f(x) = (x + \sin^5 x)^6$.

(x) $f(x) = \sin(\sin(\sin(\sin(\sin x))))$.

(xi) $f(x) = \sin((\sin^7 x^7 + 1)^7)$.

(xii) $f(x) = (((x^2 + x)^3 + x)^4 + x)^5$.

(xiii) $f(x) = \sin(x^2 + \sin(x^2 + \sin x^2))$.

(xiv) $f(x) = \sin(6 \cos(6 \sin(6 \cos 6x)))$.

(xv) $f(x) = \frac{\sin x^2 \sin^2 x}{1 + \sin x}$.

(xvi) $f(x) = \frac{1}{x - \frac{2}{x + \sin x}}$.

$$(xvii) \quad f(x) = \sin \left(\frac{x^3}{\sin \left(\frac{x^3}{\sin x} \right)} \right).$$

$$(xviii) \quad f(x) = \sin \left(\frac{x}{x - \sin \left(\frac{x}{x - \sin x} \right)} \right).$$

3. Find the derivatives of the functions $\tan$, $\cotan$, $\sec$, and cosec . (You don't have to memorize these formulas, although they will be needed once in a while; if you express your answers in the right way, they will be simple and somewhat symmetrical.)
4. For each of the following functions f , find $f'(f(x))$ (*not* $(f \circ f)'(x)$).

$$(i) \quad f(x) = \frac{1}{1+x}.$$

$$(ii) \quad f(x) = \sin x.$$

$$(iii) \quad f(x) = x^2.$$

$$(iv) \quad f(x) = 17.$$

5. For each of the following functions f , find $f(f'(x))$.

$$(i) \quad f(x) = \frac{1}{x}.$$

$$(ii) \quad f(x) = x^2.$$

$$(iii) \quad f(x) = 17.$$

$$(iv) \quad f(x) = 17x.$$

6. Find f' in terms of g' if

$$(i) \quad f(x) = g(x + g(a)).$$

$$(ii) \quad f(x) = g(x \cdot g(a)).$$

$$(iii) \quad f(x) = g(x + g(x)).$$

$$(iv) \quad f(x) = g(x)(x - a).$$

$$(v) \quad f(x) = g(a)(x - a).$$

$$(vi) \quad f(x + 3) = g(x^2).$$

7. (a) A circular object is increasing in size in some unspecified manner, but it is known that when the radius is 6, the rate of change of the radius is 4. Find the rate of change of the area when the radius is 6. (If $r(t)$ and $A(t)$ represent the radius and the area at time t , then the functions r and A satisfy $A = \pi r^2$; a straightforward use of the Chain Rule is called for.)
- (b) Suppose that we are now informed that the circular object we have been watching is really the cross section of a spherical object. Find the rate of change of the *volume* when the radius is 6. (You will clearly need to know a formula for the volume of a sphere; in case you have forgotten, the volume is $\frac{4}{3}\pi$ times the cube of the radius.)

- (c) Now suppose that the rate of change of the area of the circular cross section is 5 when the radius is 3. Find the rate of change of the volume when the radius is 3. You should be able to do this problem in two ways: first, by using the formulas for the area and volume in terms of the radius; and then by expressing the volume in terms of the area (to use this method you will need Problem 9-3).
8. Let $f(x) = x^2 \sin 1/x$ for $x \neq 0$, and let $f(0) = 0$. Suppose also that h and k are two functions such that

$$\begin{aligned} h'(x) &= \sin^2(\sin(x+1)) & k'(x) &= f(x+1) \\ h(0) &= 3 & k(0) &= 0. \end{aligned}$$

Find

- (i) $(f \circ h)'(0)$.
 - (ii) $(k \circ f)'(0)$.
 - (iii) $\alpha'(x^2)$, where $\alpha(x) = h(x^2)$. Exercise great care.
9. Find $f'(0)$ if

$$f(x) = \begin{cases} g(x) \sin \frac{1}{x}, & x \neq 0 \\ 0, & x = 0, \end{cases}$$

and

$$g(0) = g'(0) = 0.$$

10. Using the derivative of $f(x) = 1/x$, as found in Problem 9-1, find $(1/g)'(x)$ by the Chain Rule.
11. (a) Using Problem 9-3, find $f'(x)$ for $-1 < x < 1$, if $f(x) = \sqrt{1-x^2}$.
(b) Prove that the tangent line to the graph of f at $(a, \sqrt{1-a^2})$ intersects the graph only at that point (and thus show that the elementary geometry definition of the tangent line coincides with ours).
12. Prove similarly that the tangent lines to an ellipse or hyperbola intersect these sets only once.
13. If $f + g$ is differentiable at a , are f and g necessarily differentiable at a ? If $f \cdot g$ and f are differentiable at a , what conditions on f imply that g is differentiable at a ?
14. (a) Prove that if f is differentiable at a , then $|f|$ is also differentiable at a , provided that $f(a) \neq 0$.
(b) Give a counterexample if $f(a) = 0$.
(c) Prove that if f and g are differentiable at a , then the functions $\max(f, g)$ and $\min(f, g)$ are differentiable at a , provided that $f(a) \neq g(a)$.
(d) Give a counterexample if $f(a) = g(a)$.
15. Suppose that $f^{(n)}(a)$ and $g^{(n)}(a)$ exist. Prove *Leibniz's formula*:

$$(f \cdot g)^{(n)}(a) = \sum_{k=0}^n \binom{n}{k} f^{(k)}(a) \cdot g^{(n-k)}(a).$$

- *16. Prove that if $f^{(n)}(g(a))$ and $g^{(n)}(a)$ both exist, then $(f \circ g)^{(n)}(a)$ exists. A little experimentation should convince you that it is unwise to seek a formula for $(f \circ g)^{(n)}(a)$. In order to prove that $(f \circ g)^{(n)}(a)$ exists you will therefore have to devise a reasonable assertion about $(f \circ g)^{(n)}(a)$ which can be proved by induction. Try something like: " $(f \circ g)^{(n)}(a)$ exists and is a sum of terms each of which is a product of terms of the form"

17. (a) If $f(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_0$, find a function g such that $g' = f$. Find another.
(b) If

$$f(x) = \frac{b_2}{x^2} + \frac{b_3}{x^3} + \cdots + \frac{b_m}{x^m},$$

find a function g with $g' = f$.

- (c) Is there a function

$$f(x) = a_n x^n + \cdots + a_0 + \frac{b_1}{x} + \cdots + \frac{b_m}{x^m}$$

such that $f'(x) = 1/x$?

18. Show that there is a polynomial function f of degree n such that

- (a) $f'(x) = 0$ for precisely $n - 1$ numbers x .
(b) $f'(x) = 0$ for no x , if n is odd.
(c) $f'(x) = 0$ for exactly one x , if n is even.
(d) $f'(x) = 0$ for exactly k numbers x , if $n - k$ is odd.

19. (a) The number a is called a **double root** of the polynomial function f if $f(x) = (x - a)^2 g(x)$ for some polynomial function g . Prove that a is a double root of f if and only if a is a root of both f and f' .
(b) When does $f(x) = ax^2 + bx + c$ ($a \neq 0$) have a double root? What does the condition say geometrically?

20. If f is differentiable at a , let $d(x) = f(x) - f'(a)(x - a) - f(a)$. Find $d''(a)$. In connection with Problem 19, this gives another solution for Problem 9-20.

- *21. This problem is a companion to Problem 3-6. Let $a_1, \dots, a_n$ and $b_1, \dots, b_n$ be given numbers.

- (a) If $x_1, \dots, x_n$ are distinct numbers, prove that there is a polynomial function f of degree $2n - 1$, such that $f(x_j) = f'(x_j) = 0$ for $j \neq i$, and $f(x_i) = a_i$ and $f'(x_i) = b_i$. Hint: Remember Problem 19.
(b) Prove that there is a polynomial function f of degree $2n - 1$ with $f(x_i) = a_i$ and $f'(x_i) = b_i$ for all i .

- *22. Suppose that a and b are two consecutive roots of a polynomial function f , but that a and b are not double roots, so that we can write $f(x) = (x - a)(x - b)g(x)$ where $g(a) \neq 0$ and $g(b) \neq 0$.

- (a) Prove that $g(a)$ and $g(b)$ have the same sign. (Remember that a and b are consecutive roots.)
- (b) Prove that there is some number x with $a < x < b$ and $f'(x) = 0$. (Also draw a picture to illustrate this fact.) Hint: Compare the sign of $f'(a)$ and $f'(b)$.
- (c) Now prove the same fact, even if a and b are multiple roots. Hint: If $f(a) = (x - a)^m(x - b)^n g(x)$ where $g(a) \neq 0$ and $g(b) \neq 0$, consider the polynomial function $h(x) = f'(x)/(x - a)^{m-1}(x - b)^{n-1}$.

This theorem was proved by the French mathematician Rolle, in connection with the problem of approximating roots of polynomials, but the result was not originally stated in terms of derivatives. In fact, Rolle was one of the mathematicians who never accepted the new notions of calculus. This was not such a pigheaded attitude, in view of the fact that for one hundred years no one could define limits in terms that did not verge on the mystic, but on the whole history has been particularly kind to Rolle; his name has become attached to a much more general result, to appear in the next chapter, which forms the basis for the most important theoretical results of calculus.

- 23. Suppose that $f(x) = xg(x)$ for some function g which is continuous at 0. Prove that f is differentiable at 0, and find $f'(0)$ in terms of g .
- *24. Suppose f is differentiable at 0, and that $f(0) = 0$. Prove that $f(x) = xg(x)$ for some function g which is continuous at 0. Hint: What happens if you try to write $g(x) = f(x)/x$?
- 25. If $f(x) = x^{-n}$ for n in $\mathbf{N}$, prove that

$$\begin{aligned} f^{(k)}(x) &= (-1)^k \frac{(n+k-1)!}{(k-1)!} x^{-n-k} \\ &= (-1)^k n! \binom{n+k-1}{k-1} x^{-n-k}, \quad \text{for } x \neq 0. \end{aligned}$$

- *26. Prove that it is impossible to write $x = f(x)g(x)$ where f and g are differentiable and $f(0) = g(0) = 0$. Hint: Differentiate.
- 27. What is $f^{(k)}(x)$ if
 - (a) $f(x) = 1/(x-a)^n$?
 - * (b) $f(x) = 1/(x^2-1)$?

- *28. Let $f(x) = x^{2n} \sin 1/x$ if $x \neq 0$, and let $f(0) = 0$. Prove that $f'(0), \dots, f^{(n)}(0)$ exist, and that $f^{(n)}$ is not continuous at 0. (You will encounter the same basic difficulty as that in Problem 15.)
- *29. Let $f(x) = x^{2n+1} \sin 1/x$ if $x \neq 0$, and let $f(0) = 0$. Prove that $f'(0), \dots, f^{(n)}(0)$ exist, that $f^{(n)}$ is continuous at 0, and that $f^{(n)}$ is not differentiable at 0.

30. In Leibnizian notation the Chain Rule ought to read:

$$\frac{df(g(x))}{dx} = \frac{df(y)}{dy} \bigg|_{y=g(x)} \cdot \frac{dg(x)}{dx}.$$

Instead, one usually finds the following statement: "Let $y = g(x)$ and $z = f(y)$. Then

$$\frac{dz}{dx} = \frac{dz}{dy} \cdot \frac{dy}{dx}."$$

Notice that the z in dz/dx denotes the composite function $f \circ g$, while the z in dz/dy denotes the function f ; it is also understood that dz/dy will be "an expression involving y ," and that in the final answer $g(x)$ must be substituted for y . In each of the following cases, find dz/dx by using this formula; then compare with Problem 1.

- (i) $z = \sin y, \quad y = x + x^2.$
- (ii) $z = \sin y, \quad y = \cos x.$
- (iii) $z = \cos u, \quad u = \sin x.$
- (iv) $z = \sin v, \quad v = \cos u, \quad u = \sin x.$

One aim in this chapter is to justify the time we have spent learning to find the derivative of a function. As we shall see, knowing just a little about f' tells us a lot about f . Extracting information about f from information about f' requires some difficult work, however, and we shall begin with the one theorem which is really easy.

This theorem is concerned with the maximum value of a function on an interval. Although we have used this term informally in Chapter 7, it is worthwhile to be precise, and also more general.

DEFINITION

Let f be a function and A a set of numbers contained in the domain of f . A point x in A is a **maximum point** for f on A , if

$$f(x) \geq f(y) \quad \text{for every } y \text{ in } A.$$

The number $f(x)$ itself is called the **maximum value** of f on A (and we also say that f “has its maximum value on A at x ”).

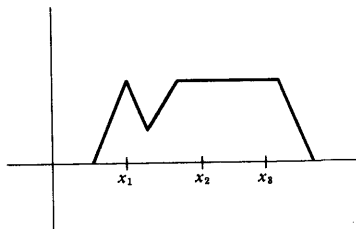


FIGURE 1

Notice that the maximum value of f on A could be $f(x)$ for several different x (Figure 1); in other words, a function f can have several different maximum points on A , although it can have at most one maximum value. Usually we shall be interested in the case where A is a closed interval $[a, b]$; if f is continuous, then Theorem 7-3 guarantees that f does indeed have a maximum value on $[a, b]$.

The definition of a minimum of f on A will be left to you. (One possible definition is the following: f has a minimum on A at x , if $-f$ has a maximum on A at x .)

We are now ready for a theorem which does not even depend upon the existence of least upper bounds.

THEOREM 1

Let f be any function defined on (a, b) . If x is a maximum (or a minimum) point for f on (a, b) , and f is differentiable at x , then $f'(x) = 0$. (Notice that we do not assume differentiability, or even continuity, of f at other points.)

PROOF

Consider the case where f has a maximum at x . (Figure 2 illustrates the simple idea behind the whole argument—secants drawn through points to the left of $(x, f(x))$ have slopes ≥ 0 , and secants drawn through points to the right of $(x, f(x))$ have slopes ≤ 0 .) Analytically, this argument proceeds as follows.

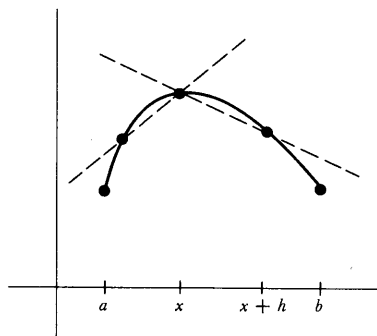


FIGURE 2

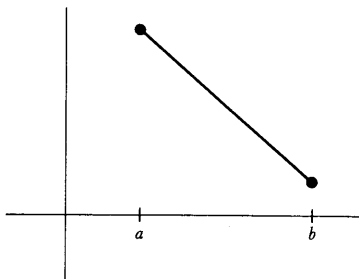


FIGURE 3

If h is any number such that $x + h$ is in (a, b) , then

$$f(x) \geq f(x + h),$$

since f has a maximum on (a, b) at x . This means that

$$f(x + h) - f(x) \leq 0.$$

Thus, if $h > 0$ we have

$$\frac{f(x + h) - f(x)}{h} \leq 0,$$

and consequently

$$\lim_{h \rightarrow 0^+} \frac{f(x + h) - f(x)}{h} \leq 0.$$

On the other hand, if $h < 0$, we have

$$\frac{f(x + h) - f(x)}{h} \geq 0,$$

so

$$\lim_{h \rightarrow 0^-} \frac{f(x + h) - f(x)}{h} \geq 0.$$

By hypothesis, f is differentiable at x , so these two limits must be equal, in fact equal to $f'(x)$. This means that

$$f'(x) \leq 0 \quad \text{and} \quad f'(x) \geq 0,$$

from which it follows that $f'(x) = 0$.

The case where f has a minimum at x is left to you (give a one-line proof). ■

Notice (Figure 3) that we cannot replace (a, b) by $[a, b]$ in the statement of the theorem (unless we add to the hypothesis the condition that x is in (a, b)).

Since $f'(x)$ depends only on the values of f near x , it is almost obvious how to get a stronger version of Theorem 1. We begin with a definition which is illustrated in Figure 4.

DEFINITION

Let f be a function, and A a set of numbers contained in the domain of f . A point x in A is a **local maximum [minimum] point** for f on A if there is some $\delta > 0$ such that x is a maximum [minimum] point for f on $A \cap (x - \delta, x + \delta)$.

THEOREM 2

If f is defined on (a, b) and has a local maximum (or minimum) at x , and f is differentiable at x , then $f'(x) = 0$.

PROOF

You should see why this is an easy application of Theorem 1. ■

The converse of Theorem 2 is definitely not true—it is possible for $f'(x)$ to be 0 even if x is not a local maximum or minimum point for f . The simplest example is provided by the function $f(x) = x^3$; in this case $f'(0) = 0$, but f has no local maximum or minimum anywhere.

Probably the most widespread misconceptions about calculus are concerned with the behavior of a function f near x when $f'(x) = 0$. The point made in the previous paragraph is so quickly forgotten by those who want the world to be simpler than it is, that we will repeat it: the converse of Theorem 2 is *not* true—the condition $f'(x) = 0$ does *not* imply that x is a local maximum or minimum point of f . Precisely for this reason, special terminology has been adopted to describe numbers x which satisfy the condition $f'(x) = 0$.

DEFINITION

A **critical point** of a function f is a number x such that

$$f'(x) = 0.$$

The number $f(x)$ itself is called a **critical value** of f .

The critical values of f , together with a few other numbers, turn out to be the ones which must be considered in order to find the maximum and minimum of a given function f . To the uninitiated, finding the maximum and minimum value of a function represents one of the most intriguing aspects of calculus, and there is no denying that problems of this sort are fun (until you have done your first hundred or so).

Let us consider first the problem of finding the maximum or minimum of f on a closed interval $[a, b]$. (Then, if f is continuous, we can at least be sure that a maximum and minimum value exist.) In order to locate the maximum and minimum of f three kinds of points must be considered:

- (1) The critical points of f in $[a, b]$.
- (2) The end points a and b .
- (3) Points x in $[a, b]$ such that f is not differentiable at x .

If x is a maximum point or a minimum point for f on $[a, b]$, then x must be in one of the three classes listed above: for if x is not in the second or third group, then x is in (a, b) and f is differentiable at x ; consequently $f'(x) = 0$, by Theorem 1, and this means that x is in the first group.

If there are many points in these three categories, finding the maximum and minimum of f may still be a hopeless proposition, but when there are only a few critical points, and only a few points where f is not differentiable, the procedure is fairly straightforward: one simply finds $f(x)$ for each x satisfying $f'(x) = 0$, and $f(x)$ for each x such that f is not differentiable at x and, finally, $f(a)$ and $f(b)$. The biggest of these will be the maximum value of f , and the smallest will be the minimum. A simple example follows.

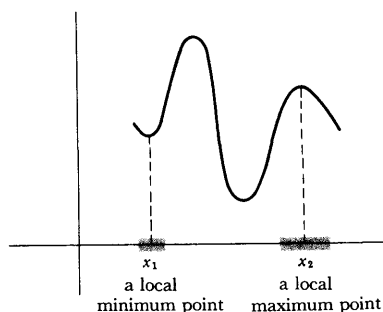


FIGURE 4

Suppose we wish to find the maximum and minimum value of the function

$$f(x) = x^3 - x$$

on the interval $[-1, 2]$. To begin with, we have

$$f'(x) = 3x^2 - 1,$$

so $f'(x) = 0$ when $3x^2 - 1 = 0$, that is, when

$$x = \sqrt{1/3} \quad \text{or} \quad -\sqrt{1/3}.$$

The numbers $\sqrt{1/3}$ and $-\sqrt{1/3}$ both lie in $[-1, 2]$, so the first group of candidates for the location of the maximum and the minimum is

$$(1) \quad \sqrt{1/3}, -\sqrt{1/3}.$$

The second group contains the end points of the interval

$$(2) \quad -1, 2.$$

The third group is empty, since f is differentiable everywhere. The final step is to compute

$$\begin{aligned} f(\sqrt{1/3}) &= (\sqrt{1/3})^3 - \sqrt{1/3} = \frac{1}{3}\sqrt{1/3} - \sqrt{1/3} = -\frac{2}{3}\sqrt{1/3}, \\ f(-\sqrt{1/3}) &= (-\sqrt{1/3})^3 - (-\sqrt{1/3}) = -\frac{1}{3}\sqrt{1/3} + \sqrt{1/3} = \frac{2}{3}\sqrt{1/3}, \\ f(-1) &= 0, \\ f(2) &= 6. \end{aligned}$$

Clearly the minimum value is $-\frac{2}{3}\sqrt{1/3}$, occurring at $\sqrt{1/3}$, and the maximum value is 6, occurring at 2.

This sort of procedure, if feasible, will always locate the maximum and minimum value of a continuous function on a closed interval. If the function we are dealing with is not continuous, however, or if we are seeking the maximum or minimum on an open interval or the whole line, then we cannot even be sure beforehand that the maximum and minimum values exist, so all the information obtained by this procedure may say nothing. Nevertheless, a little ingenuity will often reveal the nature of things. In Chapter 7 we solved just such a problem when we showed that if n is even, then the function

$$f(x) = x^n + a_{n-1}x^{n-1} + \cdots + a_0$$

has a minimum value on the whole line. This proves that the minimum value must occur at some number x satisfying

$$0 = f'(x) = nx^{n-1} + (n-1)a_{n-1}x^{n-2} + \cdots + a_1.$$

If we can solve this equation, and compare the values of $f(x)$ for such x , we can actually find the minimum of f . One more example may be helpful. Suppose we wish to find the maximum and minimum, if they exist, of the function

$$f(x) = \frac{1}{1-x^2}$$

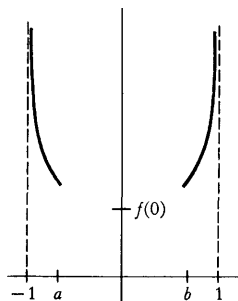


FIGURE 5

on the open interval $(-1, 1)$. We have

$$f'(x) = \frac{2x}{(1-x^2)^2},$$

so $f'(x) = 0$ only for $x = 0$. We can see immediately that for x close to 1 or -1 the values of $f(x)$ become arbitrarily large, so f certainly does not have a maximum. This observation also makes it easy to show that f has a minimum at 0. We just note (Figure 5) that there will be numbers a and b , with

$$-1 < a < 0 \quad \text{and} \quad 0 < b < 1,$$

such that $f(x) > f(0)$ for

$$-1 < x \leq a \quad \text{and} \quad b \leq x < 1.$$

This means that the minimum of f on $[a, b]$ is the minimum of f on all of $(-1, 1)$. Now on $[a, b]$ the minimum occurs either at 0 (the only place where $f' = 0$), or at a or b , and a and b have already been ruled out, so the minimum value is $f(0) = 1$.

In solving these problems we purposely did not draw the graphs of $f(x) = x^3 - x$ and $f(x) = 1/(1-x^2)$, but it is not cheating to draw the graph (Figure 6) as long as you do not rely solely on your picture to prove anything. As a matter of fact, we are now going to discuss a method of sketching the graph of a function that really gives enough information to be used in discussing maxima and minima—in fact we will be able to locate even *local* maxima and minima. This method involves consideration of the sign of $f'(x)$, and relies on some deep theorems.

The theorems about derivatives which have been proved so far, always yield information about f' in terms of information about f . This is true even of Theorem 1, although this theorem can sometimes be used to determine certain information about f , namely, the location of maxima and minima. When the derivative was first introduced, we emphasized that $f'(x)$ is not $[f(x+h) - f(x)]/h$ for any particular h , but only a limit of these numbers as h approaches 0; this fact becomes painfully relevant when one tries to extract information about f from information about f' . The simplest and most frustrating illustration of the difficulties encountered is afforded by the following question: If $f'(x) = 0$ for all x , must f be a constant function? It is impossible to imagine how f could be anything else, and this conviction is strengthened by considering the physical interpretation—if the velocity of a particle is always 0, surely the particle must be standing still! Nevertheless it is difficult even to begin a proof that only the constant functions satisfy $f'(x) = 0$ for all x . The hypothesis $f'(x) = 0$ only means that

$$\lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = 0,$$

and it is not at all obvious how one can use the information about the limit to derive information about the function.

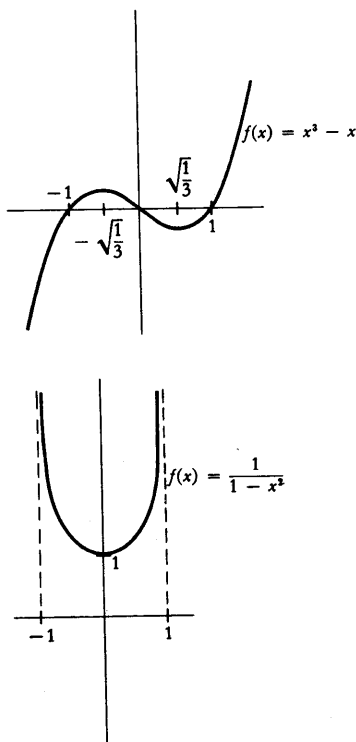


FIGURE 6

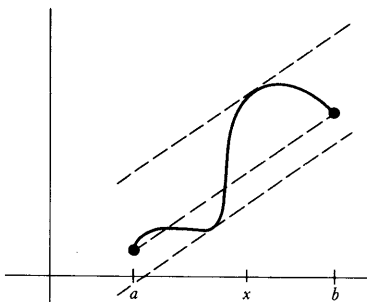


FIGURE 7

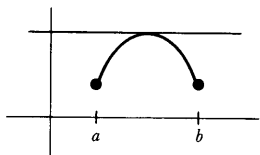


FIGURE 8

THEOREM 3 (ROLLE'S THEOREM)

If f is continuous on $[a, b]$ and differentiable on (a, b) , and $f(a) = f(b)$, then there is a number x in (a, b) such that $f'(x) = 0$.

PROOF

It follows from the continuity of f on $[a, b]$ that f has a maximum and a minimum value on $[a, b]$.

Suppose first that the maximum value occurs at a point x in (a, b) . Then $f'(x) = 0$ by Theorem 1, and we are done (Figure 8).

Suppose next that the minimum value of f occurs at some point x in (a, b) . Then, again, $f'(x) = 0$ by Theorem 1 (Figure 9).

Finally, suppose the maximum and minimum values both occur at the end points. Since $f(a) = f(b)$, the maximum and minimum values of f are equal, so f is a constant function (Figure 10), and for a constant function we can choose any x in (a, b) . ■

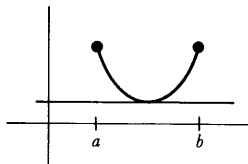


FIGURE 9

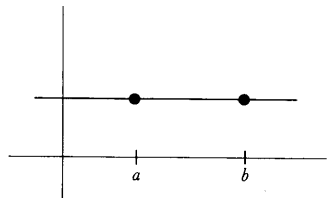


FIGURE 10

The fact that f is a constant function if $f'(x) = 0$ for all x , and many other facts of the same sort, can all be derived from a fundamental theorem, called the Mean Value Theorem, which states much stronger results. Figure 7 makes it plausible that if f is differentiable on $[a, b]$, then there is some x in (a, b) such that

$$f'(x) = \frac{f(b) - f(a)}{b - a}.$$

Geometrically this means that some tangent line is parallel to the line between $(a, f(a))$ and $(b, f(b))$. The Mean Value Theorem asserts that this is true—there is some x in (a, b) such that $f'(x)$, the instantaneous rate of change of f at x , is exactly equal to the average or “mean” change of f on $[a, b]$, this average change being $[f(b) - f(a)]/[b - a]$. (For example, if you travel 60 miles in one hour, then at some time you must have been traveling exactly 60 miles per hour.) This theorem is one of the most important theoretical tools of calculus—probably the deepest result about derivatives. From this statement you might conclude that the proof is difficult, but there you would be wrong—the hard theorems in this book have occurred long ago, in Chapter 7. It is true that if you try to prove the Mean Value Theorem yourself you will probably fail, but this is neither evidence that the theorem is hard, nor something to be ashamed of. The first proof of the theorem was an achievement, but today we can supply a proof which is quite simple. It helps to begin with a very special case.

Notice that we really needed the hypothesis that f is differentiable everywhere on (a, b) in order to apply Theorem 1. Without this assumption the theorem is false (Figure 11).

You may wonder why a special name should be attached to a theorem as easily proved as Rolle's Theorem. The reason is, that although Rolle's Theorem is a special case of the Mean Value Theorem, it also yields a simple proof of the Mean Value Theorem. In order to prove the Mean Value

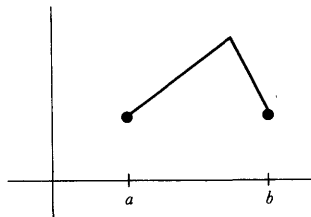


FIGURE 11

THEOREM 4 (THE MEAN VALUE THEOREM)

Theorem we will apply Rolle's Theorem to the function which gives the length of the vertical segment shown in Figure 12; this is the difference between $f(x)$, and the height at x of the line L between $(a, f(a))$ and $(b, f(b))$. Since L is the graph of

$$g(x) = \left[\frac{f(b) - f(a)}{b - a} \right] (x - a) + f(a),$$

we want to look at

$$f(x) - \left[\frac{f(b) - f(a)}{b - a} \right] (x - a) - f(a).$$

As it turns out, the constant $f(a)$ is irrelevant.

If f is continuous on $[a, b]$ and differentiable on (a, b) , then there is a number x in (a, b) such that

$$f'(x) = \frac{f(b) - f(a)}{b - a}.$$

PROOF Let

$$h(x) = f(x) - \left[\frac{f(b) - f(a)}{b - a} \right] (x - a).$$

Clearly, h is continuous on $[a, b]$ and differentiable on (a, b) , and

$$\begin{aligned} h(a) &= f(a), \\ h(b) &= f(b) - \left[\frac{f(b) - f(a)}{b - a} \right] (b - a) \\ &= f(a). \end{aligned}$$

Consequently, we may apply Rolle's Theorem to h and conclude that there is some x in (a, b) such that

$$0 = h'(x) = f'(x) - \frac{f(b) - f(a)}{b - a},$$

so that

$$f'(x) = \frac{f(b) - f(a)}{b - a}. \blacksquare$$

Notice that the Mean Value Theorem still fits into the pattern exhibited by previous theorems—information about f yields information about f' . This information is so strong, however, that we can now go in the other direction.

COROLLARY 1 If f is defined on an interval and $f'(x) = 0$ for all x in the interval, then f is constant on the interval.

PROOF Let a and b be any two points in the interval with $a \neq b$. Then there is some x

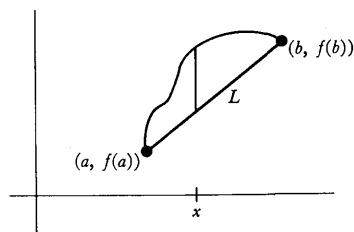


FIGURE 12

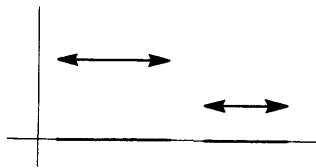


FIGURE 13

in (a, b) such that

$$f'(x) = \frac{f(b) - f(a)}{b - a}.$$

But $f'(x) = 0$ for all x in the interval, so

$$0 = \frac{f(b) - f(a)}{b - a},$$

and consequently $f(a) = f(b)$. Thus the value of f at any two points in the interval is the same, i.e., f is constant on the interval. ■

Naturally, Corollary 1 does not hold for functions defined on two or more intervals (Figure 13).

COROLLARY 2 If f and g are defined on the same interval, and $f'(x) = g'(x)$ for all x in the interval, then there is some number c such that $f = g + c$.

PROOF For all x in the interval we have $(f - g)'(x) = f'(x) - g'(x) = 0$ so, by Corollary 1, there is a number c such that $f - g = c$. ■

The statement of the next corollary requires some terminology, which is illustrated in Figure 14.

DEFINITION

A function f is **increasing** on an interval if $f(a) < f(b)$ whenever a and b are two numbers in the interval with $a < b$. The function f is **decreasing** on an interval if $f(a) > f(b)$ for all a and b in the interval with $a < b$. (We often say simply that f is increasing or decreasing, in which case the interval is understood to be the domain of f .)

COROLLARY 3 If $f'(x) > 0$ for all x in an interval, then f is increasing on the interval; if $f'(x) < 0$ for all x in the interval, then f is decreasing on the interval.

PROOF Consider the case where $f'(x) > 0$. Let a and b be two points in the interval with $a < b$. Then there is some x in (a, b) with

$$f'(x) = \frac{f(b) - f(a)}{b - a}.$$

But $f'(x) > 0$ for all x in (a, b) , so

$$\frac{f(b) - f(a)}{b - a} > 0.$$

Since $b - a > 0$ it follows that $f(b) > f(a)$.

The proof when $f'(x) < 0$ for all x is left to you. ■

Notice that although the converses of Corollary 1 and Corollary 2 are true (and obvious), the converse of Corollary 3 is not true. If f is increasing, it is easy to see that $f'(x) \geq 0$ for all x , but the equality sign might hold for some x (consider $f(x) = x^3$).

Corollary 3 provides enough information to get a good idea of the graph of a function with a minimal amount of point plotting. Consider, once more, the function $f(x) = x^3 - x$. We have

$$f'(x) = 3x^2 - 1.$$

We have already noted that $f'(x) = 0$ for $x = \sqrt{1/3}$ and $x = -\sqrt{1/3}$, and it is also possible to determine the sign of $f'(x)$ for all other x . Note that $3x^2 - 1 > 0$ precisely when

$$\begin{aligned} 3x^2 &> 1, \\ x^2 &> \frac{1}{3}, \end{aligned}$$

$$x > \sqrt{1/3} \quad \text{or} \quad x < -\sqrt{1/3};$$

thus $3x^2 - 1 < 0$ precisely when

$$-\sqrt{1/3} < x < \sqrt{1/3}.$$

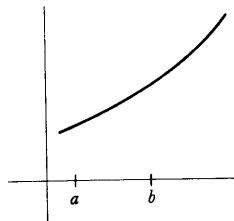
Thus f is increasing for $x < -\sqrt{1/3}$, decreasing between $-\sqrt{1/3}$ and $\sqrt{1/3}$, and once again increasing for $x > \sqrt{1/3}$. Combining this information with the following facts

- (1) $f(-\sqrt{1/3}) = \frac{2}{3}\sqrt{1/3}$,
 $f(\sqrt{1/3}) = -\frac{2}{3}\sqrt{1/3}$,
- (2) $f(x) = 0$ for $x = -1, 0, 1$,
- (3) $f(x)$ gets large as x gets large, and large negative as x gets large negative,

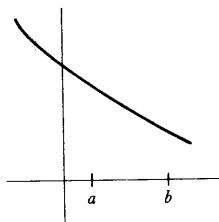
it is possible to sketch a pretty respectable approximation to the graph (Figure 15).

By the way, notice that the intervals on which f increases and decreases could have been found without even bothering to examine the sign of f' . For example, since f' is continuous, and vanishes only at $-\sqrt{1/3}$ and $\sqrt{1/3}$, we know that f' always has the same sign on the interval $(-\sqrt{1/3}, \sqrt{1/3})$. Since $f(-\sqrt{1/3}) > f(\sqrt{1/3})$, it follows that f decreases on this interval. Similarly, f' always has the same sign on $(\sqrt{1/3}, \infty)$ and $f(x)$ is large for large x , so f must be increasing on $(\sqrt{1/3}, \infty)$. Another point worth noting: If f' is continuous, then the sign of f' on the interval between two adjacent critical points can be determined simply by finding the sign of $f'(x)$ for any *one* x in this interval.

Our sketch of the graph of $f(x) = x^3 - x$ contains sufficient information to allow us to say with confidence that $-\sqrt{1/3}$ is a local maximum point, and $\sqrt{1/3}$ a local minimum point. In fact, we can give a general scheme for



(a) an increasing function



(b) a decreasing function

FIGURE 14

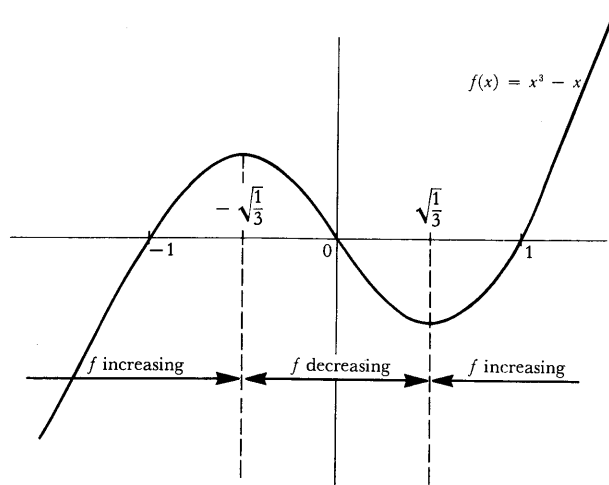


FIGURE 15

deciding whether a critical point is a local maximum point, a local minimum point, or neither (Figure 16):

- (1) if $f' > 0$ in some interval to the left of x and $f' < 0$ in some interval to the right of x , then x is a local maximum point.
- (2) if $f' < 0$ in some interval to the left of x and $f' > 0$ in some interval to the right of x , then x is a local minimum point.
- (3) if f' has the same sign in some interval to the left of x as it has in some interval to the right, then x is neither a local maximum nor a local minimum point.

(There is no point in memorizing these rules—you can always draw the pictures yourself.)

The polynomial functions can all be analyzed in this way, and it is even possible to describe the general form of the graph of such functions. To begin,

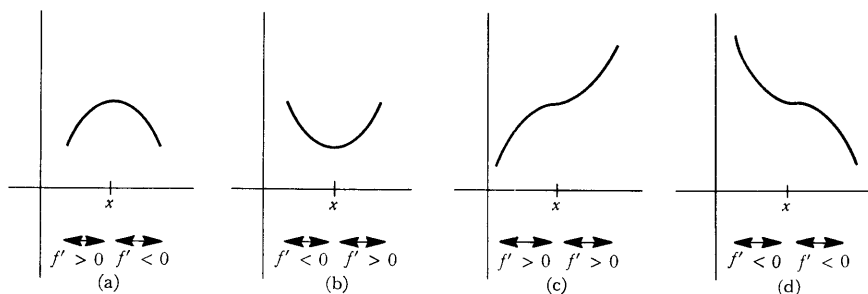


FIGURE 16

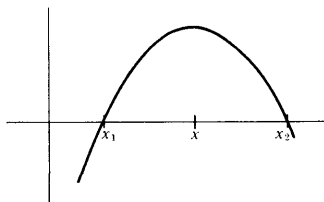


FIGURE 17

we need a result already mentioned in Problem 3-7: If

$$f(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_0,$$

then f has at most n "roots," i.e., there are at most n numbers x such that $f(x) = 0$. Although this is really an algebraic theorem, calculus can be used to give an easy proof. Notice that if x_1 and x_2 are roots of f (Figure 17), so that $f(x_1) = f(x_2) = 0$, then by Rolle's Theorem there is a number x between x_1 and x_2 such that $f'(x) = 0$. This means that if f has k different roots $x_1 < x_2 < \cdots < x_k$, then f' has at least $k - 1$ different roots: one between x_1 and x_2 , one between x_2 and x_3 , etc. It is now easy to prove by induction that a polynomial function

$$f(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_0$$

has at most n roots: The statement is surely true for $n = 1$, and if we assume that it is true for n , then the polynomial

$$g(x) = b_{n+1} x^{n+1} + b_n x^n + \cdots + b_0$$

could not have more than $n + 1$ roots, since if it did, g' would have more than n roots.

With this information it is not hard to describe the graph of

$$f(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_0.$$

The derivative, being a polynomial function of degree $n - 1$, has at most $n - 1$ roots. Therefore f has at most $n - 1$ critical points. Of course, a critical point is not necessarily a local maximum or minimum point, but at any rate, if a and b are adjacent critical points of f , then f' will remain either positive or negative on (a, b) , since f' is continuous; consequently, f will be either increasing or decreasing on (a, b) . Thus f has at most n regions of decrease or increase.

As a specific example, consider the function

$$f(x) = x^4 - 2x^2.$$

Since

$$f'(x) = 4x^3 - 4x = 4x(x - 1)(x + 1),$$

the critical points of f are -1 , 0 , and 1 , and

$$f(-1) = -1,$$

$$f(0) = 0,$$

$$f(1) = -1.$$

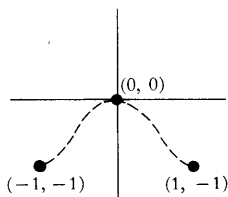


FIGURE 18

The behavior of f on the intervals between the critical points can be determined by one of the methods mentioned before. In particular, we could determine the sign of f' on these intervals simply by examining the formula for $f'(x)$. On the other hand, from the three critical values alone we can see (Figure 18) that f increases on $(-1, 0)$ and decreases on $(0, 1)$. To determine the sign of f' on $(-\infty, -1)$ and $(1, \infty)$ we can compute

$$f'(-2) = 4 \cdot (-2)^3 - 4 \cdot (-2) = -24,$$

$$f'(2) = 4 \cdot 2^3 - 4 \cdot 2 = 24,$$

and conclude that f is decreasing on $(-\infty, -1)$ and increasing on $(1, \infty)$. These conclusions also follow from the fact that $f(x)$ is large for large x and for large negative x .

We can already produce a good sketch of the graph; two other pieces of information provide the finishing touches (Figure 19). First, it is easy to determine that $f(x) = 0$ for $x = 0, \pm\sqrt{2}$; second, it is clear that f is even, $f(x) = f(-x)$, so the graph is symmetric with respect to the vertical axis. The function $f(x) = x^3 - x$, already sketched in Figure 15, is odd, $f(x) = -f(-x)$, and is consequently symmetric with respect to the origin. Half the work of graph sketching may be saved by noticing these things in the beginning.

Several problems in this and succeeding chapters ask you to sketch the graphs of functions. In each case you should determine

- (1) the critical points of f ,
- (2) the value of f at the critical points,
- (3) the sign of f' in the regions between critical points (if this is not already clear),
- (4) the numbers x such that $f(x) = 0$ (if possible),
- (5) the behavior of $f(x)$ as x becomes large or large negative (if possible).

Finally, bear in mind that a quick check, to see whether the function is odd or even, may save a lot of work.

This sort of analysis, if performed with care, will usually reveal the basic shape of the graph, but sometimes there are special features which require a little more thought. It is impossible to anticipate all of these, but one piece of information is often very important. If f is not defined at certain points (for example, if f is a rational function whose denominator vanishes at some points), then the behavior of f near these points should be determined.

For example, consider the function

$$f(x) = \frac{x^2 - 2x + 2}{x - 1},$$

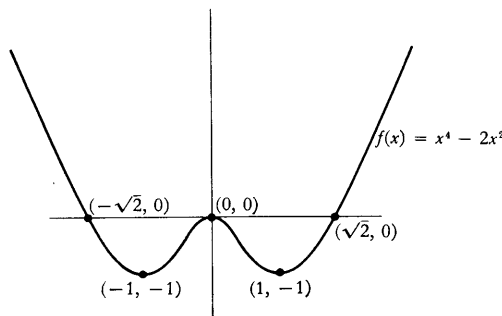


FIGURE 19

which is not defined at 1. We have

$$\begin{aligned} f'(x) &= \frac{(x-1)(2x-2) - (x^2-2x+2)}{(x-1)^2} \\ &= \frac{x(x-2)}{(x-1)^2}. \end{aligned}$$

Thus

- (1) the critical points of f are 0, 2.

Moreover,

$$\begin{aligned} (2) \quad f(0) &= -2, \\ f(2) &= 2. \end{aligned}$$

Because f is not defined on the whole interval $(0, 2)$, the sign of f' must be determined separately on the intervals $(0, 1)$ and $(1, 2)$, as well as on the intervals $(-\infty, 0)$ and $(2, \infty)$. We can do this by picking particular points in each of these intervals, or simply by staring hard at the formula for f' . Either way we find that

$$\begin{aligned} (3) \quad f'(x) &> 0 & \text{if } x < 0, \\ f'(x) &< 0 & \text{if } 0 < x < 1, \\ f'(x) &< 0 & \text{if } 1 < x < 2, \\ f'(x) &> 0 & \text{if } 2 < x. \end{aligned}$$

Finally, we must determine the behavior of $f(x)$ as x becomes large or large negative, as well as when x approaches 1 (this information will also give us another way to determine the regions on which f increases and decreases). To examine the behavior as x becomes large we write

$$\frac{x^2 - 2x + 2}{x - 1} = x - 1 + \frac{1}{x - 1};$$

clearly $f(x)$ is close to $x - 1$ (and slightly larger) when x is large, and $f(x)$ is close to $x - 1$ (but slightly smaller) when x is large negative. The behavior of f near 1 is also easy to determine; since

$$\lim_{x \rightarrow 1} (x^2 - 2x + 2) = 1 \neq 0,$$

the fraction

$$\frac{x^2 - 2x + 2}{x - 1}$$

becomes large as x approaches 1 from above and large negative as x approaches 1 from below.

All this information may seem a bit overwhelming; but there is only one way that it can be pieced together (Figure 20); be sure that you can account for each feature of the graph.

When this sketch has been completed, we might note that it looks like the

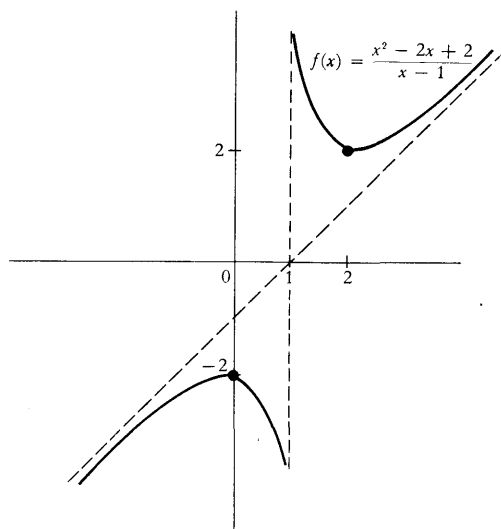


FIGURE 20

graph of an odd function shoved over 1 unit, and the expression

$$\frac{x^2 - 2x + 2}{x - 1} = \frac{(x - 1)^2 + 1}{x - 1}$$

shows that this is indeed the case. However, this is one of those special features which should be investigated only after you have used the other information to get a good idea of the appearance of the graph.

Although the location of local maxima and minima of a function is always revealed by a detailed sketch of its graph, it is usually unnecessary to do so much work. There is a popular test for local maxima and minima which depends on the behavior of the function only at its critical points.

THEOREM 5 Suppose $f'(a) = 0$. If $f''(a) > 0$, then f has a local minimum at a ; if $f''(a) < 0$, then f has a local maximum at a .

PROOF By definition,

$$f''(a) = \lim_{h \rightarrow 0} \frac{f'(a + h) - f'(a)}{h}.$$

Since $f'(a) = 0$, this can be written

$$f''(a) = \lim_{h \rightarrow 0} \frac{f'(a + h)}{h}.$$

Suppose now that $f''(a) > 0$. Then $f'(a + h)/h$ must be positive for sufficiently small h . Therefore:

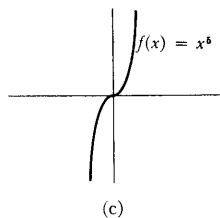
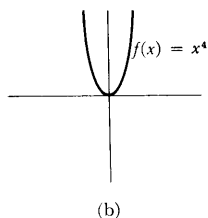
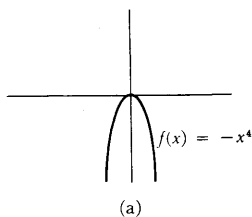


FIGURE 21

$f'(a + h)$ must be positive for sufficiently small $h > 0$
and $f'(a + h)$ must be negative for sufficiently small $h < 0$.

This means (Corollary 3) that f is increasing in some interval to the right of a and f is decreasing in some interval to the left of a . Consequently, f has a local minimum at a .

The proof for the case $f''(a) < 0$ is similar. ■

Theorem 5 may be applied to the function $f(x) = x^3 - x$, which has already been considered. We have

$$\begin{aligned} f'(x) &= 3x^2 - 1 \\ f''(x) &= 6x. \end{aligned}$$

At the critical points, $-\sqrt{1/3}$ and $\sqrt{1/3}$, we have

$$\begin{aligned} f''(-\sqrt{1/3}) &= -6\sqrt{1/3} < 0, \\ f''(\sqrt{1/3}) &= 6\sqrt{1/3} > 0. \end{aligned}$$

Consequently, $-\sqrt{1/3}$ is a local maximum point and $\sqrt{1/3}$ is a local minimum point.

Although Theorem 5 will be found quite useful for polynomial functions, the second derivative of many functions is so complicated that it is easier to consider the sign of the first derivative. Moreover, if a is a critical point of f it may happen that $f''(a) = 0$. In this case, Theorem 5 provides no information: it is possible that a is a local maximum point, a local minimum point, or neither, as shown (Figure 21) by the functions

$$f(x) = -x^4, \quad f(x) = x^4, \quad f(x) = x^5;$$

in each case $f'(0) = f''(0) = 0$, but 0 is a local maximum point for the first, a local minimum point for the second, and neither a local maximum nor minimum point for the third. This point will be pursued further in Part IV.

It is interesting to note that Theorem 5 automatically proves a partial converse of itself.

THEOREM 6 Suppose $f''(a)$ exists. If f has a local minimum at a , then $f''(a) \geq 0$; if f has a local maximum at a , then $f''(a) \leq 0$.

PROOF Suppose f has a local minimum at a . If $f''(a) < 0$, then f would also have a local maximum at a , by Theorem 5. Thus f would be constant in some interval containing a , so that $f''(a) = 0$, a contradiction. Thus we must have $f''(a) \geq 0$.

The case of a local maximum is handled similarly. ■

(This partial converse to Theorem 5 is the best we can hope for: the $\geq$ and $\leq$ signs cannot be replaced by $>$ and $<$, as shown by the functions $f(x) = x^4$ and $f(x) = -x^4$.)

The remainder of this chapter deals, not with graph sketching, or maxima and minima, but with three consequences of the Mean Value Theorem. The

first is a simple, but very beautiful, theorem which plays an important role in Chapter 15, and which also sheds light on many examples which have occurred in previous chapters.

THEOREM 7 Suppose that f is continuous at a , and that $f'(x)$ exists for all x in some interval containing a , except perhaps for $x = a$. Suppose, moreover, that $\lim_{x \rightarrow a} f'(x)$ exists. Then $f'(a)$ also exists, and

$$f'(a) = \lim_{x \rightarrow a} f'(x).$$

PROOF By definition,

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}.$$

For sufficiently small $h > 0$ the function f will be continuous on $[a, a+h]$ and differentiable on $(a, a+h)$ (a similar assertion holds for sufficiently small $h < 0$). By the Mean Value Theorem there is a number α_h in $(a, a+h)$ such that

$$\frac{f(a+h) - f(a)}{h} = f'(\alpha_h).$$

Now α_h approaches a as h approaches 0, because α_h is in $(a, a+h)$; since $\lim_{x \rightarrow a} f'(x)$ exists, it follows that

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h} = \lim_{h \rightarrow 0} f'(\alpha_h) = \lim_{x \rightarrow a} f'(x).$$

(It is a good idea to supply a rigorous ε - δ argument for this final step, which we have treated somewhat informally.) ■

Even if f is an everywhere differentiable function, it is still possible for f' to be discontinuous. This happens, for example, if

$$f(x) = \begin{cases} x^2 \sin \frac{1}{x}, & x \neq 0 \\ 0, & x = 0. \end{cases}$$

According to Theorem 7, however, the graph of f' can never exhibit a discontinuity of the type shown in Figure 22. Problem 39 outlines the proof of another beautiful theorem which gives further information about the function f' .

The next theorem, a generalization of the Mean Value Theorem, is of interest mainly because of its applications.

THEOREM 8 (THE CAUCHY MEAN VALUE THEOREM)

If f and g are continuous on $[a, b]$ and differentiable on (a, b) , then there is a number x in (a, b) such that

$$[f(b) - f(a)]g'(x) = [g(b) - g(a)]f'(x).$$

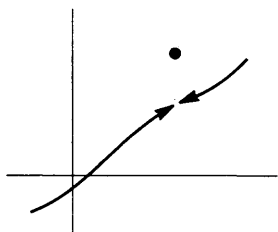


FIGURE 22

(If $g(b) \neq g(a)$, and $g'(x) \neq 0$, this equation can be written

$$\frac{f(b) - f(a)}{g(b) - g(a)} = \frac{f'(x)}{g'(x)}.$$

Notice that if $g(x) = x$ for all x , then $g'(x) = 1$, and we obtain the Mean Value Theorem. On the other hand, applying the Mean Value Theorem to f and g separately, we find that there are x and y in (a, b) with

$$\frac{f(b) - f(a)}{g(b) - g(a)} = \frac{f'(x)}{g'(y)},$$

but there is no guarantee that the x and y found in this way will be equal. These remarks may suggest that the Cauchy Mean Value Theorem will be quite difficult to prove, but actually the simplest of tricks suffices.)

PROOF Let

$$h(x) = f(x)[g(b) - g(a)] - g(x)[f(b) - f(a)].$$

Then h is continuous on $[a, b]$, differentiable on (a, b) , and

$$h(a) = f(a)g(b) - g(a)f(b) = h(b).$$

It follows from Rolle's Theorem that $h'(x) = 0$ for some x in (a, b) , which means that

$$0 = f'(x)[g(b) - g(a)] - g'(x)[f(b) - f(a)]. \blacksquare$$

The Cauchy Mean Value Theorem is the basic tool needed to prove a theorem which facilitates evaluation of limits of the form

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)},$$

when

$$\lim_{x \rightarrow a} f(x) = 0 \quad \text{and} \quad \lim_{x \rightarrow a} g(x) = 0.$$

In this case, Theorem 5-3 is of no use. Every derivative is a limit of this form, and computing derivatives frequently requires a great deal of work. If some derivatives are known, however, many limits of this form can now be evaluated easily.

THEOREM 9 (L'HÔPITAL'S RULE)

Suppose that

$$\lim_{x \rightarrow a} f(x) = 0 \quad \text{and} \quad \lim_{x \rightarrow a} g(x) = 0,$$

and suppose also that $\lim_{x \rightarrow a} f'(x)/g'(x)$ exists. Then $\lim_{x \rightarrow a} f(x)/g(x)$ exists, and

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}.$$

(Notice that Theorem 7 is a special case.)

PROOF The hypothesis that $\lim_{x \rightarrow a} f'(x)/g'(x)$ exists contains two implicit assumptions:

- (1) there is an interval $(a - \delta, a + \delta)$ such that $f'(x)$ and $g'(x)$ exist for all x in $(a - \delta, a + \delta)$ except, perhaps, for $x = a$,
- (2) in this interval $g'(x) \neq 0$ with, once again, the possible exception of $x = a$.

On the other hand, f and g are not even assumed to be defined at a . If we define $f(a) = g(a) = 0$ (changing the previous values of $f(a)$ and $g(a)$, if necessary), then f and g are continuous at a . If $a < x < a + \delta$, then the Mean Value Theorem and the Cauchy Mean Value Theorem apply to f and g on the interval $[a, x]$ (and a similar statement holds for $a - \delta < x < a$). First applying the Mean Value Theorem to g , we see that $g(x) \neq 0$, for if $g(x) = 0$ there would be some x_1 in (a, x) with $g'(x_1) = 0$, contradicting (2). Now applying the Cauchy Mean Value Theorem to f and g , we see that there is a number α_x in (a, x) such that

$$[f(x) - 0]g'(\alpha_x) = [g(x) - 0]f'(\alpha_x)$$

or

$$\frac{f(x)}{g(x)} = \frac{f'(\alpha_x)}{g'(\alpha_x)}.$$

Now α_x approaches a as x approaches a , because α_x is in (a, x) ; since $\lim_{y \rightarrow a} f'(y)/g'(y)$ exists, it follows that

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(\alpha_x)}{g'(\alpha_x)} = \lim_{y \rightarrow a} \frac{f'(y)}{g'(y)}.$$

(Once again, the reader is invited to supply the details of this part of the argument.) ■

PROBLEMS

1. For each of the following functions, find the maximum and minimum values on the indicated intervals, by finding the points in the interval where the derivative is 0, and comparing the values at these points with the values at the end points.

(i) $f(x) = x^3 - x^2 - 8x + 1$ on $[-2, 2]$.

(ii) $f(x) = x^5 + x + 1$ on $[-1, 1]$.

(iii) $f(x) = 3x^4 - 8x^3 + 6x^2$ on $[-\frac{1}{2}, \frac{1}{2}]$.

(iv) $f(x) = \frac{1}{x^5 + x + 1}$ on $[-\frac{1}{2}, 1]$.

(v) $f(x) = \frac{x + 1}{x^2 + 1}$ on $[-1, \frac{1}{2}]$.

(vi) $f(x) = \frac{x}{x^2 - 1}$ on $[0, 5]$.

2. Now sketch the graph of each of the functions in Problem 1, and find all local maximum and minimum points.

3. (a) If $a_1 < \cdots < a_n$, find the minimum value of $f(x) = \sum_{i=1}^n (x - a_i)^2$.

- *(b) Now find the minimum value of $f(x) = \sum_{i=1}^n |x - a_i|$. This is a prob-

lem where calculus won't help at all: on the intervals between the a_i 's the function f is linear, so that the minimum clearly occurs at one of the a_i , and these are precisely the points where f is not differentiable. However, the answer is easy to find if you consider how $f(x)$ changes as you pass from one such interval to another.

- *(c) Let $a > 0$. Show that the maximum value of

$$f(x) = \frac{1}{1 + |x|} + \frac{1}{1 + |x - a|}$$

is $(2 + a)/(1 + a)$. (The derivative can be found on each of the intervals $(-\infty, 0)$, $(0, a)$, and (a, ∞) separately.)

4. For each of the following functions, find all local maximum and minimum points.

$$(i) \quad f(x) = \begin{cases} x, & x \neq 3, 5, 7, 9 \\ 5, & x = 3 \\ -3, & x = 5 \\ 9, & x = 7 \\ 7, & x = 9. \end{cases}$$

$$(ii) \quad f(x) = \begin{cases} 0, & x \text{ irrational} \\ 1/q, & x = p/q \text{ in lowest terms.} \end{cases}$$

$$(iii) \quad f(x) = \begin{cases} x, & x \text{ rational} \\ 0, & x \text{ irrational.} \end{cases}$$

$$(iv) \quad f(x) = \begin{cases} 1, & x = 1/n \text{ for some } n \in \mathbf{N} \\ 0, & \text{otherwise.} \end{cases}$$

$$(v) \quad f(x) = \begin{cases} 1, & \text{if the decimal expansion of } x \text{ contains a } 5 \\ 0, & \text{otherwise.} \end{cases}$$

5. A straight line is drawn from the point $(0, a)$ to the horizontal axis, and then back to $(1, b)$, as in Figure 23. Prove that the total length is shortest when the angles α and β are equal. (Naturally you must bring a function into the picture: express the length in terms of x , where $(x, 0)$ is the point on the horizontal axis. The dashed line in Figure 23 suggests an alternative geometric proof; in either case the problem can be solved without actually finding the point $(x, 0)$.)

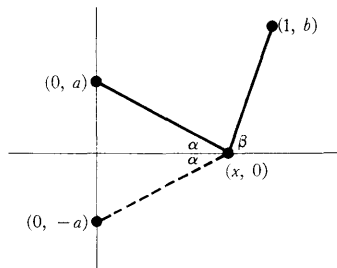


FIGURE 23

6. Prove that of all rectangles with given perimeter, the square has the greatest area.

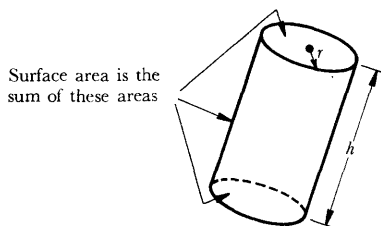


FIGURE 24

7. Find, among all right circular cylinders of fixed volume V , the one with smallest surface area (counting the areas of the faces at top and bottom, as in Figure 24).
8. Figure 25 shows the graph of the *derivative* of f . Find all local maximum and minimum points of f .

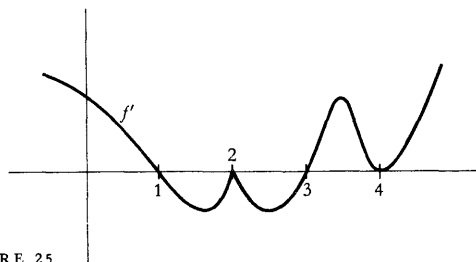


FIGURE 25

- *9. Suppose that f is a polynomial function, $f(x) = x^n + a_{n-1}x^{n-1} + \cdots + a_0$, with critical points $-1, 1, 2, 3, 4$, and corresponding critical values $6, 1, 2, 4, 3$. Sketch the graph of f , distinguishing the cases n even and n odd.
- *10. (a) Suppose that the polynomial function $f(x) = x^n + a_{n-1}x^{n-1} + \cdots + a_0$ has critical points $-1, 1, 2, 3$, and $f''(-1) = 0, f''(1) > 0, f''(2) < 0, f''(3) = 0$. Sketch the graph of f as accurately as possible on the basis of this information.
(b) Does there exist a polynomial function with the above properties, except that 3 is not a critical point?
11. Describe the graph of a rational function (in very general terms, similar to the text's description of the graph of a polynomial function).
12. (a) Prove that two polynomial functions of degree m and n , respectively, intersect in at most $\max(m, n)$ points.
(b) For each m and n exhibit two polynomial functions of degree m and n which intersect $\max(m, n)$ times.
- *13. (a) Suppose that the polynomial function $f(x) = x^n + a_{n-1}x^{n-1} + \cdots + a_0$ has exactly k critical points and $f''(x) \neq 0$ for all critical points x . Show that $n - k$ is odd.
(b) For each n , show that there is a polynomial function f of degree n with k critical points if $n - k$ is odd.
(c) Suppose that the polynomial function $f(x) = x^n + a_{n-1}x^{n-1} + \cdots + a_0$ has k_1 local maximum points and k_2 local minimum points. Show that $k_2 = k_1 + 1$ if n is even, and $k_2 = k_1$ if n is odd.
(d) Let n, k_1, k_2 be three integers with $k_2 = k_1 + 1$ if n is even, and $k_2 = k_1$ if n is odd, and $k_1 + k_2 < n$. Show that there is a polynomial function f of degree n , with k_1 local maximum points and k_2 local minimum points. Hint: Pick $a_1 < a_2 < \cdots < a_{k_1+k_2}$ and try

$$f'(x) = \prod_{i=1}^{k_1+k_2} (x - a_i) \cdot (1 + x^2)^l \text{ for an appropriate number } l.$$

14. (a) Prove that if $f'(x) \geq M$ for all x in $[a, b]$, then $f(b) \geq f(a) + M(b - a)$.
 (b) Prove that if $f'(x) \leq m$ for all x in $[a, b]$, then $f(b) \leq f(a) + m(b - a)$.
 (c) Formulate a similar theorem when $|f'(x)| \leq M$ for all x in $[a, b]$.
15. (a) Suppose that $f'(x) > g'(x)$ for all x , and that $f(a) = g(a)$. Show that $f(x) > g(x)$ for $x > a$ and $f(x) < g(x)$ for $x < a$.
 (b) Show by an example that these conclusions do not follow without the hypothesis $f(a) = g(a)$.
16. Find all functions f such that
 (a) $f'(x) = \sin x$.
 (b) $f''(x) = x^3$.
 (c) $f'''(x) = x + x^2$.
17. Although it is true that a weight dropped from rest will fall $s(t) = 16t^2$ feet after t seconds, this experimental fact does not mention the behavior of weights which are thrown upwards or downwards. On the other hand, the law $s''(t) = 32$ is always true and has just enough ambiguity to account for the behavior of a weight released from any height, with any initial velocity. For simplicity let us agree to measure heights upwards from ground level; in this case velocities are positive for rising bodies and negative for falling bodies, and all bodies fall according to the law $s''(t) = -32$.
 (a) Show that s is of the form $s(t) = -16t^2 + \alpha t + \beta$.
 (b) By setting $t = 0$ in the formula for s , and then in the formula for s' , show that $s(t) = -16t^2 + v_0 t + s_0$, where s_0 is the height from which the body is released at time 0, and v_0 is the velocity with which it is released.
 (c) A weight is thrown upwards with velocity v feet per second, at ground level. How high will it go? ("How high" means "what is the maximum height for all times.") What is its velocity at the moment it achieves its greatest height? What is its acceleration at that moment? When will it hit the ground again? What will its velocity be when it hits the ground again?
18. A cannon ball is shot from the ground with velocity v at an angle α (Figure 26) so that it has a vertical component of velocity $v \sin \alpha$ and a horizontal component $v \cos \alpha$. Its distance $s(t)$ above the ground obeys the law $s(t) = -16t^2 + (v \sin \alpha)t$, while its horizontal velocity remains constantly $v \cos \alpha$.
 (a) Show that the path of the cannon ball is a parabola (find the position at each time t , and show that these points lie on a parabola).
 (b) Find the angle α which will maximize the horizontal distance traveled by the cannon ball before striking the ground.

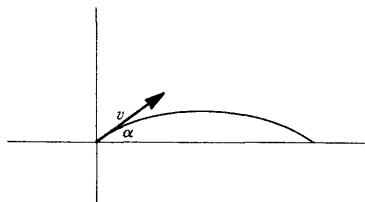


FIGURE 26

- *19. (a) Give an example of a function f for which $\lim_{x \rightarrow \infty} f(x)$ exists, but $\lim_{x \rightarrow \infty} f'(x)$ does not exist.
- (b) Prove that if $\lim_{x \rightarrow \infty} f(x)$ and $\lim_{x \rightarrow \infty} f'(x)$ both exist, then $\lim_{x \rightarrow \infty} f'(x) = 0$.
- (c) Prove that if $\lim_{x \rightarrow \infty} f(x)$ exists and $\lim_{x \rightarrow \infty} f''(x)$ exists, then $\lim_{x \rightarrow \infty} f''(x) = 0$ and also $\lim_{x \rightarrow \infty} f'(x) = 0$.
- (d) Generalize this result.
20. Suppose that f and g are two differentiable functions which satisfy $fg' - f'g = 0$. Prove that if a and b are adjacent zeros of f , and $g(a)$ and $g(b)$ are not both 0, then $g(x) = 0$ for some x between a and b . (Naturally the same result holds with f and g interchanged; thus, the zeros of f and g separate each other.) Hint: Derive a contradiction from the assumption that $g(x) \neq 0$ for all x between a and b : if a number is not 0, there is a natural thing to do with it.
21. Suppose that $|f(x) - f(y)| \leq (x - y)^n$ for $n > 1$. Prove that f is constant by considering f' . Compare with Problem 3-20.
22. Prove that if

$$\frac{a_0}{1} + \frac{a_1}{2} + \cdots + \frac{a_n}{n+1} = 0,$$

then

$$a_0 + a_1x + \cdots + a_nx^n = 0$$

for some x in $[0, 1]$.

23. Prove that the polynomial function $f_m(x) = x^3 - 3x + m$ never has two roots in $[0, 1]$, no matter what m may be. (This is an easy consequence of Rolle's Theorem. It is instructive, after giving an analytic proof, to graph f_0 and f_1 , and consider where the graph of f_m lies in relation to them.)
24. Suppose that f is continuous and differentiable on $[0, 1]$, that $f(x)$ is in $[0, 1]$ for each x , and that $f'(x) \neq 1$ for all x in $[0, 1]$. Show that there is exactly one number x in $[0, 1]$ such that $f(x) = x$. (Half of this problem has been done already, in Problem 7-11.)
- *25. Prove that if f is a twice differentiable function with $f(0) = 0$ and $f(1) = 1$ and $f'(0) = f'(1) = 0$, then $|f''(x)| \geq 4$ for some x in $[0, 1]$. In more picturesque terms: A particle which travels a unit distance in a unit time, and starts and ends with velocity 0, has at some time an acceleration ≥ 4 . Hint: Prove that either $f''(x) > 4$ for some x in $[0, \frac{1}{2}]$, or else $f''(x) < -4$ for some x in $[\frac{1}{2}, 1]$.
26. Suppose that f is a function such that $f'(x) = 1/x$ for all $x > 0$ and $f(1) = 0$. Prove that $f(xy) = f(x) + f(y)$ for all $x, y > 0$. Hint: Find $g'(x)$ when $g(x) = f(xy)$.
- *27. Suppose that f satisfies

$$f''(x) + f'(x)g(x) - f(x) = 0$$

for some function g . Prove that if f is 0 at two points, then f is 0 on the interval between them. Hint: Use Theorem 6.

28. Suppose that f is n -times differentiable and that $f(x) = 0$ for $n + 1$ different x . Prove that $f^{(n)}(x) = 0$ for some x .
29. Prove that

$$\frac{1}{9} < \sqrt{66} - 8 < \frac{1}{8}$$

(without computing $\sqrt{66}$ to 2 decimal places!).

30. Prove the following slight generalization of the Mean Value Theorem: If f is continuous and differentiable on (a, b) and $\lim_{y \rightarrow a^+} f(y)$ and $\lim_{y \rightarrow b^-} f(y)$ exist, then there is some x in (a, b) such that

$$f'(x) = \frac{\lim_{y \rightarrow b^-} f(y) - \lim_{y \rightarrow a^+} f(y)}{b - a}.$$

(Your proof should begin: "This is a trivial consequence of the Mean Value Theorem because")

31. Prove that the conclusion of the Cauchy Mean Value Theorem can be written in the form

$$\frac{f(b) - f(a)}{g(b) - g(a)} = \frac{f'(x)}{g'(x)},$$

under the additional assumptions that $g(b) \neq g(a)$ and that $f'(x)$ and $g'(x)$ are never simultaneously 0 on (a, b) .

- *32. Prove that if f and g are continuous on $[a, b]$ and differentiable on (a, b) , and $g'(x) \neq 0$ for x in (a, b) , then there is some x in (a, b) with

$$\frac{f'(x)}{g'(x)} = \frac{f(b) - f(a)}{g(b) - g(a)}.$$

Hint: Multiply out first, to see what this really says.

33. What is wrong with the following use of l'Hôpital's Rule:

$$\lim_{x \rightarrow 1} \frac{x^3 + x - 2}{x^2 - 3x + 2} = \lim_{x \rightarrow 1} \frac{3x^2 + 1}{2x - 3} = \lim_{x \rightarrow 1} \frac{6x}{2} = 3.$$

(The limit is actually -4 .)

34. Find the following limits:

(i) $\lim_{x \rightarrow 0} \frac{x}{\tan x}.$

(ii) $\lim_{x \rightarrow 0} \frac{\cos^2 x - 1}{x^2}.$

35. Find
- $f'(0)$
- if

$$f(x) = \begin{cases} \frac{g(x)}{x}, & x \neq 0 \\ 0, & x = 0, \end{cases}$$

and $g(0) = g'(0) = 0$ and $g''(0) = 17$.

36. Prove the following forms of l'Hôpital's Rule (none requiring any essentially new reasoning).

(a) If $\lim_{x \rightarrow a^+} f(x) = \lim_{x \rightarrow a^+} g(x) = 0$, and $\lim_{x \rightarrow a^+} f'(x)/g'(x) = l$, then $\lim_{x \rightarrow a^+} f(x)/g(x) = l$ (and similarly for limits from below).

(b) If $\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} g(x) = 0$, and $\lim_{x \rightarrow a} f'(x)/g'(x) = \infty$, then $\lim_{x \rightarrow a} f(x)/g(x) = \infty$ (and similarly for $-\infty$, or if $x \rightarrow a$ is replaced by $x \rightarrow a^+$ or $x \rightarrow a^-$).

(c) If $\lim_{x \rightarrow \infty} f(x) = \lim_{x \rightarrow \infty} g(x) = 0$ and $\lim_{x \rightarrow \infty} f'(x)/g'(x) = l$, then $\lim_{x \rightarrow \infty} f(x)/g(x) = l$ (and similarly for $-\infty$). Hint: Consider $\lim_{x \rightarrow 0^+} f(1/x)/g(1/x)$.

(d) If $\lim_{x \rightarrow \infty} f(x) = \lim_{x \rightarrow \infty} g(x) = 0$ and $\lim_{x \rightarrow \infty} f'(x)/g'(x) = \infty$, then $\lim_{x \rightarrow \infty} f(x)/g(x) = \infty$.

37. There is another form of l'Hôpital's Rule which requires more than algebraic manipulations: If
- $\lim_{x \rightarrow \infty} f(x) = \lim_{x \rightarrow \infty} g(x) = \infty$
- , and
- $\lim_{x \rightarrow \infty} f'(x)/g'(x) = l$
- , then
- $\lim_{x \rightarrow \infty} f(x)/g(x) = l$
- . Prove this as follows.

- (a) For every
- $\varepsilon > 0$
- there is a number
- a
- such that

$$\left| \frac{f'(x)}{g'(x)} - l \right| < \varepsilon \quad \text{for } x > a.$$

Apply the Cauchy Mean Value Theorem to f and g on $[a, x]$ to show that

$$\left| \frac{f(x) - f(a)}{g(x) - g(a)} - l \right| < \varepsilon \quad \text{for } x > a.$$

(Why can we assume $g(x) - g(a) \neq 0$?)

- (b) Now write

$$\frac{f(x)}{g(x)} = \frac{f(x) - f(a)}{g(x) - g(a)} \cdot \frac{f(x)}{f(x) - f(a)} \cdot \frac{g(x) - g(a)}{g(x)}$$

(why can we assume that $f(x) - f(a) \neq 0$ for large x ?) and conclude that

$$\left| \frac{f(x)}{g(x)} - l \right| < 2\varepsilon \quad \text{for sufficiently large } x.$$

38. To complete the orgy of variations on l'Hôpital's Rule, use Problem 37 to prove a few more cases of the following general statement (there are so many possibilities that you should select just a few, if any, that interest you):

If $\lim_{x \rightarrow []} f(x) = \lim_{x \rightarrow []} g(x) = \{ \}$ and $\lim_{x \rightarrow []} f'(x)/g'(x) = ()$, then $\lim_{x \rightarrow []} f(x)/g(x) = ()$. Here $[]$ can be a or a^+ or a^- or ∞ or $-\infty$, and $\{ \}$ can be 0 or ∞ or $-\infty$, and $()$ can be l or ∞ or $-\infty$.

- *39. (a) Suppose that f is differentiable on $[a, b]$. Prove that if the minimum of f on $[a, b]$ is at a , then $f'(a) \geq 0$, and if it is at b , then $f'(b) \leq 0$. (One half of the proof of Theorem 1 will go through.)
 (b) Suppose that $f'(a) < 0$ and $f'(b) > 0$. Show that $f'(x) = 0$ for some x in (a, b) . Hint: Consider the minimum of f on $[a, b]$; why must it be somewhere in (a, b) ?
 (c) Prove that if $f'(a) < c < f'(b)$, then $f'(x) = c$ for some x in (a, b) . (This result is known as Darboux's Theorem.) Hint: Cook up an appropriate function to which part (b) may be applied.
- *40. It is easy to find a function f such that $|f|$ is differentiable but f is not. For example, we can choose $f(x) = 1$ for x rational and $f(x) = -1$ for x irrational. In this example f is not even continuous, nor is this a mere coincidence: Prove that if $|f|$ is differentiable at a , and f is continuous at a , then f is also differentiable at a . Hint: It suffices to consider only a with $f(a) = 0$. Why? In this case, what must $|f|'(a)$ be?
- *41. (a) Let $y \neq 0$ and let n be even. Prove that $x^n + y^n = (x + y)^n$ only when $x = 0$. Hint: If $x_0^n + y^n = (x_0 + y)^n$, apply Rolle's Theorem to $f(x) = x^n + y^n - (x + y)^n$ on $[0, x_0]$.
 (b) Prove that if $y \neq 0$ and n is odd, then $x^n + y^n = (x + y)^n$ only if $x = 0$ or $x = -y$.
- **42. Use the method of Problem 41 to prove that if n is even and $f(x) = x^n$, then every tangent line to f intersects f only once.
- **43. Prove even more generally that if f' is increasing, then every tangent line intersects f only once.
- *44. Suppose that $f(0) = 0$ and f' is increasing. Prove that the function $g(x) = f(x)/x$ is increasing on $(0, \infty)$. Hint: Obviously you should look at $g'(x)$. Prove that it is positive by applying the Mean Value Theorem to f on the right interval (it will help to remember that the hypothesis $f(0) = 0$ is essential, as shown by the function $f(x) = 1 + x^2$).
- *45. Use derivatives to prove that if $n \geq 1$, then
- $$(1 + x)^n > 1 + nx \quad \text{for } -1 < x < 0 \text{ and } 0 < x.$$
- (notice that equality holds for $x = 0$).

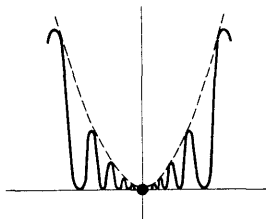


FIGURE 27

46. Let $f(x) = x^4 \sin^2 1/x$ for $x \neq 0$, and let $f(0) = 0$ (Figure 27).

- (a) Prove that 0 is a local minimum point for f .
 (b) Prove that $f'(0) = f''(0) = 0$.

This function thus provides another example to show that Theorem 6 cannot be improved. It also illustrates a subtlety about maxima and minima that often goes unnoticed: a function may not be increasing in any interval to the right of a local minimum point, nor decreasing in any interval to the left.

- *47. (a) Prove that if $f'(a) > 0$ and f' is continuous at a , then f is increasing in some interval containing a .

The next two parts of this problem show that continuity of f' is essential.

- (b) If $g(x) = x^2 \sin 1/x$, show that there are numbers x arbitrarily close to 0 with $g'(x) = 1$ and also with $g'(x) = -1$.
 (c) Suppose $0 < \alpha < 1$. Let $f(x) = \alpha x + x^2 \sin 1/x$ for $x \neq 0$, and let $f(0) = 0$ (see Figure 28). Show that f is not increasing in any open interval containing 0, by showing that in any interval there are points x with $f'(x) > 0$ and also points x with $f'(x) < 0$.

The behavior of f for $\alpha \geq 1$, which is much more difficult to analyze, is discussed in the next problem.

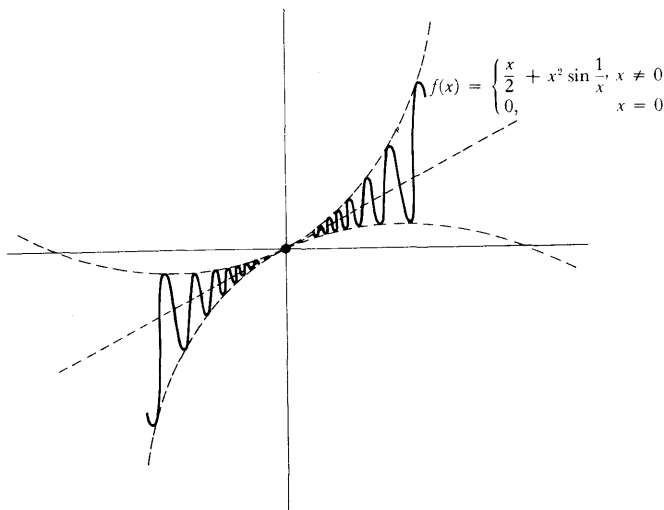


FIGURE 28

- **48. Let $f(x) = \alpha x + x^2 \sin 1/x$ for $x \neq 0$, and let $f(0) = 0$. In order to find the sign of $f'(x)$ when $\alpha \geq 1$ it is necessary to decide if $2x \sin 1/x - \cos 1/x$ is < -1 for any numbers x close to 0. It is a little more convenient to consider the function $g(y) = 2(\sin y)/y - \cos y$ for $y \neq 0$; we want to know if $g(y) < -1$ for large y . This question is quite delicate; the most

significant part of $g(y)$ is $-\cos y$, which does reach the value -1 , but this happens only when $\sin y = 0$, and it is not at all clear whether g itself can have values < -1 . The obvious approach to this problem is to find the local minimum values of g . Unfortunately, it is impossible to solve the equation $g'(y) = 0$ explicitly, so more ingenuity is required.

(a) Show that if $g'(y) = 0$, then

$$\cos y = \sin y \left(\frac{2 - y^2}{2y} \right),$$

and conclude that

$$g(y) = \sin y \left(\frac{2 + y^2}{2y} \right).$$

(b) Now show that if $g'(y) = 0$, then

$$\sin^2 y = \frac{4y^2}{4 + y^4},$$

and conclude that

$$|g(y)| = \frac{2 + y^2}{\sqrt{4 + y^4}}.$$

(c) Using the fact that $(2 + y^2)/\sqrt{4 + y^4} > 1$, show that if $\alpha = 1$, then f is not increasing in any interval around 0.

(d) Using the fact that $\lim_{y \rightarrow \infty} (2 + y^2)/\sqrt{4 + y^4} = 1$, show that if $\alpha > 1$, then f is increasing in some interval around 0.

****49.** A function f is **increasing at a** if there is some number $\delta > 0$ such that

$$f(x) > f(a) \quad \text{if} \quad a < x < a + \delta$$

and

$$f(x) < f(a) \quad \text{if} \quad a - \delta < x < a.$$

Notice that this does *not* mean that f is increasing in the interval $(a - \delta, a + \delta)$; for example, the function shown in Figure 28 is increasing at 0, but is not an increasing function in any open interval containing 0.

- Suppose that f is continuous on $[0, 1]$ and that f is increasing at a for every a in $[0, 1]$. Prove that f is increasing on $[0, 1]$. (First convince yourself that there is something to be proved.) Hint: For $0 < b < 1$, prove that the minimum of f on $[b, 1]$ must be at b .
- Prove part (a) without the assumption that f is continuous, by considering for each b in $[0, 1]$ the set $S_b = \{x: f(y) \geq f(b) \text{ for all } y \text{ in } [b, x]\}$. (This part of the problem is not necessary for the other parts.) Hint: Prove that $S_b = \{x: b \leq x \leq 1\}$ by considering $\sup S_b$.
- If f is increasing at a and f is differentiable at a , prove that $f'(a) \geq 0$ (this is easy).

- (d) If $f'(a) > 0$, prove that f is increasing at a (go right back to the definition of $f'(a)$).
- (e) Use parts (a) and (d) to show, without using the Mean Value Theorem, that if f is continuous on $[0, 1]$ and $f'(a) > 0$ for all a in $[0, 1]$, then f is increasing on $[0, 1]$.
- (f) Suppose that f is continuous on $[0, 1]$ and $f'(a) = 0$ for all a in $(0, 1)$. Apply part (e) to the function $g(x) = f(x) + \epsilon x$ to show that $f(1) - f(0) > -\epsilon$. Similarly, show that $f(1) - f(0) < \epsilon$ by considering $h(x) = \epsilon x - f(x)$. Conclude that $f(0) = f(1)$.

This particular proof that a function with zero derivative must be constant has many points in common with a proof of H. A. Schwartz, which may be the first rigorous proof ever given. Its discoverer, at least, seemed to think it was. See his exuberant letter in reference [41] of the Suggested Reading.

- **50.** If f is a constant function, then every point is a local maximum point for f . It is quite possible for this to happen even if f is not a constant function: for example, if $f(x) = 0$ for $x < 0$ and $f(x) = 1$ for $x \geq 0$. Prove, however, that if f is continuous on $[a, b]$ and every point of $[a, b]$ is a local maximum point, then f is a constant function. Hint: If the minimum of f occurs at x_0 , consider the set $\{x \text{ in } [x_0, b]: f(y) = f(x_0) \text{ for all } y \text{ in } [x_0, x]\}$.
- **51.** (a) A point x is called a **strict maximum point** for f on A if $f(x) > f(y)$ for all y in A with $y \neq x$ (compare with the definition of an ordinary maximum point). A **local strict maximum point** is defined in the obvious way. Find all local strict maximum points of the function

$$f(x) = \begin{cases} 0, & x \text{ irrational} \\ \frac{1}{q}, & x = \frac{p}{q} \text{ in lowest terms.} \end{cases}$$

It seems quite unlikely that a function can have a local strict maximum at *every* point (although the above example might give one pause for thought). Prove this as follows.

- (b) Suppose that every point is a local strict maximum point for f . Let x_1 be any number and choose $a_1 < x_1 < b_1$ with $b_1 - a_1 < 1$ such that $f(x_1) > f(x)$ for all x in $[a_1, b_1]$. Let $x_2 \neq x_1$ be any point in (a_1, b_1) and choose $a_1 \leq a_2 < x_2 < b_2 \leq b_1$ with $b_2 - a_2 < \frac{1}{2}$ such that $f(x_2) > f(x)$ for all x in $[a_2, b_2]$. Continue in this way, and use the Nested Interval Theorem (Problem 8-14) to obtain a contradiction.

APPENDIX. CONVEXITY AND CONCAVITY

Although the graph of a function can be sketched quite accurately on the basis of the information provided by the derivative, some subtle aspects of the graph are revealed only by examining the second derivative. These details were purposely omitted previously because graph sketching is complicated enough without worrying about them, and the additional information obtained is often not worth the effort. Also, correct proofs of the relevant facts are sufficiently difficult to be placed in an appendix. Despite these discouraging remarks, the information presented here is well worth assimilating, because the notions of convexity and concavity are far more important than as mere aids to graph sketching. Moreover, the proofs have a pleasantly geometric flavor not often found in calculus theorems. Indeed, the basic definition is geometric in nature (see Figure 1).

DEFINITION 1

A function f is **convex** on an interval, if for all a and b in the interval, the line segment joining $(a, f(a))$ and $(b, f(b))$ lies above the graph of f .

The geometric condition appearing in this definition can be expressed in an analytic way that is sometimes more useful in proofs. The straight line between $(a, f(a))$ and $(b, f(b))$ is the graph of the function g defined by

$$g(x) = \frac{f(b) - f(a)}{b - a} (x - a) + f(a).$$

This line lies **above** the graph of f at x if $g(x) > f(x)$, that is, if

$$\frac{f(b) - f(a)}{b - a} (x - a) + f(a) > f(x)$$

or

$$\frac{f(b) - f(a)}{b - a} (x - a) > f(x) - f(a)$$

or

$$\frac{f(b) - f(a)}{b - a} > \frac{f(x) - f(a)}{x - a}.$$

We therefore have an equivalent definition of convexity.

DEFINITION 2

A function f is **convex** on an interval if for a , x , and b in the interval with $a < x < b$ we have

$$\frac{f(x) - f(a)}{x - a} < \frac{f(b) - f(a)}{b - a}.$$

If the word “over” in Definition 1 is replaced by “under” or, equivalently,

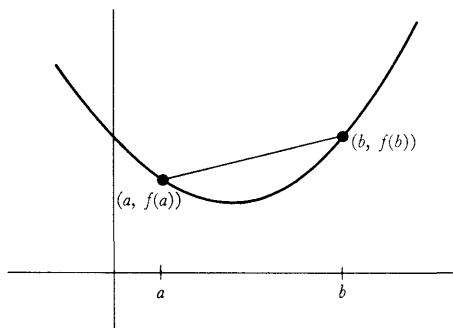


FIGURE 1

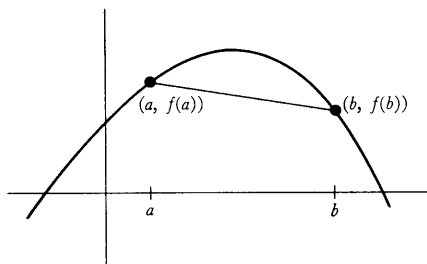


FIGURE 2

if the inequality in Definition 2 is replaced by

$$\frac{f(x) - f(a)}{x - a} > \frac{f(b) - f(a)}{b - a},$$

we obtain the definition of a **concave** function (Figure 2). It is not hard to see that the concave functions are precisely the ones of the form $-f$, where f is convex. For this reason, the next three theorems about convex functions have immediate corollaries about concave functions, so simple that we will not even bother to state them.

Figure 3 shows some tangent lines of a convex function. Two things seem to be true:

- (1) The graph of f lies above the tangent line at $(a, f(a))$ except at the point $(a, f(a))$ itself (this point is called the **point of contact** of the tangent line).
- (2) If $a < b$, then the slope of the tangent line at $(a, f(a))$ is less than the slope of the tangent line at $(b, f(b))$; that is, f' is increasing.

As a matter of fact these observations are true, and the proofs are not difficult.

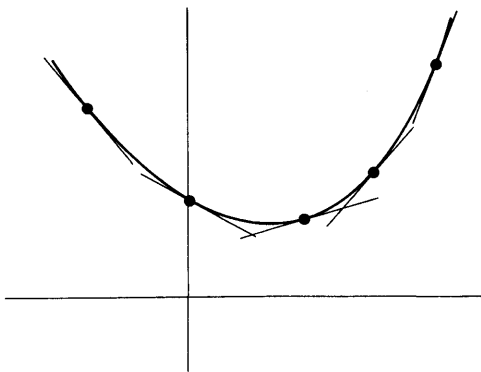


FIGURE 3

THEOREM 1 Let f be convex. If f is differentiable at a , then the graph of f lies above the tangent line through $(a, f(a))$, except at $(a, f(a))$ itself. If $a < b$ and f is differentiable at a and b , then $f'(a) < f'(b)$.

PROOF If $0 < h_1 < h_2$, then, as Figure 4 indicates,

$$(1) \quad \frac{f(a + h_1) - f(a)}{h_1} < \frac{f(a + h_2) - f(a)}{h_2}.$$

A nonpictorial proof can be derived immediately from Definition 2 applied to $a < a + h_1 < a + h_2$. Inequality (1) shows that the values of

$$\frac{f(a + h) - f(a)}{h}$$

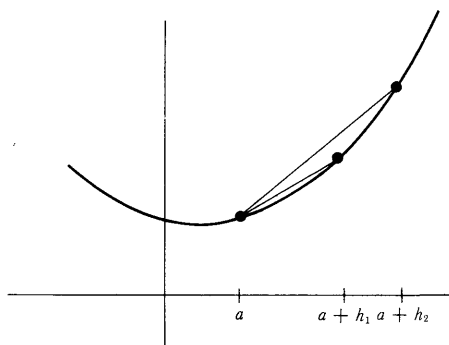


FIGURE 4

decrease as $h \rightarrow 0^+$. Consequently,

$$f'(a) < \frac{f(a+h) - f(a)}{h} \quad \text{for } h > 0$$

(in fact $f'(a)$ is the greatest lower bound of all these numbers). But this means that for $h > 0$ the secant line through $(a, f(a))$ and $(a+h, f(a+h))$ has larger slope than the tangent line, which implies that $(a+h, f(a+h))$ lies above the tangent line (an analytic translation of this argument is easily supplied).

For negative h there is a similar situation (Figure 5): if $h_2 < h_1 < 0$, then

$$\frac{f(a+h_1) - f(a)}{h_1} > \frac{f(a+h_2) - f(a)}{h_2}.$$

This shows that the slope of the tangent line is greater than

$$\frac{f(a+h) - f(a)}{h} \quad \text{for } h < 0$$

(in fact $f'(a)$ is the least upper bound of all these numbers), so that $f(a+h)$ lies above the tangent line if $h < 0$. This proves the first part of the theorem.

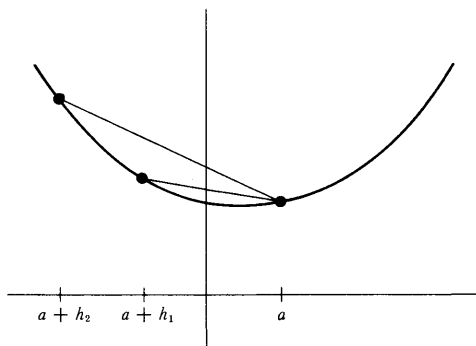


FIGURE 5

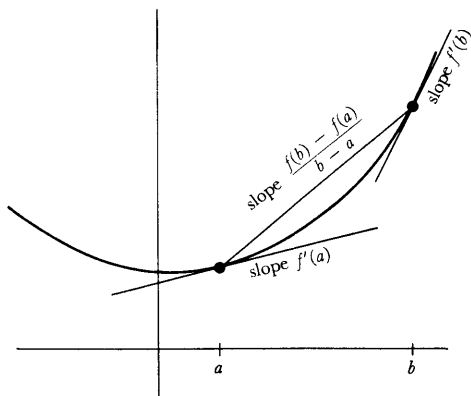


FIGURE 6

Now suppose that $a < b$. Then, as we have already seen (Figure 6),

$$\begin{aligned} f'(a) &< \frac{f(a + (b - a)) - f(a)}{b - a} \quad \text{since } b - a > 0 \\ &= \frac{f(b) - f(a)}{b - a}, \end{aligned}$$

and

$$\begin{aligned} f'(b) &> \frac{f(b + (a - b)) - f(b)}{a - b} \quad \text{since } a - b < 0 \\ &= \frac{f(a) - f(b)}{a - b} = \frac{f(b) - f(a)}{b - a}. \end{aligned}$$

Combining these inequalities, we obtain $f'(a) < f'(b)$. ■

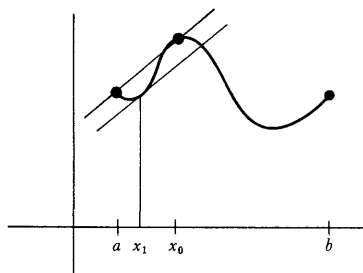


FIGURE 7

Theorem 1 has two converses. Here the proofs will be a little more difficult. We begin with a lemma that plays the same role in the next theorem that Rolle's Theorem plays in the proof of the Mean Value Theorem. It states that if f' is increasing, then the graph of f lies below any secant line *which happens to be horizontal*.

LEMMA Suppose f is differentiable and f' is increasing. If $a < b$ and $f(a) = f(b)$, then $f(x) < f(a) = f(b)$ for $a < x < b$.

PROOF Suppose first that $f(x) > f(a) = f(b)$ for some x in (a, b) . Then the maximum of f on $[a, b]$ occurs at some point x_0 in (a, b) with $f(x_0) > f(a)$ and, of course, $f'(x_0) = 0$ (Figure 7). On the other hand, applying the Mean Value Theorem to the interval $[a, x_0]$, we find that there is x_1 with $a < x_1 < x_0$ and

$$f'(x_1) = \frac{f(x_0) - f(a)}{x_0 - a} > 0,$$

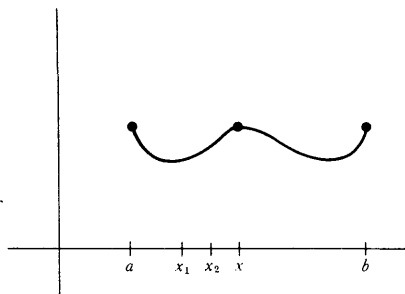


FIGURE 8

contradicting the fact that f' is increasing. This proves that $f(x) \leq f(a) = f(b)$ for $a < x < b$, and it only remains to prove that $f(x) = f(a)$ is also impossible for x in (a, b) .

Suppose $f(x) = f(a)$ for some x in (a, b) . We know that f is not constant on $[a, x]$ (if it were, f' would not be increasing on $[a, x]$) so there is (Figure 8) some x_1 with $a < x_1 < x$ and $f(x_1) < f(a)$. Applying the Mean Value Theorem to $[x_1, x]$ we conclude that there is x_2 with $x_1 < x_2 < x$ and

$$f'(x_2) = \frac{f(x) - f(x_1)}{x - x_1} > 0.$$

On the other hand, $f'(x) = 0$, since a local maximum occurs at x . Again this contradicts the hypothesis that f' is increasing. ■

We now attack the general case by the same sort of algebraic machinations that we used in the proof of the Mean Value Theorem.

THEOREM 2 If f is differentiable and f' is increasing, then f is convex.

PROOF Let $a < b$. Define g by

$$g(x) = f(x) - \frac{f(b) - f(a)}{b - a} (x - a).$$

It is easy to see that g' is also increasing; moreover, $g(a) = g(b) = f(a)$. Applying the lemma to g we conclude that

$$g(x) < f(a) \quad \text{if} \quad a < x < b.$$

In other words, if $a < x < b$, then

$$f(x) - \frac{f(b) - f(a)}{b - a} (x - a) < f(a)$$

or

$$\frac{f(x) - f(a)}{x - a} < \frac{f(b) - f(a)}{b - a}.$$

Hence f is convex. ■

THEOREM 3 If f is differentiable and the graph of f lies above each tangent line except at the point of contact, then f is convex.

PROOF Let $a < b$. It is clear from Figure 9 that if $(b, f(b))$ lies above the tangent line at $(a, f(a))$, and $(a, f(a))$ lies above the tangent line at $(b, f(b))$, then the slope of the tangent line at $(b, f(b))$ must be larger than the slope of the tangent line at $(a, f(a))$. The following argument just says this with equations.

Since the tangent line at $(a, f(a))$ is the graph of the function

$$g(x) = f'(a)(x - a) + f(a),$$

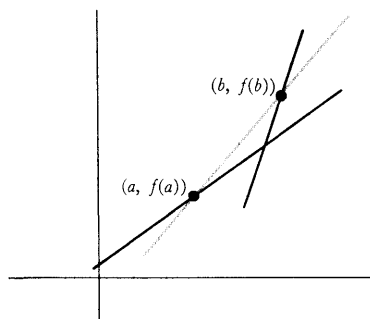


FIGURE 9

and since $(b, f(b))$ lies above the tangent line, we have

$$(1) \quad f(b) > f'(a)(b - a) + f(a).$$

Similarly, since the tangent line at $(b, f(b))$ is the graph of

$$h(x) = f'(b)(x - b) + f(b),$$

and $(a, f(a))$ lies above the tangent line at $(b, f(b))$, we have

$$(2) \quad f(a) > f'(b)(a - b) + f(b).$$

It follows from (1) and (2) that $f'(a) < f'(b)$.

It now follows from Theorem 2 that f is convex. ■

If a function f has a reasonable second derivative, the information given in these theorems can be used to discover the regions in which f is convex or concave. Consider, for example, the function

$$f(x) = \frac{1}{1 + x^2}.$$

For this function,

$$f'(x) = \frac{-2x}{(1 + x^2)^2}.$$

Thus $f'(x) = 0$ only for $x = 0$, and $f(0) = 1$, while

$$\begin{aligned} f'(x) &> 0 & \text{if } x < 0, \\ f'(x) &< 0 & \text{if } x > 0. \end{aligned}$$

Moreover,

$$\begin{aligned} f(x) &> 0 & \text{for all } x, \\ f(x) &\rightarrow 0 & \text{as } x \rightarrow \infty \text{ or } -\infty, \\ f &\text{ is even.} \end{aligned}$$

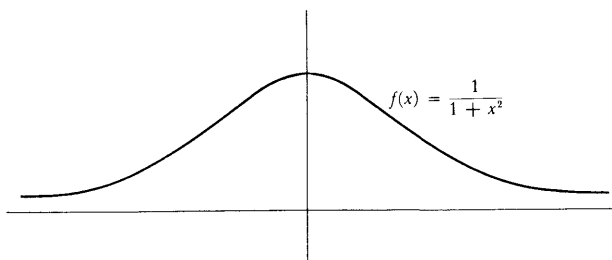


FIGURE 10

The graph of f therefore looks something like Figure 10. We now compute

$$\begin{aligned} f''(x) &= \frac{(1 + x^2)^2(-2) + 2x \cdot [2(1 + x^2) \cdot 2x]}{(1 + x^2)^4} \\ &= \frac{2(3x^2 - 1)}{(1 + x^2)^3}. \end{aligned}$$

It is not hard to determine the sign of $f''(x)$. Note first that $f''(x) = 0$ only when $x = \sqrt{1/3}$ or $-\sqrt{1/3}$. Since f'' is clearly continuous, it must keep the same sign on each of the sets

$$\begin{aligned} &(-\infty, -\sqrt{1/3}), \\ &(-\sqrt{1/3}, \sqrt{1/3}), \\ &(\sqrt{1/3}, \infty). \end{aligned}$$

Since we easily compute, for example, that

$$\begin{aligned} f''(-1) &= \frac{1}{2} > 0, \\ f''(0) &= -2 < 0, \\ f''(1) &= \frac{1}{2} > 0, \end{aligned}$$

we conclude that

$$\begin{aligned} f'' &> 0 \text{ on } (-\infty, -\sqrt{1/3}) \text{ and } (\sqrt{1/3}, \infty), \\ f'' &< 0 \text{ on } (-\sqrt{1/3}, \sqrt{1/3}). \end{aligned}$$

Since $f'' > 0$ means f' is increasing, it follows from Theorem 2 that f is convex on $(-\infty, -\sqrt{1/3})$ and $(\sqrt{1/3}, \infty)$, while on $(-\sqrt{1/3}, \sqrt{1/3})$ f is concave (Figure 11).

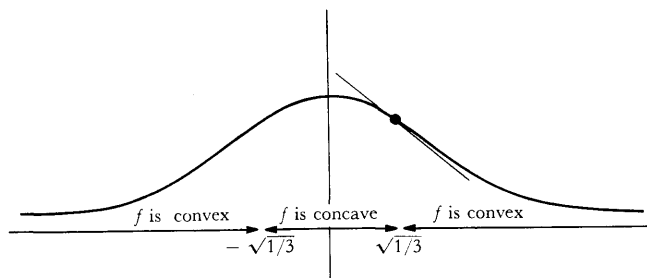


FIGURE 11

Notice that at $(\sqrt{1/3}, \frac{3}{4})$ the tangent line lies below the part of the graph to the right, since f is convex on $(\sqrt{1/3}, \infty)$, and above the part of the graph to the left, since f is concave on $(-\sqrt{1/3}, \sqrt{1/3})$; thus the tangent line crosses the graph. In general, a number a is called an **inflection point** of f if the tangent line to the graph of f at $(a, f(a))$ crosses the graph; thus $\sqrt{1/3}$ and $-\sqrt{1/3}$ are inflection points of $f(x) = 1/(1+x^2)$. Note that the condition $f''(a) = 0$ does *not* ensure that a is an inflection point of f ; for example, if $f(x) = x^4$, then $f''(0) = 0$, but f is convex, so the tangent line at $(0, 0)$ certainly doesn't cross the graph of f . In order for a to be an inflection point of a function f , it is necessary that f'' should have different signs to the left and right of a .

This example illustrates the procedure which may be used to analyze any function f . After the graph has been sketched, using the information provided

by f' , the zeros of f'' are computed and the sign of f'' is determined on the intervals between consecutive zeros. On intervals where $f'' > 0$ the function is convex; on intervals where $f'' < 0$ the function is concave. Knowledge of the regions of convexity and concavity of f can often prevent absurd misinterpretation of other data about f . Several functions, which can be analyzed in this way, are given in the problems, which also contain some theoretical questions.

There is one further fact which I cannot resist mentioning here. We have seen that convex and concave functions have the property that every tangent line intersects the graph just once, and a few drawings will probably convince you that no other functions have this property. This is indeed the case, but the only proof I know contains so many delicate points that I feel it is wiser to omit it.

PROBLEMS

1. Sketch, indicating regions of convexity and concavity and points of inflection, the functions in Problem 11-1 (consider (iv) as double starred).
2. Figure 25 in Chapter 11 shows the graph of f' . Sketch the graph of f .
3. Find two convex functions f and g such that $f(x) = g(x)$ if and only if x is an integer.
4. Show that f is convex on an interval if and only if for all x and y in the interval we have

$$f(tx + (1 - t)y) < tf(x) + (1 - t)f(y), \text{ for } 0 < t < 1.$$

(This is just a restatement of the definition, but a useful one.)

5. Prove that if f and g are convex and f is increasing, then $g \circ f$ is convex.
- *6. Let f be a twice-differentiable function with the following properties: $f(x) > 0$ for $x \geq 0$, and f is decreasing, and $f'(0) = 0$. Prove that $f''(x) = 0$ for some $x > 0$ (so that in reasonable cases f will have an inflection point at x —an example is given by $f(x) = 1/(1 + x^2)$). Every hypothesis in this theorem is essential, as shown by $f(x) = 1 - x^2$, which is not positive for all x ; by $f(x) = x^2$, which is not decreasing; and by $f(x) = 1/(x + 1)$, which does not satisfy $f'(0) = 0$. Hint: Choose $x_0 > 0$ with $f'(x_0) < 0$. We cannot have $f'(y) \leq f'(x_0)$ for all $y > x_0$. Why not? So $f'(x_1) > f'(x_0)$ for some $x_1 > x_0$. Consider f' on $[0, x_1]$.
- *7. (a) Prove that if f is convex, then $f([x + y]/2) < [f(x) + f(y)]/2$.
 (b) Suppose that f satisfies this condition. Show that $f(kx + (1 - k)y) < kf(x) + (1 - k)f(y)$ whenever k is a rational number, between 0 and 1, of the form $m/2^n$. Hint: Part (a) is the special case $n = 1$. Use induction, employing part (a) at each step.
 (c) Suppose that f satisfies the condition in part (a) and f is continuous. Show that f is convex.

***8.** Let $p_1, \dots, p_n$ be positive numbers with $\sum_{i=1}^n p_i = 1$.

(a) For any numbers $x_1, \dots, x_n$ show that $\sum_{i=1}^n p_i x_i$ lies between the smallest and the largest x_i .

(b) Show the same for $(1/t) \sum_{i=1}^{n-1} p_i x_i$, where $t = \sum_{i=1}^{n-1} p_i$.

(c) Prove *Jensen's inequality*: If f is convex, then $f\left(\sum_{i=1}^n p_i x_i\right) \leq \sum_{i=1}^n p_i f(x_i)$.

Hint: Use Problem 4, noting that $p_n = 1 - t$. (Part (b) is needed to show that $(1/t) \sum_{i=1}^{n-1} p_i x_i$ is in the domain of f if $x_1, \dots, x_n$ are.)

***9.** (a) For any function f , the right-hand derivative, $\lim_{h \rightarrow 0^+} [f(a+h) - f(a)]/h$, is denoted by $f_+'(a)$, and the left-hand derivative is denoted by $f_-'(a)$. The proof of Theorem 1 actually shows that f_+' and f_-' always exist if f is convex. Check this assertion, and also show that f_+' and f_-' are increasing, and that $f_-'(a) \leq f_+'(a)$.

****10.** (b) Show that if f is convex, then $f_+'(a) = f_-'(a)$ if and only if f_+' is continuous at a . (Thus f is differentiable precisely when f_+' is continuous.) Hint: $[f(b) - f(a)]/(b - a)$ is close to $f_-'(a)$ for $b < a$ close to a , and $f_+'(b)$ is less than this quotient.

****10.** Prove that a convex function must be continuous.

****11.** Prove that if f is convex on $\mathbf{R}$, then either f is decreasing, or else f is increasing, or else there is a number c such that f is decreasing on $(-\infty, c]$ and increasing on $[c, \infty)$. This problem requires a lot of careful bookkeeping, and is perhaps best postponed until after the next chapter, where a similar theorem is proved in detail. An important step for this problem is to establish that if $f(a) < f(b)$ for some $a < b$, then f is increasing on $[b, \infty)$, while f is decreasing on $(-\infty, a]$ if $f(a) > f(b)$.

CHAPTER 12 INVERSE FUNCTIONS

We now have at our disposal quite powerful methods for investigating functions; what we lack is an adequate supply of functions to which these methods may be applied. We have studied various ways of forming new functions from old—addition, multiplication, division, and composition—but using these alone, we can produce only the rational functions (even the sine function, although frequently used for examples, has never been defined). In the next few chapters we will begin to construct new functions in quite sophisticated ways, but there is one important method which will practically double the usefulness of any other method we discover.

If we recall that a function is a collection of pairs of numbers, we might hit upon the bright idea of simply reversing all the pairs. Thus from the function

$$f = \{(1, 2), (3, 4), (5, 9), (13, 8)\},$$

we obtain

$$g = \{(2, 1), (4, 3), (9, 5), (8, 13)\}.$$

While $f(1) = 2$ and $f(3) = 4$, we have $g(2) = 1$ and $g(4) = 3$.

Unfortunately, this bright idea does not always work. If

$$f = \{(1, 2), (3, 4), (5, 9), (13, 4)\},$$

then the collection

$$\{(2, 1), (4, 3), (9, 5), (4, 13)\}$$

is not a function at all, since it contains both $(4, 3)$ and $(4, 13)$. It is clear where the trouble lies: $f(3) = f(13)$, even though $3 \neq 13$. This is the only sort of thing that can go wrong, and it is worthwhile giving a name to the functions for which this does not happen.

DEFINITION

A function f is **one-one** (read “one-to-one”) if $f(a) \neq f(b)$ whenever $a \neq b$.

The identity function I is obviously one-one, and so is the following modification:

$$g(x) = \begin{cases} x, & x \neq 3, 5 \\ 3, & x = 5 \\ 5, & x = 3. \end{cases}$$

The function $f(x) = x^2$ is not one-one, since $f(-1) = f(1)$, but if we define

$$g(x) = x^2, \quad x \geq 0$$

(and leave g undefined for $x < 0$), then g is one-one, because g is increasing

(since $g'(x) = 2x > 0$, for $x > 0$). This observation is easily generalized: If n is a natural number and

$$f(x) = x^n, \quad x \geq 0,$$

then f is one-one. If n is odd, one can do better: the function

$$f(x) = x^n \quad \text{for all } x$$

is one-one (since $f'(x) = nx^{n-1} > 0$, for all $x \neq 0$).

It is particularly easy to decide from the graph of f whether f is one-one: the condition $f(a) \neq f(b)$ for $a \neq b$ means that no *horizontal* line intersects the graph of f twice (Figure 1).

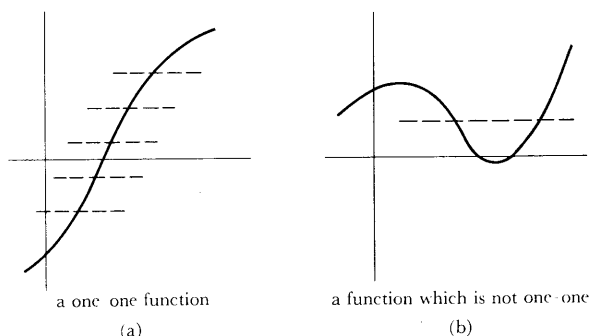


FIGURE 1

If we reverse all the pairs in (a not necessarily one-one function) f we obtain, in any case, some collection of pairs. It is popular to abstain from this procedure unless f is one-one, but there is no particular reason to do so—instead of a definition with restrictive conditions we obtain a definition and a theorem.

DEFINITION

For any function f , the **inverse** of f , denoted by f^{-1} , is the set of all pairs (a, b) for which the pair (b, a) is in f .

THEOREM 1

f^{-1} is a function if and only if f is one-one.

PROOF

Suppose first that f is one-one. Let (a, b) and (a, c) be two pairs in f^{-1} . Then (b, a) and (c, a) are in f , so $a = f(b)$ and $a = f(c)$; since f is one-one this implies that $b = c$. Thus f^{-1} is a function.

Conversely, suppose that f^{-1} is a function. If $f(b) = f(c)$, then f contains the pairs $(b, f(b))$ and $(c, f(c)) = (c, f(b))$, so $(f(b), b)$ and $(f(b), c)$ are in f^{-1} . Since f^{-1} is a function this implies that $b = c$. Thus f is one-one. ■

The graphs of f and f^{-1} are so closely related that it is possible to use the graph of f to visualize the graph of f^{-1} . Since the graph of f^{-1} consists of all pairs (a, b) with (b, a) in the graph of f , one obtains the graph of f^{-1} from the

graph of f by interchanging the horizontal and vertical axis. If f has the graph shown in Figure 2(a),

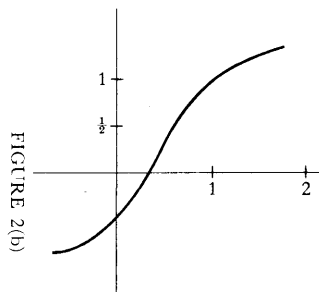


FIGURE 2(b)

FIGURE 2(a)

and you rotate this page counter clockwise through a right angle, then the graph of f^{-1} appears on your left (Figure 2(b)). The only trouble is that the numbering on the horizontal axis goes in the wrong direction, so you must flip this picture over to get the usual picture of f^{-1} , which appears on your right (Figure 3).

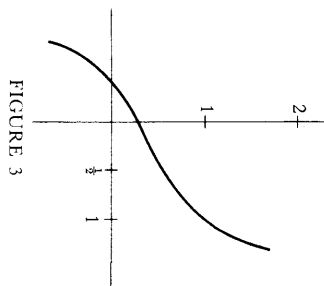


FIGURE 3

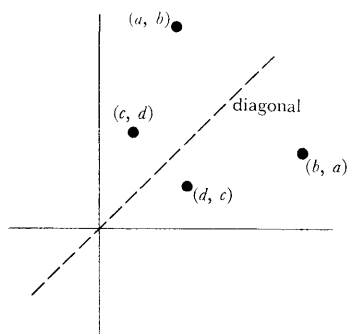


FIGURE 4

This procedure is awkward with books and impossible with blackboards, so it is fortunate that there is another way of constructing the graph of f^{-1} . The points (a, b) and (b, a) are reflections of each other through the graph of $I(x) = x$, which is called the **diagonal** (Figure 4). To obtain the graph of f^{-1} we merely reflect the graph of f through this line (Figure 5).

Reflecting through the diagonal twice will clearly leave us right back where we started; this means that $(f^{-1})^{-1} = f$, which is also clear from the definition. In conjunction with Theorem 1, this equation has a significant consequence: if f is a one-one function, then the function f^{-1} is also one-one (since $(f^{-1})^{-1}$ is a function).

There are a few other simple manipulations with inverse functions of which

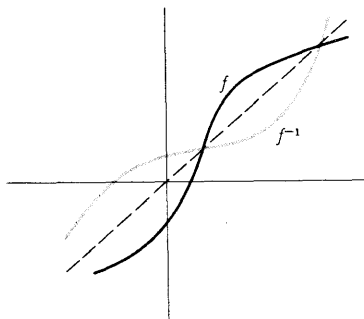


FIGURE 5

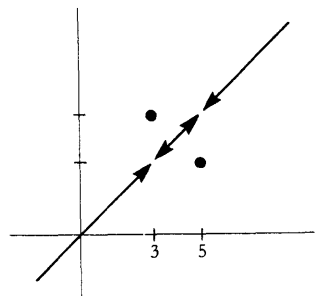


FIGURE 6

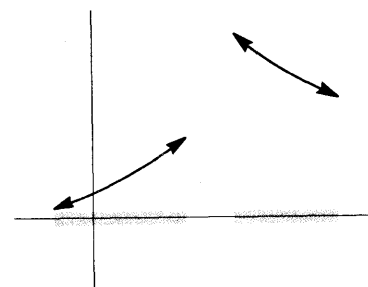


FIGURE 7

you should be aware. Since (a, b) is in f precisely when (b, a) is in f^{-1} , it follows that

$$b = f(a) \text{ means the same as } a = f^{-1}(b).$$

Thus $f^{-1}(b)$ is the (unique) number a such that $f(a) = b$; for example, if $f(x) = x^3$, then $f^{-1}(b)$ is the unique number a such that $a^3 = b$, and this number is, by definition, $\sqrt[3]{b}$.

The fact that $f^{-1}(x)$ is the number y such that $f(y) = x$ can be restated in a much more compact form:

$$f(f^{-1}(x)) = x, \text{ for all } x \text{ in the domain of } f^{-1}.$$

Moreover,

$$f^{-1}(f(x)) = x, \text{ for all } x \text{ in the domain of } f;$$

this follows from the previous equation upon replacing f by f^{-1} . These two important equations can be written

$$\begin{aligned} f \circ f^{-1} &= I, \\ f^{-1} \circ f &= I \end{aligned}$$

(except that the right side will have a bigger domain if the domain of f or f^{-1} is not all of $\mathbf{R}$).

Since many standard functions will be defined as the inverses of other functions, it is quite important that we be able to tell which functions are one-one. We have already hinted which class of functions are most easily dealt with—increasing and decreasing functions are obviously one-one. Moreover, if f is increasing, then f^{-1} is also increasing, and if f is decreasing, then f^{-1} is decreasing (the proof is left to you). In addition, f is increasing if and only if $-f$ is decreasing, a very useful fact to remember.

It is certainly not true that every one-one function is either increasing or decreasing. One example has already been mentioned, and is now graphed in Figure 6:

$$g(x) = \begin{cases} x, & x \neq 3, 5 \\ 3, & x = 5 \\ 5, & x = 3. \end{cases}$$

Figure 7 shows that there are even continuous one-one functions which are neither increasing nor decreasing. But if you try drawing a few pictures you will soon agree that every one-one continuous function defined on an interval is either increasing or decreasing. This theorem is a bureaucrat's delight; the proof uses no new ideas, but it does require careful organization. It helps to begin with two lemmas, the second being a corollary of the first.

LEMMA 1

Let f be a continuous one-one function defined on an interval, and let a, b , and c be points of the interval with $a < c < b$. If $f(a) < f(b)$, then $f(a) < f(c) < f(b)$; and if $f(a) > f(b)$, then $f(a) > f(c) > f(b)$ (as in Figure 8).

PROOF

Consider first the case where $f(a) < f(b)$. Since $c \neq a, b$ we have $f(c) \neq f(a)$,

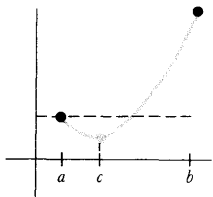


FIGURE 9

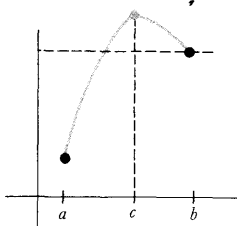


FIGURE 10

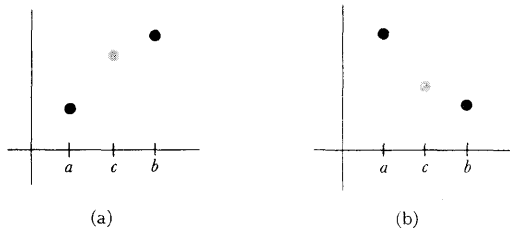


FIGURE 8

and $f(c) \neq f(b)$, so

either $f(c) < f(a)$
 or $f(a) < f(c) < f(b)$
 or $f(b) < f(c)$,

and it is only necessary to rule out the first and third possibilities.

Suppose that $f(c) < f(a)$ were true (Figure 9). Apply the Intermediate Value Theorem to the interval $[c, b]$: since $f(c) < f(a) < f(b)$ there is some x in $[c, b]$ such that $f(x) = f(a)$. But this contradicts the fact that f is one-one, since $a \neq x$. Thus $f(c) < f(a)$ is impossible.

Similarly, suppose that $f(c) > f(b)$ were true (Figure 10). Apply the Intermediate Value Theorem to the interval $[a, c]$: since $f(a) < f(b) < f(c)$ there is some x in $[a, c]$ such that $f(x) = f(b)$, again contradicting the fact that f is one-one. Thus $f(c) > f(b)$ is also false, leaving only the possibility $f(a) < f(b) < f(c)$.

The case where $f(a) > f(b)$ may be handled in precisely the same way. Or one can be elegant and apply the first case to $-f$. ■

LEMMA 2 Let f be a continuous one-one function defined on an interval, let a and b be two points of the interval, and suppose that $f(a) < f(b)$. If c is a point of the interval with $c < a$, then $f(c) < f(a)$; and similarly, if $b < c$, then $f(b) < f(c)$. (Analogous statements could be made if $f(a) > f(b)$, but we will not need them.)

PROOF Consider the case $c < a$. Suppose first that $f(c) > f(b)$. Then Lemma 1, applied to $c < a < b$, would imply that $f(a) > f(b)$, which is false. Hence $f(c) < f(b)$. But now Lemma 1 may be applied to $c < a < b$ to conclude that $f(c) < f(a) < f(b)$.

The case $b < c$ is handled similarly. ■

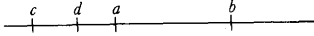
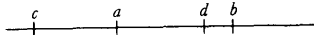
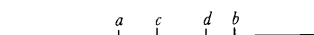
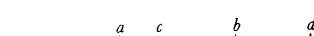
Finally, we are ready for the theorem.

THEOREM 2 If f is continuous and one-one on an interval, then f is either increasing or decreasing on that interval.

PROOF Let a and b be two numbers in the interval with $a < b$. Since $f(a) \neq f(b)$,

either $f(a) < f(b)$ or $f(a) > f(b)$. We will first concentrate on the case $f(a) < f(b)$, and show that f is increasing.

Let c and d be two points in the interval with $c < d$. If one (or both) of c, d equals a or b , then it follows immediately from Lemmas 1 and 2 that $f(c) < f(d)$. The other possibility leads to six cases:

- | | |
|---------------------|--|
| (1) $c < d < a < b$ |  |
| (2) $c < a < d < b$ |  |
| (3) $c < a < b < d$ |  |
| (4) $a < c < d < b$ |  |
| (5) $a < c < b < d$ |  |
| (6) $a < b < c < d$ |  |

Each of these cases can be treated easily, using the lemmas. For example, in case (1) first apply Lemma 2 to $c < a < b$ to conclude that $f(c) < f(a)$, and then apply Lemma 1 to $c < d < a$ to conclude that $f(c) < f(d)$. The other cases are left to you, as an easy exercise.

If $f(a) > f(b)$, one can prove similarly that f is decreasing. Or one can apply the result just proved to $-f$ (clearly the preferable method). ■

Henceforth we shall be concerned almost exclusively with continuous increasing or decreasing functions which are defined on an interval. If f is such a function, it is possible to say quite precisely what the domain of f^{-1} will be like.

Suppose first that f is a continuous increasing function on the closed interval $[a, b]$. Then, by the Intermediate Value Theorem, f takes on every value between $f(a)$ and $f(b)$. Therefore, the domain of f^{-1} is the closed interval $[f(a), f(b)]$. Similarly, if f is continuous and decreasing on $[a, b]$, then the domain of f^{-1} is $[f(b), f(a)]$.

If f is a continuous increasing function on an *open* interval (a, b) the analysis becomes a bit more difficult. To begin with, let us choose some point c in (a, b) . We will first decide which values $> f(c)$ are taken on by f . One possibility is that f takes on arbitrarily large values (Figure 11). In this case f takes on *all* values $> f(c)$, by the Intermediate Value Theorem. If, on the other hand, f does not take on arbitrarily large values, then $A = \{f(x) : c \leq x < b\}$ is bounded above, so A has a least upper bound α (Figure 12). Now suppose y is any number with $f(c) < y < \alpha$. Then f takes on some value $f(x) > y$ (because α is the least upper bound of A). By the Intermediate Value Theorem, f actually takes on the value y . Notice that f cannot take on the value α itself; for if $\alpha = f(x)$ for $a < x < b$ and we choose t with $x < t < b$, then $f(t) > \alpha$, which is impossible.

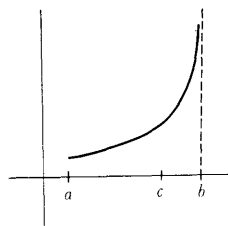


FIGURE 11

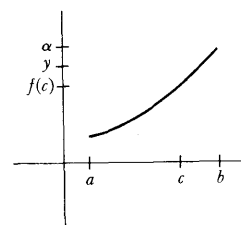


FIGURE 12

Precisely the same arguments work for values less than $f(c)$: either f takes on all values less than $f(c)$ or there is a number $\beta < f(c)$ such that f takes on all values between β and $f(c)$, but not β itself.

This entire argument can be repeated if f is decreasing, and if the domain of f is $\mathbf{R}$ or (a, ∞) or $(-\infty, a)$. Summarizing: if f is a continuous increasing, or decreasing, function whose domain is an interval having one of the forms

$$(a, b), (-\infty, b), (a, \infty), \text{ or } \mathbf{R},$$

then the domain of f^{-1} is also an interval which has one of these four forms.

Now that we have completed this preliminary analysis of continuous one-one functions, it is possible to begin asking which important properties of a one-one function are inherited by its inverse. For continuity there is no problem.

THEOREM 3 If f is continuous and one-one on an interval, then f^{-1} is also continuous.

PROOF We know by Theorem 2 that f is either increasing or decreasing. We might as well assume that f is increasing, since we can then take care of the other case by applying the usual trick of considering $-f$.

We must show that

$$\lim_{x \rightarrow b} f^{-1}(x) = f^{-1}(b)$$

for each b in the domain of f^{-1} . Such a number b is of the form $f(a)$ for some a in the domain of f . For any $\varepsilon > 0$, we want to find a $\delta > 0$ such that, for all x ,

$$\text{if } f(a) - \delta < x < f(a) + \delta, \text{ then } a - \varepsilon < f^{-1}(x) < a + \varepsilon.$$

Figure 13 suggests the way of finding δ (remember that by looking sideways you see the graph of f^{-1}): since

$$a - \varepsilon < a < a + \varepsilon,$$

it follows that

$$f(a - \varepsilon) < f(a) < f(a + \varepsilon);$$

we let δ be the smaller of $f(a + \varepsilon) - f(a)$ and $f(a) - f(a - \varepsilon)$. Figure 13 contains the entire proof that this δ works, and what follows is simply a verbal account of the information contained in this picture.

Our choice of δ ensures that

$$f(a - \varepsilon) \leq f(a) - \delta \text{ and } f(a) + \delta \leq f(a + \varepsilon).$$

Consequently, if

$$f(a) - \delta < x < f(a) + \delta,$$

then

$$f(a - \varepsilon) < x < f(a + \varepsilon).$$

Since f is increasing, f^{-1} is also increasing, and we obtain

$$f^{-1}(f(a - \varepsilon)) < f^{-1}(x) < f^{-1}(f(a + \varepsilon)),$$

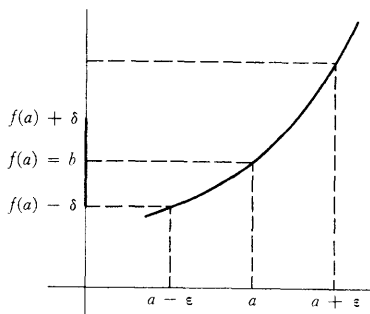


FIGURE 13

i.e.,

$$a - \varepsilon < f^{-1}(x) < a + \varepsilon,$$

which is precisely what we want. ■

Having successfully investigated continuity of f^{-1} , it is only reasonable to tackle differentiability. Again, a picture indicates just what result ought to be true. Figure 14 shows the graph of a one-one function f with a tangent line L through $(a, f(a))$. If this entire picture is reflected through the diagonal, it shows the graph of f^{-1} and the tangent line L' through $(f(a), a)$. The slope of L' is the reciprocal of the slope of L . In other words, it appears that

$$(f^{-1})'(f(a)) = \frac{1}{f'(a)}.$$

This formula can equally well be written in a way which expresses $(f^{-1})'(b)$ directly, for each b in the domain of f^{-1} :

$$(f^{-1})'(b) = \frac{1}{f'(f^{-1}(b))}.$$

Unlike the argument for continuity, this pictorial “proof” becomes somewhat involved when formulated analytically. There is another approach which might be tried. Since we know that

$$f(f^{-1}(x)) = x,$$

it is tempting to prove the desired formula by applying the Chain Rule:

$$f'(f^{-1}(x)) \cdot (f^{-1})'(x) = 1,$$

so

$$(f^{-1})'(x) = \frac{1}{f'(f^{-1}(x))}.$$

Unfortunately, this is not a proof that f^{-1} is differentiable, since the Chain Rule cannot be applied unless f^{-1} is already known to be differentiable. But this argument does show what $(f^{-1})'(x)$ will have to be if f^{-1} is differentiable, and it can also be used to obtain some important preliminary information.

THEOREM 4 If f is a continuous one-one function defined on an interval and $f'(f^{-1}(a)) = 0$, then f^{-1} is not differentiable at a .

PROOF We have

$$f(f^{-1}(x)) = x.$$

If f^{-1} were differentiable at a , the Chain Rule would imply that

$$f'(f^{-1}(a)) \cdot (f^{-1})'(a) = 1,$$

hence

$$0 \cdot (f^{-1})'(a) = 1,$$

which is absurd. ■

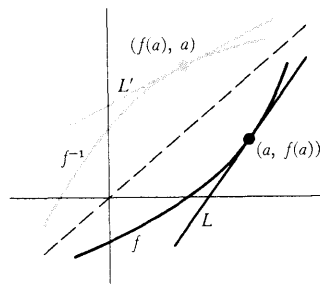


FIGURE 14

A simple example to which Theorem 4 applies is the function $f(x) = x^3$. Since $f'(0) = 0$ and $0 = f^{-1}(0)$, the function f^{-1} is not differentiable at 0 (Figure 15).

Having decided where an inverse function cannot be differentiable, we are now ready for the rigorous proof that in all other cases the derivative is given by the formula which we have already “derived” in two different ways. Notice that the following argument uses *continuity* of f^{-1} , which we have already proved.

THEOREM 5 Let f be a continuous one-one function defined on an interval, and suppose that f is differentiable at $f^{-1}(b)$, with derivative $f'(f^{-1}(b)) \neq 0$. Then f^{-1} is differentiable at b , and

$$(f^{-1})'(b) = \frac{1}{f'(f^{-1}(b))}.$$

PROOF Let $b = f(a)$. Then

$$\begin{aligned} \lim_{h \rightarrow 0} \frac{f^{-1}(b+h) - f^{-1}(b)}{h} \\ = \lim_{h \rightarrow 0} \frac{f^{-1}(b+h) - a}{h}. \end{aligned}$$

Now every number $b+h$ in the domain of f^{-1} can be written in the form

$$b+h = f(a+k)$$

for a unique k (we should really write $k(h)$, but we will stick with k for simplicity). Then

$$\begin{aligned} \lim_{h \rightarrow 0} \frac{f^{-1}(b+h) - a}{h} \\ = \lim_{h \rightarrow 0} \frac{f^{-1}(f(a+k)) - a}{f(a+k) - b} \\ = \lim_{h \rightarrow 0} \frac{k}{f(a+k) - f(a)}. \end{aligned}$$

We are clearly on the right track! It is not hard to get an explicit expression for k ; since

$$b+h = f(a+k)$$

we have

$$f^{-1}(b+h) = a+k$$

or

$$k = f^{-1}(b+h) - f^{-1}(b).$$

Now by Theorem 3, the function f^{-1} is continuous at b . This means that k

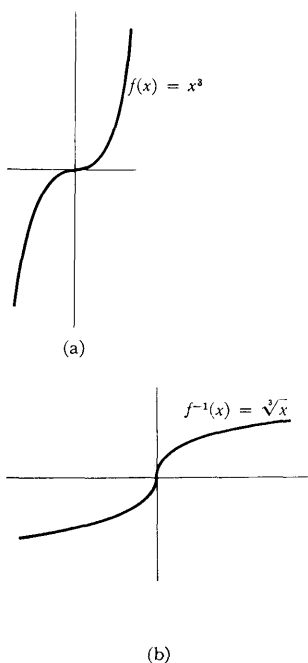


FIGURE 15

approaches 0 as h approaches 0. Since

$$\lim_{k \rightarrow 0} \frac{f(a+k) - f(a)}{k} = f'(a) = f'(f^{-1}(b)) \neq 0,$$

this implies that

$$(f^{-1})'(b) = \frac{1}{f'(f^{-1}(b))}. \blacksquare$$

The work we have done on inverse functions will be amply repaid later, but here is an immediate dividend. For n odd, let

$$f_n(x) = x^n \quad \text{for all } x;$$

for n even, let

$$f_n(x) = x^n, \quad x \geq 0.$$

Then f_n is a continuous one-one function, whose inverse function is

$$g_n(x) = \sqrt[n]{x} = x^{1/n}.$$

By Theorem 5 we have, for $x \neq 0$,

$$\begin{aligned} g_n'(x) &= \frac{1}{f_n'(f_n^{-1}(x))} \\ &= \frac{1}{n(f_n^{-1}(x))^{n-1}} \\ &= \frac{1}{n(x^{1/n})^{n-1}} \\ &= \frac{1}{n} \cdot \frac{1}{x^{1-(1/n)}} \\ &= \frac{1}{n} \cdot x^{(1/n)-1}. \end{aligned}$$

Thus, if $f(x) = x^a$, and a is an integer or the reciprocal of a natural number, then $f'(x) = ax^{a-1}$. It is now easy to check that this formula is true if a is any rational number: Let $a = m/n$, where m is an integer, and n is a natural number; if

$$f(x) = x^{m/n} = (x^{1/n})^m,$$

then, by the Chain Rule,

$$\begin{aligned} f'(x) &= m(x^{1/n})^{m-1} \cdot \frac{1}{n} \cdot x^{(1/n)-1} \\ &= \frac{m}{n} \cdot x^{[(m/n)-(1/n)] + [(1/n)-1]} \\ &= \frac{m}{n} x^{(m/n)-1}. \end{aligned}$$

Although we now have a formula for $f'(x)$ when $f(x) = x^a$ and a is rational, the treatment of the function $f(x) = x^a$ for irrational a will have to be saved for later—at the moment we do not even know the *meaning* of a symbol like $x^{\sqrt{2}}$. Actually, inverse functions will be involved crucially in the definition of x^a for irrational a . Indeed, in the next few chapters several important functions will be defined in terms of their inverse functions.

PROBLEMS

1. Find f^{-1} for each of the following f .

(i) $f(x) = x^3 + 1$.

(ii) $f(x) = (x - 1)^3$.

(iii) $f(x) = \begin{cases} x, & x \text{ rational} \\ -x, & x \text{ irrational.} \end{cases}$

(iv) $f(x) = \begin{cases} -x^2, & x \geq 0 \\ 1 - x^3, & x < 0. \end{cases}$

(v) $f(x) = \begin{cases} x, & x \neq a_1, \dots, a_n \\ a_{i+1}, & x = a_i, \quad i = 1, \dots, n-1 \\ a_1, & x = a_n. \end{cases}$

(vi) $f(x) = x + [x]$.

(vii) $f(0.a_1a_2a_3 \dots) = 0.a_2a_1a_3 \dots$ (Decimal representation is being used.)

(viii) $f(x) = \frac{x}{1-x^2}, \quad -1 < x < 1$.

2. Describe the graph of f^{-1} when

(i) f is increasing and always positive.

(ii) f is increasing and always negative.

(iii) f is decreasing and always positive.

(iv) f is decreasing and always negative.

3. Complete the proof of Theorem 2.

4. Prove that if f is increasing, then so is f^{-1} , and similarly for decreasing functions.

5. If f and g are increasing, is $f + g$? Or $f \cdot g$? Or $f \circ g$?

6. (a) Prove that if f and g are one-one, then $f \circ g$ is also one-one. Find $(f \circ g)^{-1}$ in terms of f^{-1} and g^{-1} . Hint: The answer is *not* $f^{-1} \circ g^{-1}$.

(b) Find g^{-1} in terms of f^{-1} if $g(x) = 1 + f(x)$.

7. Show that $f(x) = \frac{ax + b}{cx + d}$ is one-one if and only if $ad - bc \neq 0$, and find f^{-1} in this case.

8. On which intervals $[a, b]$ will the following functions be one-one?

- (i) $f(x) = x^3 - 3x^2$.
- (ii) $f(x) = x^5 + x$.
- (iii) $f(x) = (1 + x^2)^{-1}$.
- (iv) $f(x) = \frac{x + 1}{x^2 + 1}$.

9. Suppose that f is a one-one function and that f^{-1} has a derivative which is nowhere 0. Prove that f is differentiable. Hint: There is a one-step proof.

- *10. (a) Prove that there is a differentiable function f such that $[f(x)]^5 + f(x) + x = 0$ for all x . Hint: Show that f can be expressed as an inverse function. The easiest way to do this is to find f^{-1} . And the easiest way to do *this* is to set $x = f^{-1}(y)$.
- (b) Find f' in terms of f , using an appropriate theorem of this chapter.
- (c) Find f' in another way, by simply differentiating the equation defining f .

The function in Problem 10 is often said to be **defined implicitly** by the equation $y^5 + y + x = 0$. The situation for this equation is quite special, however. As the next problem shows, an equation does not usually define a function implicitly on the whole line, and in some regions more than one function may be defined implicitly.

11. (a) What are the two differentiable functions f which are defined implicitly on $(-1, 1)$ by the equation $x^2 + y^2 = 1$, i.e., which satisfy $x^2 + [f(x)]^2 = 1$ for all x in $(-1, 1)$? Notice that there are no solutions defined outside $[-1, 1]$.
- (b) Which functions f satisfy $x^2 + [f(x)]^2 = -1$?
- * (c) Which differentiable functions f satisfy $[f(x)]^3 - 3f(x) = x$? Hint: It will help to first draw the graph of the function $g(x) = x^3 - 3x$.

In general, determining on what intervals a differentiable function is defined implicitly by a particular equation may be a delicate affair, and is best discussed in the context of advanced calculus. If we *assume* that f is such a differentiable solution, however, then a formula for $f'(x)$ can be derived, exactly as in Problem 10(c), by differentiating both sides of the equation defining f (a process known as “implicit differentiation”):

12. (a) Apply this method to the equation $[f(x)]^2 + x^2 = 1$. Notice that your answer will involve $f(x)$; this is only to be expected, since there is more than one function defined implicitly by the equation $y^2 + x^2 = 1$.

- (b) But check that your answer works for both of the functions f found in Problem 11(a).
- (c) Apply this same method to $[f(x)]^3 - 3f(x) = x$.
13. Leibnizian notation is particularly convenient for implicit differentiation. Because y is so consistently used as an abbreviation for $f(x)$, the equation in x and y which defines f implicitly will automatically stand for the equation which f is supposed to satisfy. How would the following computation be written in our notation?

$$\begin{aligned} y^4 + y^3 + xy &= 1, \\ 4y^3 \frac{dy}{dx} + 3y^2 \frac{dy}{dx} + y + x \frac{dy}{dx} &= 0, \\ \frac{dy}{dx} &= \frac{-y}{4y^3 + 3y^2 + x}. \end{aligned}$$

14. As long as Leibnizian notation has entered the picture, the Leibnizian notation for derivatives of inverse functions should be mentioned. If dy/dx denotes the derivative of f , then the derivative of f^{-1} is denoted by dx/dy . Write out Theorem 5 in this notation. The resulting equation will show you another reason why Leibnizian notation has such a large following. It will also explain at which point $(f^{-1})'$ is to be calculated when using the dx/dy notation. What is the significance of the following computation?

$$\begin{aligned} x &= y^n, \\ y &= x^{1/n}, \\ \frac{dx^{1/n}}{dx} &= \frac{dy}{dx} = \frac{1}{\frac{dx}{dy}} = \frac{1}{ny^{n-1}}. \end{aligned}$$

15. Suppose that f is a differentiable one-one function with a nowhere zero derivative and that $f = F'$. Let $G(x) = xf^{-1}(x) - F(f^{-1}(x))$. Prove that $G'(x) = f^{-1}(x)$. (Disregarding details, this problem tells us a very interesting fact: if we know a function whose derivative is f , then we also know one whose derivative is f^{-1} . But how could anyone ever guess the function G ? Two different ways are outlined in Problems 14-10 and 18-13.)
16. Suppose h is a function such that $h'(x) = \sin^2(\sin(x+1))$ and $h(0) = 3$. Find
- $(h^{-1})'(3)$.
 - $(\beta^{-1})'(3)$, where $\beta(x) = h(x+1)$.
17. Find a formula for $(f^{-1})''(x)$.
- *18. Prove that if $f^{(k)}(f^{-1}(x))$ exists, and is nonzero, then $(f^{-1})^{(k)}(x)$ exists.

19. (a) Prove that an increasing and a decreasing function intersect at most once.
- (b) Find two continuous increasing functions f and g such that $f(x) = g(x)$ precisely when x is an integer.
- (c) Find a continuous increasing function f and a continuous decreasing function g , defined on $\mathbf{R}$, which do not intersect at all.
- *20. (a) If f is a continuous function on $\mathbf{R}$ and $f = f^{-1}$, prove that there is at least one x such that $f(x) = x$. (What does the condition $f = f^{-1}$ mean geometrically?)
- (b) Give several examples of continuous f such that $f = f^{-1}$ and $f(x) = x$ for exactly one x . Hint: Try decreasing f , and remember the geometric interpretation. One possibility is $f(x) = -x$.
- (c) Prove that if f is an increasing function such that $f = f^{-1}$, then $f(x) = x$ for all x . Hint: Although the geometric interpretation will be immediately convincing, the simplest proof (about 2 lines) is to rule out the possibilities $f(x) < x$ and $f(x) > x$.
- *21. Which functions have the property that the graph is still the graph of a function when reflected through the graph of $-I$ (the “antidiagonal”) ?
22. A function f is **nondecreasing** if $f(x) \leq f(y)$ whenever $x < y$. (To be more precise we should stipulate that the domain of f be an interval.) A **nonincreasing** function is defined similarly. Caution: Some writers use “increasing” instead of “nondecreasing,” and “strictly increasing” for our “increasing.”
- (a) Prove that if f is nondecreasing, but not increasing, then f is constant on some interval. (Beware of unintentional puns: “not increasing” is not the same as “nonincreasing.”)
- (b) Prove that if f is differentiable and nondecreasing, then $f'(x) \geq 0$ for all x .
- (c) Prove that if $f'(x) \geq 0$ for all x , then f is nondecreasing.
- *23. (a) Suppose that $f(x) > 0$ for all x , and that f is decreasing. Prove that there is a *continuous* decreasing function g such that $0 < g(x) \leq f(x)$ for all x .
- (b) Show that we can even arrange that g will satisfy $\lim_{x \rightarrow \infty} g(x)/f(x) = 0$.

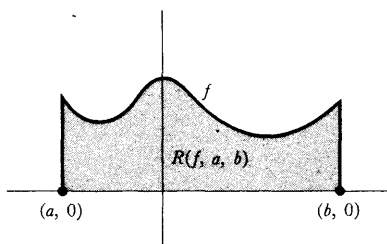


FIGURE 1

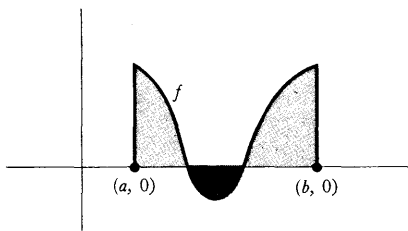


FIGURE 2

The derivative does not display its full strength until allied with the “integral,” the second main concept of Part III. At first this topic may seem to be a complete digression—in this chapter derivatives do not appear even once! The study of integrals does require a long preparation, but once this preliminary work has been completed, integrals will be an invaluable tool for creating new functions, and the derivative will reappear in Chapter 14, more powerful than ever.

Although ultimately to be defined in a quite complicated way, the integral formalizes a simple, intuitive concept—that of area. By now it should come as no surprise to learn that the definition of an intuitive concept can present great difficulties—“area” is certainly no exception.

In elementary geometry, formulas are derived for the areas of many plane figures, but a little reflection shows that an acceptable definition of area is seldom given. The area of a region is sometimes defined as the number of squares, with sides of length 1, which fit in the region. But this definition is hopelessly inadequate for any but the simplest regions. For example, a circle of radius 1 supposedly has as area the irrational number π , but it is not at all clear what “ π squares” means. Even if we consider a circle of radius $1/\sqrt{\pi}$, which supposedly has area 1, it is hard to say in what way a unit square fits in this circle, since it does not seem possible to divide the unit square into pieces which can be arranged to form a circle.

In this chapter we will only try to define the area of some very special regions (Figure 1)—those which are bounded by the horizontal axis, the vertical lines through $(a, 0)$ and $(b, 0)$, and the graph of a function f such that $f(x) \geq 0$ for all x in $[a, b]$. It is convenient to indicate this region by $R(f, a, b)$. Notice that these regions include rectangles and triangles, as well as many other important geometric figures.

The number which we will eventually assign as the area of $R(f, a, b)$ will be called the *integral* of f on $[a, b]$. Actually, the integral will be defined even for functions f which do not satisfy the condition $f(x) \geq 0$ for all x in $[a, b]$. If f is the function graphed in Figure 2, the integral will represent the difference of the area of the lightly shaded region and the area of the heavily shaded region (the “algebraic area” of $R(f, a, b)$).

The idea behind the prospective definition is indicated in Figure 3. The interval $[a, b]$ has been divided into four subintervals

$$[t_0, t_1] \quad [t_1, t_2] \quad [t_2, t_3] \quad [t_3, t_4]$$

by means of numbers t_0, t_1, t_2, t_3, t_4 with

$$a = t_0 < t_1 < t_2 < t_3 < t_4 = b$$

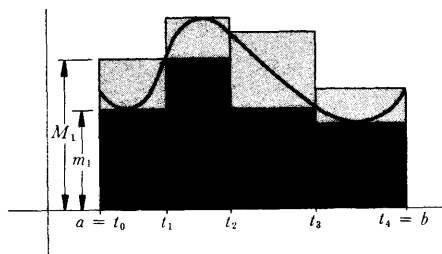


FIGURE 3

(the numbering of the subscripts begins with 0 so that the largest subscript will equal the number of subintervals).

On the first interval $[t_0, t_1]$ the function f has the minimum value m_1 and the maximum value M_1 ; similarly, on the i th interval $[t_{i-1}, t_i]$ let the minimum value of f be m_i and let the maximum value be M_i . The sum

$$s = m_1(t_1 - t_0) + m_2(t_2 - t_1) + m_3(t_3 - t_2) + m_4(t_4 - t_3)$$

represents the total area of rectangles lying inside the region $R(f, a, b)$, while the sum

$$S = M_1(t_1 - t_0) + M_2(t_2 - t_1) + M_3(t_3 - t_2) + M_4(t_4 - t_3)$$

represents the total area of rectangles containing the region $R(f, a, b)$. The guiding principle of our attempt to define the area A of $R(f, a, b)$ is the observation that A should satisfy

$$s \leq A \quad \text{and} \quad A \leq S,$$

and that this should be true, *no matter how the interval $[a, b]$ is subdivided*. It is to be hoped that these requirements will determine A . The following definitions begin to formalize, and eliminate some of the implicit assumptions in, this discussion.

DEFINITION

Let $a < b$. A **partition** of the interval $[a, b]$ is a finite collection of points in $[a, b]$, one of which is a , and one of which is b .

The points in a partition can be numbered $t_0, \dots, t_n$ so that

$$a = t_0 < t_1 < \dots < t_{n-1} < t_n = b;$$

we shall always assume that such a numbering has been assigned.

DEFINITION

Suppose f is bounded on $[a, b]$ and $P = \{t_0, \dots, t_n\}$ is a partition of $[a, b]$. Let

$$m_i = \inf \{f(x) : t_{i-1} \leq x \leq t_i\},$$

$$M_i = \sup \{f(x) : t_{i-1} \leq x \leq t_i\}.$$

The **lower sum** of f for P , denoted by $L(f, P)$, is defined as

$$L(f, P) = \sum_{i=1}^n m_i(t_i - t_{i-1}).$$

The **upper sum** of f for P , denoted by $U(f, P)$, is defined as

$$U(f, P) = \sum_{i=1}^n M_i(t_i - t_{i-1}).$$

The lower and upper sums correspond to the sums s and S in the previous

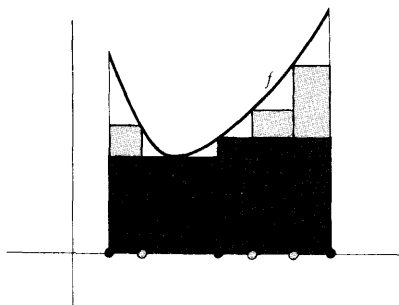


FIGURE 4

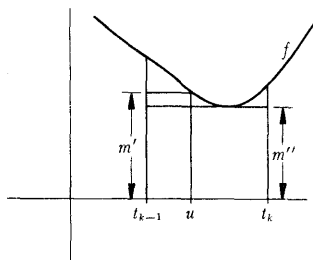


FIGURE 5

example; they are supposed to represent the total areas of rectangles lying below and above the graph of f . Notice, however, that despite the geometric motivation, these sums have been defined precisely without any appeal to a concept of “area.”

Two details of the definition deserve comment. The requirement that f be bounded on $[a, b]$ is essential in order that all the m_i and M_i be defined. Note, also, that it was necessary to define the numbers m_i and M_i as inf’s and sup’s, rather than as minima and maxima, since f was not assumed continuous.

One thing is clear about lower and upper sums: If P is any partition, then

$$L(f, P) \leq U(f, P),$$

because

$$L(f, P) = \sum_{i=1}^n m_i(t_i - t_{i-1}),$$

$$U(f, P) = \sum_{i=1}^n M_i(t_i - t_{i-1}),$$

and for each i we have

$$m_i(t_i - t_{i-1}) \leq M_i(t_i - t_{i-1}).$$

On the other hand, something less obvious *ought* to be true: If P_1 and P_2 are any two partitions of $[a, b]$, then it should be the case that

$$L(f, P_1) \leq U(f, P_2),$$

because $L(f, P_1)$ should be $\leq$ area $R(f, a, b)$, and $U(f, P_2)$ should be $\geq$ area $R(f, a, b)$. This remark proves nothing (since the “area of $R(f, a, b)$ ” has not even been defined yet), but it does indicate that if there is to be any hope of defining the area of $R(f, a, b)$, a proof that $L(f, P_1) \leq U(f, P_2)$ should come first. The proof which we are about to give depends upon a lemma which concerns the behavior of lower and upper sums when more points are included in a partition. In Figure 4 the partition P contains the points in black, and Q contains both the points in black and the points in grey. The picture indicates that the rectangles drawn for the partition Q are a better approximation to the region $R(f, a, b)$ than those for the original partition P . To be precise:

LEMMA If Q contains P (i.e., if all points of P are also in Q), then

$$\begin{aligned} L(f, P) &\leq L(f, Q), \\ U(f, P) &\geq U(f, Q). \end{aligned}$$

PROOF Consider first the special case (Figure 5) in which Q contains just one more point than P :

$$\begin{aligned} P &= \{t_0, \dots, t_n\}, \\ Q &= \{t_0, \dots, t_{k-1}, u, t_k, \dots, t_n\}, \end{aligned}$$

where

$$a = t_0 < t_1 < \dots < t_{k-1} < u < t_k < \dots < t_n = b.$$

Let

$$m' = \inf \{f(x): t_{k-1} \leq x \leq u\},$$

$$m'' = \inf \{f(x): u \leq x \leq t_k\}.$$

Then

$$L(f, P) = \sum_{i=1}^n m_i(t_i - t_{i-1}),$$

$$L(f, Q) = \sum_{i=1}^{k-1} m_i(t_i - t_{i-1}) + m'(u - t_{k-1}) + m''(t_k - u) + \sum_{i=k+1}^n m_i(t_i - t_{i-1}).$$

To prove that $L(f, P) \leq L(f, Q)$ it therefore suffices to show that

$$m_k(t_k - t_{k-1}) \leq m'(u - t_{k-1}) + m''(t_k - u).$$

Now the set $\{f(x): t_{k-1} \leq x \leq t_k\}$ contains all the numbers in $\{f(x): t_{k-1} \leq x \leq u\}$, and possibly some smaller ones, so the greatest lower bound of the first set is *less than or equal to* the greatest lower bound of the second; thus

$$m_k \leq m'.$$

Similarly,

$$m_k \leq m''.$$

Therefore,

$$m_k(t_k - t_{k-1}) = m_k(u - t_{k-1}) + m_k(t_k - u) \leq m'(u - t_{k-1}) + m''(t_k - u).$$

This proves, in this special case, that $L(f, P) \leq L(f, Q)$. The proof that $U(f, P) \geq U(f, Q)$ is similar, and is left to you as an easy, but valuable, exercise.

The general case can now be deduced quite easily. The partition Q can be obtained from P by adding one point at a time; in other words, there is a sequence of partitions

$$P = P_1, P_2, \dots, P_\alpha = Q$$

such that P_{j+1} contains just one more point than P_j . Then

$$L(f, P) = L(f, P_1) \leq L(f, P_2) \leq \dots \leq L(f, P_\alpha) = L(f, Q),$$

and

$$U(f, P) = U(f, P_1) \geq U(f, P_2) \geq \dots \geq U(f, P_\alpha) = U(f, Q). \blacksquare$$

The theorem we wish to prove is a simple consequence of this lemma.

THEOREM 1 Let P_1 and P_2 be partitions of $[a, b]$, and let f be a function which is bounded on $[a, b]$. Then

$$L(f, P_1) \leq U(f, P_2).$$

PROOF There is a partition P which contains both P_1 and P_2 (let P consist of all points in both P_1 and P_2). According to the lemma,

$$L(f, P_1) \leq L(f, P) \leq U(f, P) \leq U(f, P_2). \blacksquare$$

It follows from Theorem 1 that any upper sum $U(f, P')$ is an upper bound for the set of all lower sums $L(f, P)$. Consequently, any upper sum $U(f, P')$ is greater than or equal to the *least* upper bound of all lower sums:

$$\sup \{L(f, P): P \text{ a partition of } [a, b]\} \leq U(f, P'),$$

for every P' . This, in turn, means that $\sup \{L(f, P)\}$ is a lower bound for the set of all upper sums of f . Consequently,

$$\sup \{L(f, P)\} \leq \inf \{U(f, P)\}.$$

It is clear that both of these numbers are between the lower sum and upper sum of f for *all* partitions:

$$\begin{aligned} L(f, P') &\leq \sup \{L(f, P)\} \leq U(f, P'), \\ L(f, P') &\leq \inf \{U(f, P)\} \leq U(f, P'), \end{aligned}$$

for all partitions P' .

It may well happen that

$$\sup \{L(f, P)\} = \inf \{U(f, P)\};$$

in this case, this is the *only* number between the lower sum and upper sum of f for all partitions, and this number is consequently an ideal candidate for the area of $R(f, a, b)$. On the other hand, if

$$\sup \{L(f, P)\} < \inf \{U(f, P)\},$$

then every number x between $\sup \{L(f, P)\}$ and $\inf \{U(f, P)\}$ will satisfy

$$L(f, P') \leq x \leq U(f, P')$$

for all partitions P' .

It is not at all clear just when such an embarrassment of riches will occur. The following two examples, although not as interesting as many which will soon appear, show that both phenomena are possible.

Suppose first that $f(x) = c$ for all x in $[a, b]$ (Figure 6). If $P = \{t_0, \dots, t_n\}$ is any partition of $[a, b]$, then

$$m_i = M_i = c,$$

so

$$L(f, P) = \sum_{i=1}^n c(t_i - t_{i-1}) = c(b - a),$$

$$U(f, P) = \sum_{i=1}^n c(t_i - t_{i-1}) = c(b - a).$$

In this case, all lower sums and upper sums are equal, and

$$\sup \{L(f, P)\} = \inf \{U(f, P)\} = c(b - a).$$

Now consider (Figure 7) the function f defined by

$$f(x) = \begin{cases} 0, & x \text{ irrational} \\ 1, & x \text{ rational.} \end{cases}$$

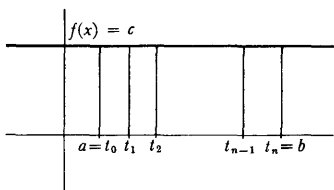


FIGURE 6

If $P = \{t_0, \dots, t_n\}$ is any partition, then

$m_i = 0$, since there is an irrational number in $[t_{i-1}, t_i]$,

and

$M_i = 1$, since there is a rational number in $[t_{i-1}, t_i]$.

Therefore,

$$L(f, P) = \sum_{i=1}^n 0 \cdot (t_i - t_{i-1}) = 0,$$

$$U(f, P) = \sum_{i=1}^n 1 \cdot (t_i - t_{i-1}) = b - a.$$

Thus, in this case it is certainly *not* true that $\sup \{L(f, P)\} = \inf \{U(f, P)\}$. The principle upon which the definition of area was to be based provides insufficient information to determine a specific area for $R(f, a, b)$ —any number between 0 and $b - a$ seems equally good. On the other hand, the region $R(f, a, b)$ is so weird that we might with justice refuse to assign it any area at all. In fact, we can maintain, more generally, that whenever

$$\sup \{L(f, P)\} \neq \inf \{U(f, P)\},$$

the region $R(f, a, b)$ is too unreasonable to deserve having an area. As our appeal to the word “unreasonable” suggests, we are about to cloak our ignorance in terminology.

DEFINITION

A function f which is bounded on $[a, b]$ is **integrable** on $[a, b]$ if $\sup \{L(f, P) : P \text{ a partition of } [a, b]\} = \inf \{U(f, P) : P \text{ a partition of } [a, b]\}$. In this case, this common number is called the **integral** of f on $[a, b]$ and is denoted by

$$\int_a^b f.$$

(The symbol $\int$ is called an *integral sign* and was originally an elongated s , for “sum;” the numbers a and b are called the *lower* and *upper limits of integration*.) The integral $\int_a^b f$ is also called the **area** of $R(f, a, b)$ when $f(x) \geq 0$ for all x in $[a, b]$.

If f is integrable, then according to this definition,

$$L(f, P) \leq \int_a^b f \leq U(f, P) \quad \text{for all partitions } P \text{ of } [a, b].$$

Moreover, $\int_a^b f$ is the *unique* number with this property.

This definition merely pinpoints, and does not solve, the problem discussed before: we do not know which functions are integrable (nor do we know how to find the integral of f on $[a, b]$ when f is integrable). At present we know only

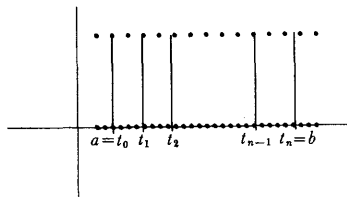


FIGURE 7

two examples:

$$(1) \text{ if } f(x) = c, \text{ then } f \text{ is integrable on } [a, b] \text{ and } \int_a^b f = c \cdot (b - a).$$

(Notice that this integral assigns the expected area to a rectangle.)

$$(2) \text{ if } f(x) = \begin{cases} 0, & x \text{ irrational} \\ 1, & x \text{ rational,} \end{cases} \text{ then } f \text{ is not integrable on } [a, b].$$

Several more examples will be given before discussing these problems further. Even for these examples, however, it helps to have the following simple criterion for integrability stated explicitly.

THEOREM 2 If f is bounded on $[a, b]$, then f is integrable on $[a, b]$ if and only if for every $\epsilon > 0$ there is a partition P of $[a, b]$ such that

$$U(f, P) - L(f, P) < \epsilon.$$

PROOF Suppose first that for every $\epsilon > 0$ there is a partition P with

$$U(f, P) - L(f, P) < \epsilon.$$

Since

$$\begin{aligned} \inf \{U(f, P')\} &\leq U(f, P), \\ \sup \{L(f, P')\} &\geq L(f, P), \end{aligned}$$

it follows that

$$\inf \{U(f, P')\} - \sup \{L(f, P')\} < \epsilon.$$

Since this is true for all $\epsilon > 0$, it follows that

$$\sup \{L(f, P')\} = \inf \{U(f, P')\};$$

by definition, then, f is integrable. The proof of the converse assertion is similar. If f is integrable, then

$$\sup \{L(f, P)\} = \inf \{U(f, P)\}.$$

This means that for each $\epsilon > 0$ there are partitions P', P'' with

$$U(f, P'') - L(f, P') < \epsilon.$$

Let P be a partition which contains both P' and P'' . Then, according to the lemma,

$$\begin{aligned} U(f, P) &\leq U(f, P''), \\ L(f, P) &\geq L(f, P'); \end{aligned}$$

consequently,

$$U(f, P) - L(f, P) \leq U(f, P'') - L(f, P') < \epsilon. \blacksquare$$

Although the mechanics of the proof take up a little space, it should be clear that Theorem 2 amounts to nothing more than a restatement of the definition

of integrability. Nevertheless, it is a very convenient restatement because there is no mention of sup's and inf's, which are often difficult to work with. The next example illustrates this point, and also serves as a good introduction to the type of reasoning which the complicated definition of the integral necessitates, even in very simple situations.

Let f be defined on $[0, 2]$ by

$$f(x) = \begin{cases} 0, & x \neq 1 \\ 1, & x = 1. \end{cases}$$

Suppose $P = \{t_0, \dots, t_n\}$ is a partition of $[0, 2]$ with

$$t_{j-1} < 1 < t_j$$

(see Figure 8). Then

$$m_i = M_i = 0 \quad \text{if } i \neq j,$$

but

$$m_j = 0 \quad \text{and} \quad M_j = 1.$$

Since

$$\begin{aligned} L(f, P) &= \sum_{i=1}^{j-1} m_i(t_i - t_{i-1}) + m_j(t_j - t_{j-1}) + \sum_{i=j+1}^n m_i(t_i - t_{i-1}), \\ U(f, P) &= \sum_{i=1}^{j-1} M_i(t_i - t_{i-1}) + M_j(t_j - t_{j-1}) + \sum_{i=j+1}^n M_i(t_i - t_{i-1}), \end{aligned}$$

we have

$$U(f, P) - L(f, P) = t_j - t_{j-1}.$$

This certainly shows that f is integrable: to obtain a partition P with

$$U(f, P) - L(f, P) < \epsilon,$$

it is only necessary to choose a partition with

$$t_{j-1} < 1 < t_j \quad \text{and} \quad t_j - t_{j-1} < \epsilon.$$

Moreover, it is clear that

$$L(f, P) \leq 0 \leq U(f, P) \quad \text{for all partitions } P.$$

Since f is integrable, there is only *one* number between all lower and upper sums, namely, the integral of f , so

$$\int_0^2 f = 0.$$

Although the discontinuity of f was responsible for the difficulties in this example, even worse problems arise for very simple continuous functions. For example, let $f(x) = x$, and for simplicity consider an interval $[0, b]$, where $b > 0$. If $P = \{t_0, \dots, t_n\}$ is a partition of $[0, b]$, then (Figure 9)

$$m_i = t_{i-1} \quad \text{and} \quad M_i = t_i$$

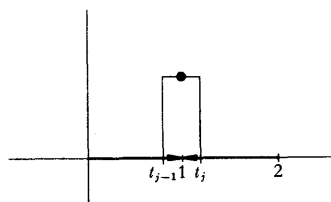


FIGURE 8

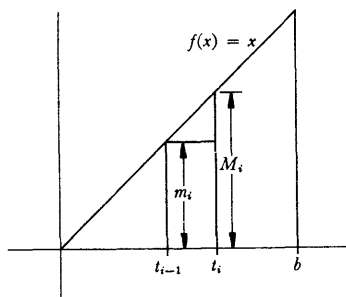


FIGURE 9

and therefore

$$\begin{aligned}
 L(f, P) &= \sum_{i=1}^n t_{i-1}(t_i - t_{i-1}) \\
 &= t_0(t_1 - t_0) + t_1(t_2 - t_1) + \cdots + t_{n-1}(t_n - t_{n-1}), \\
 U(f, P) &= \sum_{i=1}^n t_i(t_i - t_{i-1}) \\
 &= t_1(t_1 - t_0) + t_2(t_2 - t_1) + \cdots + t_n(t_n - t_{n-1}).
 \end{aligned}$$

Neither of these formulas is particularly appealing, but both simplify considerably for partitions $P_n = \{t_0, \dots, t_n\}$ into n equal subintervals. In this case, the length $t_i - t_{i-1}$ of each subinterval is b/n , so

$$\begin{aligned}
 t_0 &= 0, \\
 t_1 &= \frac{b}{n}, \\
 t_2 &= \frac{2b}{n}, \text{ etc;}
 \end{aligned}$$

in general,

$$t_i = \frac{ib}{n}.$$

Then

$$\begin{aligned}
 L(f, P_n) &= \sum_{i=1}^n t_{i-1}(t_i - t_{i-1}) \\
 &= \sum_{i=1}^n \left\{ \frac{(i-1)b}{n} \right\} \cdot \frac{b}{n} \\
 &= \left(\sum_{j=0}^{n-1} j \right) \frac{b^2}{n^2}.
 \end{aligned}$$

Remembering the formula

$$1 + \cdots + k = \frac{k(k+1)}{2},$$

this can be written

$$\begin{aligned}
 L(f, P_n) &= \frac{(n-1)(n)}{2} \cdot \frac{b^2}{n^2} \\
 &= \frac{n-1}{n} \cdot \frac{b^2}{2}.
 \end{aligned}$$

Similarly,

$$\begin{aligned}
 U(f, P_n) &= \sum_{i=1}^n t_i(t_i - t_{i-1}) \\
 &= \sum_{i=1}^n \frac{ib}{n} \cdot \frac{b}{n} \\
 &= \frac{n(n+1)}{2} \cdot \frac{b^2}{n^2} \\
 &= \frac{n+1}{n} \cdot \frac{b^2}{2}.
 \end{aligned}$$

If n is very large, both $L(f, P_n)$ and $U(f, P_n)$ are close to $b^2/2$, and this remark makes it easy to show that f is integrable. Notice first that

$$U(f, P_n) - L(f, P_n) = \frac{2}{n} \cdot \frac{b^2}{2}.$$

This shows that there are partitions P_n with $U(f, P_n) - L(f, P_n)$ as small as desired. By Theorem 2 the function f is integrable. Moreover, $\int_0^b f$ may now be found with only a little work. It is clear, first of all, that

$$L(f, P_n) \leq \frac{b^2}{2} \leq U(f, P_n) \quad \text{for all } n.$$

This inequality shows only that $b^2/2$ lies between certain special upper and lower sums, but we have just seen that $U(f, P_n) - L(f, P_n)$ can be made as small as desired, so there is *only one* number with this property. Since the integral certainly has this property, we can conclude that

$$\int_0^b f = \frac{b^2}{2}.$$

Notice that this equation assigns area $b^2/2$ to a right triangle with base and altitude b (Figure 10). Using more involved calculations, or appealing to Theorem 4, it can be shown that

$$\int_a^b f = \frac{b^2}{2} - \frac{a^2}{2}.$$

The function $f(x) = x^2$ presents even greater difficulties. In this case (Figure 11), if $P = \{t_0, \dots, t_n\}$ is a partition of $[0, b]$, then

$$m_i = f(t_{i-1}) = (t_{i-1})^2 \quad \text{and} \quad M_i = f(t_i) = t_i^2.$$

Choosing, once again, a partition $P_n = \{t_0, \dots, t_n\}$ into n equal parts, so that

$$t_i = \frac{i \cdot b}{n},$$

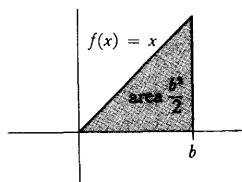


FIGURE 10

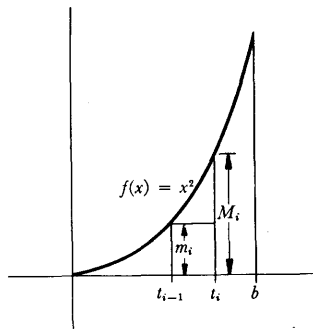


FIGURE 11

the lower and upper sums become

$$\begin{aligned}
 L(f, P_n) &= \sum_{i=1}^n (t_{i-1})^2 \cdot (t_i - t_{i-1}) \\
 &= \sum_{i=1}^n (i-1)^2 \frac{b^2}{n^2} \cdot \frac{b}{n} \\
 &= \frac{b^3}{n^3} \cdot \sum_{j=0}^{n-1} j^2, \\
 U(f, P_n) &= \sum_{i=1}^n t_i^2 \cdot (t_i - t_{i-1}) \\
 &= \sum_{i=1}^n i^2 \frac{b^2}{n^2} \cdot \frac{b}{n} \\
 &= \frac{b^3}{n^3} \sum_{j=1}^n j^2.
 \end{aligned}$$

Recalling the formula

$$1^2 + \cdots + k^2 = \frac{1}{6}k(k+1)(2k+1)$$

from Problem 2-1, these sums can be written as

$$\begin{aligned}
 L(f, P_n) &= \frac{b^3}{n^3} \cdot \frac{1}{6} (n-1)(n)(2n-1), \\
 U(f, P_n) &= \frac{b^3}{n^3} \cdot \frac{1}{6} n(n+1)(2n+1).
 \end{aligned}$$

It is not too hard to show that

$$L(f, P_n) \leq \frac{b^3}{3} \leq U(f, P_n),$$

and that $U(f, P_n) - L(f, P_n)$ can be made as small as desired, by choosing n sufficiently large. The same sort of reasoning as before then shows that

$$\int_0^b f = \frac{b^3}{3}.$$

This calculation already represents a nontrivial result—the area of the region bounded by a parabola is not usually derived in elementary geometry. Nevertheless, the result was known to Archimedes, who derived it in essentially the same way. The only superiority we can claim is that in the next chapter we will discover a much simpler way to arrive at this result.

Some of our investigations can be summarized as follows:

$$\begin{aligned}\int_a^b f &= c \cdot (b - a) && \text{if } f(x) = c \text{ for all } x, \\ \int_a^b f &= \frac{b^2}{2} - \frac{a^2}{2} && \text{if } f(x) = x \text{ for all } x, \\ \int_a^b f &= \frac{b^3}{3} - \frac{a^3}{3} && \text{if } f(x) = x^2 \text{ for all } x.\end{aligned}$$

This list already reveals that the notation $\int_a^b f$ suffers from the lack of a convenient notation for naming functions defined by formulas. For this reason an alternative notation,* analogous to the notation $\lim_{x \rightarrow a} f(x)$, is also useful:

$$\int_a^b f(x) \, dx \quad \text{means precisely the same as} \quad \int_a^b f.$$

Thus

$$\begin{aligned}\int_a^b c \, dx &= c \cdot (b - a), \\ \int_a^b x \, dx &= \frac{b^2}{2} - \frac{a^2}{2}, \\ \int_a^b x^2 \, dx &= \frac{b^3}{3} - \frac{a^3}{3}.\end{aligned}$$

Notice that, as in the notation $\lim_{x \rightarrow a} f(x)$, the symbol x can be replaced by any other letter (except f , a , or b , of course);

$$\int_a^b f(x) \, dx = \int_a^b f(t) \, dt = \int_a^b f(\alpha) \, d\alpha = \int_a^b f(y) \, dy = \int_a^b f(c) \, dc.$$

The symbol dx has no meaning in isolation, any more than the symbol $x \rightarrow$ has any meaning, except in the context $\lim_{x \rightarrow a} f(x)$. In the equation

$$\int_a^b x^2 \, dx = \frac{b^3}{3} - \frac{a^3}{3},$$

the *entire* symbol $x^2 \, dx$ may be regarded as an abbreviation for:

the function f such that $f(x) = x^2$ for all x .

* The notation $\int_a^b f(x) \, dx$ is actually the older, and was for many years the only, symbol for the integral. Leibniz used this symbol because he considered the integral to be the sum (denoted by f) of infinitely many rectangles with height $f(x)$ and “infinitely small” width dx . Later writers used $x_0, \dots, x_n$ to denote the points of a partition, and abbreviated $x_i - x_{i-1}$ by Δx_i . The integral was defined as the limit as Δx_i approaches 0 of the sums $\sum_{i=1}^n f(x_i) \Delta x_i$ (analogous to lower and upper sums). The fact that the limit is obtained by changing Σ to f , $f(x_i)$ to $f(x)$, and Δx_i to dx , delights many people.

This notation for the integral is as flexible as the notation $\lim_{x \rightarrow a} f(x)$. Several examples may aid in the interpretation of various types of formulas which frequently appear; we have made use of Theorems 5 and 6.*

$$\begin{aligned}
 (1) \quad \int_a^b (x + y) \, dx &= \int_a^b x \, dx + \int_a^b y \, dx = \frac{b^2}{2} - \frac{a^2}{2} + y(b - a). \\
 (2) \quad \int_a^x (y + t) \, dy &= \int_a^x y \, dy + \int_a^x t \, dy = \frac{x^2}{2} - \frac{a^2}{2} + t(x - a). \\
 (3) \quad \int_a^b \left(\int_a^x (1 + t) \, dz \right) dx &= \int_a^b (1 + t)(x - a) \, dx \\
 &= (1 + t) \int_a^b (x - a) \, dx \\
 &= (1 + t) \left[\frac{b^2}{2} - \frac{a^2}{2} - a(b - a) \right]. \\
 (4) \quad \int_a^b \left(\int_c^d (x + y) \, dy \right) dx &= \int_a^b \left[x(d - c) + \frac{d^2}{2} - \frac{c^2}{2} \right] dx \\
 &= \left(\frac{d^2}{2} - \frac{c^2}{2} \right) (b - a) + (d - c) \int_a^b x \, dx \\
 &= \left(\frac{d^2}{2} - \frac{c^2}{2} \right) (b - a) + (d - c) \left(\frac{b^2}{2} - \frac{a^2}{2} \right).
 \end{aligned}$$

The computations of $\int_a^b x \, dx$ and $\int_a^b x^2 \, dx$ may suggest that evaluating integrals is generally difficult or impossible. As a matter of fact, the integrals of most functions *are* impossible to determine exactly (*although they may be computed to any degree of accuracy desired by calculating lower and upper sums*). Nevertheless, as we shall see in the next chapter, the integral of many functions can be computed very easily.

Even though most integrals cannot be computed exactly, it is important at least to know when a function f is integrable on $[a, b]$. Although it is possible to say precisely which functions are integrable, the criterion for integrability is a little too difficult to be stated here, and we will have to settle for partial results. The most useful result will be stated now, but proved at the end of the next chapter.

THEOREM 3 If f is continuous on $[a, b]$, then f is integrable on $[a, b]$.

(Notice that it is unnecessary to assume that f is bounded on $[a, b]$, because of continuity.)

* Lest chaos overtake the reader when he reads other books, equation (1) requires an important qualification. This equation interprets $\int_a^b y \, dx$ to mean the integral of the function f such that each value $f(x)$ is the number y . But classical notation often uses y for $y(x)$, so $\int_a^b y \, dx$ might mean the integral of some arbitrary function y .